

Department of Software Engineering
Faculty of Science and Information Technology



Daffodil
International
University

OBE Curriculum of B. Sc. (Hons.) in
Software Engineering (SWE)

Daffodil International University (DIU)

Dhaka, Bangladesh

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B. Sc. (Hons.) in Software Engineering

PART A: INTRODUCTION

1. Title of the Academic Program:

Software Engineering (SWE)

2. Name of the University:

Daffodil International University

3. Vision of the University:

A center of excellence through up-to-date, technology-based and student centric education of global standard, research and internationalization.

4. Mission of the University:

- Providing students with educational opportunities and practices that prepare them for a rapidly changing world;
- Promoting a world-class education system through a culture of innovation and internationalization;
- Contributing to the advancement of knowledge and sustainable development through research and innovation;
- Empowering students and researchers to make a positive impact on society and to lead the way towards a brighter future for all; and
- Maintaining industry-academia linkage for learning and research leading to successful career

5. Name of the Degree:

B. Sc. (Hons.) in Software Engineering

6. Name of the Faculty offering the program:

Faculty of Science and Information Technology (FSIT)

7. Name of the Department offering the program:

Software Engineering

8. Vision of the Program:

To become a tertiary-level academic and research center of excellence in the field of Software Engineering for creating global leaders with high self-esteem who are able to accept and overcome challenges in a fast-changing environment.

9. Mission of the Program:

- To empower aspiring software engineers with a comprehensive education, fostering innovation, critical thinking, and ethical leadership to drive technological advancements.

- To cultivate a diverse community of software engineers equipped with technical expertise, problem-solving abilities, and a commitment to creating impactful solutions that positively transform industries and communities.
- To prepare skilled graduates embracing lifelong learning and adaptability that are aligned with National Development Agenda (NDA), Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs).

10. Description of the Program

The Department of Software Engineering offers the program of B.Sc. (Hons.) in Software Engineering. The program is designed to produce skilled manpower for the global IT market in order to satisfy the growing demands of software engineering professionals throughout the world. The central goal is to create knowledgeable, efficient, and skilled software engineering graduates so that they are able to make themselves competent to work not only with the software industries in their home country but also with any giant technological organization of the world.

11. Program Educational Objectives (PEO)

No.	PEO Statement
PEO1	Professionalism: Graduates will establish themselves as practicing global professionals with high esteem in Software Engineering or related fields for accepting and overcoming challenges in a fast-changing environment.
PEO2	Continuous Personal Development: Graduates will engage in lifelong pursuit of knowledge and interdisciplinary learning appropriate for research, industrial and academic careers.
PEO3	Social Engagement: Graduates will contribute to the well-being of people and society by producing research, innovation, and entrepreneurship that are aligned with the national development agenda, Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs).

12. Program Learning Outcomes (PLO)

No.	Program Learning Outcome
PLO1	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex software engineering problems.
PLO2	Identify, formulate, research and analyze complex software engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.
PLO3	Design solutions for complex software engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal and environmental concerns.

PLO4	Conduct investigations of complex problems, considering experimental design, data analysis and interpretation and information synthesis to provide valid conclusions.
PLO5	Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex software engineering activities with an understanding of their limitations.
PLO6	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PLO7	Understand the impact of professional software engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
PLO8	Apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PLO9	Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PLO10	Communicate effectively about complex software engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PLO11	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PLO12	Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

13. Generic Skills:

Hard Skill			
No.	Generic Skills	Courses	Assessments
1	Should be able to gather or explore, analyze, document, verify and validate requirements of a software system.	SE 231, SE331, SE422	Lab Report, Oral, Report Presentation and Viva
2	Should be able to design a software system using a specific methodology such as object-oriented methodology.	SE133, SE 231, SE331, SE422	Report, Oral, Presentation, Project and Viva
3	Should be able to implement a software system based on the design.	SE133, SE 231, SE331, SE422	Report, Oral, Presentation and Viva
4	Should be able to test, maintain and assure quality of a software system.	SE331, SE422	Report, Oral , Presentation and Viva
5	Should be able to carry out research and development in the field of software engineering and other future technologies	SE 599	Report, Oral, Presentation and Viva

	such as robotics, data science, cyber security etc.		
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Soft Skill			
No.	Generic Skills	Courses	Assessments
1	Should be able to read and understand technical documents.	SE 422	Report, Oral, Presentation and Viva
2	Should be able to write and present technical reports.	SE133, SE 231, SE331, SE422	Report, Oral, Presentation and Viva, Project
3	Should be able to listen to technical presentations and identify the key points.	All Theory Courses, SE 422	Viva
4	Should be able to work effectively as an individual and also in a team.	All Courses	Class Test, Mid Exam, Final Exam, Presentation/Viva, Assignment, and Project
5	Should be able to analyze and interpret experimental results.	SE 422	Report, Oral , Presentation and Viva
6	Should be able to critically think and be creative in solving engineering problems.	All Courses	Class Test, Mid Exam, Final Exam, Presentation/Viva, Assignment, and Project

14. Mapping University's and Program's Vision and Mission vs PEO

No.	PEO Statement	Vision SE	Mission SE	Vision DIU	Mission DIU
PEO1	Professionalism: Graduates will establish themselves as practicing global professionals having high esteem in Software Engineering or related fields for accepting and overcoming challenges in a fast-changing environment.	X		X	
PEO2	Continuous Personal Development: Graduates will engage in lifelong pursuit of knowledge and interdisciplinary learning appropriate for research, industrial and academic careers.		X		

PEO3	Social Engagement: Graduates will contribute to the well-being of people and society by producing research, innovation and entrepreneurship that are aligned with national development agenda, Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs).		X		X
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15. Mapping PEO vs PLO

PEO/PLO	PEO1	PEO2	PEO3
PLO1	X		
PLO2	X		
PLO3	X		
PLO4	X		
PLO5	X		
PLO6			X
PLO7			X
PLO8			X
PLO9	X		
PLO10	X		
PLO11	X		
PLO12		X	

PART B: STRUCTURE OF THE CURRICULUM

1. Duration of the program:

4 years

2. Admission Requirements:

General Student:

Students having a minimum 2.5 GPA both in Secondary School Certificate (SSC) and Higher Secondary Certificate (HSC) 12 class passed equivalent from Science with a minimum C grade in Physics, Mathematics and English may apply for admission.

Diploma Holder:

Students having a minimum 2.5 GPA both in SSC (Science) and Diploma from Computer Technology may apply for admission.

English Medium Student:

Students completing five O-level subjects and at least two A-level subjects may apply. Out of these 7 subjects, applicants must have a minimum 4 "B" grades & 3 "C" grades. The applicants must have Physics and Mathematics both at O-level and A-level.

NOTE:

- Students having below 'B' grade in English in HSC/Equivalent level must have to enroll in an English Language Remedial Course immediately after admission.
- A maximum gap of two (02) years is accepted.

3. Total credit requirement:

147

4. Minimum CGPA requirements for graduation:

2.00

5. Category of Courses:

- General Education courses: 12
- Core course: 18
- Lab course: 12
- Project course: 3
- PGC: 10
- Final year project thesis: 1

6. Degree offer with Major:

Four major course courses are offered with the degree also.

- Major in Data science
- Major in Cyber security
- Major in Robotics
- Major in Testing

Requirements to enroll the major

Sl	Title of the major	Completed credit before taking the major	Prerequisite course	Credits need to enroll
1	Major in Data science	117	Introduction to Machine Learning	12
2	Major in Cyber security		Information System Security	
3	Major in Robotics		Introduction to Robotics	
4	Major in Testing		Software Quality Assurance & Testing	

			Software Quality Assurance & Testing Lab	
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Course for Major in Data science

Credit	Course Code	Title
2	DS 331	Introduction to Data Science and Data Management & Analysis
1	DS 332	Introduction to Data Science and Data Management & Analysis Lab (ds major)
2	DS 411	Statistical Data Analysis
1	DS 412	Statistical Data Analysis Lab (ds-major)
2	DS 421	Machine Learning Driven Data Analysis I(DS Major)
1	DS 422	Machine Learning Driven Data Analysis Lab I(DS Major)
2	DS 423	Machine Learning Driven Data Analysis II and Communicating Data Insights (DS Major)
1	DS 424	Machine Learning Driven Data Analysis II and Communicating Data Insights Lab (DS Major)

Course for Major in Cyber security

Credit	Course Code	Title
3	CS418	Network and Communication Security
3	CS211	Cyber Security Fundamentals
3	CS422	Digital Forensic
1	CS334	Ethical Hacking & Countermeasures
2	CS335	Ethical Hacking & Countermeasures LAB

Course for Major in Robotics

Credit	Course Code	Title
2	RE 421	Robotics process Automation Design & Development
1	RE 422	Robotics process Automation Design & Development Lab
2	RE 411	Embedded System Design & Development
1	RE 412	Embedded System Design & Development Lab
2	RE 331	Embedded Programming
1	RE 332	Embedded Programming Lab
2	RE 423	Advanced Robotics
1	RE 424	Advanced Robotics Lab

Course for Major in Testing

Credit	Course Code	Title
2	ST 411	Agile Testing/Agile Test Engineering
1	ST 412	Agile Testing Lab
2	ST 413	Test Automation & Test Management
1	ST 414	Test Automation & Test Management Lab
2	ST 421	Testing with Generative AI
1	ST 422	Testing with Generative AI Lab
2	ST 423	Software Performance Engineering & Security Testing

1	ST 424	Software Performance Engineering & Security Testing Lab
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7. Notional Hour:

For 1st Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	12	12*40= 480 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 4 theory courses (3*4 credit) 42*4=168 hours	At least 4 hours
Lab	1	1*60= 60 Hours	For 1 lab course (1 credit) 1.5 Hour/week * 14 week=21 hours	
Total	13	540 Hours	189 Hours	

For 2nd Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	12	12*40=480 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 4 theory courses (3*4 credit) 42*4=168 hours	At least 4 hours
Lab	1	1*60=60 Hours	For 1 lab course (1 credit) 1.5 Hour/week * 14 week=21 hours	
Total	13	540 Hours	189 Hours	

For 3rd Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	12	12*40=480 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 4 theory courses (3*4 credit) 42*4=168 hours	At least 4 hours
Lab	1	1*60=60 Hours	For 1 lab course (1 credit) 1.5 Hour/week * 14 week=21 hours	
Total	13	540 Hours	189 Hours	

For 4th Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	12	12*40=480 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 4 theory courses (3*4 credit) 42*4=168 hours	At least 4 hours
Lab	1	1*60=60 Hours	For 1 lab course (1 credit) 1.5 Hour/week * 14 week=21 hours	
Total	13	540 Hours	189 Hours	

For 5th Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	9	9*40=360 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 3 theory courses (3*3 credit) 42*3=126 hours	At least 4.45 hours.
Lab	2	2*60=120 Hours	For 1 lab course (1 credit) 1.5 Hour/week * 14 week=21 hours For 2 lab course (1*2 credit) 21*2=42 hours	
Total	11	480 Hours	168 Hours	

For 6th Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	9	9*40=360 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 3 theory courses (3*3 credit) 42*3=126 hours	At least 6 hours
Lab	5	5*60=300 Hours	For 1 lab course (1 credit) 1.5 Hour/week * 14 week=21 hours For 5 lab course (1*5 credit) 21*5=105 hours	
Total	14	660 Hours	231 Hours	

For 7th Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	12	12*40=480 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 4 theory courses (3*4 credit) 42*4=168 hours	At least 4 hours
Lab	2	2*60=120 Hours	For 1 lab course (1 credit) 1.5 Hour/week * 14 week=21 hours For 2 lab course (1*2 credit)	

			21*2=42 hours	
Total	14	500 Hours	210 Hours	

For 8th Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	12	12*40=480 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 4 theory courses (3*4 credit) 42*4=168 hours	At least 4 hours
Lab	1	1*60=60 Hours	For 1 lab course (1 credit) 1.5 Hour/week * 14 week=21 hours	
Total	13	540 Hours	189 Hours	

For 9th Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	9	9*40=360 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 3 theory courses (3*3 credit) 42*3=126 hours	At least 5.5 hours
Lab	4	4*60=240 Hours	For 1 lab course (1 credit) 1.5 Hour/week * 14 week=21 hours For 4 lab course (1*4 credit) 21*4=84 hours	
Total	13	600 Hours	210 Hours	

For 10th Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	9	9*40=360 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 3 theory courses (3*3 credit) 42*3=126 hours	At least 3 hours
Total	9	360 Hours	126 Hours	

For 11th Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	9	9*40=360 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 3 theory courses (3*3 credit) 42*3=126 hours	At least 3 hours
Total	9	360 Hours	126 Hours	

For 12th Semester

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	6	6*40=240 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours For 3 theory courses (3*2 credit) 42*2=126 hours	At least 2 hours
Total	6	240 Hours	84 Hours	

PART C: DESCRIPTION OF ALL COURSES



**Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)**

SE111 Computer Fundamentals

Course Code: SE111	CIE Marks: 60
ISCED: 0611-111	SEE Marks: 40

Course Title: Computer Fundamentals		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Basic computer system, computer codes, convert number systems, logic gates, Application of logic gate, Basic concepts on microprocessors and microcomputers. Data and information, variables, Loops, Condition, identify errors in logical flow in flowcharts, correct the logic of the control structures of simple programs in programming language. Various parts of a computer system, input/output devices, Memory hierarchy, Types of memory, Memory operation, Data communication, Types of networks.

Rationale of the Course

The course provides students with essential knowledge of computer systems, digital electronics, and basic programming concepts. It introduces computer codes, number systems, and logic gates, which form the foundation of modern computing and digital circuit design. The course also explains microprocessors, microcomputers, memory, and input/output devices, helping learners understand how computers function internally. Through topics such as variables, loops, conditions, and flowcharts, students develop problem-solving and logical thinking skills..

Course Objectives

The goal of this course is to introduce the students to the concept of basic logic operation and the basics of computer function and operation. The main objectives of this course are,

- To integrate the basic components and functions of a computer system.
- To construct the digital circuits using logic gates and other electronic components.
- To apply the principles of software systems to solve problems and design solutions.
- To evaluate the security of computer systems and networks.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Integrate the basic components and functions of a computer system.	PLO2	C2	Fundamental Domain
CLO2	Construct the digital circuits using logic gates and other electronic components.	PLO1	C3	Fundamental Domain
CLO3	Apply the principles of software systems to solve problems and design solutions.	PLO1	C3	Fundamental Domain

CLO4	Determine the security of computer systems and networks.	PLO2	C3	Fundamental Domain
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Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2	√											
CLO3	√											
CLO4		√										

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview on Course Outline, Discuss on Course outcome, Program Outcome & marks distribution.	Syllabus Walkthrough, Interactive Q&A	Class Test, Midterm exam	CLO1
	Lesson 2: Data & Information, Data Processing, Characteristics of a computer, Block Diagram of a computer, Basic functions & components of a computer	Interactive Lecture, Whiteboard Diagramming	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview on Input/Output Devices, Processing Unit, Memory Unit	Visual Presentation, Object Demonstration	Class Test, Midterm exam	CLO1
	Lesson 2: Hardware and Software Integration, Operating System Fundamentals, System Bus	Interactive Lecture, Real-world Analogy Discussion	Class Test, Assignment, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of number system conversion, Binary, Decimal, Octal and Hexadecimal Number Systems conversion, Complements	Whiteboard Demonstration	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Practice on Number System Conversion Class Test -1	Guided Practice, Peer Problem-Solving	Class Test, Assignment, Midterm exam	CLO1
Week-4	Lesson 1: Logic Gate, Different types of logic gate.	Visual Lecture, Concept Mapping	Class Test, Assignment,	CLO2

Lesson 1 & 2 [3 Hours]			Midterm exam	
	Lesson 2: Logic Circuit Design, Boolean Algebra, Boolean Laws.	Whiteboard Demonstration, Rule-based Practice	Class Test, Assignment, Midterm exam	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Combinational circuits, Converting expressions into circuits, Circuit analysis	Step-by-step Circuit Demonstration, Collaborative Design	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Simplification techniques using Boolean algebra and Boolean laws.	Guided Problem-Solving, Whiteboard Practice	Class Test, Assignment, Midterm exam	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Practice on Converting Expression to logic circuit and Logic circuit to expression, Simplification	Hands-on Practice, Peer Review of expressions	Midterm exam	CLO2
	Lesson 2: Half Adder, Full Adder, Multiplexer, Demultiplexer, Encoder, Decoder Class Test -2	Interactive Lecture, Diagram Analysis	Midterm exam	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Basic Database Concepts, Database Management Systems	Interactive Lecture, Real-world Case Study	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Fundamental part of Database Structures, Database Access and Development	Diagram Analysis, Guided Discussion	Class Test, Assignment, Final exam	CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Basic Concepts of Networking and Data Communication	Interactive Lecture, Visual Metaphors	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Network Topology, Different Types of Topologies, Network Design & Implementation	Concept Mapping, Collaborative Network Design	Class Test, Assignment, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Logical Thinking and Problem-Solving Techniques, Logic Development	Brainstorming Sessions, Logic Puzzles	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Flowchart Symbols and Notations, Flowchart Development for Problem Solving	Whiteboard Demonstration, Guided Flowchart Drawing	Assignment, Final exam	CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to C programming	Live Coding Demonstration, Syntax Walkthrough	Assignment, Final exam	CLO3
	Lesson 2: Conditional Statements and Looping Techniques for Problem Solving Class Test -3	Hands-on Coding Practice, Code Tracing	Assignment, Final exam	CLO3

Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview on common Software Vulnerabilities (e.g., Injection, XSS, CSRF), Secure Coding Practices	Case Study Analysis (Real-world vulnerabilities)	Class Test, Assignment, Final exam	CLO4
	Lesson 2: Malware and Common Cyber Attacks, Virus, Worm, Trojan Horse, Spyware, Ransomware, Phishing attacks, social engineering basics	Threat Scenario Discussion, Visual Presentation	Assignment, Final exam	CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Network Security Fundamentals	Interactive Lecture, Diagram Walkthrough	Assignment, Final exam	CLO4
	Lesson 2: Encryption and Authentication Techniques rules	Interactive Lecture, Diagram Walkthrough	Assignment, Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Security Protection and Best Practices, Data backup and recovery	Collaborative Discussion, Scenario-based Problem Solving	Assignment, Final exam	CLO4
	Lesson 2: Software updates and patching, Safe browsing practices, Ethical and legal issues in cybersecurity	Collaborative Discussion, Scenario-based Problem Solving	Assignment, Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on Final Syllabus	Integrative Review, Q&A Session	Presentation, Final exam	CLO3,CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's		Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	
Attendance	--	--	--		7			
Class Test 1	5(15)				AVG 15			
Class Test 2		5(15)						
Class Test 3			5(15)					
Assignment					5			
Presentation					8			
Midterm Exam	15	10			25	10	15	25
Final exam			30	10	40	30	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Priti Sinha, Pradeep K. Sinha, Computer Fundamentals: Concepts, Systems & Applications, 8th Edition, BPB Publications, 2021, ISBN: 978-9388176774

Reference Books/ Other Supplementary Readings

Pradeep K. Sinha, Priti Sinha, Computer Fundamentals: Concepts, Systems & Applications, 8th Edition, BPB Publications, 2021. ISBN: 9789389949328

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

CodeBlocks, Draw.io

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 112 Computer Fundamentals Lab

Course Code: SE 112		CIE Marks: 60
ISCED: SE112 /0611-112		LF Marks: 40
Course Title: Computer Fundamentals Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

The Computer Fundamentals Lab is designed to provide students with a hands-on understanding of core concepts in computing, programming, and essential software tools. Throughout this course, students will delve into the fundamentals of data, programming logic, and software applications, enabling them to build a strong foundation in computer science and information technology. Emphasis will be placed on practical implementation, ensuring students gain real-world skills that are applicable in various academic and professional settings.

Rationale of the Course

In today's digital age, understanding the core principles of computing is essential for success in both academic and professional realms. The Computer Fundamentals Lab aims to equip students with a solid foundation in key concepts of computing, programming, and essential software tools. Below are the rationales for the core components of this course: Understanding Data and Information Implemented in Programming, Learning Program Execution Sequence with Practical Examples, Exploring Usability of Conditions through Practical Applications, Understanding Various Types of Programming Conditions, Describing Program Flow Sequentially Using Flowcharts, Skill Development in Microsoft Office and Basic Computer Tools.

Course Objectives

By the end of course through lectures, readings, home works, lab assignments and exams, students will be taught:

- To Understand the concepts of data and information and their relevance in programming and hands-on experience in creating websites using Google Sites;
- To understand basic computer tools like Draw.io, Canva, and various Google tools;
- To apply programming concepts through practical examples and exercises;
- To apply essential skills in using Microsoft Office tools such as Word, Excel, and PowerPoint.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Apply basic computer tools, office software, and educational platforms effectively, including Word, PowerPoint, Excel, Google Apps, Canva, and AI-based tools.	PLO2	C3	Fundamental Domain
CLO2	Apply knowledge of Microsoft Office and Google productivity tools in practical scenarios.	PLO5	C3, P2	Fundamental Domain
CLO3	Conduct investigations and solve problems using computational thinking, flowcharts, and basic coding logic.	PLO4	C3, P3	Fundamental Domain
CLO4	Write effective reports, documentation, and presentations for any given problem.	PLO10	A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2					√							
CLO3				√								
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Discussion on course outline, course learning outcomes, assessment methods, deadlines, and other administrative information.	Syllabus Walkthrough, Interactive Q&A	None	None
Week-2 [1.5 Hours]	Introduction to Microsoft Word and document formatting. This topic focuses on familiarizing learners with the basic features of Microsoft Word, including typing, formatting, paragraph settings, page setup, and document preparation.	Interactive Demonstration, Guided Hands-on Practice	CLO1, CLO2	Lab Performance
Week-3 [1.5 Hours]	Advanced features and tools in Microsoft Word, along with email writing and email template preparation. Students will practice creating professional documents and effective email communication.	Interactive Demonstration, Hands-on Practice, Peer Review	CLO1, CLO2, CLO4	Lab Performance, Viva
Week-4 [1.5 Hours]	Introduction to PowerPoint as a foundational topic aimed at familiarizing learners with the essential features and functions of Microsoft	Visual Demonstration,	CLO1, CLO2, CLO4	Lab Performance

	PowerPoint. This topic helps students create visually engaging slideshows and develop basic presentation skills.	Guided Hands-on Practice		
Week-5 [1.5 Hours]	Advanced PowerPoint focuses on designing and delivering dynamic presentations using hyperlinks, multimedia, animations, transitions, and custom navigation to improve audience engagement and understanding.	Interactive Lecture, Hands-on Design Practice	CLO1, CLO2, CLO4	Lab Performance, Viva
Week-6 [1.5 Hours]	Spreadsheet/Sheet fundamentals. Students will understand the spreadsheet interface, enter and format data, use common formulas and functions such as SUM, AVERAGE, and COUNT, and create simple charts.	Live Software Demonstration, Hands-on Data Practice	CLO1, CLO2	Lab Performance, Viva
Week-7 [1.5 Hours]	Lab Test 1 based on previous lessons.	Practical Examination	CLO1, CLO2, CLO4	Lab Test 1, Lab Performance, Viva
Week-8 [1.5 Hours]	AI tools and BLC. Students will understand the basic use of AI-based educational tools and Basic BLC functionality for academic activities.	Interactive Demonstration, Hands-on Tool Implementation	CLO1, CLO2	Lab Performance, Viva
Week-9 [1.5 Hours]	Logical thinking and problem-solving. This topic introduces students to basic computational thinking, step-by-step problem analysis, and simple logic building.	Logic Puzzles, Active Problem-Solving, Interactive Discussion	CLO3	Lab Performance, Viva
Week-10 [1.5 Hours]	Flowchart fundamentals. Students will learn how to solve given problems using sequence, condition, loop, and flowchart-based program flow.	Whiteboard Demonstration, Guided Flowchart Drawing	CLO3, CLO4	Lab Performance, Viva, Lab Final
Week-11 [1.5 Hours]	Basic and advanced functionality of Draw.io. Students will use Draw.io to design flowcharts, diagrams, and documentation for given problems.	Live Software Demonstration, Collaborative Design Practice	CLO3, CLO4	Lab Performance, Viva, Lab Final
Week-12 [1.5 Hours]	Functionality of Canva basic and professional. Students will learn to create simple academic and professional visual content, presentation materials, and design-based documentation.	Visual Demonstration, Hands-on Design Practice	CLO1, CLO4	Lab Performance, Viva
Week-13 [1.5 Hours]	Industry session.	Guest Lecture / Industry Expert Talk, Interactive Q&A	CLO1, CLO2, CLO3, CLO4	Lab Performance
Week-14 [1.5 Hours]	Lab Final. Students will solve given problems in the lab using the tools and concepts covered throughout the course.	Summative Practical Assessment	CLO1, CLO2, CLO3, CLO4	Viva, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		

Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO1
Lab Final	40	CLO2	PLO5
		CLO3	PLO4
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

1. Peter Weverka, Office 365 All-in-One For Dummies, 2nd Edition, For Dummies, 2022, ISBN 10: 978-1119830702
2. John Hubbard, Schaum's Outline of Programming with C, 2nd Edition, McGraw-Hill, 2020, ISBN 10: 0071823062.

Reference Books/ Other Supplementary Readings

1. John Hubbard, Schaum's Outline of Programming with C, 2nd Edition, McGraw-Hill, 2020, ISBN 10: 0071823062.
2. Lab Manual

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Microsoft Office, Canva, BLC, AI Tools

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 113 Introduction to Software Engineering

Course Code: SE 113	CIE Marks: 60
ISCED: 0613-113	SEE Marks: 40
Course Title: Introduction to Software Engineering	Total Marks: 100
Semester:	
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)

Prerequisite: (If any)
Course Type: Core

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Software engineering fundamentals, software process, software process models, methodologies; prototyping, iterative process models, incremental software development, agile software development, extreme programming, Kanban, and SCRUM; Software requirement Engineering: functional and non-functional requirement, requirement engineering process, requirement elicitation, specification, validation, and change; software design: software modeling, UML diagrams; software testing, different types of testing, test-case design, white box testing: basis path testing with cyclomatic complexity, black box testing: interface testing, equivalence partitioning, boundary value analysis, object-oriented software testing: class testing, behavioral testing, Halstead's complexity, Decision table; maintenance, documentation, reliability engineering, software quality and security, reverse engineering, software risk management, software project management: phases of software project management, estimation techniques, scheduling techniques, COCOMO models.

Rationale of the Course

The course provides students with fundamental knowledge of software development processes, methodologies, and best practices. It covers the Software Development Life Cycle (SDLC), software process models, requirements analysis, design principles, and UML modeling techniques. Students gain an understanding of software testing, quality assurance, dependability, and security to develop reliable software systems. The course also introduces software maintenance, project management, risk management, and estimation techniques essential for successful project execution. By exploring both students develop the skills needed to design, develop, manage, and maintain high-quality software effectively.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of Software Engineering. More specifically,

- To apply fundamental software engineering principles and practices to real-world scenarios.
- To compare and contrast different SDLC models (waterfall, agile, iterative, etc.) and their suitability for various projects. Also, core agile methodologies like Scrum and Kanban for flexible and adaptive project management.
- To analyze user needs and translate them into clear, concise, and measurable requirements, including various requirement elicitation techniques (interviews, surveys, observation) to capture user feedback.
- To create and interpret UML diagrams (use case, class, sequence, activity) to visualize software structure and behavior.
- To understand software security and quality through different approaches.
- To apply fundamental project management principles (planning, monitoring, control) to software projects and utilize project cost estimation techniques like Functional Point Analysis and COCOMO models.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skills
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CLO1	Apply software engineering and design principles in building software, systems and appropriate process models.	PLO1	C3	Fundamental Domain
CLO2	Identify common causes of project failure and explain strategies for effective project planning, task organization, prototyping, and the use of AI tools to improve project success.	PLO5	C2	Fundamental Domain
CLO3	Elicit and analyze software requirements from stakeholders and select, implement suitable testing techniques and tools depending on the development stage, and differentiate among various test strategies for application in software projects.	PLO3	C3, C4	Fundamental Domain
CLO4	Apply cost estimation methods to determine the total budget for software projects.	PLO10	C3	Fundamental Domain
CLO5	Utilize development and maintenance metrics to ensure delivery of high-quality software.	PLO7	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PL O1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2					√							
CLO3			√									
CLO4										√		
CLO5							√					

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE, Introduction to Software Engineering. Software, Types of software and Characteristics of a quality Software.	Interactive Lecture, Real-world Case Discussion	Class Test, Midterm exam	CLO1
	Lesson 2: Importance of Software Engineering, challenges of Software Engineering. The History of Software Engineering, software process, software process models, software development practice, activities and task.	Interactive Lecture, Collaborative Timeline Review	Class Test, Midterm exam	CLO1

Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Phases of the Software Development Life Cycle (SDLC): Differentiation Among Phases, Activities, and Tasks.	Visual Presentation, Guided Phase Mapping	Class Test, Midterm exam	CLO1
	Lesson 2: Relationship Between Software Process Models and Software Development Methodologies. Comparison Between Process Models and Methodologies Role of SDLC in Software Development.	Comparative Discussion, Concept Mapping	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Study of the Waterfall Classical Model and V-Model: Structure, Applications, Advantages, and Disadvantages.	Interactive Lecture, Scenario-based Application	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Prototype Model and Iterative model: Structure, Applications, Advantages, and Disadvantages.	Interactive Lecture, Diagram Analysis	Class Test, Assignment, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Spiral Model: Structure, Applications, Advantages, and Disadvantages. Comparison Between SDLC Models,	Visual Demonstration, Comparative Analysis	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Agile Methodologies: Agile Manifesto, Core Values, Principles, Practices, Benefits, and Challenges, Frameworks.	Interactive Lecture, Agile Values Brainstorming	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Class Test -1			
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Study of the Scrum Model: Concepts, Roles, Practices, Applications, Advantages, and Disadvantages	Interactive Lecture, Role-play Scenario (Scrum Roles)	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Study of the Extreme Programming (XP) Model: Concepts, Practices, Applications, Advantages, and Disadvantages	Collaborative Discussion, Concept Mapping	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hs]	Lesson 1: Study of the Kanban Model: Concepts, Applications, Advantages, Disadvantages, Practices	Visual Demonstration, Workflow Simulation	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Class Test -2			
	Lesson 2: Comparative Analysis of Scrum, XP, and Kanban. Workflow Management in Scrum, XP, and Kanban	Comparative Group Discussion, Diagramming	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Artificial Intelligence (AI). Basic Concepts of AI in Software Engineering. Importance of AI in Modern Software Development. Pros and Cons of Using AI in Software Development. Future Scope of AI in Software Engineering	Interactive Lecture, Real-world AI Case Studies	Class Test, Assignment, Final exam	CLO2

	and Basic Examples of AI Applications in Software Engineering			
	Lesson 2: Introduction to AI Tools. Importance of AI Tools in Software Engineering. AI Tools for Coding Assistance. AI Tools for Writing, Slide Making, Documentation and Diagram Generation. Pros and cons of using AI Tools.	Live Tool Demonstration, Hands-on Tool Practice	Class Test, Assignment, Final exam	CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Basic of Software Requirements, Software Requirements Elicitation and Analysis, Verification and validation, software testing. Important of testing and different types of testing with definition	Interactive Lecture, Scenario-based Discussion	Class Test, Assignment, Final exam	CLO3, CLO5
	Lesson 2: White box testing: basis path testing with cyclomatic complexity, black box testing: interface testing, equivalence partitioning, boundary value analysis.	Whiteboard Demonstration, Guided Problem-Solving	Class Test, Assignment, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Definition of Software Quality. Importance of Software Quality, Characteristics of Good Quality Software. Software Reliability, Difference Between Quality and Reliability. Common Software Errors and Defects. Basic Concepts of QA and QC.	Interactive Lecture, Concept Mapping	Class Test, Final exam	CLO5, CLO3
	Lesson 2: Software quality, Quality factors, Qualitative quality assessment, Quantitative quality assessment, Software quality dilemma, Achieving software quality	Collaborative Discussion, Analytical Case Study	Class Test, Final exam	CLO5
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Security and dependability, Security terminology, Security risk assessment, Security requirements	Interactive Lecture, Threat Scenario Discussion	Class Test, Final exam	CLO5, CLO2
	Lesson 2: Software maintenance, Steps in a maintenance process, Strategies of software maintenance. Importance of software maintenance	Interactive Lecture, Process Flow Analysis	Class Test, Final exam	CLO5
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Decomposition and Estimation techniques: Software Sizing, FP Based Estimation, Related mathematics	Mathematical Demonstration, Guided Calculation Practice	Class Test, Final exam	CLO4

	Lesson 2: COCOMO models (Basic and Intermediate) and COCOMO Model related mathematics, Practice Class Test -3	Whiteboard Demonstration, Peer Problem-Solving	Class Test, Final exam	CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Software project scheduling, basic principles of scheduling, task set for project planning. Scheduling related mathematics: Critical Path Analysis.	Mathematical Demonstration, Collaborative Diagramming (CPA)	Assignment, Final exam	CLO4, CLO2
	Lesson 2: Software project management, Project management in Waterfall model based projects, Project management in agile model based projects, Team management of software projects	Comparative Discussion, Scenario-based Planning	Assignment, Final exam	CLO2, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Software risks, Seven principles of risk management, Risk identification, assessing overall project risk, risk estimation, Risk mitigation, Risk monitoring and Risk management	Interactive Lecture, Risk Assessment Workshop	Assignment, Final exam	CLO2, CLO5
	Lesson2: Introduction to Work Breakdown Structure (WBS). Importance of WBS in Software Projects. Pros and Cons of WBS and Basic Examples of WBS. Software Prototype, Types of Prototype, Basic Principles of User Interface and Basic Layout Design	Visual Demonstration, Guided Hands-on Design	Assignment, Final exam	CLO2
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4, CLO5
	Lesson 2: Review class on final exam syllabus. Presentation on a given topic	Integrative Review, Student Presentations	Presentation, Open Discussion	CLO1, CLO2, CLO3, CLO4, CLO5

Overall Assessment Scheme

Assessment Task	CLO's					Mark (Total =100)	PLO's					Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4	CLO5		PLO1	PLO2	PLO3	PLO4	PLO5	
Attendance	--	--			--	7						
Class Test 1	5(15)					AVG 15						
Class Test 2		5(15)										

Class Test 3				5(15)		5						
Assignment		5				10	5					5
Presentation						15						
Midterm Exam	15	10				10	15	10				25
Final exam		5	10	15	10	40		5	10	15	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Roger S., Pressman, Software Engineering: A Practitioner's Approach, 9th Edition, McGraw-Hill, 2019, ISBN 10: 1259872971

Reference Books/ Other Supplementary Readings

Ian Sommerville, Software Engineering, 9th edition, Pearson, 2021, ISBN: 9780133943030

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Lucidchart, Draw.io, AI tools

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

ENG 101 English I

Course Code: ENG 101/GED 104		CIE Marks: 60
ISCED: 0611-101		SEE Marks: 40
Course Title: ENGLISH I/English Reading, Writing & Public Speaking		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)		Contact Hours: 3 (Total weeks: 14)
Prerequisite: N/A		
Course Type: GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Overview of Basic Skills of English: Speaking, Basic Grammar: Use of Articles, Identifying Parts of Speech, Syntax: Run on, Fragments, speaking: Self Introduction, Social English: Greeting, Answering to Greeting, Agreeing & Disagreeing, Small Talks etc., Relating and Demonstrating Ideas on Selective Topics; Introduction to IELTS Speaking: Part I and II, Reading: Reading techniques (Skimming and Scanning), Reading Comprehension: Practices from IELTS Reading Comprehension: (True/False/Not Given, Flow chart, Matching Heading, Matching

Features, Multiple Choice Question, Short Answer etc.), Writing: Formal Letter Writing, Cover Letter, CV, Resume, Video Resume, E-mail Writing, Paragraph Writing, Essay Writing.

Rationale of the Course

This curriculum provides a comprehensive foundation in English proficiency across speaking, grammar, reading, and writing to cultivate the linguistic competencies required for global success. It enhances verbal fluency through practical social interactions and targeted IELTS Speaking preparation while solidifying core grammar and syntax for effective communication. The reading module emphasizes efficient retrieval techniques, like skimming and scanning, alongside rigorous IELTS Reading practices. Furthermore, versatile writing exercises ranging from CVs and formal emails to academic essays equip learners for real-world demands, ensuring they are fully prepared for academic, professional, and everyday communication scenarios.

Course Objectives

- To improve the basic skills of English language to communicate confidently and naturally, more specifically,
- To enhance the ways to communicate opinions and information on everyday topics and common experiences, and express and justify opinions.
 - To improve the ability to follow conversations, to retain information and assimilate factual details, and to draw conclusions from information gathered through effective listening.
 - To develop the reading techniques for general sense, main ideas, and details, and also to understand the author's inferences and opinions
 - To make the writing skill for writing a response, organize ideas, and use wide-ranging vocabulary and grammar well and accurately.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Understand the linguistic and grammatical aspects of contextual source materials in target language	PLO1	C1	Fundamental Domain
CLO2	Apply speaking skills to communicate in various context of everyday life, how to summarize	PLO9	C2, A2	Social, Thinking, Personal Domain.
CLO3	Analyze various reading techniques to infer and comprehend a text; apply and respond to various question formats on the IELTS	PLO2	C4 A2	Fundamental Domain
CLO4	Create practical samples of formal/informal emails, formal letters, cover letter, curriculum vitae, resume, video resume,	PLO10	C3 P5 A4	Social, Thinking, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2									√			
CLO3		√										
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Correspo nding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of the course [Text Book – 1, 2, 3 and 4]	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Basic Grammar – Article, Identifying Parts of Speech Practice session on Article and Preposition [Text Book – 1 and 2], Speaking – IELTS Part I (Self Introduction) [Cambridge IELTS, Practice I]	Interactive Lecture, Peer-to-Peer Speaking Practice	Class Test, Midterm exam	CLO1,C LO2
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Basic Grammar – Understanding and Applying Different Types of Tense Practice session on Parts of Speech [Text Book – 1 and 2]	Interactive Lecture, Whiteboard Demonstration	Class Test, Midterm exam	CLO1
	Lesson 2: Reading Techniques – Skimming and Scanning [Cambridge IELTS, Practice II]	Brainstorming, Guided Reading Practice	Class Test, Midterm exam	CLO3
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Basic Grammar – Subject-Verb Agreement Practice session on Tense [Text Book – 1 and 2]	Interactive Lecture, Rule-based Practice	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Reading Comprehension – Understanding the Meaning of Unknown Words [Cambridge IELTS, Practice III]	Active Reading, Vocabulary Deduction Practice	Class Test, Assignment, Midterm exam	CLO3
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Basic Grammar - Right Form of Verb Practice session on Subject-Verb Agreement [Text Book – 1 and 2]	Whiteboard Demonstration, Guided Practice	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Reading Comprehension – True/False/Not Given, Multiple Choice Question [Cambridge IELTS, Practice IV and V] Class Test -1	Guided Comprehension Practice, Peer Review	Class Test, Assignment, Midterm exam	CLO3
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Reading Comprehension – Matching Information Practice session on Reading Comprehension (True/False/Not Given, Multiple Choice	Scenario-based Reading, Group Analysis	Class Test, Assignment, Midterm exam	CLO3

	Question) [Cambridge IELTS, Practice V and VI].			
	Lesson 2: Writing - Introduction to Academic Writing Practice session on Reading Comprehension (Matching Information) [Text Book - 3]	Interactive Lecture, Guided Writing Practice	Class Test, Assignment, Midterm exam	CLO4
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Writing - Email Writing (Formal and Informal), Formal Letter Writing [Text Book - 3]	Visual Demonstration, Hands-on Drafting	Class Test, Assignment, Midterm exam	CLO4
	Lesson 2: Speaking – Social English: Greeting, Answering to Greeting, Agreeing & Disagreeing etc. Class Test -2	Role-play, Interactive Dialogue Practice	Class Test, Assignment, Midterm exam	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Writing - Paragraph Writing Practice session on email Writing [Text Book -3]	Guided Writing Practice, Peer Review	Midterm exam	CLO4
	Lesson 2: Speaking – Social English: Thanking and Apologizing, Small Talks etc.	Scenario-based Role-play	Midterm exam	CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Problem-Solving Class on Basic Grammar (Parts of Speech, Tense, Right Form Verb, Subject-Verb Agreement) [Text Book - 1 and 2], Problem-Solving Class on Reading Comprehension and Academic Writing [Cambridge IELTS, Practice I, II, III, IV and V]	Collaborative Problem-Solving, Review Session	Class Test, Assignment, Final exam	CLO1,C LO3
	Lesson 2: Writing - Cover Letter, CV, Resume, Writing - Video Resume [Text Book - 3], Basic Grammar - Conditional Sentences Speaking - Relating and Demonstrating Ideas on Selective Topics (IELTS Part II), Syntax - Run on and Fragments Practice session on Conditional Sentences [Text Book - 1 and 2]	Hands-on Drafting, Visual Examples	Class Test, Assignment, Final exam	CLO1,C LO4
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Reading Comprehension – Short Answer, Explaining Table Practice session on Run on and Fragments [Cambridge IELTS, Practice VI]	Guided Reading, Diagram Analysis	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Reading Comprehension - Matching Heading, and Explaining Graph and Charts (IELTS Part I) Cambridge IELTS, Practice VII]	Visual Demonstration, Collaborative Analysis	Class Test, Assignment, Final exam	CLO3

Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Dependability, principal dependability properties, Attributes of dependable processes	Interactive Lecture, Concept Mapping	Assignment, Final exam	CLO3
	Lesson 2: Reading Comprehension - Matching Heading, and Explaining Graph and Charts (IELTS Part II) [Cambridge IELTS, Practice VIII and IX]	Guided Comprehension Practice	Assignment, Final exam	CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1 Writing - Essay Writing (Structure and IELTS-based Types), Summary Writing [Text Book - 3]	Step-by-step Writing Demonstration, Hands-on Practice	Assignment, Final exam	CLO1, CLO2
	Lesson 2: Basic Grammar - Review Class on Article and Tense (Problem-Solving) [Text Book - 1 and 2] Class Test -3	Problem-Solving Workshop	Assignment, Final exam	CLO1
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Reading Comprehension - Q/A, Matching Headings, and Explaining Tables and Diagrams, Part II [Cambridge IELTS, Practice IX and X]	Analytical Reading Practice	Assignment, Final exam	CLO3
	Lesson 2: Speaking - Part II and III [Cambridge IELTS, Practice IV and V]	Peer-to-Peer Speaking Practice, Mock Interviews	Assignment, Final exam	CLO1, CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Reading Comprehension - Practices from IELTS Reading Comprehension (Part III and IV) [Cambridge IELTS, Practice VI, VII, VIII, IX AND X]	Timed Reading Practice, Review Analysis	Assignment, Final exam	CLO3
	Lesson2: Writing - Practicing Free-hand Writing (Essay Writing) [Text Book - 3]	Independent Writing Practice, Instructor Feedback	Assignment, Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Speaking - Part II and III, Reading Comprehension - Matching Heading and Explaining Graph and Charts [Cambridge IELTS, Practice VI, VII, VIII, IX and X], Review class on Final Syllabus topic	Integrative Review, Mock Practice	Assignment, Final exam	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

	CLO's	Mark	PLO's	Mark
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Assessment Task	CLO1	CLO2	CLO3	CLO4	(Total=100)	PLO1	PLO2	PLO9	PLO10	(Total =70)
Attendance	--	--	--	--	7	--				
Class Test 1	8		7		Avg (15)					
Class Test 2		5(15)								
Class Test 3				5(15)						
Assignment				5	5				5	5
Presentation		8			8					
Midterm Exam	10		7	8	25		17		8	25
Final exam	10		15	15	40		25		15	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

- 1.Murphy, R. (2019). English Grammar in Use (5th ed.). Cambridge University Press. ISBN: 978-1108586627.
- 2.Stephen Bailey, Academic Writing: A Handbook for International Students, 4th Edition, Routledge, 2018, ISBN 10: 1138658820.
- 3.Cambridge University Press, Cambridge IELTS 15 -19, Cambridge University Press, 2020-2024

Reference Books/ Other Supplementary Readings

Raymond Murphy, Essential english grammar: a self-study reference and practice book for elementary students of english : with answers, 2nd Edition, Cambridge University Press, 1998, ISBN: 0486216497

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU) (Version 3)

BNS 101 Bangladesh Studies

Course Code: BNS 101		CIE Marks: 60
ISCED: 0314-101		SEE Marks: 40
Course Title: Bangladesh Studies		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: (If any)		
Course Type: GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Geography location, borders, topography, resources, climate; Demographic traits: demographic dividend, human development; Society and culture: Social stratification, traditions, values, festivals; Ethnic identity of people of Bangladesh; Origin and development of the name of Bangladesh; Origin and development of Bangla language; Bangladesh in international affairs: principles, objectives and determinants of foreign policy, achievements and challenges; Constitution of Bangladesh: concept, essentials, principles, rights, and amendments; Organs and functions of government: Digitalization and Service Sector in Bangladesh; Historical background of Bangladesh: Language movement, Six-points demands, Liberation war, constitutional development in post -liberation; Economic profile of Bangladesh: macro perspective, blue economy, gig economy; Development approaches: Vision 2041, SDGs, Delta plan; Agricultural Productivity and Rural development; Urbanization: push-pull model of migration; Environmental degradation and climate change; Industrial sector: overview of major industries, ICT & Software sector of Bangladesh.

Rationale of the Course

Bangladesh Studies is a multidisciplinary course designed to equip students with knowledge on focal themes relating to Bangladesh. The first theme is the inevitability of the emergence of Bangladesh as a state entity in the context of a long historical background, and the second theme draws attention to the experience of Bangladesh in governance and democratization. In the light of long historical epistemology, students will connect with the benchmark symbol for globally connected and locally engaged students along with being competent to meet the challenges of the 21st century. It will also enhance students' inner vigilant excellence on current phenomena in the country.

Course Objectives

This course aims to introduce the students to the multi-faceted concept of modern Bangladesh, along with its long historical background and inevitable emergence as a state entity in the modern world. The main objectives of this course are:

- To introduce students to the long political history of Bangladesh and the necessity of its emergence by discussing the actual events and facts.
- To make students' understanding clearer, authentic, and rational in the nation-building and state-building process.
- To prepare students for competitive exams by providing them with basic knowledge and realities about the origin and surrounding environment.
- To increase the sense of patriotism and aspirations among the learners that can strengthen their commitment to nation-building.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skills
CLO1	Apply SWOT (Strengths, Weaknesses, Opportunities, and Threats) analysis to the geography, demography, economy, society, culture, government, and politics of Bangladesh.	PLO12	C3	Fundamental Domain
CLO2	Analyze the national and global factors behind the advancement of Bangladesh to construct specific development approaches.	PLO2	C4	Fundamental Domain
CLO3	Demonstrate the historical, ethical, normative, and legal paradigms affecting contemporary Bangladesh with questioning popular perceptions and evaluating by pragmatism.	PLO6	C4	Fundamental Domain
CLO4	Explain the diverse nature of prevailing socio-economic and environmental concerns through rigorous analysis to propose initiatives for sustainable development.	PLO2	C5	Fundamental Domain
CLO5	Develop rational and practical solutions both individually and collectively through a sense of ownership to meet the critical challenges of contemporary Bangladesh.	PLO9	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1												√

CLO2		√										
CLO3						√						
CLO 4		√										
CLO5									√			

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to course, Course Approach, Marks Distribution, Assessment and Evaluation, Geographical Traits of Bangladesh e.g., Location, Border & Boundary, Topography, Rivers, Resources, Climate	Syllabus Walkthrough, Map Demonstration & Analysis	Class Test, Midterm exam	CLO1,CLO3
	Lesson 2: Demographic Traits of Bangladesh e.g., Population of Bangladesh, Labor Force, Demographic Dividend, Social Traits and Cultural Traits of Bangladesh, Cultural Challenges; Formations of Government, Sovereignty, and its features	Demographic Data Analysis, Open Discussion	Class Test, Midterm exam	CLO1 CLO4
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Bangladesh according to sociological perspective, Bangladesh according to anthropological perspective	Sociological Case Study, Interactive Lecture	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Origin and Identity of Bangalee; Major races come to Bengal; Mixed Race: An explanation.	Historical Timeline Review, Storytelling	Class Test, Midterm exam	CLO2, CLO3
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Origin of the name of Bangladesh, Name of this region in Pre-Muslim Era, Name of this region in Muslim Era, Name of this region in post-Muslim Era	Interactive Lecture, Conceptual Mapping	Class Test, Assignment, Midterm exam	CLO2, CLO3
	Lesson 2: Origin of Bangla Language; Development of Bangla Language, Contribution of Various Nations and Peoples	Interactive Lecture, Collaborative Discussion	Class Test, Assignment, Midterm exam	CLO2, CLO3
Week-4 Lesson 1	Lesson 1: Constitution as a Concept; Essentials of a Good Constitution, Bangladesh Constitution: Making	Legal Framework Analysis,	Class Test, Assignment, Midterm exam	CLO4, CLO5

& 2 [3 Hours]	Process, Basic Facts and Figures and Features.	Interactive Lecture		
	Lesson 2: Bangladesh Constitution: Fundamental Principles, Fundamental Rights, Shortcomings and Amendments Class Test -1	Scenario-Based Discussion (Constitutional Rights)	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Types of Government, Organs of government, Functions of Legislative, Functions of Executive organ, E-Governance	Visual Diagramming (Government Organs)	Class Test, Assignment, Midterm exam	CLO3, CLO5
	Lesson 2: Functions of Judiciary, Independence of Judiciary, Digitalization and Service Sector in Bangladesh	Interactive Lecture, Real-world Service Sector Examples	Class Test, Assignment, Midterm exam	CLO4, CLO5
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Bangladesh in International Affairs, Foreign Policy as a concept, Bangladesh Foreign Policy: Principles, Objectives, Determinants	Policy Analysis, Interactive Debate	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Bangladesh Foreign Policy: Making Bodies, Strengths, Weaknesses and Present status Class Test -2	Collaborative Brainstorming (SWOT Analysis)	Class Test, Assignment, Midterm exam	CLO4, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Bangladesh in International Affairs, Strengths, and Weaknesses of Bangladesh in Foreign Relations, Feedback regarding Foreign relations of Bangladesh, Slight Review of all topics Week-1 – Week-7	Integrative Review, Interactive Q&A	Midterm exam	CLO1, CLO2, CLO5
	Lesson 2: Historical Background of Bangladesh: key movements in British period (Sanyasi and fakir rebellion (1763-1800), Faraizi Movement (1819), Swadeshi Movement (1905), Non-cooperation movement (1920-1922), Quit India movement (1942), Language Movement of 1952, Stages of Language Movement, Outcomes of language movement, Background of Bangladesh: Discriminations against East Pakistan, Social-Political-Economic Disparity	Multimedia Historical Presentation, Timeline Mapping	Class Test, Assignment, Final exam	CLO1, CLO2

Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Six Points Movements, Outcome of Six Points Movement, Mass Upsurge in 1969, National Election in 1970	Historical Event Sequencing, Group Discussion	Class Test, Assignment, Final exam	CLO2, CLO3
	Lesson 2: Liberation War of Bangladesh, Historical Speech of Sheikh Mujibur Rahman, Role of Bangabandhu, Emergence of Bangladesh	Multimedia Presentation, Analytical Discussion	Class Test, Assignment, Final exam	CLO2, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Economy of Bangladesh; Economy as a concept; Macro and Micro Economy; Nature of Bangladesh Economy	Economic Indicator Analysis, Interactive Lecture	Class Test, Assignment, Final exam	CLO3, CLO5
	Lesson 2: Approaches towards Development, Vision 2041, SDGs and realities of Bangladesh, Delta Plan	Goal-mapping (SDGs), Strategic Document Review	Class Test, Assignment, Final exam	CLO4, CLO5
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: National Budget, Role of Financial Institutions and Role of ECNEC, NEC, Present vs Past Comparison, Review class on topic discussed in Week-9 to Week-15	Comparative Financial Analysis, Review Session	Class Test, Final exam	CLO1, CLO4, CLO5
	Lesson 2: Blue Economy; Gig Economy; Different reviews on Bangladesh Economy, Economic Prospects and Sector-wise Development, Present Economic Trend, Economic Problems and Steps need to accelerate the Development.	Trend Analysis, Collaborative Problem-Solving	Class Test, Final exam	CLO1, CLO2, CLO5
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Environment, Ecosystem, Climate Change and Environmental Degradation, Factors of Environmental Degradation, Impact of Climate Change	Scenario-Based Discussion (Climate Impact)	Final exam	CLO1, CLO2
	Class Test - 3 Lesson 2: Common Environmental Problems in Bangladesh; Initiatives for environmental protection: Bangladesh and International; Role of Individual in protecting the Environment	Case Study Analysis (Local Initiatives), Interactive Lecture	Final exam	CLO1, CLO5
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Urbanization and Migration as a Concept, Trend of Urbanization: Bangladesh and International Perspective; Impact of Urbanization; Problems and Solution of Urbanization	Urban Trend Analysis, Brainstorming Solutions	Assignment, Final exam	CLO1, CLO5

Hours]	Lesson 2: Rural Development; Indicators of Rural Development; History of Rural Development; BARD: Comilla Model; Local Government and Role of NGOs in rural Development Class Test -3	Model Comparison (Comilla Model), Interactive Lecture	Assignment, Final exam	CLO2, CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Industrial Sector in Bangladesh, Industrial Growth Trend: Major Industries in Bangladesh, SME, ICT Industry in Bangladesh, Development of ICT Industry	Industry Trend Mapping, Interactive Discussion	Assignment, Final exam	CLO1, CLO4,CLO5
	Lesson 2: Status of Bangladesh in ICT sector, Hi-Tech Park, and Its contribution; Software Park and Its Contribution, Prospect of Software Industry, Govt. Initiatives for the promotion of ICT; Problems of ICT Industry and Prospects of ICT Industry	SWOT Analysis (ICT Sector), Visual Demonstration	Assignment, Final exam	CLO1, CLO3,CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: SME; Major Industries in Bangladesh; Contribution of Industry in GDP; Prospects and Challenges of the Industry Sector in Bangladesh	Economic Case Study, Interactive Q&A	Assignment, Final exam	CLO1, CLO2
	Lesson 2: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4, CLO5

Overall Assessment Scheme

Assessment Task	CLO's					Mark (Total=100)	PLO's				Mark (Total =70)
	CLO 1	CL O2	CL O3	C L O 4	C L O 5		PLO 12	PLO 2	PLO6	PLO9	
Attendance	--	--	--			7					
Class Test 1	5(15)					AV G 15					
Class Test 2		5(15)									
Class Test 3			5(15)								

Assignment		5				5	5				5
Presentation						8					
Midterm Exam	15	10				25	15	10			25
Final exam			16	16	8	40		16	16	8	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Bangladesh Historical Studies, Dhaka: Bangladesh History Association, 2015.

Reference Books/ Other Supplementary Readings

Md. Abul Halim, Constitution, Constitutional Law and Politics: Bangladesh Perspectives, 3rd Edition, Sams Publication, 2006,

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



**Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)**

MAT101 Mathematics-I

Course Code: MAT101		CIE Marks: 60
ISCED: 0541-101		SEE Marks: 40
Course Title: Mathematics-I		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Function, Domain, Range, Graph, Limit, Continuity & Differentiability, Derivatives; Differentiation and Successive Differentiation of various types of function; Leibnitz's Theorem, Roll's Theorem; Mean value Theorem; Taylor's and McLaurin's Theorem in finite and infinite forms, Euler's Theorem; Maximum and minimum values of functions of single variable, Expansion of functions; Evaluation of indeterminate forms; Partial differentiation, Integration by the method of substitutions, successive reduction; Integration by parts; Standard integrals; Definite integrals: properties and use in summing series; Walli's Formula, Improper integrals, beta function and gamma function; Jacobian; Multiple integral and its application, Area and volume between curves and axes.

Rationale of the Course

This foundational calculus course equips students with essential mathematical tools for scientific and engineering disciplines. It explores differential calculus—including limits, derivatives, optimization, and key theorems like Taylor's and Mean Value—to rigorously analyze function behavior and rates of change. The curriculum then advances to multivariable systems through partial differentiation and Jacobians. Furthermore, it covers comprehensive integral calculus, utilizing definite and improper integrals, reduction formulas, and Beta/Gamma

functions to compute areas, volumes, and accumulation. Ultimately, this course builds the advanced problem-solving and analytical modeling skills crucial for success in modern STEM fields.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of calculus. More specifically,

- To learn the basic concepts of function, its domain, range, and graph of a function.
- To learn various kinds of methods of differentiation and partial differentiation.
- To learn different kinds of methods of indefinite, definite, and multiple integrations.
- To learn the concepts and techniques to find the area and volume of surfaces.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Interpret the basic concepts of calculus and the application of calculus to solve engineering problems.	PLO2	C2	Fundamental Domain
CLO2	Apply these analytical concepts from the principles and knowledge as a systematic way for problem-solving in scientific design and analysis.	PLO1	C3	Fundamental Domain
CLO3	Examine the appropriate method to solve the issues based on differentiation and integration.	PLO2	C4	Fundamental Domain
CLO4	Implement the concepts and techniques to solve real-life problems.	PLO1	C3, A2, P1, P3, P4	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2	√											
CLO3		√										
CLO4	√											

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Understanding about the course and mark distribution, Understanding the cognitive levels and CO-PO mapping, Overview on course outline and grading system.	Syllabus Walkthrough, Interactive Q&A	None	None

	Lesson 2: Overview of calculus, Application, Calculus, Application and properties.	Interactive Lecture, Real-World Application Discussion	None	CLO1, CLO2, CLO3
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Calculus, Application and properties.	Mathematical Demonstration, Guided Practice	Class Test, Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Function, Domain, Range	Whiteboard Diagramming, Peer Problem-Solving	Class Test, Midterm exam	CLO1, CLO2, CLO3
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Graph of piecewise functions, Graph of functions	Whiteboard Graphing Demonstration, Guided Visual Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Graph of piecewise functions, Classwork on domain, range and graph	In-class Guided Problem Solving, Peer Review	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Limit, Continuity & Differentiability	Mathematical Demonstration, Limit Evaluation Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Overview of differentiation and its application Class Test -1	Rule-based Practice, Real-world Rate of Change Examples	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Differentiate sums, product and quotients of composite polynomial,	Step-by-Step Formula Demonstration, Guided Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Differentiate trigonometric, exponential, and logarithmic functions.	Mathematical Demonstration, Collaborative Board Work	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Classwork on various methods of differentiation	Hands-on Problem Solving, Instructor Feedback	Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Successive Differentiation of various types of function; Leibnitz's Theorem. Class Test -2	Theorem Proof Walkthrough, Guided Calculation	Midterm exam	CLO1, CLO2, CLO3
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Euler's Theorem; Maximum and minimum values of functions of a single variable, Classwork on Maximum and minimum values of functions.	Optimization Modeling, Visual Curve Analysis	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
	Lesson 2: Expansion of functions; Evaluation of indeterminate forms; Partial differentiation, Classwork on Partial differentiation.	Step-by-Step Indeterminate Forms Demo, Guided Practice	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Review class on a topic discussed in Week-1 – Week-4.	Integrative Review, Concept Clarification Q&A	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3

	Lesson 2: Review class on a topic discussed in Week-5 – Week-8.	Integrative Review, Advanced Problem-Solving	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of Integration and its application.	Visual Demonstration (Area under curve), Interactive Lecture	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
	Lesson 2: Integration by the method of substitutions	Mathematical Demonstration, Rule-based Practice	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Successive reduction; Integration by parts	Step-by-Step Formula Application, Guided Practice	Assignment, Final exam	CLO1, CLO2, CLO3
	Lesson 2: Classwork on the method of substitutions and Integration by parts	Peer Problem-Solving, In-class Guided Execution	Assignment, Final exam	CLO1, CLO2, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Integration using partial fractions	Algebraic Decomposition Demo, Collaborative Practice	Assignment, Final exam	CLO1, CLO2, CLO3
	Lesson 2: Classwork on various methods of integration Class Test -3	Hands-on Problem Solving Session	Assignment, Final exam	CLO1, CLO2, CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Standard integrals; Definite integrals: properties and use in summing series; Walli's Formula.	Formula Derivation Walkthrough, Guided Calculation	Assignment, Final exam	CLO1, CLO2, CLO3
	Lesson 2: Class work on definite integrals: properties and use in summing series; Walli's Formula.	Collaborative Problem-Solving, Whiteboard Practice	Assignment, Final exam	CLO1, CLO2, CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Multiple integral and its application, Improper integrals, beta function and gamma function; Jacobian.	Advanced Mathematical Demonstration, Concept Mapping	Assignment, Final exam	CLO1, CLO2, CLO3
	Lesson2: Evaluate area and volume of surfaces, Classwork on the real-life problem of finding area and volume, Classwork on the real-life problem of finding area and volume.	3D Visual Modeling, Applied Physics/Engineering Calculations	Assignment, Final exam	CLO1, CLO2, CLO3, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Applications of Math in Tech, Interactive Q&A	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on Final Syllabus Topic	Integrative Review, Mock Final Problems	Open discussion	CLO1, CLO2, CLO3

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's		Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	
Attendance	--	--	--		7			

Class Test 1	5(15)				Avg 15			
Class Test 2	5(15)							
Class Test 3		5(15)						
Assignment			5		5		5	5
Presentation				8	8		8	8
Midterm Exam	10	15			25	15	10	25
Final exam	15	15	10		40	15	25	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Howard Anton, Irl C. Bivens, Stephen Davis, Calculus: Early Transcendentals, 10th Edition, Wiley, 2013, ISBN: 9781118092408

Reference Books/ Other Supplementary Readings

Khosh Mohammad, P.K. Bhattacharjee, Md. Abdul Latif, A Textbook on Differential Calculus, 10th Edition, Gonith Prokashon, 2010

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)

(Version 3)

SE 121 Structured Programming

Course Code: SE 121		CIE Marks: 60
ISCED: 0613-121		SEE Marks: 40
Course Title: Structured Programming		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Overview of C: Basic structure of C programs, Importance of learning C programming, Execution of a C program. Constant, Variables, and Data Types: Different data types and their specifiers, declaration of variables, Assigning values to variables, taking input using scanf() function, Operators and Expressions: Arithmetic operator, Relational operator, Logical operator, Increment Decrement operator, Conditional operator, Conditional Statements: simple if statement, if...else statement, Nesting if... else statement, if...else if ladder, switch statement. Loops: while loop, do..while loop, for loop, nested for loop, continue and break, Array: Declaration and Initialization of 1D and 2D arrays. Strings: Declaration and Initialization of String variables, arithmetic operation on characters, string handling functions, User-defined Functions: Definition of user-defined function, Elements of user-defined functions, Category of user-defined functions, Structures: Defining a structure, declaring structure variables, Accessing structure members, Structure initialization.

Rationale of the Course

This foundational course provides a comprehensive introduction to C programming, highlighting its widespread industry relevance and core role in computer science education. It guides students through basic syntax, data types, and dynamic input/output mechanisms. The curriculum covers arithmetic and logical expressions alongside essential control structures, including conditionals and loops. Students will master data organization using arrays and strings, while implementing user-defined functions for modularity. By concluding with custom data structures, this course establishes the essential problem-solving skills and programming architecture required for advanced software development

Course Objectives

To provide a solid conceptual understanding of the high-level, general-purpose Structured

Programming Language. More specifically,

- To learn preprogramming steps like writing algorithms, and drawing flowcharts.
- To learn the syntax and semantics of C programming.
- To learn different control structures like decision control, and loop control.
- To Understand the limitations of basic data types and learn the concepts of derived data types and user-defined data types (Array and Structure).
- To use strings and learn about the string handling functions.
- To learn the concepts and benefits of using user-defined functions.
- To develop logical thinking and being able to analyze the execution of a C program.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLO	CLO Description	Mapping with PLOs	Learning Domains (C, P, A)	BNQF Skill
CLO1	Describe number theory concepts, core computer principles, essential programming terminology, and structured programming fundamentals.	PLO1	C1, C2	Fundamental Domain
CLO2	Address complex computational challenges by designing algorithms, creating flowcharts, and implementing solutions in C using diverse constructs, tools, and strategies.	PLO3	C3	Fundamental Domain
CLO3	Write C programs to tackle computational and engineering-related problems across various scientific domains.	PLO2	C3	Fundamental Domain
CLO4	Collaborate effectively in teams and work independently to analyze, break down, and solve large-scale computational problems, ensuring strong teamwork and communication.	PLO4	C4	Fundamental Domain
CLO5	Transfer skills and knowledge from learning C to quickly adapt to other programming languages.	PLO12	C4	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2			√									
CLO3		√										
CLO4				√								
CLO5												√

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Outcome-Based Education (OBE), learning objectives, knowledge levels, learning domains, and assessment frameworks.	Classroom discussion, Open discussion.	None	None

	Lesson 2: Introduction to programming concepts, structured programming fundamentals, program organization, execution process, basic computational thinking, and problem-solving concepts.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes, Open discussion.	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Programming fundamentals including tokens, data types, identifiers, variables, constants, and basic syntax structures.	Brainstorming sessions, Lecture video, Lecture notes,	Class Test, Midterm exam	CLO1
	Lesson 2: Input-output operations, variable initialization, data handling, and solving basic computational problems through programming.	Brainstorming sessions, Lecture video, Lecture notes, and open discussion.	Class Test, Midterm exam	CLO2, CLO3
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Problem-solving techniques using algorithms and flowcharts, introduction to operators, and expression handling in programming.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Advanced expression evaluation, operator precedence, debugging techniques, and implementation of logical problem-solving strategies.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes	Class Test, Assignment, Midterm exam	CLO2, CLO3
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Decision-making concepts using conditional structures and implementation of logical branching techniques.	Classroom discussions, Voice over PPT, Lecture video, Lecture note	Class Test, Assignment, Midterm exam	CLO2, CLO3
	Lesson 2: Advanced conditional statements, multi-way decision structures, nested decision-making, and practical problem-solving applications.	Classroom discussions, Voice over PPT, Lecture video, Lecture note	Class Test, Assignment, Midterm exam	CLO2, CLO3
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Iterative problem-solving techniques using looping constructs and repetition-based program design.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes, Open discussion.	Class Test, Assignment, Midterm exam	CLO2, CLO3
	Lesson 2: Nested iteration, pattern generation concepts, loop control mechanisms, and structured implementation of repetitive computational tasks.	Classroom discussion, Lecture notes, Open discussion	Class Test, Assignment, Midterm exam	CLO2, CLO3
Week-6	Lesson 1: Declaration of arrays and 2D arrays, initialization	Classroom discussion, Hands-on coding, Team-	Presentation, Mid term	CLO3, CLO4

Lesson 1 & 2 [3 Hours]	techniques, implementation using arrays, debugging, and error correction practices.	based problem-solving activities, Peer collaboration, Lecture notes, Open discussion.		
	Lesson 2: Array operations including insertion, deletion, and manipulation using 1D and 2D arrays.	Collaborative coding sessions, Group tasks, Brainstorming sessions, Lecture video, Open discussion.	Presentation, Mid term	CLO3, CLO4
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to strings, reading and writing string data.	Classroom discussion, Pair programming, Team exercises, Lecture notes, Open discussion.	Class Test, Assignment, presentation Final exam	CLO3, CLO4
	Lesson 2: Character manipulation, string operations, concatenation, and determining string length through programming practices, problem-solving using pseudocode, brute force solution design.	Collaborative practice sessions, Group coding activities, Lecture video, Open discussion.	Class Test, Assignment, presentation Final exam	CLO2, CLO3, CLO4, CLO5
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: String manipulation techniques including concatenation and length evaluation.	Team discussion, Collaborative exercises, Lecture notes, Brainstorming sessions.	Class Test, Assignment, presentation Final exam, presentation	CLO3, CLO4
	Lesson 2: String comparison techniques and usage of standard string handling functions, problem-solving using pseudocode, brute force solution design.	Group-based coding activities, Peer learning, Open discussion, Lecture video.	Class Test, Assignment, presentation Final exam	CLO2, CLO3, CLO4, CLO5
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Pointer, Understanding memory address and pointer variable declaration	Hands-on collaborative coding, Classroom discussion, Peer learning.	Class Test, Assignment, presentation Final exam	CLO3, CLO4
	Lesson 2: Pointer arithmetic, relation between pointers and arrays and dereferencing concept	Team-based code review, Collaborative debugging sessions, Open discussion.	Class Test, Assignment, presentation Final exam	CLO2, CLO3, CLO4, CLO5
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to user-defined functions and their significance; components, structure, declarations, return types, and function call mechanisms; practical implementation, pseudocode, and brute-force design.	Classroom discussion, Team-based code reviews, Group activities, Collaborative debugging sessions, Peer learning, Lecture notes/videos.	Assignment, Final exam	CLO3, CLO4
	Lesson 2: Categories of user-defined functions; solving	Team coding practice, Collaborative exercises,	Assignment,	CLO2, CLO3, CLO4, CLO5

	computational problems using modularity; introduction to recursion and recursive problem-solving strategies.	Lecture video, Open discussion.	Final exam	
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to structures, comparison between arrays and structures, defining structures, and accessing structure members.	Classroom discussion, Team exercises, Collaborative problem-solving activities.	Assignment, Final exam	CLO3, CLO4
	Lesson 2: Structure initialization, operations on structures, copying structure members, and arrays of structures, problem-solving using pseudocode, brute force solution design.	Group projects, Team coding sessions, Lecture notes, Open discussion.	Assignment, Final exam	CLO2, CLO3, CLO4, CLO5
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to file handling, types of files, opening and closing files and sequential file access handling	Hands-on collaborative coding, Classroom discussion, Peer learning.	Assignment, Final exam	CLO3, CLO4
	Lesson 2: Reading and writing data from/to files using standard input/output file streams along with essential error handling and validation techniques	Team-based code review, Collaborative debugging sessions, Open discussion.	Assignment, Final exam	CLO2, CLO3, CLO4, CLO5
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry session focusing on collaborative problem-solving and professional programming practices.	Industry expert interaction, Team discussion, Interactive session.	Assignment, Final exam	CLO3, CLO4
	Lesson 2: Comprehensive review and collaborative discussion of concepts covered from Week 1 to Week 7.	Group revision activities, Peer learning, Open discussion, Collaborative Q&A session.	Assignment, Final exam	CLO3, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Interactive Lectures and Presentations, Expert Talks, Hands-On Workshops, Site Visits and Virtual Tours	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4, CLO5
	Lesson 2: Review of Final term	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note, Open discussion	None	CLO3, CLO4, CLO5

Overall Assessment Scheme

Assessment Task	CLO's					Mark (Total=100) (Avg)	PLO's					Mark (Total=70)
	CLO 1	CLO2	CLO3	CLO4	CLO5		PLO 1	PLO 2	PLO 3	PLO 4	PLO1 2	
Attendance	--	--			--	7						
Class Test 1	5(15)					(Avg)						

Class Test 2		5(15)				15						
Class Test 3			5(15)									
Assignment				5		5						
Presentation					8	8						
Midterm Exam	10	15				25	10		15	25		25
Final exam			10	20	10	40		10		20	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Yashavant Kanetkar, Let Us C, 15th Edition, BPB Publications, 2002, ISBN: 8183331637 | 8176566217

Reference Books/ Other Supplementary Readings

Balaguruswamy, E., Programming in ANSI C, 8th edition, Tata McGraw-Hill Education Private Limited, 2019, ISBN: 9789353165130

Zed A. Shaw, Learn C the Hard Way: A Clear & Direct Introduction to Modern C Programming, 1st Edition, Addison-Wesley Professional, 2015, ISBN: 9780321884923

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

CodeBlocks, Visual Studio, Online Compiler

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 122 Structured Programming Lab

Course Code: SE 122		CIE Marks: 60
ISCED: 0613-122		LF Marks: 40
Course Title: Structured Programming Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: SE 111, SE112		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

Overview of C: Basic structure of C programs, Importance of learning C programming, Execution of a C program. Constant, Variables, and Data Types: Different data types and their specifiers, declaration of variables, Assigning values to variables, taking input using scanf() function, Operators and Expressions: Arithmetic operator, Relational operator, Logical operator, Increment Decrement operator, Conditional operator, Conditional Statements: simple if statement, if...else statement, Nesting if...else statement, if...else if ladder, switch statement. Loops: while loop, do..while loop, for loop, nested for loop, continue and break, Array: Declaration and Initialization of 1D and 2D arrays. Strings: Declaration and Initialization of String variables, arithmetic operation on characters, string handling functions, User-defined Functions: Definition of user defined function, Elements of user defined functions, Category of user defined functions, Structures: Defining a structure, declaring structure variables, Accessing structure members, Structure initialization.

Rationale of the Course

This foundational course introduces structured programming paradigms essential for developing efficient, readable, and maintainable software. The curriculum emphasizes core control structures—including sequence, selection, and iteration—alongside modularization and systematic algorithm design. By mastering these concepts, students learn to construct logical execution flows, simplify debugging, and solve complex computational problems systematically. Ultimately, this course instills universal programming principles applicable across various languages and domains, providing the essential architectural groundwork necessary for students to confidently pursue advanced development methodologies and professional software engineering careers.

Course Objectives

The goal of this course is to introduce the students about the concept of structured programming.

The main objectives of this course are,

- . To analyze a problem and develop the logical thinking
- . To design, implement, and evaluate an algorithm
- . To apply knowledge of C programming in solving problems
- . To become enabled to determine errors from a code and correct it.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Analyze a given problem properly to figure out the solution.	PLO2	C4	Fundamental Domain
CLO2	Implement a solution for a given problem using proper syntax and semantics.	PLO3	C3	Fundamental Domain
CLO3	Determine the errors of a code and rewrite it correctly.	PLO4	C4	Fundamental Domain
CLO4	Write effective reports and design documentation for the any given problem	PLO10	C3, A1,P4	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2			√									
CLO3				√								
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Discussion on OBE of Structured Programming Lab, Introduction and Simple programming problems with I/O	Syllabus Walkthrough, Live Coding Demonstration	CLO1, CLO2, CLO3	Lab Performance, Lab Report
Week-2 [1.5 Hours]	Simple programming problem solving with I/O and using different operators	Guided Hands-on Coding, Peer Problem-Solving	CLO1, CLO2, CLO3, CLO4	Lab Performance, Lab Report
Week-3 [1.5 Hours]	Simple programming problem solving using conditional statements like simple if, if...else, nested if...else	Scenario-based Logic Building, Hands-on Coding	CLO1, CLO2, CLO3, CLO4	Lab Performance, Lab Report, Viva
Week-4 [1.5 Hours]	Simple programming problem solving using conditional statements like simple if, if...elseif, switch	Guided Problem Solving, Code Tracing	CLO1, CLO2, CLO3, CLO4	Lab Performance, Lab Report
Week-5 [1.5 Hours]	Simple programming Problem solving using loops i.e. while loop, do while loop, for loop, nested for loop, using of break and continue in coding etc.	Live Coding (Loops), Collaborative Practice	CLO1, CLO2, CLO3, CLO4	Lab Performance, Lab Report
Week-6 [1.5 Hours]	Simple programming problem solving using array i.e. taking input in array, accessing values of array	Visual Demonstration (Memory), Hands-on Array Practice	CLO1, CLO2, CLO3, CLO4	Lab Performance, Lab Report
Week-7 [1.5 Hours]	Simple programming problem solving using insertion, deletion, and other problems related to array.	Guided Hands-on Coding (string)	CLO1, CLO2, CLO3, CLO4	Lab Performance, Lab Report
Week-8 [1.5 Hours]	Simple programming problem solving using string i.e. reading and writing string, concatenating, comparing two strings.	Live Coding (pointer), Hands-on Practice	CLO1, CLO2, CLO3, CLO4	Lab Performance, Lab Report
Week-9 [1.5 Hours]	Programming problem solving using pointers, i.e. introduction to pointer, understanding memory addresses, pointer variable declarations, pointer arithmetic, relationship between pointers and arrays	Code Modularity Demonstration, Peer Review	CLO1, CLO2, CLO3, CLO4	Lab Performance, Lab Report, Lab Assignment
Week-10 [1.5 Hours]	Programming problem solving using user defined functions	Complex Data Visualization, Guided Implementation	CLO1, CLO2, CLO3, CLO4	Lab Performance, Lab Report, Lab Assignment

Week-11 [1.5 Hours]	Programming problem solving using structure and array of structure	Visual Diagramming (Pointers), Memory Tracing	CLO1, CLO2, CLO3, CLO4	Lab Performance
Week-12 [1.5 Hours]	Introduction to file handling, types of files, opening and closing files, sequential of file access, reading and writing data from I/O, Error handling and validation techniques.	Live Demonstration (File I/O Streams), Collaborative Practice	CLO1, CLO2, CLO3, CLO4	Lab Performance
Week-13 [1.5 Hours]	Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	CLO1, CLO2, CLO3, CLO4	Assignment, Presentation
Week-14 [1.5 Hours]	Practice programming problems from online judge platforms i.e. codeforces, URI etc	Competitive Programming Simulation, Instructor Feedback	CLO1, CLO2, CLO3, CLO4	Lab Performance

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO1	PLO2
		CLO2	PLO3
		CLO3	PLO4
		CLO4	PLO10
Lab Performance	25	CLO1	PLO2
		CLO2	PLO3
		CLO3	PLO4
Lab Final	40	CLO1	PLO2
		CLO2	PLO3
		CLO3	PLO4
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Yashavant Kanetkar, Let Us C, 15th Edition, BPB Publications, 2002, ISBN: 8183331637 | 8176566217

Reference Books/ Other Supplementary Readings

1. Zed A. Shaw, Learn C the Hard Way: A Clear & Direct Introduction to Modern C Programming, 1st Edition, Addison-Wesley Professional, 2015, ISBN: 9780321884923.

2.Lab Manual

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

CodeBlocks, Visual Studio, Online Compiler

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 123 Discrete Mathematics

Course Code: SE 123		CIE Marks: 60
ISCED: 0613-123		SEE Marks: 40
Course Title: Discrete Mathematics		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

The syllabus of the course has been divided into four core sections. These are propositional & predicate logic, set theory & probability, function & relation, and graph theory & binary tree. These four sections have been expanded with necessary subsections. The propositional & predicate logic starts with a simple and compound logical statement. After that, with the discussion and application of logical connectives and the truth table, the students learn to apply the concept to solve relevant problems. The set theory and probability contain the basic terminologies, sample space formation, and probability concepts in appropriate contexts. The function and relation section introduces different types of functions, their scope of application, and the comparison between function and relation. Finally, graph theory and binary tree cover the terminologies of graph theory, different types of graphs, and the essentials of the binary tree.

Rationale of the Course

This course provides a robust foundation in essential mathematical concepts and problem-solving techniques across four core areas. It begins with propositional and predicate logic, utilizing truth tables and connectives to build logical reasoning. Students then explore set theory and probability, mastering sample spaces and context-driven applications. The curriculum further examines the distinct properties and practical uses of functions and

relations. Finally, it introduces graph theory and binary trees, establishing key structural terminologies and principles. Ultimately, this structured approach equips learners with the analytical tools necessary to navigate and apply mathematical theories effectively across diverse domains.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of Discrete Mathematics, more specifically,

- Express a logic sentence in terms of predicates, quantifiers, and logical connectives.
- Apply the rules of inference and methods of proof including direct and indirect proof forms, proof by contradiction, and mathematical induction.
- Use tree and graph algorithms to solve problems.
- Evaluate Boolean functions and simplify expressions using the properties of Boolean algebra.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Interpret the concept and scope of application of propositional and predicate logic.	PLO2	C2	Fundamental Domain
CLO2	Apply the set theory and probability in different application related to software logic development.	PLO1	C3	Fundamental Domain
CLO3	Analyze the mathematical problems in discrete domains to understand problem Statements and plan solutions.	PLO2	C4	Fundamental Domain
CLO4	Evaluate logical solutions using graph theory and binary trees with programming concepts based on real-life examples.	PLO4	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2	√											
CLO3		√										
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
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Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Discrete Mathematics, Proposition and its condition, negative proposition.	Interactive Lecture, Conceptual Mapping	Class Test, Midterm exam	CLO1
	Lesson 2: Compound proposition, logical operators, Logic and Proofs, Logical equivalences.	Mathematical Demonstration, Truth Table Practice	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Different logics: AND, OR, Exclusive OR, Conditional, Bi-conditional logic, etc. Identifying characteristics of those logics and practicing Truth tables in class	Guided Practice, Peer Problem-Solving	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Understand logical equivalence and solve using the truth table	Mathematical Demonstration, Collaborative Board Work	Class Test, Midterm exam	CLO1, CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Use logical equivalence to solve problems	Hands-on Problem Solving, Instructor Feedback	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Learn predicate and nested quantifier syntax, apply them to different scenarios, and practice in class	Interactive Lecture, Scenario-based Application	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: The basic concept of probability	Interactive Lecture, Real-world Analogy Discussion	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Sample space derivation for any situation Class Test -1	Mathematical Demonstration, Visual Diagramming	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Problem-solving related set theory, Sets and Subsets	Guided Problem-Solving, Whiteboard Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Comparison among number sets, Venn diagrams	Visual Diagramming (Venn), Peer Review	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Set Operations.	Mathematical Demonstration, Rule-based Practice	Midterm exam	CLO1, CLO2
	Lesson 2: Practicing problems related to set theory Class Test -2	Hands-on Problem Solving Session	Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Revision before midterm	Integrative Review, Interactive Q&A	Assignment, Final exam	CLO3
	Lesson 2: Understand basic Functions	Interactive Lecture, Conceptual Mapping	Assignment, Final exam	CLO3
Week-8	Lesson 1: Different types of functions	Interactive Lecture, Conceptual Mapping	Class Test, Assignment, Final exam	CLO3

Lesson 1 & 2 [3 Hours]	Lesson 2: Application of Functions	Mathematical Demonstration, Truth Table Practice	Class Test, Assignment, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Understand Sequences and Sums and compare them with programming basics	Guided Practice, Peer Problem-Solving	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Comparative analysis between function and relation, Number Theory: The division algorithm, divisibility and the Euclidian algorithm, prime numbers, congruence, application of congruence	Mathematical Demonstration, Collaborative Board Work	Class Test, Assignment, Final exam	CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to relation	Hands-on Problem Solving, Instructor Feedback	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Application of functions and relations with real-world examples, Induction and Recursion: Mathematical Induction, Recursive Definitions and Structural induction, program correctness.	Interactive Lecture, Scenario-based Application	Class Test, Assignment, Final exam	CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Graph Theory Class Test -3	Interactive Lecture, Real-world Analogy Discussion	Assignment, Final exam	CLO3
	Lesson 2: Graph Terminology	Mathematical Demonstration, Visual Diagramming	Assignment, Final exam	CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Graph construction, Different types of graphs,	Guided Problem-Solving, Whiteboard Practice	Assignment, Final exam	CLO3
	Lesson 2: Application of graph theory, Graph theory related problem solving, transitivity, reflexivity, equivalence classes, congruence, closure of relations and functions	Visual Diagramming (Venn), Peer Review	Assignment, Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to binary tree, Different types of binary trees	Mathematical Demonstration, Rule-based Practice	Assignment, Final exam	CLO4
	Lesson 2: Application of binary trees, Different Algorithms of Graph Theory	Hands-on Problem Solving Session	Assignment, Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Application of Graph Theory, Euler and Hamilton Paths.	Integrative Review, Interactive Q&A	Assignment, Final exam	CLO4
	Lesson 2: Industry Session	Interactive Lecture, Conceptual Mapping	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO4	
Attendance	--	--		--	7				
Class Test 1	7.5	7.5			AVG 15				
Class Test 2	7.5	7.5							
Class Test 3		5	5	5					
Assignment		5			5				
Presentation				8	8				
Midterm Exam	10	15			25	15	10		25
Final exam			20	20	40		20	20	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Kenneth H. Rosen, Kamala Krithivasan, Discrete mathematics and its applications: with combinatorics and graph theory, 8th Edition, Tata McGraw-Hill Education, 2021, ISBN: 9789390727353

Reference Books/ Other Supplementary Readings

J. P. Tremblay, R. Manohar, Discrete mathematical structures with applications to computer science, 7th Edition, McGraw-Hill, 1975, ISBN: 0070651426

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 212 Software Requirements Specification & Analysis

Course Code: SE212		CIE Marks: 60
ISCED: 0613-212		SEE Marks: 40
Course Title: Software Requirements Specification & Analysis		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Basic software requirements, Requirements Engineering, Functional and Non-Functional requirements, Feasibility Study, Stakeholders for a system, design user profile, Requirements Elicitation Techniques, Requirements Representation, SRS documentation, Use Case, Requirements Modeling, Actor of a use case, Include and Extended Use Case, Case description Requirements Prioritization Techniques, Requirements Traceability Matrix, Prototyping, Usability and UX, SUS calculation.

Rationale of the Course

Software requirements engineering is the cornerstone of successful project development. This course provides a comprehensive framework for systematically eliciting, analyzing, documenting, and managing stakeholder needs throughout the software lifecycle. Students will explore feasibility studies, user-centric profiles, and the critical distinction between functional and non-functional requirements. The curriculum covers formal documentation via Software Requirements Specifications (SRS), structural modeling using use cases, and strategic resource allocation through prioritization. By integrating requirements traceability matrices, early prototyping, and standardized usability evaluation (SUS), this course equips future engineers to mitigate design risks and deliver successful, high-quality software solutions.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of data communications. More specifically,

- To learn the basic concepts of the Requirement Engineering Process.
- To learn different lifecycle models and explain their importance.
- To learn many approaches to gather requirements from users and design with proper documents.
- To learn the design requirements for developers.
- To learn the concepts and techniques in error detection and correction.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO 1	Analyse the software requirements engineering process, including functional and non-functional requirements, user needs, feasibility considerations and sustainable system practices.	PLO2	C2	Fundamental Domain
CLO 2	Apply appropriate requirements elicitation techniques to collect, organise and document stakeholder requirements in a structured Software Requirements Specification	PLO4	C2	Fundamental Domain
CLO 3	Apply appropriate software and IT tools to prepare UML-based requirement models and related SRS artefacts for a target system.	PLO5	C3	Fundamental Domain
CLO 4	Analyse software requirements, use cases and traceability relationships to determine their accuracy, completeness and consistency.	PLO2	C4	Fundamental Domain
CLO 5	Analyse the requirements of a target software system and identify sustainability requirements that address environmental, social, and long-term impacts of software solutions.	PLO7	A3	Fundamental Domain
CLO 6	Demonstrate effective teamwork by collaboratively analysing and documenting the software requirements of a target system.	PLO9	A3	Fundamental Domain
CLO 7	Apply appropriate communication techniques to document and present software requirements effectively to technical and non-technical stakeholders.	PLO10	A2	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2				√								
CLO3					√							
CLO4		√										
CLO5							√					
CLO6									√			
CLO7										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to course outline, Introduction to CLO-PLO mapping	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Overview of Software Engineering process, Importance, Different types of SDLC [Waterfall, V-model]	Interactive Lecture, Lifecycle Mapping	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Different types of SDLC [Agile, Iterative, Spiral]	Visual Presentation, Comparative Analysis	Class Test, Midterm exam	CLO1
	Lesson 2: Different types of SDLC [Agile, Iterative, Spiral]	Group Discussion, Concept Mapping	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Requirements Engineering, What are the Requirements? Types and Examples, Types of requirements, Functional requirements, Non-functional requirements.	Interactive Lecture, Real-world Case Discussion	Class Test, Assignment, Presentation, Documentation Midterm exam	CLO1, CLO5, CLO6, CLO7
	Lesson 2: User requirements, System requirements, Overview of the Feasibility study, types of feasibility, how to define FS.	Feasibility Scenario Analysis, Brainstorming	Class Test, Assignment, Presentation, Documentation Midterm exam	CLO1, CLO5, CLO6, CLO7
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Understanding User, identify the user of a system, characteristics of the user, and components that affect user and stakeholder.	Persona Mapping Exercise, Collaborative Discussion	Class Test, Assignment, Presentation, Documentation Midterm exam	CLO1, CLO5, CLO6, CLO7
	Lesson 2: Design user profile, identify implied requirements, and identify user pattern. Class Test -1	Guided Persona Design, Peer Review	Class Test, Assignment, Presentation, Documentation Midterm exam	CLO1, CLO5, CLO6, CLO7
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Understand Stakeholder, overview of elicitation technique (ET), importance and need of ET.	Interactive Lecture, Role-play Scenario	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO2, CLO5, CLO6, CLO7
	Lesson 2: Learn different elicitation techniques with implementation, advantages, and drawbacks.	Comparative Discussion, Strategy Selection	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO2, CLO5, CLO6, CLO7
Week-6 Lesson 1 & 2	Lesson 1: Learn different elicitation techniques with implementation, advantages and drawbacks.	Interactive Lecture, Case Study Analysis	Class Test, Assignment, Presentation, Documentation	CLO2, CLO5, CLO6, CLO7

[3 Hours]			Midterm exam, Final exam	
	Lesson 2: Classwork on elicitation techniques. Class Test -2	Mock Interview/Survey Design, Peer Evaluation	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO2, CLO5, CLO6, CLO7
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of Software requirement specification and raw requirements.	Interactive Lecture, Document Review	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO2, CLO5, CLO6, CLO7
	Lesson 2: Writing process of software requirement specifications.[Textbook: Chapter-6, Page (161, 168-172)]	Guided SRS Drafting, Collaborative Writing	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO3, CLO5, CLO6, CLO7
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of Use Case, symbol to design, define case name, design basic use case from SRS	Visual Diagramming (UML), Concept Mapping	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO3, CLO5, CLO6, CLO7
	Lesson 2: Types of actor, actor representation, include and extend, Generalization.	Interactive Lecture, Guided Use Case Drawing	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO3, CLO5, CLO6, CLO7
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Case description, format to write, define attribute of case description, Trigger of a case.	Step-by-Step Template Review, Hands-on Writing	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO3, CLO5, CLO6, CLO7
	Lesson 2: Main flow and alternative flow, Derive description from use case.	Scenario Walkthrough, Peer Review	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO3, CLO5, CLO6, CLO7
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Requirements Engineering Process, Overview of Requirements Prioritization Techniques [PT], Needs of PT	Interactive Lecture, Prioritization Debate	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO3, CLO5, CLO6, CLO7
	Lesson 2: Design different Prioritization Techniques, Implement Prioritization, Techniques, 100\$ test, Moscow	Moscow Method Workshop, Collaborative Allocation	Class Test, Assignment, Presentation, Documentation	CLO3, CLO5, CLO6, CLO7

			Midterm exam, Final exam	
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Requirements Validation, Importance of validation, Traceability Matrix, Requirement Traceability Matrix.	Interactive Lecture, Visual Matrix Demonstration	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO3, CLO5, CLO6, CLO7
	Lesson 2: Why RTM, Goals of Traceability Matrix, Parameters to include in Requirement Traceability Matrix, Parameters to include in Requirement Traceability Matrix, Build a Traceability Matrix. Class Test -3	Guided RTM Construction, Cross-Checking Exercise	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO3, CLO5, CLO6, CLO7
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Classwork on SRS, Use case, Case description.	Hands-on Documentation Practice, Instructor Feedback	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO2, CLO3, CLO5, CLO6, CLO7
	Lesson 2: Classwork on Prioritization, Traceability Matrix.	Interactive Workshop, Peer Problem-Solving	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO2, CLO3, CLO5, CLO6, CLO7
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Over of Prototyping, Prototyping Development, Low fidelity prototype using PowerPoint, Paper Prototyping.	Visual Demonstration, Hands-on Paper Prototyping	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO4, CLO5, CLO6, CLO7
	Lesson2: Usability and UX, System usability scale (SUS), QUIS (Questionnaire for user interaction satisfaction), SUS questionnaire, SUS result, Problem Solving	Usability Evaluation Exercise, Survey Analysis	Class Test, Assignment, Presentation, Documentation Midterm exam, Final exam	CLO4, CLO5, CLO6, CLO7
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	None	None
	Lesson 2: Review class	Integrative Review, Open Q&A	None	None

Overall Assessment Scheme

	CLO's			
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Assessment Task								Mark (Total=100)	PLO's						Mark (Total=78)
	CL O1	CL O2	CL O3	CL O4	CL O5	CL O6	CL O7		PL O2	PL O4	PL O5	PL O7	PL O9	PL O10	
Attendance	--	--	--					7							
Class Test 1	5(15)							Avg 15							
Class Test 2		5(15)													
Class Test 3			5(15)												
Assignment					5			5				5			5
Presentation						4	4	8					4	4	8
Midterm Exam	11	9	5					25	11	5	9				25
Final exam		10	23	7				40	7	23	10				40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Karl Wieggers, Software Requirements Essentials: Core Practices for Successful Business Analysis, 1st Edition, Addison-Wesley Professional, 2023, ISBN: 978-0138190286.

Reference Books/ Other Supplementary Readings

Karl Eugene Wieggers, Joy Beatty, Software Requirements, 3rd edition, New York: Microsoft Press, c2013, ISBN: 9780735679665.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 213 Digital Electronics & Logic Design

Course Code: SE 213		CIE Marks: 60
ISCED: 0714-213		SEE Marks: 40
Course Title: Digital Electronics & Logic Design		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: (If any)		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Foundation of Digital Electronics, Fundamental concepts, digital and analog quantities, advantages of digital electronics, Introduction to Number Systems, Conversion Number Systems, Expression, truth table, universal properties of NAND & NOR gates, Simplification of expression using Boolean algebra, Binary Subtraction using 1's and 2's complements, Design of Combinational Logic using TT, Boolean algebra, Adder Circuit, Half Adder, Full Adder, Parallel Adder, Introduction to K-Map, SOP, POS, MINTERM, MAXTERM, 3-4 variable K- MAP, construction of map, implementation of expression using map, grouping and simplification, Advanced Combinational Circuits, Decoder (Binary to Decimal, Octal, Hexadecimal, Advanced Combinational Circuits,

Converters (BCD to XS- 3), Introduction to Multiplexers and De- multiplexer, Design of Multiplexer Circuits, Design of De- multiplexer Circuits, Introduction to Parity, Design of Parity Generator Circuits, Design of Parity Checker Circuit to detect error during transfer or transmission of Data, Introduction to comparator circuit using complex combinational Logic, Introduction to Computer Memory, Introduction to Flip Flops, Design of J-K, S-R, Latche.

Rationale of the Course

This foundational course in Digital Electronics introduces the core principles of digital systems alongside analog quantities. Students will master number systems, universal gates, and expression simplification using Boolean algebra and Karnaugh Maps (K-Maps). The curriculum details combinational logic design—including adders, decoders, multiplexers, and parity circuits for error detection—before advancing to sequential logic. This later phase covers computer memory elements, latches, and flip-flops (J-K, S-R). Ultimately, this course builds the analytical and hardware design skills required to understand, evaluate, and construct complex digital circuits and modern computational architectures.

Course Objectives

The objective of this course is to deliver insights into digital electronics and logical designs as the circuits in the real-world works, specifically:

- To learn to represent numerical values in various number systems,
- To learn to demonstrate knowledge of logic gates, Boolean algebraic manipulation
- To learn to apply theorems and mappings for simplification of expressions
- To learn to design and implement types of decoders, encoders, multiplexers, and De-multiplexers.
- To learn to apply circuits for developing flip-flops and memory registers

Course Learning Outcomes (CLO) with Mappings

At the end of the course, students will be able to:

CLO's	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Estimate the concept of representation of numerical values in various number systems.	PLO2	C2	Fundamental Domain
CLO2	Apply the knowledge of logic gates, Boolean algebra for implementation and manipulation of various logic circuits using simplification techniques i.e. K-map.	PLO1	C3	Fundamental Domain
CLO3	Design combinational circuits for decoders, encoders, multiplexers, de-multiplexers and others complex circuits.	PLO3	C3	Fundamental Domain
CLO4	Construct sequential circuits for flip-flops, registers, and memory of computing devices.	PLO3	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2	√											
CLO3			√									
CLO4			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Correspo nding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Foundation of Digital Electronics, Fundamental concepts, digital and analog quantities, advantages of digital electronics.	Interactive Lecture, Conceptual Metaphors	Class Test, Midterm exam	CLO1
	Lesson 2: Introduction to Number Systems, Conversion between different Number Systems	Mathematical Demonstration, Guided Practice	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Digital Logic & Gates	Visual Diagramming, Hardware Components Demo	Class Test, Midterm exam	CLO1
	Lesson 2: Expression, truth table, universal properties of NAND & NOR gates	Truth Table Construction, Circuit Analysis	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Boolean Algebra, Theorems and postulates, proof of theorem.	Theorem Derivation, Step-by-Step Proofs	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Simplification of expression using Boolean algebra.	Guided Problem-Solving, Whiteboard Practice	Class Test, Assignment, Midterm exam	CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Binary Addition.	Mathematical Demonstration	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Binary Subtraction using 1's and 2's complements. Class Test -1	Guided Calculation Practice, Peer Review	Class Test, Assignment, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Design of Combinational Logic using TT, Boolean algebra.	Step-by-step Circuit Demonstration, Collaborative Design	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Adder Circuit, Half Adder, Full Adder, Parallel Adder.	Diagram Analysis, Visual Logic Mapping	Class Test, Assignment, Midterm exam, Final exam	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to K-Map, SOP, POS,	Visual Lecture, Concept Mapping	Class Test, Assignment, Midterm exam, Final exam	CLO2
	Lesson 2: Introduction to	Visual Lecture, Concept Mapping	Class Test,	CLO2

	MINTERM, MAXTERM.		Assignment, Midterm exam, Final exam	
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: 3-4 variable K- MAP, construction of map	Whiteboard K-Map Drawing, Guided Grouping Practice	Class Test, Assignment, Final exam	CLO2
	Lesson 2: implementation of expression using map, grouping and simplification	Whiteboard K-Map Drawing, Guided Grouping Practice	Class Test, Assignment, Final exam	CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Advanced Combinational Circuits, Encoder (Decimal, Octal, Hexadecimal to Binary).	Interactive Lecture, Circuit Tracing	Class Test, Assignment, Presentation, Final exam	CLO3
	Lesson 2: Advanced Combinational Circuits, Decoder (Binary to Decimal, Octal, Hexadecimal)	Diagram Analysis, Peer Problem-Solving	Class Test, Assignment, Presentation, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Advanced Combinational Circuits, Converters (BCD to XS-3)	Whiteboard Demonstration, Rule-based Practice	Class Test, Assignment, Presentation, Final exam	CLO3
	Lesson 2: Introduction to Multiplexers and Demultiplexer.	Visual Presentation, Real-world Use Case Discussion	Class Test, Assignment, Presentation, Final exam	CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Design of Multiplexer Circuits,	Collaborative Design, Circuit Simulation	Class Test, Assignment, Presentation, Final exam	CLO3
	Lesson 2: Design of De-Multiplexer Circuits.	Guided Hands-on Design	Class Test, Assignment, Presentation, Final exam	CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to comparator circuit using complex combinational Logic. Class Test -3	Visual Diagramming, Guided Practice	Assignment, Presentation, Final exam	CLO3
	Lesson 2: Multiplexer circuit for representation of Boolean expression.	Synthesis Exercise, Peer Review	Assignment, Presentation, Final exam	CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Sequential Circuit, Introduction to Latches & Flip Flops	Interactive Lecture, State Machine Visualization	Assignment, Final exam	CLO4

	Lesson 2: Design of J-K, S-R, D, T Flip flop.	Timing Diagram Analysis, Step-by-Step Demonstration	Assignment, Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Basics of Register, Designing of various types of registers i.e. SISO, SIPO, PISO, PIPO etc.	Visual Data Flow Modeling, Collaborative Architecture Design	Assignment, Final exam	CLO4
	Lesson2: Introduction to Computer Memory, Designing of PLA, RAM etc.	Conceptual Mapping, Hardware Demonstration	Assignment, Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Assignment, Presentation	CLO1 , CLO2 , CLO3 , CLO4
	Lesson 2: Review class on final syllabus topic	Integrative Review, Mock Practice Problems	None	CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO2	PLO1	PLO3	
Attendance	--	--	--		7				
Class Test 1	5(15)				Avg 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment			5		5				
Presentation		4	4		8				
Midterm Exam	10	15			25	10	15		25
Final exam		10	15	15	40		10	30	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40

Total	100
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Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Ronald J. Tocci, Neal S. Widmer, Gregory L. Moss, Digital Systems: Principles and Applications, 10th Edition, Prentice-Hall , 2013, ISBN: 9788131727249

Reference Books/ Other Supplementary Readings

Mano, M. Morris. Digital Logic & Computer Design, 4th edition, Prentice Hall, 2004, ISBN: 0-13-306551-3.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

PHY101 Physics I

Course Code: PHY101		CIE Marks: 60
ISCED: 0533-101		SEE Marks: 40
Course Title: Physics I		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Mechanics: Overview; history; classical mechanics; Newton's laws of motion; Motion in two dimensions: Projectile motion and its application; Force and friction force, Work & Energy, Momentum and Moment of inertia
 Waves and oscillations: Concepts of waves & oscillations; Simple harmonic motion: energy, composition, damped and forced vibration; types of waves, stationary and progressive wave and their applications. Optics: Concepts of the historical background of optics, classifications: Geometrical optics; reflection, refraction, total internal reflection; Physical optics: Interference, diffraction, polarization, their uses. Atom and Modern Physics: Atomic structure and atomic development, atomic nucleus and nuclear energy. Basic concepts of modern physics: Theory of relativity; Einstein's photoelectric effect, Compton effect, pair production, X-Ray. Radioactivity.

Rationale of the Course

This foundational course explores a broad spectrum of physics principles essential for scientific and technological advancement. It begins with classical mechanics, covering Newton's laws and two-dimensional projectile motion to govern force and motion. The curriculum transitions into wave phenomena and oscillations, introducing simple harmonic motion alongside geometrical and physical optics to elucidate light and sound behaviors. Concluding with modern physics, students delve into atomic structure, relativity, the photoelectric effect, and radioactivity.

Ultimately, this comprehensive syllabus builds the critical thinking and analytical skills required for rigorous scientific inquiry and practical application.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of Software Engineering. More specifically,

- To grasp the foundational concepts and development of mechanics, analyzing two-dimensional motion including projectile motion and forces such as friction.
- To explore key principles in work, energy, momentum, and moment of inertia.
- To Understand waves and oscillations, investigating simple harmonic motion and various wave types.
- To explore optics, covering historical context, geometrical optics, and physical optics.
- To examine the atomic structure, the nucleus, and the concept of nuclear energy.
- To introduce fundamental principles in modern physics, including the theory of relativity, the photoelectric effect, the Compton effect, pair production, X-rays, and radioactivity.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skills
CLO1	Recognize and list the distinctions between various terms and conditions used in physics-related topics	PLO2	C2	Fundamental Domain
CLO2	Describe how theories are applied for various practical purposes in real-world scenarios within the field of physics.	PLO2	C2	Fundamental Domain
CLO3	Apply the theories of Physics to solve real-world problems or design practical solutions for specific purposes.	PLO1	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3	√											

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: To introduce each other, OBE system and identify the course expectations, History and Scope of mechanics and its significance in science and technology.	Syllabus Walkthrough, Interactive History Discussion	None	None

	Lesson 2: Motion in One Dimension, Kinematics review, Newton's three laws of motion and their applications.	Interactive Lecture, Real-world Vector Application	Class Test, Midterm exam	CLO1, CLO2, CLO3
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Problem-solving: Practice solving problems involving motion, force, and acceleration., Visualization: Use of diagrams, animations, or online simulations to visualize concepts.	Guided Problem-Solving, Online Simulation Analysis	Class Test, Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Motion in Two Dimensions: Projectile Motion: Analyze projectile motion in two dimensions, considering factors like angle and air resistance.Applications: Discussion of real-world applications of projectile motion (e.g., sports, ballistics).	Whiteboard Derivation, Ballistics Case Discussion	Class Test, Midterm exam	CLO1, CLO2, CLO3
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Mathematical problem solving on projectile, Forces: Introduce different types of forces, including friction and its impact on motion. Work and Energy: Explain the concepts of work, kinetic energy, potential energy, and conservation of energy.	Mathematical Demonstration, Free-Body Diagramming	Class Test, Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Momentum: Introduce the concept of momentum and its conservation principle. Problem-solving: Practice solving problems involving force, work, energy, and momentum calculations, Rotational motion: Introduction of key terms of circular and rotational motion. Visualization: Use of diagrams, animations, or online simulations to visualize these concepts.	Interactive Visual Simulation, Collaborative Problem-Solving	Class Test, Assignment Midterm exam	CLO1, CLO2, CLO3
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Mathematical problem solving on rotational motion.	Hands-on Calculation Practice, Board Work	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Simple Harmonic Motion and its energy aspects, Differentiate Equation, general form. Energy Transformations are SHM (kinetic and potential). Damped and Forced Vibrations Class Test -1	Differential Equation Walkthrough, Spring-Mass Animation	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Wave: Types of Waves and Applications (2 classes), Stationary vs. Progressive Waves: Differentiate between stationary and progressive waves and their properties.	Interactive Lecture, Visual Wave Comparison	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3

	Lesson 2: Applications: Discuss applications of different types of waves (e.g., sound in communication, light in vision).	Acoustic & Optical Scenario Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Kanban Model. Usages, advantages & disadvantages of the model.	Interactive Lecture, Workflow Simulation	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Practice problems involving oscillations and wave	Guided Formula Application, Peer Review	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
	Class Test -2			
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Optics: Historical Background, Laws of Reflection and Laws of Refraction	Interactive Lecture, Optical Bench Concept Demo	Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Refractive index, Total Internal Reflection, Problems solving	Ray Diagramming, Mathematical Demonstration	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Interface design elements, component-level design elements, deployment-level design elements, Basic definitions: Coherent sources, Phase Difference and Path Difference, Interference and superposition of light, Young's double slit experiment	Visual Demonstration, Light Superposition Mapping	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
	Lesson 2: Expression for fringe width, Sustained interference, Applications and problems, Diffraction of Light, Fresnel/Franhouffer. Single slit, Fresnel distance, Grating, Applications and problems, Basics of Polarization: Characteristics of Polarized and Unpolarized Light	Mathematical Derivation, Grating Pattern Analysis	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: object-oriented software testing: class testing, behavioral testing, Halstead's complexity, Decision table	Interactive Lecture, Concept Mapping	Class Test, Final exam	CLO1, CLO2, CLO3
	Lesson 2: Polarization Techniques, Applications	Optical Bench Visualization, Industry Case Study	Class Test, Final exam	CLO1, CLO2, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Problem solving on physical optics	Collaborative Problem-Solving Session	Final exam	CLO1, CLO2, CLO3
	Lesson 2: Reliability Engineering (availability and reliability, reliability requirements, fault-tolerant architectures, programming for reliability, reliability measurement)	Interactive Lecture, System Architecture Review	Final exam	CLO1, CLO2, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Atomic Structure and Atomic Models, Development of Atomic Theory, Atomic Nucleus and Nuclear Energy, nuclear mass defect and binding energy, their calculation; isotopes, isotones and isobars, Nuclear	Concept Mapping, Binding Energy Calculation Practice	Final exam	CLO1, CLO2, CLO3

	fission and fusion and production of nuclear energy			
	Lesson 2: Introduction to Basic Concepts of Modern Physics. Einstein's special theory of relativity, Theory of Relativity: Photoelectric Effect. Compton Effect and Pair Production Class Test -3	Relativity Proof Walkthrough, Quantum Effect Discussion	Assignment, Final exam	CLO1, CLO2, CLO3
Week-12	Lesson 1: X-Rays and their uses	Visual Presentation, Real-world Medical/Industrial Cases	Assignment, Final exam	CLO1, CLO2, CLO3
Lesson 1 & 2 [3 Hours]	Lesson 2: Einstein's mass-energy equation, relative mass, length and time and their applications Radioactivity and its phenomena	Mathematical Demonstration, Decay Chain Mapping	Assignment, Final exam	CLO1, CLO2, CLO3
Week-13	Lesson 1: Applications of Radioactivity	Nuclear Application Discussion, Safety Review	Assignment, Final exam	CLO3
Lesson 1 & 2 [3 Hours]	Lesson 2: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Assignment, Presentation	CLO1, CLO2, CLO3
Week-14	Lesson 1: Presentation	Student Presentations, Peer Feedback	Presentation	CLO1, CLO2, CLO3
Lesson 1 & 2 [3 Hours]	Lesson 2: Review class on topic discussed in Previous class	Integrative Review, Open Discussion	None	None

Overall Assessment Scheme

Assessment Task	CLO's			Mark (Total=100)	PLO's		Mark (Total =70)
	CLO1	CLO2	CLO3		PLO1	PLO2	
Attendance	--	--	--	7			
Class Test 1	5(15)			AVG 15			
Class Test 2		5(15)					
Class Test 3			5(15)				
Assignment		5		5			5
Presentation				8			
Midterm Exam	5	10	10	25	10	15	25
Final exam	10	15	15	40	15	25	40

Marks Distribution

Class attendance	7
Assignment	5

Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

David Halliday, Robert Resnick, Jearl Walker, A student's companion to fundamentals of physics : fundamentals of physics, 7th edition, John Wiley, 2005., ISBN: 0471470627

Arthur Beiser, Concepts of Modern Physics, 7th Edition, McGraw-Hill Education, 2016, ISBN: 978-1259586688.

Reference Books/ Other Supplementary Readings

N. Subrahmanyam, Brij Lal, Jivan Seshan, Atomic and Nuclear Physics, 1st Edition, S. Chand & Company, 2007, ISBN: 978-8121904148

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

MATLAB

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)

Daffodil International University (DIU)
(Version 3)

MAT 102 Mathematics II

Course Code: MAT 102		CIE Marks: 60
ISCED: 0541-102		SEE Marks: 40
Course Title: Mathematics II		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: MAT 101		
Course Type: GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Linear Algebra: Basic of Matrix and Matrix Algebra, Types of Matrix, Determinant of Matrix (Higher Order), Inverse Matrix, Rank of Matrix, RREF & NF of a Matrix, System of Linear Equations, Eigenvalues and Eigenvectors, Cayley-Hamilton Theorem; Complex Variables: Basic of Complex Numbers, Modulus, Argument, Different form of Complex Number, Cauchy Riemann Equation, Laplace's Equation, Harmonic Function, Euler Theorem, De Moivre's Theorem; Fourier Analysis: Fourier series, Fourier Transformation.

Rationale of the Course

Mathematics II provides a comprehensive foundation in Linear Algebra, Complex Variables, and Fourier Analysis, emphasizing practical applications across science and engineering. The Linear Algebra section covers matrix theory and systems of linear equations, which are essential for structural modeling in physics, economics, and computing. The Complex Variables module explores the core principles of complex numbers and analytic functions. Concluding with Fourier Analysis, the course introduces Fourier transforms to decompose signals into primitive components, simplify complex integrals, and uncover hidden data structures.

Course Objectives

To provide a solid conceptual understanding of Linear Algebra, Complex Variables, and Fourier analysis. More specifically,

- To learn to solve problems using linear algebra
- Demonstrate comprehension of operations on matrices.
- To learn how to find Eigenvalues and Eigenvectors of a matrix
- To be able to use linear algebra in real-life applications.
- To learn the complex number system.
- To learn about analytic functions of complex variables.
- To learn about the Euler Theorem, and De Moivre's Theorem.
- To learn about the Fourier series & Fourier Transformation.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
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CLO1	Identify the concept of Matrix Algebra, Complex Variables and Fourier Analysis.	PLO2	C1, C2	Fundamental Domain
CLO2	Analyze the mathematical knowledge and formulation in different ways.	PLO2	C4	Fundamental Domain
CLO3	Apply acquired knowledge, facts, and techniques to solve different types of mathematical problems.	PLO1	C3	Fundamental Domain
CLO4	Explain the solution procedure of mathematical problems for development to the professional engineering practice.	PLO2	C4	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3	√											
CLO4		√										

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of the Mathematics II, Importance of Math II in SWE, Basic Concept, Different Types of Matrix.	Interactive Lecture	Class Test, Midterm exam	CLO1
	Lesson 2: Different types of Matrix	Whiteboard Demonstration	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Determinant of Matrix	Collaborative Problem-Solving	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Determinant of Matrix, Cofactor, Minor	Guided Peer Practice	Class Test, Midterm exam	CLO1, CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Finding Inverse.	Step-by-Step Algorithm Walkthrough	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Rank, REF, RREF, NF	Structured Calculation Drills	Class Test, Assignment, Midterm exam	CLO3
Week-4	Lesson 1: Rank, REF, RREF, NF	Peer-Group Practice Exercises	Class Test, Assignment, Midterm exam	CLO3

Lesson 1 & 2 [3 Hours]	Lesson 2: SLE, Canonical method for Inverse matrix. Class Test -1	Problem-Solving Workshop	Class Test, Assignment, Midterm exam	CLO2, CLO3
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Higher dimension matrix manipulation.	Guided Calculation Inquiry	Midterm exam	CLO2, CLO3
	Lesson 2: Eigenvalues and Eigenvectors.	Whiteboard Derivation	Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Eigenvalues and Eigenvectors.	Targeted Practice Drills	Class Test, Assignment, Final exam	CLO1, CLO2
	Lesson 2: Cayley-Hamilton Theorem. Class Test -2	Case Study Application	Class Test, Assignment, Final exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Basic Concept of Complex number, Modulus, Argument.	Visual Mapping	Class Test, Assignment, Final exam	CLO1
	Lesson 2: Different forms of Complex numbers, Complex variables.	Comparative Problem Modeling	Class Test, Assignment, Final exam	CLO1
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Cauchy Riemann Equation, Euler Theorem, De Moivre's Theorem.	Concept Mapping	Class Test, Assignment, Final exam	CLO2
	Lesson 2: Cauchy Riemann Equation, Euler Theorem, De Moivre's Theorem, Laplace Equation, Harmonic Function	Integrated Problem Solving	Class Test, Assignment, Final exam	CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Laplace Equation, Harmonic Function Class Test -3	Analytical Discussion	Assignment, Final exam	CLO2
	Lesson 2: Review class on topic discussed in Week-10 – Week-12	Q&A Session	Assignment, Final exam	None
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Basic Concept of Fourier Analysis.	Conceptual Discussion	Assignment, Final exam	CLO1
	Lesson 2: Fourier Series (Basic Formula Discussion).	Pattern Analysis	Assignment, Final exam	CLO1
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1:: Fourier Series	Collaborative Problem Solving	Assignment, Final exam	CLO3
	Lesson 2: Fourier Series Class Test -3	Applied Exercise	Assignment, Final exam, Presentation	CLO1, CLO3
Week-12	Lesson 1:: Fourier Transform	Signal Processing Logic Demo	Assignment, Final exam, Presentation	CLO3

Lesson 1 & 2 [3 Hours]	Lesson 1:: Fourier Transform	Applied Problem Solving	Assignment, Final exam, Presentation	CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Problem solving on topic discussed in Week-11 – Week-16	Focused Review Session	None	None
	Lesson 2: Industry Session	Industry Interaction & Q&A	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Review class on topic discussed in Week-11 – Week-16	Q&A Session	None	None
	Lesson 2: Review class on a topic discussed in Week-11 – Week-16.	Targeted Remediation	None	None

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's		Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	
Attendance	--	--	--	--	7			
Class Test 1	5	5	7		15			
Class Test 2	5	5	7					
Class Test 3	5	5	7					
Assignment			5		5		5	5
Presentation					8			
Midterm Exam	5	3	12	5	25	12	13	25
Final exam	8	10	12	10	40	12	28	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Mohammad Salek Parvez, Linear Algebra: A First Course, 1st Edition, CreateSpace Independent Publishing Platform, 2018, ISBN: 978-1985593585.

Reference Books/ Other Supplementary Readings

Howard Anton, Chris Rorres, Elementary Linear Algebra, 12th Edition, Wiley, 2019, ISBN: 978-1119379560.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 131 Data Structure

Course Code: SE 131		CIE Marks: 60
ISCED: 0612-131		SEE Marks: 40
Course Title: Data Structure		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)		Contact Hours: 3 (Total weeks: 14)
Prerequisite: SE121, SE122, SE123		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Overview of Data Structures, Algorithm and Complexity, Time Space Tradeoff, Recursion, Iteration, Record, Pointer, Implementation of Memory, Array: Traverse, Insert, Insert at any position, Delete at any position, Linear Search, Linear Search complexity, Binary Search, Binary Search Complexity, Searching- Bubble Sort, Factorial and Tower of Hanoi Problem; Marge, Stack: Stack data Insertion, Stack Deletion, Stack : Search, Prefix, Infix and Postfix Expressions, Queue: Queue data Insertion, Queue Deletion, Queue: Search, Double Ended Queue, Priority Queue, Hashing: Hash Indices and Hash Functions, Static Linked List: List ADTs, Linked List data Insertion, Deletion, Search, Double Way Linked List: data Insertion, Deletion, Search, Circular Way Linked List: data Insertion, Deletion, Search, Tree, Tree terminologies, Binary Search Tree: Insertion, Deletion, Search, AVL Tree, Heaps, Heap Sort, Graph, Graph Terminologies, Adjacency matrix, Adjacency List, Graph data Insertion, Graph Deletion, Graph: Search.

Rationale of the Course

The course provides a systematic exploration of foundational computer science concepts, bridging algorithmic thinking with efficient data organization. Students gain hands-on experience starting with array manipulation and search algorithms before mastering linear structures like stacks, queues, and linked lists. The curriculum then advances to non-linear structures, including trees, graphs, and hashing techniques, alongside essential sorting algorithms. Crucially, the course emphasizes time and space complexity analysis, empowering students to make optimized architectural decisions. Ultimately, this course builds the rigorous problem-solving skills required to tackle complex programming challenges confidently.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of data Structure, More specifically,

- To provide the knowledge of basic data structures and their implementations.
- To understand the importance of data structures in the context of writing efficient programs.
- To develop skills to apply appropriate data structures in problem-solving.
- To learn how data are stored in our computer memory
- Be familiar with writing recursive methods

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Describe the basics of Data Structure	PLO1	C1	Fundamental Domain
CLO2	Interpret the performance and complexity of various data structure	PLO1	C2	Fundamental Domain
CLO3	Apply the static and dynamic data structure, linear data structure	PLO2	C3	Fundamental Domain
CLO4	Apply the concept of non-linear data structure	PLO2	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2	√											
CLO3		√										
CLO4		√										

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
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Lesson (hour)				
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Data Structure	Interactive Concept Mapping; Guided Discussion	Class Test, Midterm exam	CLO1
	Lesson 2: Data Structure main concept, basic operations, Algorithm, complexity.	Whiteboard Derivation; Comparative Analysis	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Time Space Trade-off, Recursion, Iteration.	Code Tracing Exercise; Step-by-Step Logic Walkthrough	Class Test, Midterm exam	CLO1
	Lesson 2: Iteration. Recursion, Array – Traverse, Insert, Insert at any position.	Hands-on Coding Drill; Algorithm Visualizing	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Array Deletion Operation	Algorithmic Flowcharting; Dry-Run Exercises	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Linear Search & Binary Search with complexity	Comparative Algorithm Efficiency Demo; Speed Drills	Class Test, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Sorting- Bubble Sort with complexity	Peer-Led Logic Demonstration; Complexity Analysis	Class Test, Midterm exam	CLO2
	Lesson 2: Prefix, Infix, Operation Class Test -1	Interactive Notation Conversion Drills	Class Test, Assignment, Midterm exam	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Postfix Operation.	Stack-Based Evaluation Workshop	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Review class on topic discussed in Week 1- Week 5, Presentation on a given topic.	Student-Led Mini Presentations; Targeted Q&A	Presentation, Open discussion	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Stack – Stack data Insertion, Stack Deletion, Stack – Search.	Simulation of LIFO Logic; Implementation Practice	Class Test, Midterm exam	CLO1, CLO3
	Lesson 2: Static Linked List – List ADTs, Linked List data Insertion. Class Test -2	Linked Structure Visualization; Pointer Manipulation Drill	Class Test, Midterm exam	CLO1, CLO3
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Linked List Deletion, Linked List – Search.	Pointer Re-linking Exercises; Debugging Workshop	Assignment, Midterm exam, Final exam	CLO1, CLO3
	Lesson 2: Double Way Linked List – Double Way Linked List data Insertion.	Visual Diagramming; Node Insertion Practice	Assignment, Midterm exam, Final exam	CLO1, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Circular Way Linked List Deletion, Circular Way Linked List – Search, Insertion.	Structural Comparison Drill; Circular Logic Simulation	Assignment, Midterm exam Final exam	CLO1, CLO3
	Lesson 2: Concept of Queue, Queue Data Storage in Memory.	Conceptual Mapping; Memory Layout Sketching	Class Test, Midterm exam	CLO1, CLO3

			Final exam	
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Queue Traversing	Algorithmic Step-by-Step Tracing	Class Test, Midterm exam Final exam	CLO1, CLO3
	Lesson 2: Queue insertion and deletion.	Hands-on Implementation; FIFO Simulation	Class Test, Midterm exam Final exam	CLO1, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Searching in Queue.	Comparative Efficiency Debate; Logic Walkthrough	Class Test, Midterm exam Final exam	CLO1, CLO3
	Lesson 2: Queue – Double Ended Queue, Priority Queue.	Case-Based Scenario Solving	Class Test, Midterm exam Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Concept of Heap, Heap Sort.	Hierarchical Structure Building; Sort Simulation	Class Test, Final exam	CLO1, CLO3
	Lesson 2: Tree- Concept of Tree, Binary Search Tree. Class Test -3	Tree Drawing Workshop; Recursive Logic Practice	Class Test, Final exam	CLO1, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Binary Search Tree – Tree data Insertion, Tree Deletion, Tree – Search, AVL	Tree Balancing Exercises; Insertion/Deletion Drills	Assignment, Final exam	CLO1, CLO4
	Lesson 2: Hashing - Hash Indices and Hash Functions.	Collision Resolution Simulation	Assignment, Final exam	CLO1, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Concept and Implementation of Recursion.	Code Pattern Recognition; Problem Solving	Assignment, Final exam	CLO1, CLO4
	Lesson 2: Graph, Graph Terminologies, Adjacency matrix, Adjacency List, data Insertion, Graph Deletion.	Matrix vs. List Modeling Exercise	Assignment, Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Professional Q&A; Applied Scenario Discussion	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on Final Syllabus Topic	Problem-Set Challenge; Targeted Remediation	Presentation, Open Discussion	CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's		Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	
Attendance	--	--	--		7			
Class Test 1	5(15)				Average 15			
Class Test 2		5(15)						
Class Test 3			5(15)					
Assignment		5			5		5	5
Presentation					8			
Midterm Exam	4	4	17		25	8	17	25

Final exam			20	20	40		40	40
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Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Behrouz A. Forouzan and Richard F. Gilberg, Computer Science: A Structured Programming Approach in C, 3rd Edition, Cengage Learning, 2007, ISBN : 9788131503638

Reference Books/ Other Supplementary Readings

Aaron M. Tenenbaum, Yedidiah Langsam, Moshe J. Augenstein, Data structures using C, Prentice Hall, c1990., ISBN: 0131997467

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 132 Data Structure Lab

Course Code: SE 132		CIE Marks: 60
ISCED: 0613-132		LF Marks: 40
Course Title: Data Structure Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: SE121, SE122		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

Array – Traverse, Insert, Insert at any position, Delete at any position, Linear Search, Linear Search complexity, Binary Search, Searching- Bubble Sort, Stack – Stack data Insertion, Stack Deletion, Stack – Search, Prefix, Infix and Postfix Expressions, Queue – Queue data Insertion, Queue Deletion, Queue – Search, Double Ended Queue, Priority Queue, Static Linked List – List ADTs, Linked List data Insertion, Linked List Deletion, Linked List – Search, Double Way Linked List – Double Way Linked List data Insertion, Double Way Linked List Deletion, Double Way Linked List – Search, Circular Way Linked List – Circular Way Linked List data Insertion, Circular Way Linked List Deletion, Circular Way Linked List – Search, Tree, Binary Search Tree – Tree data Insertion, Tree Deletion, Tree – Search, AVL Tree, Heaps, Heap Sort, Graph, Graph Terminologies, Adjacency matrix, Adjacency List, Graph data Insertion, Graph Deletion, Graph – Search.

Rationale of the Course

The course on data structures aims to equip students with the fundamental concepts and practical skills necessary for efficient data organization and retrieval. Covering a diverse range of topics including arrays, stacks, queues, linked lists, trees, heaps, and graphs, the course offers a comprehensive understanding of various data structures and their associated operations. By exploring algorithms for insertion, deletion, searching, and sorting, students learn to analyze and implement solutions to computational problems effectively. Through hands-on exercises and theoretical discussions, the course prepares students to make informed decisions regarding the selection and utilization of data structures in software development, fostering their ability to design and optimize algorithms for real-world applications.

Course Objectives

The three main objectives of the Data Structure Lab Course are:

- **Hands-on Implementation:** Enable students to gain practical experience in implementing various data structures and algorithms studied in theoretical courses, thereby reinforcing their understanding of fundamental concepts through practical application.

- **Algorithmic Analysis and Optimization:** Develop students' ability to analyze the time and space complexities of different data structures and algorithms, fostering critical thinking skills and enabling them to optimize solutions for improved performance.
- **Problem-Solving Proficiency:** Enhance students' problem-solving skills by challenging them to apply appropriate data structures and algorithms to solve diverse computational problems, preparing them to tackle real-world challenges in software development and beyond.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Apply the knowledge of linear and nonlinear data structure to solve the problems related to computer programming	PLO1	C3	Fundamental Domain
CLO2	Analyze computer programming related problems and reach substantiated conclusions using the principles of data structure operations.	PLO2	C4, A5	Fundamental Domain
CLO3	Apply appropriate techniques for constructing the programming solution.	PLO5	C3, P2	Fundamental Domain
CLO4	Write effective reports and design documentation for the data structure applications	PLO10	C3, A1, P4	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3					√							
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Array: Concept, Library Functions, User Defined Functions, Iterations, Array Traverse, Insert, Delete	Live Coding Demonstration; Guided Algorithm Implementation	CLO1	Lab Report, Lab Final
Week-2 [1.5 Hours]	Array Linear Search, Binary Search, Bubble Sort	Algorithmic Tracing; Debugging & Complexity Analysis	CLO1, CLO4, CLO3, CLO2	Lab Report, Lab Assignment, Lab Final

Week-3 [1.5 Hours]	Recursion	Visual Stack Tracing; Recursive Logic Practice	CLO1, CLO4, CLO3, CLO2	Lab Report, Lab Assignment, Lab Final
Week-4 [1.5 Hours]	Dynamic Memory Allocation	Memory Allocation Exercises; Pointer Manipulation Practice	CLO1, CLO4, CLO2	Lab Report, Lab Assignment, Lab Final
Week-5 [1.5 Hours]	Linked List: Linked List creation Singly Linked List traverse, insert, delete, search and Singly Linked List related problem solving	Node Structure Visualization; Pointer Re-linking Drills	CLO1, CLO4	Lab Report, Lab Assignment,
Week-6 [1.5 Hours]	Doubly Linked list & Circular Linked List creation, traverse, insert, delete, search, Doubly and Circular Linked List related problem solving	Structural Logic Mapping; Complex Pointer Re-linking Exercises	CLO1, CLO4, CLO2	Lab Report, Lab Assignment, Lab Final
Week-7 [1.5 Hours]	Stack: Stack creation by using array, Stack traverse, insert, delete, Stack related problem solving	LIFO Simulation; Implementation of Stack-based Algorithms	CLO1, CLO4	Lab Report
Week-8 [1.5 Hours]	Queue: Queue construction, insertion, deletion, search, Priority Queue, Queue related problem solving	FIFO/Priority Logic Coding; Queue Implementation Exercise	CLO1, CLO4, CLO3, CLO2	Lab Report, Lab Assignment, Presentation, Lab Final
Week-9 [1.5 Hours]	Concept of Binary Search Tree, BST creation, traverse, insert, delete, search	Tree Visualization ; Recursive Traversal Coding	CLO1, CLO4, CLO3, CLO2	Lab Report, Lab Assignment, Presentation, Lab Final
Week-10 [1.5 Hours]	Heap	Self-Balancing Simulation; Heap Property Verification Drills	CLO1, CLO4, CLO2	Lab Report, Lab Assignment, Presentation, Lab Final
Week-11 [1.5 Hours]	Hashing	Collision Resolution Simulation; Hash Function Implementation	CLO1, CLO4, CLO2	Lab Report, Lab Assignment, Presentation, Lab Final
Week-12 [1.5 Hours]	Graph: Graph creation, Graph Traverse, insert, delete, Graph related problem solving	Adjacency Representation; Graph Traversal Implementation	CLO1, CLO2, CLO3	Assignment, Presentation
Week-13 [1.5 Hours]	Industry Session	Applied Problem-Solving Workshop; Expert Scenario Analysis	CLO1, CLO2, CLO3, CLO4	Presentation
Week-14	Lab Final	Independent Implementation & Performance Assessment	None	None

[1.5 Hours]				
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Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO1
Lab Final	40	CLO2	PLO2
		CLO3	PLO5
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Behrouz A. Forouzan and Richard F. Gilberg, Computer Science: A Structured Programming Approach in C, 3rd Edition, Cengage Learning, 2007, ISBN : 9788131503638

Reference Books/ Other Supplementary Readings

Aaron M. Tenenbaum, Yedidyah Langsam, Moshe J. Augenstein, Data structures using C, Prentice Hall, c1990., ISBN: 0131997467

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

CodeBlocks, Visual Studio, Online C Compiler

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 133 Software Development Capstone Project

Course Code: SE 133	CIE Marks: 60
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ISCED: 0613-133		SEE Marks: 40
Course Title: Software Development Capstone Project		Total Marks: 100
Semester: Spring 2025		
Credit Value: 3 (Project)	Contact Hours: 4.5 (Total week : As per BNQF)	
Prerequisite: SE121, SE122		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

The “Software Development Capstone Project” course provides hands-on experience and practical application of key programming concepts using structured programming language. Students will be guided through a systematic approach to develop a project, emphasizing problem-solving skills and logical thinking. The course covers a range of topics from problem formulation to the evaluation of solutions, ensuring that students acquire a comprehensive understanding of the software development process. The course encompasses fundamental programming elements, placing particular emphasis on data types, variables, condition statements, loop control structures, switch cases, functions, arrays, pointers, file operations, and string manipulation. These core concepts form the backbone of the curriculum, providing students with a comprehensive understanding of essential programming principles.

Rationale of the Course

The "Software Development Capstone Project" course provides an immersive, hands-on experience in applying structured programming concepts to solve tangible problems. Moving systematically from problem formulation to solution evaluation, students cultivate critical analysis, logical thinking, and end-to-end project development skills. The curriculum reinforces foundational programming mechanics, thoroughly utilizing control structures, functions, arrays, pointers, string manipulation, and file operations. By bridging theoretical principles with active project execution, this capstone course equips learners with a robust, industry-ready skill set, ensuring they can confidently analyze, design, and implement software solutions to address complex, real-world challenges.

Course Objectives

The goal of this course is to introduce the students to the development process of a project using structured programming language. The main objectives of this course are-

- To develop a course project from scratch using the C programming language.
- Learning advanced C programming and multi-dimensional usage of that.
- To write code for controlling processes and threads in Modern environment.
- To formulate the requirement specification, through design of the logical solution and implementation.
- To learn the evaluation of the solution with documentation and writing a report for the course project.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
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CLO1	Integrate knowledge and skill for identifying a real life complex engineering problem as the capstone project.	PLO1	C2, C4	Fundamental Domain
CLO2	Analyze various aspects of the objectives for designing a solution of the capstone project.	PLO2	C3	Fundamental Domain
CLO3	Design solutions for the capstone project that meet safety, cultural, societal, and environmental considerations.	PLO3	C2, C3	Fundamental Domain
CLO4	Examine student's participation both individually and within the team context throughout the project implementation process.	PLO9	A2	Social, Thinking, Personal Domain
CLO5	Present the implemented project through a well-structured final documentation.	PLO10	A1, C3	Social, Thinking, Personal Domain
CLO6	Demonstrate the engineering knowledge by showing the outcomes of the implemented project as a team member or a leader.	PLO11	A2,C3	Fundamental, Social, Thinking, Personal Domain

Knowledge Profile, Complex Engineering Problem & Activities mapping

Criteria	Mapping
Knowledge Profile	K3, K4, K6
Complex Engineering Problem	CEP1, CEP3, CEP6
Complex engineering activities	CEA1

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3			√									
CLO4									√			
CLO5										√		
CLO6											√	

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Introduction class ,Explain course outline, CLO, PLO, Explain objectives, assessment plan, Complex Engineering Problem	Live IDE Configuration; Guided Syntax Diagnostic Testing	-	
	Lesson 2: Review of C programming and showing installation of different IDE, discuss about different real life complex engineering problems and probable solution	Algorithmic Tracing; Code Dry-Run Workshops	-	
	Lesson 3:Review Data types, Operators, Conditional Statements, basic Loops	Array Manipulation Labs; Real-time Efficiency Debugging	Class Performance, Group Activities, Final Project Demonstration	CLO2, CLO3, CLO4, CLO6
Week-2 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Give Problem List about Condition and Loops as Class Work	String Manipulation Drills; Memory Buffer Practice	-	
	Lesson 2: Give Feedback to the given tasks and Review the Solution	Requirement Engineering Scenarios; Peer-Design Reviews	-	
	Lesson 3:Review Nested Loops	Modular Code Refactoring; Interface Design Exercises	Class Performance, Group Activities, Final Project Demonstration	CLO2, CLO3, CLO4, CLO6
Week-3 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Give Problem List about Various Loops as Class Work	Collaborative Integration Labs; Proposal Defense Coaching	-	
	Lesson 2: Give Feedback to the given tasks and Review the Solution	Iterative Pitch & Feedback Cycles; Milestone Definition	-	
	Lesson 3: Teach the concept of 1D, 2D Arrays	Memory-Safe Coding Workshops; Pointer-Refinement Sessions	Class Performance, Group Activities, Final Project Demonstration	CLO2, CLO3, CLO4, CLO6
Week-4 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Give Problem List from different Arrays as Class Work	Stream Manipulation Labs; Data Persistence Simulations	-	
	Lesson 2: Give Feedback to the given tasks and Review the Solution	Complex Data Handling Integration; Error-Handling Drills	-	
	Lesson 3: Teach the core concept of String	Professional QA Testing; Technical Writing Workshops	Class Performance, Group Activities, Final Project Demonstration	CLO2, CLO3, CLO4, CLO6
Week-5 Lesson 1,2 & 3	Lesson 1: Teach the students process of problem identifying and requirement analysis	Simulated Industry Demo; Critical Peer Feedback Sessions	Group Activities	CLO4

[4.5 Hours]	Lesson 2: Assign Groups and give task in group to identify a problem, Analyze requirements and solve the problem Group Activities	Post-Project Reflective Analysis; Industry Competency Review	Group Activities	CLO4
	Lesson 3: Teach the concept of Structure	Live IDE Configuration; Guided Syntax Diagnostic Testing	-	
Week-6 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Give Problem List from Condition, Loops, Array, Structure and String Class Performance 1	Algorithmic Tracing; Code Dry-Run Workshops	Class Performance	CLO2, CLO3
	Lesson 2: Give Feedback to the given tasks and Review the Solution	Array Manipulation Labs; Real-time Efficiency Debugging	-	
	Lesson 3: Teach User Defined Function	String Manipulation Drills; Memory Buffer Practice	Class Performance, Group Activities, Final Project Demonstration	CLO2, CLO3, CLO4, CLO6
Week-7 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Give Problem List that needs to be solved using Function	Requirement Engineering Scenarios; Peer-Design Reviews	-	
	Lesson 2: Give Feedback to the given tasks and Review the Solution	Modular Code Refactoring; Interface Design Exercises	-	
	Lesson 3: Discuss about the work process of a real life project in industry and Teach how to prepare and present the project proposal	Collaborative Integration Labs; Proposal Defense Coaching	Viva, Project Presentation	CLO1, CLO4, CLO5
Week-8 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Present Group wise Project Proposal	Iterative Pitch & Feedback Cycles; Milestone Definition	-	
	Lesson 2: Present Group wise Project Proposal	Memory-Safe Coding Workshops; Pointer-Refinement Sessions	-	
	Lesson 3: Give Feedback on students group wise project requirements	Stream Manipulation Labs; Data Persistence Simulations	-	
Week-9 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Engaging the students in group work to design and develop the project step by step	Complex Data Handling Integration; Error-Handling Drills	-	
	Lesson 2: Engaging the students in group work to design and develop the project step by step	Professional QA Testing; Technical Writing Workshops	-	
	Lesson 3: Lesson on utilizing pointers	Simulated Industry Demo; Critical Peer Feedback Sessions	Class Performance, Group Activities, Final Project Demonstration	CLO2, CLO3, CLO4, CLO6

Week-10 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Give Problem List from previous all topics to solve the problems Class Performance 2	Post-Project Reflective Analysis; Industry Competency Review	Class Performance	CLO2
	Lesson 2: Give Feedback to the given tasks and Review the Solution	Live IDE Configuration; Guided Syntax Diagnostic Testing	-	
	Lesson 3: Lesson on File Handling (File reading and writing)	Algorithmic Tracing; Code Dry-Run Workshops	Class Performance, Group Activities, Final Project Demonstration	CLO2, CLO3, CLO4, CLO6
Week-11 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Take update of their group Project work	Array Manipulation Labs; Real-time Efficiency Debugging	-	
	Lesson 2: Discuss various problems that students faces during Project Implementation	String Manipulation Drills; Memory Buffer Practice	-	
	Lesson on File Handling (Insertion and deletion of data in file)	Requirement Engineering Scenarios; Peer-Design Reviews	Class Performance, Group Activities, Final Project Demonstration	CLO2, CLO3, CLO4, CLO6
Week-12 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Give Problem List from previous all topics to solve the problems Class Performance 3	Modular Code Refactoring; Interface Design Exercises	Class Performance	CLO2
	Lesson 2: Give Feedback to the given tasks and Review the Solution	Collaborative Integration Labs; Proposal Defense Coaching	-	
	Lesson 3: Explain project demonstration process, Provide sample project documents	Iterative Pitch & Feedback Cycles; Milestone Definition	Viva, Project Presentation, Final Documentation	CLO1, CLO4, CLO5
Week-13 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Final project demonstration showing and group presentation	Memory-Safe Coding Workshops; Pointer-Refinement Sessions	Project Presentation	CLO4, CLO5
	Lesson 2: Final project demonstration showing and group presentation	Stream Manipulation Labs; Data Persistence Simulations		
	Lesson 3: Industry Session	Complex Data Handling Integration; Error-Handling Drills	-	
Week-14 Lesson 1,2 & 3 [4.5 Hours]	Lesson 1: Discussion about preparing Final Document for the implemented project	Professional QA Testing; Technical Writing Workshops	Final Documentation	CLO4, CLO5
	Lesson 2: Engaging the students to prepare their Group-wise Final Documentation	Simulated Industry Demo; Critical Peer Feedback Sessions		
	Lesson 3: Review class on students learning about real life project and complex engineering problem	Post-Project Reflective Analysis;	-	

		Industry Competency Review		
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Overall Assessment Scheme

Criteria	Assigned Marks	Sub Criteria	Sub Criteria Marks	CLO	PLO
Attendance	10				
Lab Report	25	Viva	10	CLO1	PLO1
		Final Documentation	15	CLO5	PLO10
Lab Performance	25	Class Performance	15	CLO2	PLO2
		Group Activities	10	CLO4	PLO9
Lab Final	40	Project Demonstration	25	CLO3, CLO6	PLO3, PLO11
		Project Presentation	15	CLO4, CLO5	PLO9, PLO10
Total	100		90		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Daniel Gookin, Tiny C Projects, 1st Edition, Manning Publications, December 2022, ISBN 978-1633439825.

Reference Books/ Other Supplementary Readings

1. Balaguruswamy, E. , Programming in ANSI C, 8th edition, Tata McGraw-Hill Education Private Limited, 2019, ISBN: 9789353165130
2. Yashavant Kanetkar, Let Us C, 15th Edition, BPB Publications, 2002, ISBN: 8183331637 | 8176566217

3.Lab Manual

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

CodeBlocks, Visual Studio Code, Online compiler

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 216 Object Oriented Programming

Course Code: SE 216		CIE Marks: 60
ISCED: 0613-216		SEE Marks: 40
Course Title: Object Oriented Programming		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE 121, SE122		
Course Type: Core		

CIE: Continuous Internal Evaluation,

SEE: Semester End Examination

Course Content (from syllabus)

Basic Concepts of Object-Oriented Programming, Classes and Objects, Characteristics of Objects, Relationships and Methods, Advantages of OOP over structured programming, Access specifiers, Static and non-static members, Constructors and Its types, Destructors, Array of objects, object pointers, and object references,

Abstraction, Encapsulation, Inheritance, Polymorphism, Abstract classes and methods, Abstract methods, Abstract layer, Basic principles of Interfaces, Extending Interfaces, Multiple Interfaces, Interface Implementation, Definition of Refactoring, Different Types of Code Smells, Refactoring Techniques, Functions and Modules, Library like List, set, stack, queue, Tuple, Dictionary etc., Exception Handling, File Handling, Software Engineering Life Cycle, Modeling Problem and Solution, Moving to Code, Internal Component Definition, Design for Reuse, Design Class Diagrams, Iterating the Design, Refactoring, Design Best Practices, Cohesion, Complexity, Coupling, Congruence, Singleton pattern, Factory pattern, Observer pattern, Strategy pattern, Use Case Model, Use Case Diagram, Use Case Description, Main Success Scenario and Alternate Paths, Use Case Relationships (Generalization, <<extends>>, <<includes>>), Extension Points and Packages, Overview and Basic Concepts, Classifiers and Well-Formedness Rules, Basic Notation, Classes, Objects, Relationships, and Methods, UML Diagrams.

Rationale of the Course

This Object-Oriented Programming (OOP) course provides a comprehensive journey from foundational principles to advanced methodologies for modern software development. Students explore classes, objects, and relationships before mastering abstraction, encapsulation, and polymorphism. The curriculum covers efficient code management via access specifiers, constructors, and destructors, alongside critical design patterns like Singleton and Observer. Transitioning into the software engineering lifecycle, the course highlights design for reuse, use case modeling, and UML diagrams. Ultimately, combining OOP paradigms with software design principles empowers students to architect and deploy robust, maintainable, industry-standard software solutions.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of object-oriented programming, More specifically,

- To understand key OOP concepts, including classes, objects, and advanced topics like Abstraction and Inheritance
- To learn effective code structuring with access specifiers, constructors, and advanced features like static members and object pointers.
- To apply Singleton, Factory, and Observer patterns. Understand refactoring for code improvement.
- To navigate the Software Engineering Life Cycle, focusing on modeling, internal components, and design principles.
- To gain a concise understanding of UML diagrams and Use Case Modeling for streamlined software development.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Describe the understanding of writing object-oriented programs that combine functions and data.	PLO2	C2	Fundamental Domain
CLO2	Explain the four pillars of OOP	PLO2	C2	Fundamental Domain
CLO3	Apply the object-oriented programming concept to develop software, including programs utilizing multiple classes.	PLO1	C3	Fundamental Domain

CLO4	Design and develop software using best practices in Object oriented programming	PLO3	C3	Fundamental Domain
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Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3	√											
CLO4			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Object Oriented Programming	Students analyze a real-world software requirement document to map CLOs to project outcomes.	-	CLO1
	Lesson 2: Introduction and Overview Principle of Software Engineering based on OOP. Principle of Code Reusing and Extending Code	Students take legacy non-OOP code and rewrite sections using Encapsulation and Inheritance.	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Characteristics of Objects, Relationships and Methods	Students draw relationship diagrams for a Library System, differentiating between Association and Composition.	Class Test, Midterm exam	CLO1
	Lesson 2: Strings. Input and Output, Access Specifier	Students identify and correct intentionally broken code snippets containing incorrect access modifiers.	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Static, Non-Static Member	Students manually trace the execution order of static versus instance members on the whiteboard.	Class Test, Midterm exam	CLO1, CLO2

	Lesson 2: Constructor and Destructor	Students implement custom constructors and destructors to track memory allocation during object lifecycles.	Class Test, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Array of objects Object pointers, and Object references	Students read a student data CSV file and map each row into an array of Student objects.	Class Test, Midterm exam	CLO2
	Lesson 2: State and behavior of the object. Class Test -1	Students define a Bank Account object, listing all its valid states and corresponding transition behaviors.	Class Test, Midterm exam	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Abstraction: Abstract classes and methods, Abstract methods, Abstract layer, Basic principles of Interfaces, Extending Interfaces, Multiple Interfaces, Interface Implementation	Students convert a set of high-level requirements into a base class and interface skeleton.	Class Test, Midterm exam	CLO2
	Lesson 2: Encapsulation: Introduction to Encapsulation, Data Hiding, Access Modifiers, Encapsulation in Classes, Getters and Setters, Encapsulation in Real-world Scenarios, Benefits and Importance, Encapsulation in Software Architecture.	Students implement data attributes as private and manage them strictly via public getters and setters.	Class Test, Midterm exam	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Inheritance, Types of Inheritance, Base and Derived Classes, Inheriting Attributes and Methods, Method Overriding, Abstract Classes, Interfaces in Inheritance.	Students refactor a flat class structure into a hierarchical model, such as a Vehicle, Car, and SUV tree.	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Polymorphism, Compile-time Polymorphism, Run-time Polymorphism, Polymorphic Variables Class Test -2	Students implement a single function that executes different logic based on the passed object type.	Class Test, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Polymorphism with Interfaces, Polymorphism in Inheritance, Operator Overloading	Students overload the '+' operator to define custom addition logic for complex number objects.	Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Association, aggregation, and composition.	Students create a UML diagram that explicitly labels Aggregation versus Composition relationships.	Assignment, Midterm exam	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Review class	Students conduct a peer-led quiz focusing on object lifecycle stages and inheritance constraints.	Assignment, Midterm exam	CLO1, CLO2

	Lesson 2: Definition of Refactoring, Different Types of Code Smells	Students identify three distinct code smells in provided legacy code and apply clean-code fixes.	Assignment, Final exam	CLO1, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Refactoring Technique.	Students break down a large, monolithic 100-line method into smaller, specialized functions.	Assignment, Final exam	CLO2, CLO3
	Lesson 2: Functions and Modules, Libraries like List, set, stack, queue, Tuple, Dictionary, etc.	Students build a system that stores User objects in a Dictionary and retrieves them using unique IDs.	Assignment, Final exam	CLO2, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Exception Handling.	Students implement try-catch blocks to handle file I/O errors and invalid user-supplied data inputs.	Class Test, Assignment, Final exam	CLO2, CLO3
	Lesson 2: Software Engineering Life Cycle, Modeling Problem and Solution	Students map their current project tasks to the specific phase of the software life cycle.	Class Test, Assignment, Final exam	CLO2, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: File Handling	Students serialize current object states to a file and implement functionality to reload them at runtime.	Class Test, Assignment, Final exam	CLO4
	Lesson 2: Moving to Code, Internal Component Definition, Design for Reuse, Design Class Diagrams Class Test -3	Students implement a functional 3-class structure based strictly on a provided technical class diagram.	Class test, Assignment, Final exam	CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Iterating the Design, Refactoring, Design Best Practices, Cohesion, Complexity, Coupling, and Congruence	Students measure their current project code for Cohesion and Coupling levels using IDE metrics.	Assignment, Final exam	CLO4
	Lesson 2: Singleton pattern, factory pattern, observer pattern, strategy pattern.	Students refactor a database connection class to implement the Singleton pattern.	Assignment, Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Use Case Model, Use Case Diagram, Use Case Description, Main Success Scenario and Alternate Paths, Use Case Relationships, Extension Points and Packages	Students map a main success scenario to a sequence of function calls within their existing codebase.	Assignment, Final exam	CLO4
	Lesson 2: Overview and Basic Concepts, Classifiers and Well-Formedness Rules, Basic Notation, Classes, Objects, Relationships, and Methods, UML Diagrams	Students document their project structure by generating both class and sequence diagrams.	Assignment, Final exam	CLO4

Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Students perform an architectural audit on the OOP structure of an open-source project.	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on Final Syllabus, Presentation on a given topic	Students present a critical design flaw discovered in their final project and demonstrate their fix.	Presentation, Open Discussion	CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO3	
Attendance	--	--	--	--	7				
Class Test 1	5(15)				Avg 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment		5			5		5		5
Presentation			8		8				
Midterm Exam	10	15			25		2		25
Final exam			20	20	40	20		20	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Herbert Schildt, Java: The Complete Reference, 9th Edition, McGraw-Hill Education, 2014, ISBN: 9789339212094.

Reference Books/ Other Supplementary Readings

CT-oriented programming with Java, 3rd Edition, McGraw-Hill, 2004., ISBN: 0073040959

.Robert C. Martin, Clean Code: A Handbook of Agile Software Craftsmanship, 1st Edition, Prentice Hall, 2008, ISBN: 978-0132350884.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 217 Object Oriented Programming Lab

Course Code: SE 217		CIE Marks: 60
ISCED: 0613-217		LF Marks: 40
Course Title: Object Oriented Programming Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: SE 121, SE 122		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

Setting Up the Development Environment, Introduction to Classes and Objects, Implementing Access Specifiers, Working with Static and Non-Static Members, Constructor and Destructor Implementation, Handling Arrays of Objects and Object Pointers, Implementing Abstraction, Encapsulation in Action, Working with Inheritance, Exploring Polymorphism, Abstract Classes and Methods Implementation, Interface Implementation, Applying Refactoring Techniques, Creating and Using Libraries, Exception Handling in Practice, File Handling, Software Engineering Life Cycle, Transition from Design to Code, Design Class Diagrams, Refining Design Practices, Assessing Cohesion, Complexity, and Coupling, Implementing Design Patterns, Creating Use Case Diagrams, Use Case Descriptions and Relationships, UML Diagrams in Action.

Rationale of the Course

This comprehensive OOP course equips students with practical software development skills, from environment setup to advanced architecture. Learners master foundational concepts like classes, objects, encapsulation, and constructors before advancing to inheritance, polymorphism, and interfaces. Hands-on exercises cover file manipulation, exception handling, and refactoring techniques. Crucially, the curriculum bridges design and code, teaching use case modeling, UML diagrams, and design patterns. By assessing design quality through cohesion, coupling, and complexity analysis, this course prepares students to build robust, scalable, industry-standard software solutions.

Course Objectives

The goal of this course is to introduce the students to the concept of database principles in practice. The main objectives of this course are,

- . To have a comprehensive understanding of object-oriented programming concepts.
- . To use access specifiers to control the visibility of class members
- . To implement design patterns in the code.
- . To create and use libraries in their code.
- . To apply refactoring techniques to their code.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Implement the basic knowledge of Object programming.	PLO1	C3	Fundamental Domain
CLO2	Analyze various types of Object-Oriented Concepts.	PLO2	C4, A5	Fundamental Domain
CLO3	Apply appropriate techniques for constructing the programming solution.	PLO5	C3, P2	Fundamental Domain
CLO4	Write effective reports and design documentation for the data structure applications	PLO10	C3, A1, P4	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12

CLO1	√											
CLO2		√										
CLO3					√							
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning, and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Introduction to the OBE of the course	Analyze a software requirement document to map CLOs to project outcomes.	CLO1	None
Week-2 [1.5 Hours]	Setting Up the Development Environment & Introduction to Classes and Objects	Live IDE configuration; Implement basic class-object structures for a Student Record System.	CLO1, CLO4, CLO3, CLO2	Lab Report, Lab Performance, Lab Final
Week-3 [1.5 Hours]	Implementing Access Specifiers Working with Static and Non-Static Members, Constructor and Destructor Implementation Handling Arrays of Objects and Object Pointers	Implement custom constructors and destructors to monitor object lifecycle memory allocation.	CLO1, CLO4, CLO3, CLO2	Lab Report, Lab Performance, Lab Final
Week-4 [1.5 Hours]	Exploring abstraction, encapsulation	Transform a flat requirement into a base class skeleton with private members and public getters/setters.	CLO1, CLO4, CLO2	Lab Report, Lab Performance, Lab Final
Week-5 [1.5 Hours]	Exploring inheritance, Polymorphism	Refactor a flat structure into a logical hierarchy (e.g., Person -> Employee) with method overriding.	CLO1, CLO4	Lab Report, Lab Performance,
Week-6 [1.5 Hours]	Applying Refactoring Techniques	Identify three code smells in existing code and apply specific refactoring fixes to improve readability.	CLO1, CLO4, CLO2	Lab Report, Lab Assignment, Lab Performance, Lab Final
Week-7 [1.5 Hours]	Creating and Using Libraries Exception Handling in Practice	Implement a custom library that safely handles file I/O errors using try-catch blocks.	CLO1, CLO2, CLO4	Lab Report, Lab Performance, Lab Final

Week-8 [1.5 Hours]	File Handling Software Engineering Life Cycle	Serialize object states to a file and map current coding tasks to the SDLC phase.	CLO1, CLO4, CLO3, CLO2	Lab Report, Lab Assignment, Lab Performance, Lab Final
Week-9 [1.5 Hours]	The transition from Design to Code, Design Class Diagrams	Translate a provided technical class diagram into functional C++ class skeletons.	CLO1, CLO4, CLO3, CLO2	Lab Report, Lab Assignment, Lab Performance, Lab Final
Week-10 [1.5 Hours]	Refining Design Practices Assessing Cohesion, Complexity, and Coupling	Measure project code for Cohesion and Coupling levels using professional IDE metric tools.	CLO1, CLO4, CLO2	Lab Report, Lab Assignment, Lab Performance, Lab Final
Week-11 [1.5 Hours]	Implementing Design Patterns, Creating Use Case Diagrams, Use Case Descriptions and Relationships, UML Diagrams in Action	Refactor a database class using the Singleton pattern and document it with a sequence diagram.	CLO1, CLO4, CLO2	Lab Report, Lab Assignment, Lab Performance Lab Final
Week-12 [1.5 Hours]	Project Presentation	Present project progress with a critique of a major design choice made during development.	CLO1	Presentation, Lab Performance
Week-13 [1.5 Hours]	Industry Session	Perform an architectural audit on the OOP structure of an open-source project.	CLO1, CLO2, CLO3, CLO4	Interactive Lectures and Presentations, Expert Talks, Hands-On Workshops, Site Visits and Virtual Tours
Week-14 [1.5 Hours]	Lab Final	Demonstrate independent implementation of a complex engineering problem from scratch.	CLO2, CLO3	Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO5
Lab Performance	25	CLO1	PLO1
Lab Final	40	CLO2	PLO2
		CLO3	PLO5
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Herbert Schildt, Java: The Complete Reference, 9th Edition, McGraw-Hill Education, 2014, ISBN: 9789339212094.

Reference Books/ Other Supplementary Readings

C. Thomas Wu, An introduction to object-oriented programming with Java, 3rd Edition, McGraw-Hill, 2004., ISBN: 0073040959

.Robert C. Martin, Clean Code: A Handbook of Agile Software Craftsmanship, 1st Edition, Prentice Hall, 2008, ISBN: 978-0132350884.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Netbeans, Eclipse, Java Doctor

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 222 Computer Architecture

Course Code: SE 222		CIE Marks: 60
ISCED: 0612-222		SEE Marks: 40
Course Title: Computer Architecture		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)		Contact Hours: 3 (Total weeks: 14)
Prerequisite: SE 213		
Course Type: Core		

CIE: Continuous Internal Evaluation,

SEE: Semester End Examination

Course Content (from syllabus)

Instruction Set Architecture, Evolution of Computer, Processor performance (Moore's law), Benchmarks (Ahmdahl's law), clock speed and instruction per second, General Purpose Processor and its instruction patterns, Function of GPP, Hypothetical Processor, Instruction fetch and execute, Logical Units, Instruction Cycle States,

Interrupt and Interrupt handling, Interconnection structure, Bus interconnection, Bus design (Address bus, control bus, data bus, multiple bus), Traditional bus architecture, high performance architecture, Elements of bus design, Method of arbitration, Timing, performance of memory, The memory hierarchy, Cache memory principles, Cache address, Mapping mechanism, Direct mapping, Full and Set associative cache, Replacement algorithms, Write policy, Line size, Number of cache, Interconnection structure, Data flow, Arithmetic logic unit (ALU), Number representation, Addition subtraction technique, Multiplication Technique (1 and 2's complement), Floating point representations, Machine instructions characteristics, Elements of machine instruction, Instruction representation, Instruction type, Number of address, Instruction set design, Operand types, Operation types, Branch instructions, Skip instructions, Procedure call instructions, Addressing technique, CPU internal structure, Register Organization, User visible registers (flags), Condition codes, Control and Status registers, Data flow, Instruction pipelining, Pipeline performance, Pipeline hazards, Dealing with branches.

Rationale of the Course

This course in Computer Architecture and Organization explores the foundational principles underpinning modern computing systems. It examines processor performance and technological evolution through Moore's and Amdahl's Laws. Students will analyze instruction set architectures, fetch-execute cycles, and general-purpose processor design. The curriculum details memory hierarchies, cache performance, and bus interconnections alongside arithmetic logic units and number representation. Crucially, it covers CPU internal structures, register organization, and instruction pipelining to optimize system efficiency. By blending theory with practical insights, this course empowers students to design, analyze, and optimize complex computer architectures.

Course Objectives

- To provide a solid conceptual understanding of the fundamentals of Computer architecture, More specifically,
- Illustrate the organizations, architectures, structures and functions of modern digital computer systems.
 - Organize the basic knowledge required to understand the architectures and organizations of modern digital computer systems.
 - Conclude the hardware level arithmetic operation algorithms.
 - Interpret parallelism both in terms of a single processor and multiple processors.
 - Distinguish a wide variety of memory technologies both internal and external.
 - Determine the system and memory performance measurements under measurement metrics.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Classify the basic architecture of a computer, parallelism both in terms of a single processor and multiple processors and processor performance.	PLO2	C2	Fundamental Domain
CLO2	Explain how a modern computer works, how it supports the execution of software, how to design alternatives affect performance and cost.	PLO3	C4	Fundamental Domain
CLO3	Investigate the major components of a computer including CPU, memory, I/O, and storage	PLO2	C4	Fundamental Domain
CLO4	Apply an instruction set design	PLO1	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2			√									
CLO3		√										
CLO4	√											

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Computer Architecture	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Overview of Instruction Set Architecture, Evolution of Computers.	Interactive Lecture, Whiteboard Diagramming	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Processor performance (Moore' law), Benchmarks (Ahmdahl's law), clock speed and instruction per second.	Guided Problem-Solving, Whiteboard Practice	Class Test, Midterm exam	CLO1
	Lesson 2: General Purpose Processor and its Instruction Patterns.	Visual Lecture, Concept Mapping	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Function of GPP, Hypothetical Processor.	Diagram Analysis, Guided Discussion	Class Test, Midterm exam	CLO1
	Lesson 2: Instruction fetch and execute, Logical Units, Instruction Cycle States Class Test -1	Step-by-step Process Walkthrough, Interactive Q&A	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Interrupt and Interrupt Handling.	Real-world Analogy Discussion, Flowchart Tracing	Class Test, Midterm exam	CLO1
	Lesson 2: Generic Bus interconnection, Bus design(Address bus).	Interactive Lecture, Diagram Walkthrough	Class Test, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Generic Bus interconnection, Bus design(Control bus).	Whiteboard Demonstration, Signal Mapping Practice	Class Test, Midterm exam	CLO1
	Lesson 2: data bus, multiple bus Traditional bus architecture, Elements of bus design, Method of arbitration.high-performance architecture	Collaborative Discussion, Comparative Case Analysis	Class Test, Midterm exam	CLO1
Week-6	Lesson 1: Timing, performance of memory, The memory hierarchy.	Visual Presentation, Hierarchy Architecture Mapping	Class Test, Midterm exam, Final exam	CLO2

Lesson 1 & 2 [3 Hours]	Lesson 2: Cache memory principles, Cache address. Class Test -2	Interactive Lecture, Address Field Dissection	Class Test, Midterm exam, Final exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Mapping mechanism, Direct mapping.	Whiteboard Demonstration, Rule-based Practice	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Full and Set associative cache, Replacement algorithms.	Guided Practice, Peer Problem-Solving	Class Test, Assignment, Final exam	CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Write policy, Line size, Number of cache, Interconnection structure.	Scenario-based Discussion, Design Matrix Analysis	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Data flow and Arithmetic logic unit (ALU) description.	Step-by-step Circuit Demonstration, Collaborative Design	Class Test, Assignment, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: through using the technique. Number of process executions: Addition subtraction, Multiplication Technique (1 and 2's complement)	Hands-on Binary Practice, Code/Logic Tracing	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Floating point representations, Machine instructions characteristics.	Hands-on Bit Dissection, Error Diagnostics Workshop	Class Test, Assignment, Final exam	CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Machine instructions, characteristics, Elements of machine instruction, Instruction representation	Interactive Lecture, Syntax Walkthrough	Class Test, Assignment, Final exam	CLO4
	Lesson 2: Instruction type, Number of addresses, Instruction set design, Operand types, Operation types.	Comparative Analysis, Collaborative Framework Design	Class Test, Final exam	CLO4
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Branch instructions, Skip instructions, Procedure call instruction.	Whiteboard Demonstration, Guided Flowchart Drawing	Class Test, Final exam	CLO4
	Lesson 2: Discussion on Addressing technique. Class Test - 3	Concept Mapping, Interactive Questioning	Class Test, Final exam	CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: CPU internal structure and Register Organization.	Interactive Lecture, Structural Diagram Walkthrough	Assignment, Final exam	CLO4
	Lesson 2: User visible registers (flags). and Condition codes	Logic Puzzles, Logic State Tracing Exercise	Assignment, Final exam	CLO4

Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Control and Status registers, Data flow.	Visual Presentation, Data Flow Diagram Analysis	Assignment, Final exam	CLO4
	Lesson 2: Instruction on pipelining and its performance hazards and dealing with branches.	Threat Scenario Discussion, Hazard Resolution Workshop	Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on final exam syllabus	Integrative Review, Board Problem Solving	Presentation, Open discussion	CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO3	
Attendance	--	--	--		7				
Class Test 1	5(15)				Average 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment		5			5	5			5
Presentation					8				
Midterm Exam	15	10			25		15	10	25
Final exam		10	20	10	40	10	20	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

William Stallings, Computer Organization and Architecture: Designing for Performance, 10th Edition, Pearson, 2017, ISBN: 9789332570405.

Reference Books/ Other Supplementary Readings

Linda Null, Julia Lobur, The Essentials of Computer Organization and Architecture, 5th Edition, Jones & Bartlett Learning, 2018, ISBN: 978-1284123036.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule.



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

STA 101 Statistics I

Course Code: STA 101/ GED 239	CIE Marks: 60
ISCED: 0613-101	SEE Marks: 40
Course Title: Statistics I / Probability & Statistics in Software Engineering	Total Marks: 100
Semester:	
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)
Prerequisite: SE 221	
Course Type: GED	

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Instructor Details

Name:	
Employee ID:	
Designation:	
Department:	
Office Address:	
Telephone/Extension:	
Mobile:	
Website:	
GTA/UTA(If Any):	

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 10:00 AM	10:00 AM- 11:30 AM	11:30 AM- 01:00 PM	01:00 PM- 02:30 PM	2:30 PM- 4:00 PM	4:00 PM- 5:30 PM
Saturday						
Sunday						
Monday						
Tuesday						
Wednesday						
Thursday						

Course Content (from syllabus)

Meaning and Definition of Statistics, Types of statistics, Population and sample, Parameter and statistic, Variable and types of variable; Characteristics Levels of data, Sampling techniques, Define data presentation, Constructing frequency distribution and relative frequency, percentage frequency and cumulative frequency distribution both for Qualitative and quantitative data; Graphic presentation of a quantitative frequency distribution (Histogram, Frequency polygon, Ogive curve) with their merits and demerits; Graphic presentation of a qualitative frequency distribution (Bar Chart, Pie Chart) with their merits and demerits, Time series data, Line diagram; Define Measures of Central Tendency, applications, Arithmetic Mean, Geometric Mean, Harmonic Mean, Weighted Mean for ungroup data and grouped data with their merits and demerits; Calculation of Median, mode both for group and ungroup data, Application of measures of central tendency for different types of level of measurements. (Definition of Measures of Location, concepts of Quartiles, Percentiles, and Deciles, Procedure of calculating quartiles, deciles, and percentiles for ungrouped data. Measures of Location (Procedures of calculating Quartiles, Percentiles, and Deciles for group data), mathematical problem solving with interpretations. Calculation of IQR with interpretation, Drawing Box and Whisker Plot with its applications. Define measures of dispersion, applications, types of dispersion, Absolute measures, Relative Measures, and Calculation of Mean deviation. Calculation of Population variance, Population Standard Deviation, Sample variance, Sample Standard Deviation, and Coefficient of Variation for un group data with mathematical problem-solving. Calculation of Population variance, Population Standard Deviation, Sample variance, Sample Standard Deviation, and Coefficient of Variation for group data with mathematical problem-solving. Define the shape of the distribution, Concepts of Skewness and Kurtosis with graphical presentation. Calculation of Coefficient of skewness and coefficient of Kurtosis with their interpretations. Define correlation Analysis, Types of Correlation, positive correlation, negative correlation, simple correlation, partial correlation, multiple correlation, linear correlation, non-linear correlation, Scatter diagram with interpretation. Calculation of Coefficient of correlation with interpretation, mathematical problem-solving. Define Regression Analysis, Types of Regression, Simple regression analysis, Multiple regression analysis, Calculation of Simple regression coefficients. Interpretation of regression coefficients, Calculation of the standard error of the regression coefficients, and Calculation of coefficient of determination with interpretation. Define Probability, equally likely outcomes, mutually exclusive outcomes, Sample Space, Tree diagram, Venn diagram, Laws of probability, Additional rules. Marginal probability, Joint probability, Conditional probability, multiplication rules complement rule. Bayes theorem with related math. Random variable, Discrete Random variable, Continuous random variable, Mean and variance and standard deviation of random variables. Random variable, Discrete Random variable, Continuous random variable, Mean and variance and standard deviation of random variables. Define Probability distribution, Basic idea of Discrete Probability Distribution/ Probability Mass Function (PMF) and Continuous Probability Distribution/ Probability Density Function (PDF). Probability Distribution (Bernoulli distribution and Binomial Distribution with related maths, Geometric and Poisson Probability distribution with related math, Normal and exponential probability distribution with related math). Basic concepts of Test of hypothesis, Null hypothesis, Alternative hypothesis, Test Statistics, One-tailed test, Two-tailed test. Basic idea of One sample test and two sample test, Z-test, t-test, critical value, Acceptance area, rejection area, Decision rule. One sample Z-test and t-Test with related math and interpretation. Two sample Z-test and t-Test with related math and interpretation

Rationale of the Course

This comprehensive statistics course provides a solid foundation in theoretical concepts and practical data analysis. It covers fundamental definitions, sampling techniques, and data organization via frequency distributions and graphical charts. Students will master descriptive statistics, including measures of central tendency, location, dispersion, skewness, and kurtosis to interpret dataset characteristics. The curriculum then advances to predictive and inferential methods, covering correlation, regression, and probability theory. Concluding with hypothesis testing via Z-tests and t-tests, this course equips learners with the essential analytical tools to evaluate and interpret data effectively across various fields.

Course Objectives

The goal of this course is to introduce the students to the concept of modern operating system principles. The main objectives of this course are,

- To develop a solid foundation in statistical concepts.
- To acquire proficiency in constructing frequency distributions for both qualitative and quantitative data.
- To apply statistical methods to real-world problem-solving.
- To develop proficiency in quantitative and mathematical problem-solving related to statistical concepts, reinforcing theoretical knowledge with practical applications.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Explain the terminology of statistics.	PLO2	C2	Fundamental Domain
CLO2	Apply the Fundamental knowledge about descriptive statistics and their applications.	PLO1	C3	Fundamental Domain
CLO3	Apply appropriate statistical methods for making decisions.	PLO1	C3	Fundamental Domain
CLO4	Analyze appropriate statistical tools and their skills in practical problems.	PLO5	C4	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO 1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2	√											
CLO3	√											
CLO4					√							

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Statistics	Syllabus Walkthrough, Interactive Q&A	Class Test	CLO1
	Lesson 2: Definition and types of statistics, the concept of population, sample and parameter	Interactive Lecture, Conceptual Mapping	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Characteristics of variable and level of data sampling techniques	Real-world Analogy Discussion, Case Analysis	Class Test Midterm exam	CLO1
	Lesson 2: Define data presentation, Constructing frequency distribution and relative frequency, percentage frequency and cumulative frequency distribution both for Qualitative and quantitative data	Whiteboard Demonstration, Rule-based Practice	Class Test, Midterm exam	CLO1, CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Graphic presentation of a quantitative frequency distribution (Histogram, Frequency polygon, Bar Chart, Pie Chart with their merits and demerits, Time series data, Line diagram)	Chart Construction Workshop, Diagram Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Measures of Central Tendency (Define Measures of Central Tendency, applications, Arithmetic Mean, Geometric Mean, Harmonic Mean, Weighted Mean for ungrouped data with their merits and demerits)	Guided Formula Practice, Comparative Discussion	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Measures of Central Tendency Arithmetic Mean, Geometric Mean, Harmonic Mean, Weighted Mean for group data with their applications and merits and demerits, Median, mode both for group and ungroup data Class Test -1	Whiteboard Demonstration, Peer Problem-Solving	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Measures of Location (Definition, concepts of Quartiles, Percentiles and Deciles , Procedure of calculating quartiles, deciles and percentiles for ungrouped data	Step-by-step Calculation, Structured Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Procedures of calculating Quartiles, Percentiles and Deciles for grouped data), mathematical problem solving with interpretations	Guided Problem-Solving, Peer Review of expressions	Class Test, Assignment, Midterm exam	CLO1, CLO2

	Lesson 2: Calculation of IQR with interpretation, Drawing Bow and Whisker Plot with its applications	Visual Presentation, Data Plotting Exercise	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Define measures of dispersion, applications, types of dispersion, Absolute measures, Relative Measures, Calculation of Mean deviation	Interactive Lecture, Formula Walkthrough	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Calculation of Population variance, Population Standard Deviation, Sample variance, Sample Standard Deviation, Coefficient of Variation for ungrouped data with mathematical problem solving Class Test -2	Guided Problem-Solving, Whiteboard Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Concept and Calculation of Coefficient of skewness, Coefficient of Kurtosis with their interpretations	Visual Metaphors, Distribution Curve Analysis	Midterm exam	CLO1, CLO3
	Lesson 2: Define correlation Analysis, Types of Correlation, Scatter diagram with interpretation (Calculation of Coefficient of correlation with interpretation, mathematical problem solving)	Scatter Plot Analysis, Guided Interpretation	Midterm exam, Final exam	CLO1, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Regression Analysis (Define Regression Analysis, Types of Regression, Simple regression analysis, Multiple regression analysis, Calculation of Simple regression coefficients)	Interactive Lecture, Trendline Diagramming	Class Test, Assignment, Final exam	CLO1, CLO3, CLO4
	Lesson 2: Regression Analysis (Interpretation of regression coefficients, Calculation of the standard error of the regression coefficients, Calculation of coefficient of determination with interpretation)	Diagnostic Case Study, Collaborative Verification	Class Test, Assignment, Final exam	CLO1, CLO3, CLO4
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Probability (Define Probability, equally likely outcomes, mutually exclusive outcomes, Sample Space, Tree diagram, Venn diagram, Laws of probability, Additional rules)	Visual Lecture, Venn Diagram Walkthrough	Class Test, Assignment, Final exam	CLO1, CLO3, CLO4
	Lesson 2: Introduction to Probability (Marginal probability, Joint probability, Conditional probability, multiplication rules complement rule)	Logic Puzzles, Contingency Table Practice	Class Test, Assignment, Final exam	CLO1, CLO3, CLO4
Week-10	Lesson 1: Introduction to Probability (Bayes theorem with related mathematical problems)	Step-by-step Demonstration, Tree Diagram Drawing	Class Test, Assignment, Final exam	CLO1, CLO3, CLO4

Lesson 1 & 2 [3 Hours]	Lesson 2: Introduction to Probability (Random variable, Discrete Random variable, Continuous random variable, Mean and variance and standard deviation of random variables)	Concept Mapping, Guided Sample Computations	Class Test, Assignment, Final exam	CLO1, CLO3, CLO4
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Probability Distribution Definition and concept of PMF, CDF, PDF	Interactive Lecture, Function Graph Analysis	Final exam	CLO1, CLO3, CLO4
	Lesson 2: Bernouli and Binomial Distribution Class Test - 3	Problem-Solving Workshop, Peer Review	Final exam	CLO1, CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Normal and Exponential Distribution	Visual Metaphors, Z-Score Table Walkthrough	Final exam	CLO1, CLO3
	Lesson 2: Basic concept of test of hypothesis, Process of Testing, six steps for testing hypothesis	Six-Step Process Framework, Case Walkthrough	Final exam	CLO1, CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Null hypothesis, Alternative hypothesis, Test Statistics, One-tailed test, Two tailed test	Threat Scenario Discussion, Decision Error Analysis	Final exam	CLO1, CLO3
	Lesson 2: One sample and two sample Z-test and t-Test with related maths and interpretation	Structured Testing Lab, Output Interpretation	Final exam	CLO1, CLO3
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on final exam syllabus.	Integrative Review, Board Problem Solving	Presentation, Open discussion	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =78)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO5	
Attendance	--	--	--	--	7				
Class Test 1	5(15)				Average 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment			5		5		5		5
Presentation				8	8			8	8

Midterm Exam	5	13	7		25	20	5		25
Final exam		5	15	20	40	20		20	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Barbara Illowsky, Susan Dean, Introductory Statistics, 2nd Edition, OpenStax College, XanEdu Publishing Inc, 2023, ISBN: 978-1-947172-05-0.

Reference Books/ Other Supplementary Readings

Barbara Illowsky, Susan Dean, Introductory Statistics, 2nd Edition, OpenStax College, XanEdu Publishing Inc, 2023, ISBN: 978-1-947172-05-0.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

AOL 101 Art of Living

Course Code: AOL 101		CIE Marks: 60
ISCED: 0613-131		SEE Marks: 40
Course Title: Art Of Living		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: GED		

CIE: Continuous Internal Evaluation,
SEE: Semester End Examination

Course Content (from syllabus)

It has been perceived from different evaluation that intellectual ability needs to be exercised in an effective way through scheduled practice and performance. Success comes not just from the knowledge and skills but also from behavior which can create distance between success and failure. And we can achieve expected behavior only by learning and unlearning ways of thinking that finally craft body language and ensure effective living in this beautiful world. This course assists the students to achieve the ability to understand life positively in personal, family and professional levels.

Rationale of the Course

The "Art of Living" is an interdisciplinary course combining psychology, philosophy, and practical wisdom to empower students to lead purposeful, resilient, and fulfilling lives. The curriculum uniquely bridges personal well-being and professional growth, covering self-discovery, emotional intelligence, mindfulness, and stress transformation alongside essential 21st-century life skills like time, financial, and career management. Emphasizing universal human values and ethical conduct, the course utilizes experiential learning and introspection to foster authentic relationships. Ultimately, it equips individuals with a holistic toolkit to navigate modern complexities with compassion, integrity, and harmony.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of personal growth and well-being, More specifically,

- Self Confidence: All learners will have a strong belief that we all are inborn genius and that any dream is possible to achieve to create a difference.
- Boldness: They will become strong with critical thinking & creativity to accept the unacceptable in the face of unavoidable challenges in daily life.
- Equity with Empathy: They will have fundamental human & life skills including conscious self-awareness to learn and find how to do things in an empathetic and equitable way - a way that pleases the Creator.
- Questioning: They will not hesitate to ask any question to unleash inner genius and the genius of local & global collectives, towards building a Just Daffodil, Bangladesh, and World.

- Values: Learners will set, feel, and realize core human values and act with or reflect those in their daily life.
- Learning to learn: They will learn how to learn & will be able to criticize themselves.
- Peace & Mutual Happiness: All learners will achieve Emotional Intelligence (EQ) to create harmonious relationships with parents, family, friends & community in order to experience a good, peaceful, and mutually happy life.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Understand the basic concepts or theories related to the course.	PLO7	C2, A4	Fundamental, Social, Thinking, Personal Domain
CLO2	Discover growth mindset, emotional intelligence, life skills, leadership skills, humanitarian values and ethics, and professional values.	PLO8	C4, A3	Fundamental, Social, Thinking, Personal Domain
CLO3	Evaluate the impact of communication, leadership skills, goal setting, and emotional intelligence in personal and professional life.	PLO11	C5, A2	Fundamental, Social, Thinking, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1							√					
CLO2								√				
CLO3											√	

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Human values and Ethics a) Introduction to values b) Types of values c) Introduction to ethics d) The purpose of ethics e) Personal ethics f) Codes of ethics g) Ethics vs morals	Quiz Game, Interactive Lecture Head-to-Head Discussion, Concept Mapping	Class Test, Assignment	CLO1, CLO2, CLO3

Week-2 Lesson 1 & 2 [3 Hours]	Benefit of Ethical living a) Importance of ethics and values b) Living an ethical life c) Advantages of ethical behavior	Scenario Analysis, Reflective Dialogue Small Group Study, Case Presentation	Class Test, Assignment, Presentation, Midterm exam	CLO1, CLO2, CLO3
Week-3 Lesson 1 & 2 [3 Hours]	Self esteem and You a) Self esteem and growth mindset b) Self-esteem and self-acceptance c) Types of self-esteem and their characteristics d) Elements of self esteem e) Importance of self esteem f) Building healthy self esteem	Self-Assessment Exercise, Interactive Lecture Collaborative Reflection, Group Discussion	Class Test, Assignment, Presentation, Midterm exam	CLO1, CLO2, CLO3
Week-4 Lesson 1 & 2 [3 Hours]	Coming Out of the Box a) Concept of coming out of the box b) Conventional thinking c) Success beyond the box d) Divergent thinking Class Test-1	Brainstorming Session, Cognitive Mapping Dilemma Analysis, Creative Problem-Solving	Class Test, Assignment, Presentation, Midterm exam	CLO1, CLO2, CLO3
Week-5 Lesson 1 & 2 [3 Hours]	Etiquette a) Basics of etiquette b) Types of etiquette c) Positive effect of etiquette	Behavioral Role-play, Interactive Presentation Case Study Review, Open Discussion	Class Test, Assignment, Presentation, Midterm exam	CLO1, CLO2, CLO3
Week-6 Lesson 1 & 2 [3 Hours]	Parents Life and Living a) Most precious relationship in the world b) Valuing our parents c) Bonding with parents d) Understanding self parenting e) Rules of self parenting f) Necessity of parenting g) Rules of parenting Class Test-2	Real-world Analogy Discussion, Narrative Review Conceptual Framework Walkthrough, Q&A	Class Test, Assignment, Presentation, Midterm exam	CLO1, CLO2, CLO3
Week-7 Lesson 1 & 2 [3 Hours]	Equity, Leadership & You a) Types of social responsibility b) Psychosocial support c) Equity focused leadership d) Moral Leadership e) Be a changemaker	Threat Scenario Discussion, Concept Mapping Framework Modeling, Collaborative Discussion	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3

Week-8 Lesson 1 & 2 [3 Hours]	Persons, Moral values & Good life a) Variety of moral issues b) Role of moral values in everyday life c) Empathy d) Respect for others e) Self control f) Caring g) Sharing	Value Dissection Workshop, Peer Dialogue Case-based Problem Solving, Video Analysis	Class Test, Assignment, Midterm exam Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Communication Ethics a) Ethical communication and its importance b) Principles of ethical communication	Scenario Walkthrough, Interactive Q&A Matrix Analysis, Peer Review of Expressions	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Orientation of Organization/ Corporate Culture and Technological Skills for Industries	Corporate Case Study, Structure Diagramming Visual Lecture, Skill Blueprint Mapping	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Defining Digital, Media, Technology Literacy, and Innovation and Social Entrepreneurship in the IT Industry Class Test-3	Media Deconstruction Lab, Critical Review Collaborative Venture Design, Brainstorming	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Etiquette, Work Ethics, and Integrity in the Digital Age	Real-world Vulnerabilities Analysis, Debate Ethical Dilemma Modeling, Guided Discussion	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Presentation Skill and Public Speaking, Adaptation to New technology, How to tackle interpersonal & Daily situation, Industrial ethics and values	Live Speaking Practice, Peer Feedback Role-play Simulation, Conflict Resolution Lab	Assignment, Presentation Final exam	CLO1, CLO2, CLO3
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Review class week 6 – week -10 Lesson 2: Review class week 10 – week -17	Terminology Matrix Review, Cross-Q&A Session	Assignment, Final exam	CLO1, CLO2, CLO3

Overall Assessment Scheme

Assessment Task	CLO's	Mark	PLO's	Mark
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	CLO1	CLO2	CLO3	(Total=100)	PLO7	PLO8	PLO11	(Total =65)
Attendance	--	--	--	7				
Class Test 1	5(15)			Avg 15				
Class Test 2		5(15)						
Class Test 3			5(15)					
Assignment	5			5				
Presentation	4	4		8				
Midterm Exam	10	8	7	25	10	8	7	25
Final exam	10	15	15	40	10	15	15	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

1. Dr. Md. Sabur Khan, Syed Mizanur Rahman, Art of Living Workbook, Daffodil International University, 2012.
2. Dr. Md. Sabur Khan, Syed Mizanur Rahman, Art of Living Textbook, Daffodil International University, 2012.
3. Mark Getlein, Gilbert's living with art, 6th Edition, Boston: McGraw-Hill, c2001. ISBN: 0072317264

Reference Books/ Other Supplementary Readings

1. Lickona, Thomas., Education for Character: How Our Schools Can Teach Respect and Responsibility, Bantam Books, New York, 1992, ISBN: 9780553370522.
2. Mackenzie, John S., A Manual of Ethics, Surjeet Publication, June 2004, ISBN: 9788121901035.
3. Newman, Fred, and Phyllis Goldberg., Performance of a Lifetime: A Practical Philosophical Guide to the Joyous Life, Castillo International Inc., 1996, ISBN-13: 978-0962862175

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 211 Object Oriented Concepts

Course Code: SE 211		CIE Marks: 60
ISCED: 0612-211		SEE Marks: 40
Course Title: Object Oriented Concepts		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)		Contact Hours: 3 (Total weeks: 14)
Prerequisite: SE121, SE122		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

The four pillars of object-oriented programming – Abstraction, Encapsulation, Inheritance, and Polymorphism. Abstract class and interface. Definition of Refactoring and Code Smells. Different types of code smells and refactoring techniques. Use of different refactoring techniques for different purposes. Introduction to Design Patterns. Definition and classification of design patterns. Definition, Use Case, Code examples of Creational Patterns – Singleton, Factory Method, Abstract Factory, Prototype, Builder. Definition, Use Case, Code examples of Creational Patterns – Adapter Pattern, Bridge Pattern, Composite Pattern, Decorator Pattern, Facade Pattern, Flyweight Pattern, Proxy Pattern. Definition, Use Case, Code examples of Creational Patterns – Chain of Responsibility Pattern, Command pattern, Interpreter pattern, Iterator pattern, Mediator pattern, Memento Pattern, Observer Pattern, State pattern, Strategy Pattern, Template pattern, Visitor pattern.

Rationale of the Course

The comprehensive course in Object-Oriented Concepts provides a solid foundation in advanced programming principles and scalable software design. Students will master the four pillars of OOP—Abstraction, Encapsulation, Inheritance, and Polymorphism—using abstract classes and interfaces to construct modular, extensible systems. The curriculum strictly addresses code quality by teaching students to identify code smells and implement effective refactoring techniques. Additionally, it introduces an extensive catalog of creational, structural, and behavioral design patterns. Ultimately, this course equips future developers with the vital architectural toolkit to build robust, maintainable, and industry-standard software solutions.

Course Objectives

- To provide a solid conceptual understanding of the fundamentals of object oriented. More specifically,
- To provide the knowledge of model of object-oriented programming: abstract data types, encapsulation, inheritance, and polymorphism.
 - To know the fundamental features of an object-oriented language like Java: object classes and interfaces, exceptions, and libraries of object collections.
 - To develop skills in how to take the statement of a business problem and from this determine suitable logic for solving the problem; then be able to proceed to code that logic as a program written in Java.

To learn how to test, document, and prepare a professional-looking package for each business project using Javadocs.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Elaborate the understanding of writing object-oriented programs that combine functions and data.	PLO2	C2	Fundamental Domain
CLO2	Analyze a problem statement to develop a mental model of objects necessary to create software architecture.	PLO2	C4	Fundamental Domain
CLO3	Apply the object-oriented programming language to develop software, including programs utilizing multiple Classes.	PLO3	C3	Fundamental Domain
CLO4	Apply abstract concepts to Classes in software.	PLO3	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3			√									
CLO4			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Object Oriented Concepts	Syllabus Walkthrough, Interactive Q&A	-	CLO1
	Lesson 2: Introduction and Overview Principle of Software Engineering based on OOP. Principle of Code Reusing and Extending Code	Interactive Lecture, Architecture Diagramming	Class Test, Midterm exam	CLO1
Week-2	Lesson 1: Time Space Trade-off, Recursion, Iteration. Review of Fundamentals of Procedural	Comparative Analysis, Trace Table Practice	Class Test, Midterm exam	CLO1

Lesson 1 & 2 [3 Hours]	Programming Data types and variables, operators			
	Lesson 2: Strings. Input and Output.	Live Coding Demonstration, Syntax Walkthrough	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: If-else, Switch Case.	Whiteboard Demonstration, Logic Puzzles	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: For loop, While loop, Do While loop.	Hands-on Coding Practice, Code Tracing	Class Test, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: String Type Arrays	Live Coding Demonstration, Array Mapping	Class Test, Midterm exam	CLO2
	Lesson 2: State and behavior of the object. Class Test -1	Object Demonstration, Memory Visualisation	Class Test, Midterm exam	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Identify objects and their structure.	Real-world Case Study, Object Modeling	Class Test, Midterm exam	CLO2
	Lesson 2: Develop class(es) from objects and References.	Whiteboard Demonstration, Diagram Walkthrough	Class Test, Midterm exam	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: The basic syntax of constructor, Constructor chaining.	Step-by-step Process Walkthrough, Code Tracing	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Code reusability, Parent-child relationship. Class Test -2	Visual Presentation, Hierarchy Mapping	Class Test, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Super, assert, break, continue keyword.	Syntax Walkthrough, Code Execution Tracking	Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Association, aggregation and composition.	Diagrammatic Analysis, Real-world Case Study	Assignment, Midterm exam	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Data Abstraction, Information Hiding & Encapsulation and Abstraction.	Concept Mapping, Visual Metaphors	Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Class Methods and Information Hiding.	Live Coding Demonstration, Access Restriction Practice	Assignment, Midterm exam	CLO1, CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Access Modifier and Accessibility of Access Modifier.	Whiteboard Matrix Analysis, Group Discussion	Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Polymorphism: Statics and Dynamic, Basic Principle of Polymorphism	Dynamic Memory Tracing, Coding Workshop	Assignment, Midterm exam	CLO1, CLO2

Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Abstract Classes and Methods. Basic Principles of Interfaces, Extending interface.	Blueprint Modeling, Structural Walkthrough	Class Test, Assignment, Final exam	CLO2, CLO3
	Lesson 2: Multiple Interface and Interface Implementation.	Collaborative Class Design, Live Coding	Class Test, Assignment, Final exam	CLO2, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Propagation of Exception and Exceptions Handling, Compile time error, Run time error.	Error Diagnostic Workshop, Hazard Identification	Assignment, Final exam	CLO4
	Lesson 2: Exception Handling, User-defined exceptions. Class Test -3	Hands-on Coding Practice, Code Tracing	Assignment, Final exam	CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Analyze a real-life story, Analyze a user story, and Extract objects.	User Story Dissection, Collaborative Modeling	Assignment, Final exam	CLO4
	Lesson 2: Design classes and Implement those classes	Guided Class Drawing, Collaborative Design	Assignment, Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Propagation of Exception.	Step-by-step Stack Trace, Diagram Analysis	Assignment, Final exam	CLO4
	Lesson 2: Practice on Exception Handling.	Peer Code Debugging, Hands-on Practice	Assignment, Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on final syllabus, Presentation on a given topic	Student Presentations, Integrative Review	Presentation, Open Discussion	CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's		Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO2	PLO3	
Attendance	--	--	--		7			
Class Test 1	5(15)				Average 15			
Class Test 2		5(15)						
Class Test 3			5(15)					

Assignment		5			5			
Presentation					8			
Midterm Exam	10	15			25	25		25
Final exam			20	20	40		40	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Herbert Schildt, Java: The Complete Reference, 9th Edition, McGraw-Hill Education, 2014, ISBN: 9789339212094.

Reference Books/ Other Supplementary Readings

C. Thomas Wu, An introduction to object-oriented programming with Java, 3rd Edition, McGraw-Hill, 2004., ISBN: 0073040959

.Robert C. Martin, Clean Code: A Handbook of Agile Software Craftsmanship, 1st Edition, Prentice Hall, 2008, ISBN: 978-0132350884.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 221 Object Oriented Design

Course Code: SE 221		CIE Marks: 60
ISCED: 0613-221		SEE Marks: 40
Course Title: Object Oriented Design		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE 211		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

The Object-Oriented Paradigm (Basic Concepts Classes and Objects, Characteristic of Objects, Relationships Methods), Object-oriented programming (Programming as a tool, More than one programming as a language, Data-types, Variables, Basic operators, Basic I/O, Boolean values, Conditional execution loops, List and list processing, Logical and bitwise operations, functions, tuples, dictionaries and data processing, modules, packages, string, list methods, exceptions), Object oriented approach (Classes, Methods, Objects, exception handling, working with files.), Analysis, Design, Modeling (Systems and Software Quality Software Engineering Life Cycle Properties, Analysis and Design, Structured vs. Object Approach Modeling, Modeling Problem and Solution, Modeling Perspective), Processes and Approaches to OOA&D (Unified Process, Rational Unified Process, Agile

Approaches, Other Approaches), Design Phase (Moving to Code, Internal Component Definition, Design for Reuse, Design Class Diagrams, Operations, Methods), Designing Quality into Modules (Iterating the Design, Refactoring, Design Best Practices, Cohesion, Complexity, Coupling, Congruence), Physical Design, (UML Implementation Diagrams, Component Diagram, Application of the Component Diagram, Component Diagram Notation and Syntax, Deployment diagram, Application of the Deployment Diagram, Deployment Diagram Notation and Syntax), Domain Modeling (The Information Model (Structural Modeling, Objects, Attributes, and Associations, Links and Messages, Information Modeling in the UML, Basic Constructs, Relationships, Stereotypes, Packages), Use Case and the Behavior Model (Use Case Model, Use Cases in the Life Cycle, Use Case Diagram, Use Case Description, Main Success Scenario, Alternate Paths, Preconditions and Post-Conditions, Exception Paths, Use Case Extensions, Use Case Relationships, Generalization, The <<extends>> Relationship, The <<includes>> Relationship, Extension Points, Packages), Basic Unified Modeling Language (Overview, Basic Concepts, Classifiers, Well-Formedness Rules, Basic Notation, Classes and Objects, Characteristic of Objects, Relationships and Methods, Other Notation, UML Diagrams)

Rationale of the Course

This course integrates Object-Oriented Programming (OOP) with the Software Engineering Life Cycle to provide a solid foundation in analysis, design, and system modeling. Beginning with OOP fundamentals, exception handling, and file operations, the curriculum transitions into structural OOA&D methodologies, including the Unified Process and Agile frameworks. It emphasizes design for reuse, internal component definition, and quality assurance via refactoring. By mastering Unified Modeling Language (UML) for domain, use case, and behavior modeling, students learn to effectively visualize complex systems. Ultimately, this holistic approach empowers learners to build high-quality software solutions in real-world development scenarios.

Course Objectives

The goal of this course is to introduce the students to the concept of object-oriented design. The main objectives of this course are,

- To viewing a system as a collection of objects collaborating to fulfill the system's functionality
- To advocates of object-oriented techniques, highlight the advantages of flexibility, extensibility, modularity, abstraction, and encapsulation in such systems.
- Apply object-oriented analysis and design principles to real-world scenarios and case studies, ensuring that students can translate theoretical knowledge into practical solutions for developing software systems.
- Gain a comprehensive understanding of the Object-Oriented Analysis and Design (OOA&D) processes, including industry-standard methodologies like the Unified Process, and appreciate the benefits of adopting an object-oriented approach in software development.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Show the programming and problem-solving skills from a object-oriented software developer	PLO4	C3	Fundamental Domain
CLO2	Apply object-oriented design principles using an Object Oriented Language on real life problems.	PLO3	C2	Fundamental Domain
CLO3	Compute domain modeling for efficient object-oriented design	PLO2	C2	Fundamental Domain

CLO4	Build applications using the best practices in Object oriented programming.	PLO5	C3, C5	Fundamental Domain
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Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1				√								
CLO2			√									
CLO3		√										
CLO4					√							

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Object Oriented Design	Syllabus Walkthrough, Interactive Q&A	N/A	CLO1
	Lesson 2: Basic Concepts Classes and Objects, Characteristics of Objects, Relationships Method.	Interactive Lecture, Whiteboard Diagramming	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Object Oriented Programming	Visual Presentation, Object Demonstration	Class Test, Midterm exam	CLO1
	Lesson 2: Programming as a tool, More than one programming as a language, Data-types, Variables, Basic operators, Basic I/O, Boolean values, Conditional execution loops	Live Coding Demonstration, Syntax Walkthrough	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: List and list processing, Logical and bitwise operations, functions, tuples, dictionaries and data processing	Hands-on Practice, Code Tracing	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Modules, packages, string, list methods, exceptions	Step-by-step Process Walkthrough, Interactive Q&A	Class Test, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Object Oriented Approach	Interactive Lecture, Real-world Analogy Discussion	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Classes, Methods, Objects, exception handling, working with files. Class Test -1	Hands-on Coding Practice, Error Diagnostics Workshop	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-5	Lesson 1: Analysis, Design, Modeling	Concept Mapping, Visual Lecture	Class Test, Assignment, Midterm exam	CLO1, CLO2

Lesson 1 & 2 [3 Hours]	Lesson 2: Systems and Software Quality Software Engineering Life Cycle Properties	Case Study Analysis, Group Discussion	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Analysis and Design, Structured vs. Object Approach Modeling	Comparative Analysis, Diagrammatic Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Modeling Problem and Solution, Modeling Perspective Class Test -2	Problem-Solving Workshop, Peer Review	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Processes and approaches to OOA & D.	Interactive Lecture, Framework Modeling	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Unified Process, Rational Unified Process, Agile Approaches, Other Approaches	Comparative Matrix Analysis, Collaborative Discussion	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Design Phase	Creative Brainstorming, Architecture Mapping	Midterm exam	CLO1, CLO2
	Lesson 2: Moving to Code, Internal Component Definition, Design for Reuse	Structural Blueprint Walkthrough, Interactive Presentation	Midterm exam	CLO1, CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Design Class Diagrams, Operations, Methods	Whiteboard Demonstration, Diagram Drawing Practice	Assignment, Final exam	CLO1, CLO2
	Lesson 2: Designing Quality into Modules	Metric Dissection Workshop, Peer Review of Expressions	Assignment, Final exam	CLO1, CLO2
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Iterating the Design, Refactoring, Design Best Practices, Cohesion, Complexity, Coupling, Congruence	Code Refactoring Lab, Logic Puzzles	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: UML Implementation Diagrams, Component Diagram, Application of the Component Diagram, Component Diagram Notation and Syntax	Diagram Walkthrough, Collaborative Design	Class Test, Assignment, Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Deployment diagram, Application of the Deployment Diagram, Deployment Diagram Notation and Syntax	Visual Presentation, Topology Blueprint Mapping	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Domain Modeling and The Information Model Structural Modeling, Objects, Attributes, and Associations, Links and Messages Class Test -3	Entity Relationship Analysis, Step-by-step Flowchart Drawing	Class Test, Assignment, Final exam	CLO1, CLO3

Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Information Modeling in the UML, Basic Constructs, Relationships, Stereotypes, Packages	Interactive Lecture, Notation Walkthrough	Assignment, Final exam	CLO4
	Lesson 2: Use Case and the Behavior Model (Use Case Model, Use Cases in the Life Cycle, Use Case Diagram, Use Case Description, Main Success Scenario, Alternate Paths, Preconditions and Post-Condition.	User Story Dissection, Guided Diagramming	Assignment, Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Exception Paths, Use Case Extensions, Use Case Relationships, Generalization, The <<extends>> Relationship, The <<includes>> Relationship, Extension Points, Packages	Scenario-based Problem Solving, Boundary Condition Mapping	Assignment, Final exam	CLO4
	Lesson 2: Basic Unified Modeling Language (Overview, Basic Concepts, Classifiers, Well-Formedness Rules, Basic Notation, Classes and Objects, Characteristic of Objects, Relationships and Methods, Other Notation, UML Diagrams)	Comprehensive System Walkthrough, Matrix Review	Assignment, Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Presentation, Open discussion	CLO4
	Lesson 2: Evaluating presentation and designed model. Overview of final exam syllabus	Integrative Review, Board Problem Solving	Presentation, Open discussion	CLO1, CLO2

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO2	PLO3	PLO4	PLO5	
Attendance	--	--	--		7					
Class Test 1	5(15)				Average 15					
Class Test 2		5(15)		5						
Class Test 3			5(15)	5						
Assignment				5	5	5				5
Presentation					8					
Midterm Exam	10	10	05		25	05	10	10		25
Final exam		10	23	07	40	23	10		07	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Herbert Schildt, Java: The Complete Reference, 9th Edition, McGraw-Hill Education, 2014, ISBN: 9789339212094.

Reference Books/ Other Supplementary Readings

C. Thomas Wu, An introduction to object-oriented programming with Java, 3rd Edition, McGraw-Hill, 2004., ISBN: 0073040959

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 214 Algorithm Design & Analysis

Course Code: SE 214		CIE Marks: 60
ISCED: 0613-214		SEE Marks: 40
Course Title: Algorithm Design & Analysis		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE131, SE132		
Course Type: Core		

CIE: Continuous Internal Evaluation,

SEE: Semester End Examination

Course Content (from syllabus)

Basic strategies of algorithm design: top-down design, divide and conquer, average and worst-case criteria, asymptotic costs. Simple recurrence relations for asymptotic costs. Choice of appropriate data structures: arrays, lists, stacks, queues, trees, heaps, priority queues, graphs. Applications to sorting and searching, matrix algorithms, shortest-path and spanning tree problems. Introduction to discrete optimization algorithms: dynamic programming, greedy algorithms. Graph algorithms: depth-first and breadth-first search.

Rationale of the Course

This course provides a comprehensive introduction to algorithmic design and analysis for efficient problem-solving. Starting with top-down design, students will evaluate algorithm performance using average or worst-case criteria, asymptotic costs, and recurrence relations. The curriculum covers core design paradigms, including divide-and-conquer, greedy algorithms, and dynamic programming for discrete optimization. Students will strategically apply data structures such as heaps, trees, and graphs to fundamental problems, including sorting, searching, shortest paths, and minimum spanning trees.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of algorithm, More specifically,

- To apply knowledge of computing and mathematics to algorithm design.
- To analyze a problem and identify the computing requirements appropriate for its solution.
- To design, implement, and evaluate an algorithm to meet desired needs.
- To apply mathematical foundations, algorithmic principles, and computer science theory to the modeling and design of computer-based systems in a way that demonstrates comprehension of the trade-offs involved in design choices.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Analyze time complexity and space complexity of various algorithms or any written code using mathematical formula and comparison of algorithms.	PLO2	C4	Fundamental Domain
CLO2	Interpret the output of iterative and recursive codes with the analysis of the problem definition.	PLO2	C2	Fundamental Domain
CLO3	Analyze which algorithm is listed under which algorithmic paradigm.	PLO2	C4	Fundamental Domain
CLO4	Decide appropriate algorithms to implement real-life problems.	PLO3	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3		√										
CLO4			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: OBE Discussion, Course outline brief discussion	Syllabus Walkthrough, Interactive Q&A	Class Test, Midterm exam	CLO1
	Lesson 2: Introduction to algorithm terminologies	Interactive Lecture, Conceptual Mapping	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: to Big O Notation, Time & Space complexity analysis.	Guided Problem-Solving, Complexity Tracing	Class Test, Midterm exam	CLO1
	Lesson 2: Introduction to Divide & Conquer Approach Algorithms and Analysis of Binary search Algorithm's Applications and complexity analysis	Comparative Analysis, Step-by-step Trace Table	Class Test, Midterm exam	CLO1
Week-3	Lesson 1: Introduction to Brute Force Algorithms algorithm and complexity analysis	Real-world Analogy Discussion, Algorithm Dissection	Class Test, Midterm exam	CLO1, CLO2

Lesson 1 & 2 [3 Hours]	Lesson 2: Selection Sort algorithm and complexity analysis	Whiteboard Demonstration, Trace Table Practice	Class Test, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Insertion Sort and Radix Sort algorithm and complexity analysis	Hands-on Array Sorting, Visual Demonstration	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Merge Sort and Quick Sort algorithm and complexity analysis Class Test - 1	Collaborative Execution, Recursion Tree Mapping	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Greedy Algorithm Concept	Interactive Lecture, Optimization Analogy Discussion	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Coin Change Problem	Greedy Strategy Modeling, Counter-example Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Fractional Knapsack Problem	Guided Problem-Solving, Value-to-Weight Matrix	Assignment, Midterm exam	CLO2, CLO3
	Lesson 2: Huffman Coding Problem Class Test - 2	Step-by-step Tree Construction, Binary Encoding	Assignment, Midterm exam	CLO2, CLO3
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Optimization Problem Analysis By Bin Packing Algorithm	Heuristic Strategy Analysis, Whiteboard Practice	Class Test, Assignment, Midterm exam	CLO2, CLO3
	Lesson 2: Basics of Dynamic programming, Optimal Substructure & Overlapping sub-problems	Dynamic Programming Matrix Modeling, Comparison	Class Test, Assignment, Final exam	CLO2, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Factorial Number Problem using optimal substructure of DP	Memoization vs Tabulation, Logic Tracing	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Knapsack problem with DP Solution	2D Array State Design, Step-by-step Demonstration	Class Test, Assignment, Final exam	CLO1, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Coin Change and Fibonacci Series with DP	Recursive Tree Deconstruction, State Mapping	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Longest Common Subsequence and Longest Increasing Subsequence Problem-Solving	Grid-based Derivation, Row-by-row Execution	Class Test, Assignment, Final exam	CLO1, CLO3
Week-10	Lesson 1: Introduction to Graph Algorithm, Graph Representation, Adjacency matrix, Adjacency List	Visual Graph Translation, Data Structure Mapping	Presentation, Final exam	CLO1, CLO3

Lesson 1 & 2 [3 Hours]	Lesson 2: Applications of Graph, Graph Coloring, Cycling, Strongly Connected Component Algorithm	Constraint Logic Puzzles, Component Extraction	Presentation, Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Breadth First Search and Depth First Search Algorithm analysis.	Queue vs Stack Tracing, Graph Traversal Simulation	Presentation, Final exam	CLO3, CLO4
	Lesson 2: Prim's Algorithm for Minimum Spanning Tree (MST) Class Test - 3	Vertex-based Growing Demonstration, Step-by-step Trace	Presentation, Final exam	CLO3, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Kruskal Algorithm for Minimum Spanning Tree (MST)	Edge-based Disjoint Set Tracking, Cycle Detection	Presentation, Final exam	CLO3, CLO4
	Lesson 2: Floyd Warshall Algorithm	All-Pairs Shortest Path Matrix Tracing, Multi-step Update	Presentation, Final exam	CLO3, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Bellman Ford Algorithm	Edge Relaxation Iteration, Negative Cycle Evaluation	Final exam	CLO3, CLO4
	Lesson 2: Dijkstra Algorithm	Priority Queue Node Selection, Distance Update Lab	Final exam	CLO3, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Discussion on Final Exam topic	Integrative Review, Board Problem Solving	Final exam	CLO1, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO3	
Attendance	--	--	--		7				
Class Test 1	5(15)				Average 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment		5			5		5		5

Presentation					8				
Midterm Exam	5	15	5		25		25		25
Final exam			20	20	40		20	20	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms, 3rd Edition, MIT Press, 2022, ISBN:9788120340077.

Reference Books/ Other Supplementary Readings

1. Robert Sedgewick, Algorithms in C, 3rd Edition, Addison-Wesley Professional, 1998, ISBN:020175608

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 215 Algorithm Design and Analysis Lab

Course Code: SE 215		CIE Marks: 60
ISCED: 0613-215		LF Marks: 40
Course Title: Algorithm Design and Analysis Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: SE131, SE132		

Course Type: Core**CIE:** Continuous Internal Evaluation**LF:** Lab Final**Course Content (from syllabus)**

Big O Notation, Space and time complexity. Function and Recursion. Choice of appropriate data structures: Arrays, Lists, Stacks, Queues, Trees, Heaps, Priority Queues, Graphs. Sorting algorithms: Bubble Sort, Improved Bubble Sort, Selection Sort, Merge Sort, Quick Sort, Radix Sort. Search algorithms: Linear Search, Binary Search. Matrix Algorithms. Introduction to discrete optimization algorithms: dynamic programming techniques, Fractional Number, Knapsack, Longest Common Subsequence, Greedy algorithms: Coin Change Problem, Huffman Coding. Graph representations: Adjacency List, Adjacency Matrix. Graph algorithms: Depth First Search, Breadth First Search, Shortest Path, Minimum Spanning Tree, Floyd Warshall, Dijkstra Algorithm, Bellman Ford, Kruskal's Algorithm, Prim's Algorithm.

Rationale of the Course

The Algorithm Design and Analysis Lab fosters a deeper understanding of the theoretical concepts learned in the accompanying course. Through practical implementation, students gain the ability to: design efficient algorithms using fundamental strategies like divide and conquer, analyze algorithm complexity in terms of asymptotic costs and data structures, select appropriate data structures for various problem types, including arrays, trees, and graphs, apply algorithms to real-world problems like sorting, searching, and graph traversal, and explore optimization techniques like dynamic programming and greedy algorithms. This hands-on experience equips students with the critical skills to become effective problem solvers and algorithm designers.

Course Objectives

By the end of course through lectures, readings, home works, lab assignments and exams, students will be taught:

- To apply knowledge of computing and mathematics to algorithm design;
- To analyze a problem and identify the computing requirements appropriate for its solution;
- To design, implement, and evaluate an algorithm to meet desired needs;
- To apply mathematical foundations, algorithmic principles, and computer science theory to the modeling and design of computer-based systems in a way that demonstrates comprehension of the trade-offs involved in design choices.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Apply mathematical principles to select appropriate data structures and algorithms suitable for specific scenarios.	PLO1	C3	Fundamental Domain
CLO2	Apply modern programming languages and libraries to implement, analyze, and visualize algorithm performance, showcasing an understanding of their strengths and limitations in various scenarios.	PLO5	C3, P2	Fundamental Domain
CLO3	Investigate algorithmic problems to formulate clear and concise problem statements, develop and analyze	PLO4	C3, P3	Fundamental Domain

	alternative solutions, evaluate strengths and weaknesses of different approaches, and justify their design choices.			
CLO4	Communicate the design and analysis of algorithms, both orally and in writing, through presentations, reports, and code documentation, adhering to professional standards and clarity.	PLO10	C5, A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2					√							
CLO3				√								
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Discussion on course outline, course learning outcomes, assessment methods, deadlines, and other administrative information. Setting up a programming environment, and introduction to different algorithmic terminologies.	Syllabus Walkthrough, Environment Setup Lab	CLO1, CLO2, CLO3	Lab Performance
Week-2 [1.5 Hours]	Designing Divide & Conquer techniques in real life problems. Analyzing time complexity by Implementing	Live Coding Demonstration, Complexity Profiling	CLO1	Lab Performance
Week-3 [1.5 Hours]	Designing Brute Force algorithms. Implementing Selection Sort and Insertion Sort algorithm and complexity analysis.	Hands-on Coding Practice, Code Tracing	CLO2, CLO3	Lab Performance
Week-4 [1.5 Hours]	Implementing Merge Sort and Quick sort algorithm and complexity analysis.	Recursive Tree Walkthrough, Peer Code Debugging	CLO2, CLO3	Viva, Lab Final
Week-5 [1.5 Hours]	Implementing Bin Packing Problem in different case scenario.	Step-by-step Implementation, Test-Case Analysis	CLO3, CLO4	Viva, Lab Final
Week-6 [1.5 Hours]	Designing and implementing greedy algorithms. Solving Fraction Knapsack and Coin Change problems.	Logic Framework Tracing, Hands-on Programming	CLO2, CLO3	Lab Performance, Viva, Lab Final

Week-7 [1.5 Hours]	Implementing Huffman Coding algorithm and review of class on the topics covered in previous weeks.	Tree Construction Practice, Midterm Code Review	CLO1	Lab Performance
Week-8 [1.5 Hours]	Designing dynamic algorithms: identifying optimal substructures & overlapping sub-problems. Solving Factorial Number problem, Knapsack Problem, Fibonacci Series, and Longest Common Subsequence problems.	Matrix-based State Tracking, Hands-on Coding	CLO1	Lab Performance, Viva, Lab Final
Week-9 [1.5 Hours]	Designing Graphs: Adjacency List, Adjacency Matrix, etc. Implementing Graph searching algorithms such as Breadth First Search, Depth First Search, Applications on Graph etc.	Visual Matrix Mapping, Traversal Debugging	CLO2, CLO3	Lab Performance, Viva, Lab Final
Week-10 [1.5 Hours]	Constructing minimum spanning trees using Prim's algorithm and Kruskal's algorithm.	Interactive Graph Tracing, Edge Relaxation Lab	CLO4	Lab Performance, Viva, Lab Final
Week-11 [1.5 Hours]	Implementing Dijkstra's algorithm and Bellman Ford algorithm to find shortest path between any nodes in a graph.	Shortest Path Simulation, Boundary Case Debugging	CLO2, CLO3	Lab Performance, Viva, Lab Final
Week-12 [1.5 Hours]	Implementing Floyd Warshall algorithm to find shortest path between all pairs of nodes in a graph.	Multi-dimensional Array Mapping, Code Optimization	CLO2, CLO4	Lab Performance
Week-13 [1.5 Hours]	Industry Session	Industry Expert Talk, Production Code Review	CLO1, CLO2, CLO3, CLO4	Lab Performance, presentation
Week-14 [1.5 Hours]	Lab Final	Independent Practical Exam, Code Execution Task	CLO1, CLO2, CLO3, CLO4	Viva, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO1
Lab Final	40	CLO2	PLO5
		CLO3	PLO4
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms, 3rd Edition, MIT Press, 2022, ISBN:9788120340077.

Reference Books/ Other Supplementary Readings

Robert Sedgewick, Algorithms in C, 3rd Edition, Addison-Wesley Professional, 1998, ISBN:0201756080

Lab Manual

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

CodeBlocks, Visual Studio, Netbeans, Eclipse, Online Compiler

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 235 Desktop & Web Programming

Course Code: SE 235		CIE Marks: 60
ISCED: 0613-235		SEE Marks: 40
Course Title: Desktop & Web Programming		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE 121, SE216		
Course Type: Core		

CIE: Continuous Internal Evaluation,
SEE: Semester End Examination

Course Content (from syllabus)

The Desktop & Web Programming course offers a comprehensive exploration of software development, covering essential concepts and practical skills necessary for crafting applications across desktop and web platforms. Encompassing programming fundamentals, frontend and backend development techniques, application security, and project deployment strategies, this course equips students with a robust understanding of creating functional, user-centric, and secure software solutions in both desktop and web environments. Additionally, the course touches upon emerging trends, providing insights into the future landscape of software development.

Rationale of the Course

The Desktop & Web Programming course is a comprehensive exploration of software development encompassing both desktop and web-based applications. Beginning with foundational programming concepts and languages pertinent to desktop and web development, the course branches into frontend and backend aspects of web development. It covers HTML, CSS, responsive design, JavaScript frameworks, server-side scripting, and database integration. Emphasizing security measures in web applications and practical projects, students gain hands-on experience in creating desktop applications and deploying web solutions. The course culminates by delving into emerging trends, ensuring students are equipped to develop secure, user-centric applications across diverse platforms.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of Desktop & Web programming, More specifically,

- To introduce fundamental principles of desktop and web-based application development.
- To develop programming proficiency in languages for both desktop (Java, C#, Python) and web (JavaScript, PHP) environments.
- To attain mastery in frontend development, covering HTML, CSS, responsive design, and JavaScript frameworks.
- To gain competence in backend technologies, including server-side scripting, databases, and backend frameworks.
- To apply knowledge through practical projects, creating desktop applications and deploying web solutions.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcome s (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Describe the comprehensive understanding acquired in programming languages utilized for desktop and web development, emphasizing syntax, structures, and fundamental concepts.	PLO2	C2	Fundamental Domain
CLO2	Analysis of diverse application architectures employed in desktop and web programming, evaluating frontend and backend integration and functionalities.	PLO2	C4	Fundamental Domain
CLO3	Apply the concept of backend technologies, including server-side scripting, databases, and backend frameworks to implement problem-solving tasks.	PLO3	C3	Fundamental Domain
CLO4	Apply and utilize version control and collaborative tools to efficiently manage and deploy projects in desktop and web programming.	PLO3	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3			√									
CLO4			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
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Lesson (hour)				
Week-1 Lesson 1 & 2 [2.5 Hours]	Lesson 1: OBE Discussion, Course outline brief discussion	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Introduction to Course terminologies	Interactive Lecture, Conceptual Mapping	None	None
Week-2 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Introduction of Basic Concepts of Object-Oriented Programming, Classes and Objects	Visual Presentation, Object Demonstration	Class Test, Midterm exam	CLO1
	Lesson 2: Characteristics of Objects and Relationships and Methods	Diagrammatic Analysis, Real-world Analogy Discussion	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Advantages of OOP over structured programming and Access specifiers	Comparative Analysis, Access Matrix Walkthrough	Class Test, Midterm exam	CLO1
	Lesson 2: Static and non-static members Class Test - 1	Whiteboard Demonstration, Memory Allocations Tracing	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Control Structures (conditional Statement & Loops)	Live Coding Demonstration, Syntax Walkthrough	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Array of objects, Object pointers, and Object references	Hands-on Code Tracing, Pointer Mapping Practice	Class Test, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Abstraction, Abstract classes and methods, Abstract layer	Concept Mapping, Interface Blueprint Design	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Basic principles of Inheritance and polymorphism	Hierarchy Diagramming, Dynamic Binding Workshop	Class Test, Midterm exam	CLO2
Week-6 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Introduction to HTML5 and its structure	Live Tag Walkthrough, Document Tree Analysis	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Introduction to CSS and its structures	Style Sheet Dissection, Rules-based Practice	Class Test, Midterm exam	CLO2
Week-7 Lesson 1 & 2 [2.5 Hours]	Lesson 1: CSS3 fundamentals for styling web content	Layout Box-Model Workshop, Visual Presentation	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Building basic web pages using HTML and CSS Class Test -2	Collaborative Layout Design, Hands-on Coding	Class Test, Midterm exam, Final exam	CLO2
Week-8 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Fundamentals of JavaScript language	Scripting Syntax Walkthrough, Console Practice	Class Test, Assignment, Midterm exam, Final exam	CLO2
	Lesson 2: Introduction to Document Object Model (DOM)	DOM Tree Visualisation, Structural Walkthrough	Class Test, Assignment, Midterm exam	CLO2
Week-9	Lesson 1: Document Object Model (DOM) and its manipulation	Live Code Manipulation, Element Selection Lab	Class Test, Assignment, Midterm exam,	CLO1, CLO2

Lesson 1 & 2 [2.5 Hours]				
	Lesson 2: Basic JavaScript functions and events handling	Event Listener Wiring, Peer Problem-Solving	Class Test, Midterm exam,	CLO1, CLO2
Week-10 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Introduction to frontend frameworks	Architecture Mapping, Component Paradigm Discussion	Class Test, Final exam	CLO3
	Lesson 2: React and Angular framework	Comparative Matrix Analysis, Visual Presentation	Class Test, Final exam	CLO3
Week-11 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Vue.js framework, State management and component-based architecture	Component Flow Diagramming, State Tracking Lab	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Authentication & Encryption, Secure coding practices and avoiding vulnerabilities	Case Study Analysis (Real-world vulnerabilities), Threat Scenario Discussion	Class Test, Final exam	CLO3
Week-12 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Server-side scripting, Node js, Python, PHP	Request-Response Cycle Mapping, Backend Setup Walkthrough	Class Test, Assignment, Presentation, Final exam	CLO3
	Lesson 2: Introduction to Desktop application frameworks (JavaFX, .NET), GUI design principles	UI Layout Prototyping, Wireframe Analysis	Class Test, Final exam	CLO3
Week-13 Lesson 1 & 2 [2.5 Hours]	Lesson 1: local data management	Schema Mapping, Query Construction Practice	Final exam	CLO4
	Lesson 2: Project management, version control, and collaborative development practices	Git Branching Simulation, Team Workflow Mapping	Final exam	CLO4
Week-14 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on topic discussed in Week-14- Week-17	Integrative Review, Board Problem Solving	Final exam	CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO3	
Attendance	--	--	--	--	7				
Class Test 1	5(15)				Avg				

Class Test 2		5(15)			15				
Class Test 3			5(15)						
Assignment		5			5		5		5
Presentation			8		8				
Midterm Exam	10	15			25		25		25
Final exam			20	20	40			40	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Gerti Kappel, Web engineering : the discipline of systematic development of web applications, John Wiley & Sons, c2006., ISBN: 9780470015544

Reference Books/ Other Supplementary Readings

Nixon, Robin. Learning PHP, MySQL & JavaScript with jQuery, CSS & HTML5, 4th Edition, O'Reilly Media, 2015. ISBN: 9789352130153.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 236 Desktop & Web Programming Lab

Course Code: SE 236		CIE Marks: 60
ISCED: 0613-236		LF Marks: 40
Course Title: Desktop & Web Programming Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: SE 216, SE 217		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

Setting up the development environment for Java, Basic Java syntax, and programming concepts, Hands-on exercises on variables, data types, and operators, Control structures in Java (loops and conditionals), Object-Oriented Programming (OOP) principles in Java. Implementing classes, objects, inheritance, and polymorphism in Java, Creating a basic HTML5 webpage, Introduction to CSS3 and styling web content, Building basic web pages using HTML and CSS, Fundamentals of JavaScript language, Manipulating the Document Object Model (DOM). Basic JavaScript functions and events handling, Introduction to a frontend framework (React, Angular, or Vue.js), Hands-on development using the selected framework, Implementing state management and component-based architecture, Server-side scripting with Node.js, Python, or PHP, Integrating databases (SQL, NoSQL) in backend development, Creating RESTful APIs for web applications, Identifying and mitigating common security threats, Implementation of security measures (authentication, encryption), Hands-on exercises on secure coding practices and avoiding vulnerabilities, Introduction to desktop application frameworks (JavaFX, .NET), GUI design principles and local data management, Project management, version control, and collaborative development practices

Rationale of the Course

This comprehensive course bridges foundational software engineering with modern full-stack web and desktop development. Students begin with Java syntax and Object-Oriented Programming (OOP) before mastering frontend essentials—HTML5, CSS3, JavaScript, and component-based frameworks like React. The curriculum then advances to server-side scripting, database integration, and building secure RESTful APIs with robust authentication and encryption. Additionally, it covers desktop GUI design using JavaFX and .NET. Emphasizing secure coding, version control, and collaborative project management, this course equips learners with industry-standard skills to deploy scalable, multi-platform software solutions in dynamic professional environments.

Course Objectives

The goal of this course is to introduce the students to the concept of database principles in practice. The main objectives of this course are,

- . To have a comprehensive understanding of object-oriented programming concepts.
- . To acquire knowledge of web programming applications.
- . To implement frontend frameworks and backend technologies.

To identify the errors and security vulnerabilities of software.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Demonstrate the basic knowledge of Object-Oriented Programming	PLO2	P2	Fundamental Domain
CLO2	Implement essential skills in web development.	PLO2	C3	Fundamental Domain
CLO3	Apply and Implement Advanced Framework and Backend Development	PLO3	C3, P2	Fundamental Domain
CLO4	Design and develop software using best practices in real life requirements	PLO5	C6, P5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3			√									
CLO4					√							

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning, and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.15 Hours]	Discussion on OBE & Introduction to OOP	Syllabus Walkthrough, Environment Setup Lab	N/A	N/A
Week-2 [1.15 Hours]	Setting Up the Development Environment, Exercise on variables, data types and operators	Live Coding Demonstration, Console Syntax Practice	CLO1	Lab Performance
Week-3 [1.15 Hours]	Control structures and OOP principles, Implementing Class, Object Concept	Hands-on Coding Practice, Object Structure Tracking	CLO1	Lab Performance

Week-4 [1.15 Hours]	Create a basic HTML5 webpage and Introduction to CSS3 and styling web content	Live Tag Walkthrough, Visual Layout Styling	CLO2	Lab Report, Lab Final
Week-5 [1.15 Hours]	Building basic web page using HTML and CSS, Fundamental of JavaScript and DOM	DOM Tree Selection Lab, Live Code Manipulation	CLO2	Lab Report, Lab Final
Week-6 [1.15 Hours]	Exploring javascript functions and event handling, Introduction to a frontend framework (React, Angular, or Vue.js) and Hands-on development using the selected framework	Event Listener Wiring, Component Execution Lab	CLO3	Lab Performance, Lab Report, Lab Final
Week-7 [1.15 Hours]	Implementing state management and component-based architecture, Server-side scripting with Node.js, Python, or PHP	Component Flow Diagramming, Backend Setup Practice	CLO3	Lab Performance, Lab Report, Lab Final
Week-8 [1.15 Hours]	Integrating databases (SQL, NoSQL) in backend development and Creating RESTful APIs for web applications	Query Construction Practice, API Route Mapping	CLO4	Lab Performance, Lab Report, Lab Final
Week-9 [1.15 Hours]	Identifying and mitigating common security threats	Threat Scenario Discussion, Vulnerability Analysis	CLO4	Lab Performance, Lab Report, Lab Final
Week-10 [1.15 Hours]	Introduction to a frontend framework (React, Angular, or Vue.js) and Hands-on development using the selected framework	Live Framework Demonstration, Component Building	CLO3	Lab Performance
Week-11 [1.15 Hours]	Implementation of security measures (authentication, encryption) and Hands-on exercises on secure coding practices and avoiding vulnerabilities	Secure Coding Lab, Cipher Scripting Exercises	CLO4	Lab Performance, Lab Report, Lab Final
Week-12 [1.15 Hours]	Introduction to desktop application frameworks (JavaFX, .NET)	UI Layout Prototyping, Wireframe Coding	CLO4	Lab Performance
Week-13 [1.15 Hours]	Industry Session	Industry Expert Talk, Production Code Review	CLO1, CLO2, CLO3, CLO4	Presentation, Assignment
Week-14 [1.15 Hours]	Lab Final	Independent Practical Exam, Code Execution Task	CLO1, CLO2, CLO3, CLO4	Lab Report, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO5
Lab Performance	25	CLO1	PLO2
Lab Final	40	CLO2	PLO2
		CLO3	PLO3
Total	100		

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Gerti Kappel, Web engineering : the discipline of systematic development of web applications, John Wiley & Sons, c2006., ISBN: 9780470015544

Reference Books/ Other Supplementary Readings

Nixon, Robin. Learning PHP, MySQL & JavaScript with jQuery, CSS & HTML5, 4th Edition, O'Reilly Media, 2015. ISBN: 9789352130153.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Visual Studio Code, PHP, Laravel, XAMPP, Git, Web browser (Chrome/Firefox), Firebase

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 223 Database System

Course Code: SE 223		CIE Marks: 60
ISCED: 0613-223		SEE Marks: 40
Course Title: Database System		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Databases and Database Users: Introduction - An Example, Characteristics of the Database Approach, Advantages of Using the DBMS Approach. Database System Concepts and Architecture: Data Models, Schemas and Instances. Three- Schema Architecture and Data Independence, Database Languages and Interfaces, The Database System Environment, Centralized and Client/Server Architectures for DBMSs. Entity-Relationship (ER) Model :Using High-Level Conceptual Data Models for Database Design, An Example Database Application, Entity Types, Entity Sets, Attributes, and Keys - Relationship Types, Relationship Sets, Roles, and Structural Constraints, Weak Entity Types, Refining the ER Design for the COMPANY Database - ER Diagrams, Naming Conventions, and Design Issues The Relational Algebra and Relational Calculus: Unary Relational Operations: SELECT and PROJECT, Relational Algebra Operations from Set Theory. SQL Data Definition and Data Types, Specifying Constraints in SQL, Schema Change Statements in SQL, Basic Queries in SQL INSERT, DELETE, and UPDATE Statements in SQL, Views (Virtual Tables) in SQL. Functional Dependencies and Normalization for Relational Databases: Informal Design Guidelines for Relation Schemas, Functional Dependencies, Normal Forms Based on Primary Keys - General Definitions of Second and Third Normal Forms, Boyce-Codd Normal Form. Introduction to Transaction Processing, Transaction and System Concepts, Desirable Properties of Transactions, Characterizing Schedules Based on Recoverability, Characterizing Schedules Based on Serializability, Concurrency Control Techniques: Two-Phase Locking Techniques for Concurrency Control.

Rationale of the Course

This course provides a comprehensive introduction to the principles, architecture, and practical application of database management systems. Students begin with core concepts like the three-schema architecture and data independence before mastering conceptual design using the Entity-Relationship (ER) model. The curriculum establishes a formal foundation in relational algebra and calculus, transitioning directly into practical SQL operations, data definitions, and constraints. To ensure structural integrity and efficiency, the course covers normalization techniques alongside advanced topics in transaction processing and concurrency control mechanics like two-phase locking.

Course Objectives

The main objectives of this course are, to present an introduction to database management systems, with an emphasis on how to organize, maintain and retrieve - efficiently, and effectively - information from a DBMS. More specifically,

- To learn the basic concepts of databases.

- To learn the relational model.
- To learn database design.
- To learn transactions of data in a database.
- To learn the concepts and techniques for developing databases using database language.
- To learn normalization concept to reduce anomalies in database
- To learn the basic concept of a relational database.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLO's	CLO Descriptions	PLO's	Learning Domains (C,P,A)	BNQF Skill
CLO1	Describe the basic concepts and uses of databases with different applications/Systems.	PLO2	C4	Fundamental Domain
CLO2	Apply different stages of database development using different data models	PLO3	C4	Fundamental Domain
CLO3	Investigate information from data and database objects through the use of SQL.	PLO4	C3	Fundamental Domain
CLO4	Transform the database design by normalization.	PLO3	C2, C4 P3 A2	Fundamental Domain, Thinking Domain, Personal Domain
CLO5	Generalize transaction processing, concurrency control and recovery techniques	PLO 2	C2	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2			√									
CLO3				√								
CLO4			√									
CLO5		√										

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Course outline, CLO-PLO mapping	Syllabus Walkthrough, Interactive Q&A	N/A	CLO1
	Lesson 2: Databases and Database Users: Introduction - An Example, Characteristics of the Database Approach, Advantages of Using the DBMS Approach, File processing system, database architecture, database view	Comparative Analysis, Schema Architecture Mapping	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Database System Concepts and Architecture: Data Models, and Instances. Three- Schema Architecture and Data Independence, Database Languages and Interfaces	Three-Schema Walkthrough, Interactive Lecture	Class Test, Midterm exam	CLO1
	Lesson 2: Database administrator concepts, Components of database, The Database System Environment, Centralized and Client/Server Architectures for DBMSs.	Visual Presentation, Infrastructure Blueprint Mapping	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Basics relational algebra operations: introduction, Selection and projection.	Whiteboard Demonstration, Operator Execution Practice	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Set operations, Renaming, Joins, Basics relational algebra operations: syntax, semantics, Operators, grouping and ungrouping relational comparison.	Relational Expressions Tracing, Peer Problem-Solving	Class Test, Assignment, Midterm exam	CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Structure of Relational Databases, Database Schema, Keys, Schema Diagrams, Relational Query Languages.	Schema Diagram Analysis, Key Constraints Identification	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Entity-Relationship Model, Constraints, Diagrams- Unary, Binary, ternary, Aggregation. Case Study of ERD. Class Test - 1	ERD Notation Modeling, Case Study Review	Class Test, Presentation, Assignment, Midterm exam	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of the SQL Query Language, SQL Data Definition.	Live Query Demonstration, Syntax Walkthrough	Class Test, Assignment, Midterm exam	CLO3
	Lesson 2: Basic Structure of SQL Queries	Hands-on Writing Practice, Query Execution Tracing	Class Test, Assignment, Midterm exam	CLO3
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Additional Basic Operations, Set Operations, Aggregate Functions.	Interactive Lecture, Multi-table Join Breakdown	Assignment, Midterm, Final Examination	CLO3
	Lesson 2: Data Manipulation Language Class Test -2	DML Logic Exercises, Live Verification	Assignment, Midterm, Final Examination	CLO3

Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Nested sub queries	Step-by-step Query Deconstruction, Matrix Evaluation	Assignment, Midterm Final exam	CLO3
	Lesson 2: Summarize SQL concepts and implementation	Integrative System Tracing, Diagnostic Review	Assignment, Midterm Final exam	CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Features of Good Relational Designs, Basics of normalization, anomalies	Anomaly Diagnostics Workshop, Scenario Analysis	Class Test, Assignment, Final exam	CLO4
	Lesson 2: Atomic Domains and First Normal Form, Functional Dependencies, Closure set of Functional dependencies	Dependency Closure Calculation, Board Practice	Class Test, Assignment, Final exam	CLO4
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Procedure for Computing F+, Boyce Code Normal form, BCNF Decomposition Algorithm	Algorithm Step-by- step Execution, Table Split Tracing	Class Test, Final exam	CLO4
	Lesson 2: Third Normal Form, Third Normal Form, Decomposition Algorithm	Comparative Matrix Analysis, Dependency Mapping	Class Test, Final exam	CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Normalization to ERD	Reverse Engineering Lab, Visual Diagram Alignment	Class Test, Presentation, Final exam	CLO4
	Lesson 2: Case study of database design and normalization	Collaborative Design, Corporate Case Analysis	Class Test, Final exam	CLO5
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Transaction Concept, A Simple Transaction Model	State Transition Diagramming, Real- world Analogy Discussion	Class Test, Final exam	CLO5
	Lesson 2: Storage Structure, Transaction Atomicity and Durability, Serializability Class Test - 3	Conflict Serializability Graphing, Logic Puzzles	Class Test, Final exam	CLO5
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Concurrency Problems & Solutions	Deadlock Scenario Analysis, Locking Protocol Simulation	Final exam	CLO5
	Lesson 2: File Organization	Storage Structure Walkthrough, Diagrammatic Analysis	Final exam	CLO5
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Recovery System: Failure Classification, Storage	Threat Scenario Discussion, Log- based System Review	Final exam	CLO5
	Lesson 2: Recovery and Atomicity, Recovery Algorithm, ARIES, Remote Backup Systems	Fail-Safe Execution Flowcharting, ARIES Phase Analysis	Final exam	CLO5

Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Assignment	CLO1, CLO2, CLO3, CLO4, CLO5
	Lesson 2: Review class on the final exam syllabus topic	Integrative Review, Board Problem Solving	Open discussion	CLO2, CLO3, CLO4, CLO5

Overall Assessment Scheme

Assessment Task	CLO's					Mark (Total=100)	PLO's			Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4	CLO5		PLO2	PLO3	PLO4	
Attendance			--	--	--	7				
Class Test 1	5(15)					Avg 15				
Class Test 2		5(15)								
Class Test 3			5(15)							
Assignment		5				5				
Presentation				8		8				
Midterm Exam	10	10	5			25	10	10	5	25
Final exam		5	10	15	10	40	5	20	15	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Abraham Silberschatz, Henry Korth, S. Sudarshan, Database System Concepts, 7th Edition, McGraw-Hill Education, 2020, ISBN: 978-0078022159.

Reference Books/ Other Supplementary Readings

Raghu Ramakrishnan, Johannes Gehrke, Database Management Systems, 3rd Edition, Tata McGraw-Hill, 2003, ISBN: 9780071231510

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Xampp Server, MySQL Server

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 224 Database Systems Lab

Course Code: SE 224		CIE Marks: 60
ISCED: 0613-224		LF Marks: 40
Course Title: Database Systems Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

The "Database Systems Lab" course provides students with a comprehensive exploration of fundamental introduction to DBMS, Purpose of DBMS, Relational Model, Data Abstraction, Database Architecture, Keys, ER Model, ER Model Scenario and ER Model, Relational Schema, ER Model to Schema Practice Example, Relational Algebra, Basic SQL Queries, Join Operation, More SQL Queries, Sub-queries in DBMS, Different Operation on Table, Stored Procedure and Views, Transaction, More about Transaction, Trigger, Normalization, Join Operation etc.

Rationale of the Course

This lab course bridges database architecture with hands-on engineering. Students translate Entity-Relationship diagrams into concrete schemas, execute complex SQL queries, and apply normalization to eliminate

redundancy. The curriculum also covers critical backend mechanics like transaction management, triggers, and stored procedures. By balancing design theory with active code execution, this course builds the precise technical proficiency required to enforce data integrity and manage scalable database systems

Course Objectives

The goal of this course is to introduce the students to the concept of database principles in practice. The main objectives of this course are,

- To learn the implementation of the basic concept of a database.
- To learn database language.
- To learn various database tools.
- To implement SQL programming through a variety of database problems.
- To Develop database applications using front-end tools and back-end DBMS.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Perform a wide range of data manipulation activities using a query language for solving database queries	PLO1	C3	Fundamental Domain
CLO2	Apply appropriate techniques for representing the database-oriented application.	PLO5	C3, P2	Fundamental Domain
CLO3	Conduct investigations of database-oriented applications considering design to provide valid conclusions.	PLO4	C3, P3	Fundamental Domain
CLO4	Write effective reports and design documentation for the database oriented applications.	PLO10	C5, A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2					√							
CLO3				√								
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning, and assessment strategy mapped with CLOs:

Week/	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
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Lesson (hour)				
Week-1 [1.5 Hours]	Discussion on OBE of Database System Lab & Introduction to SQL, SQL*Plus	Syllabus Walkthrough, Environment Setup Lab	CLO1	Lab Performance
Week-2 [1.5 Hours]	Road way travels of E-R Diagrams and various Data Types & Key Constraints	Database Schema Mapping, Constraints Analysis	CLO1	Lab Performance
Week-3 [1.5 Hours]	Software Installations	Guided Installation, Environment Verification	CLO1	NONE
Week-4 [1.5 Hours]	Implementation of DDL commands of SQL with suitable examples: Create table, Alter table, Drop Table	Live Coding Demonstration, Schema Modification Practice	CLO1 CLO2, CLO4	Lab Performance, Lab Final, Lab Report
Week-5 [1.5 Hours]	Implementation of DML commands of SQL with suitable examples, Insert, Update Delete	Hands-on Data Manipulation, Row Modification Lab	CLO1, CLO2, CLO4	Lab Performance, Lab Final, Lab Report
Week-6 [1.5 Hours]	Implementation of different types of function with suitable examples, Number function, Aggregate Function, Character	Query Scripting Practice, Data Aggregation Exercises	CLO1, CLO3, CLO4	Lab Performance, Lab Final, Lab Report
Week-7 [1.5 Hours]	Implementation of different types of operators in SQL Arithmetic Operators, Logical Operators, Comparison Operator, Special Operator, Set Operation.	Logic Expression Testing, Dataset Intersection Lab	CLO1	Lab Performance, Lab Report
Week-8 [1.5 Hours]	Implementation of different types of Joins: Inner Join, Outer Join, Natural Join etc.	Multi-table Data Merging, Visual Join Analysis	CLO1, CLO3, CLO4	Lab Performance, Lab Final, Lab Report
Week-9 [1.5 Hours]	Study & Implementation of Sub queries & Views and different types of constraints.	Nested Query Deconstruction, Virtual Table Creation	CLO1, CLO3, CLO4	Lab Performance, Lab Final, Lab Report
Week-10 [1.5 Hours]	Study & Implementation of Database Backup & Recovery commands. Study & Implementation of Rollback, Commit, Savepoint.	Transaction Control Simulation, Fail-Safe Recovery Lab	CLO1, CLO3, CLO4	Lab Performance, Lab Final, Lab Report
Week-11 [1.5 Hours]	Creating Database, Table Space Managing Users: Create User, Delete User, Managing roles: Grant, Revoke.	DBA Administration Practice, Access Privilege Control	CLO4	Lab Report
Week-12 [1.5 Hours]	Study and Implementation of Group By & having clause, Order by clause, Indexing and Subprograms SQL-procedure PL/ S	PL/SQL Blocks Scripting, Query Performance Tuning	CLO2, CLO3	Lab Performance, Lab Final, Lab Report
Week-13 [1.5 Hours]	Industry Session	Industry Expert Talk, Production Database Review	CLO4	Lab Report
Week-14	Lab Final	Independent Practical Exam, Practical Task Execution	CLO2, CLO3	Lab Final

[1.5 Hours]				
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Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO1
Lab Final	40	CLO2	PLO5
		CLO3	PLO4
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Michael J. Hernandez, Database Design for Mere Mortals: A Hands-On Guide to Relational Database Design, 3rd Edition, Pearson Education, 2021, ISBN-13: 9780133122275.

Reference Books/ Other Supplementary Readings

1. Raghu Ramakrishnan, Johannes Gehrke, Database Management Systems, 3rd Edition, Tata McGraw-Hill, 2003, ISBN: 9780071231510

2. Lab Manual

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

XAMPP, MySQL

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 232 Operating System and System Programming

Course Code: SE 232		CIE Marks: 60
ISCED: 0613-232		SEE Marks: 40
Course Title: Operating System and System Programming		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)		Contact Hours: 3 (Total weeks: 14)
Prerequisite: SE 222		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End

Examination

Course Content (from syllabus)

This course introduces modern operating systems. It focuses on UNIX-based operating systems, though alternative operating systems, including Windows, are introduced. This course is beginning with an overview of the structure of modern operating systems. Throughout the subsequent units, discuss the history of modern computers, analyze in detail each of the major components of an operating system (from processes to threads), and explore more advanced topics in the field, including concurrency (synchronization, mutual exclusion, deadlock, starvation), memory (both primary and secondary) management and input/output file management and organization (segmentation, paging, swapping), file systems, and operating system support for distributed systems. Different CPU scheduling algorithms and disk scheduling algorithms have also been discussed in detail.

Rationale of the Course

This course explores the operating system as a vital abstraction layer between hardware and application software. Students examine how the OS manages core resources through CPU scheduling, process concurrency, and synchronization mechanisms to prevent data conflicts. The curriculum covers memory management strategies, explicitly introducing virtual memory and paging to optimize limited physical resources. Additionally, it details file system organization for efficient data storage and retrieval.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of Software Engineering. More specifically,

- To explore the processes and threads roles, states, components, scheduling, and concurrency of the modern operating system.
- To demonstrate the process scheduler.
- To explain and analyze the concurrency problem including synchronization, mutual exclusion, deadlock and starvation.
- To explain and solve problems on memory management (primary and secondary) and virtual memory management systems and file management systems.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Identify the basic functions and principles of operating systems.	PLO2	C1	Fundamental Domain
CLO2	Apply different algorithms for the solution of major components of operating systems	PLO1	C3	Fundamental Domain
CLO3	Analyze the performance of control manager, memory manager, I/O manager, and file manager	PLO2	C4	Fundamental Domain
CLO4	Evaluate the best technique for a specific problem related to the function of the operating system.	PLO2	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2	√											
CLO3		√										
CLO4		√										

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Course outline, CLO-PLO mapping	Syllabus Walkthrough, Interactive Q&A	N/A	CLO1
	Lesson 2: Operating System Overview Operating System Objectives and Functions. Characteristics of modern Operating System, The evolution of Operating System	Interactive Lecture, Timeline Jigsaw	Class Test, Midterm exam	CLO1
Week-2	Lesson 1: Operating System Structure	Visual Presentation, Kernel Architecture Diagramming	Class Test, Midterm exam	CLO1, CLO2

Lesson 1 & 2 [3 Hours]	Lesson 2: Different types of Operating System with their importance.	Comparative Analysis, Group Discussion	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Process Description and Control. Process concept and state.	State Transition Diagramming, Interactive Lecture	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Process Description Process Control UNIX FreeBSD Process Management	Structure Walkthrough, Step-by-step Process Tracing	Class Test, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Threads, SMP, and Microkernels Processes and Threads Symmetric Multiprocessing (SMP).	Whiteboard Comparative Analysis, Visual Lecture	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Microkernels Windows Vista Thread and SMP Management Linux Process and Thread Management Class Test -1	Case Study Analysis, Architectural Walkthrough	Class Test, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Uniprocessor Scheduling Scheduling Algorithms	Whiteboard Demonstration, Gantt Chart Drawing	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Scheduling Algorithms	Guided Problem-Solving, Peer Problem-Solving	Class Test, Assignment, Midterm exam	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Concurrency: Mutual Exclusion and Synchronization	Real-world Analogy Discussion, Interactive Lecture	Final exam	CLO3
	Lesson 2: Principles of Concurrency, Mutual Exclusion: Hardware Support Class Test -2	Logic Analysis, Hardware Control Tracing	Final exam	CLO3
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Semaphore, Monitors	Step-by-step Process Walkthrough, Logic Puzzles	Final exam	CLO2, CLO3
	Lesson 2: Message Passing Readers/Writers Problem	Classical Concurrency Modeling, Conflict Analysis	Final exam	CLO2, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Concurrency: Deadlock and Starvation Principles of Deadlock	Interactive Lecture, Resource Allocation Graphing	Class Test, Final exam	CLO2, CLO3
	Lesson 2: Deadlock Prevention Deadlock Avoidance	Safe State Derivation, Rule-based Practice	Class Test, Final exam	CLO2, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Deadlock Detection An Integrated Deadlock Strategy	Matrix-based Calculation, Strategy Matrix Design	Class Test, Final exam	CLO2, CLO3
	Lesson 2: Dining Philosophers Problem UNIX Concurrency Mechanisms Linux Kernel Concurrency Mechanisms	Paradigm Simulation, Kernel Code Walkthrough	Class Test, Final exam	CLO2, CLO3

Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Memory Management Memory Management Requirements	Visual Metaphors, Conceptual Mapping	Class Test, Presentation, Final exam	CLO3, CLO4
	Lesson 2: Memory Partitioning, Paging Segmentation	Address Translation Practice, Whiteboard Practice	Class Test, Presentation, Final exam	CLO3, CLO4
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Virtual Memory Hardware and Control Structures	Diagram Analysis, Page Table Walkthrough	Class Test, Presentation, Final exam	CLO3, CLO4
	Lesson 2: Operating System Software Linux Memory Management Class Test -3	Page Replacement Workshop, Comparative Analysis	Class Test, Presentation, Final exam	CLO3, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: I/O Management and Disk Scheduling, I/O Devices, Organization of the I/O Function	Interactive Lecture, Data Flow Mapping	Presentation, Final exam	CLO3, CLO4
	Lesson 2: Operating System Design Issues, I/O Buffering Disk Scheduling RAID, Linux I/O	Request Math Exercises, RAID Matrix Analysis	Presentation, Final exam	CLO3, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: File Management Overview File Organization and Access File Directories	Structural Blueprint Mapping, Interactive Lecture	Presentation, Final Exam	CLO3, CLO4
	Lesson 2: File Sharing Record Blocking Secondary Storage Management Linux File Management	Bitmask Allocation Practice, Indirection Tree Tracing	Presentation, Final Exam	CLO3, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on final exam syllabus	Integrative Review, Board Problem Solving	Presentation, Final Exam	CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's		Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	
Attendance	--	--	--		7			
Class Test 1	5(15)				Avg 15	5		
Class Test 2		5(15)					5	
Class Test 3			5(15)				5	
Assignment					5			5

Presentation					8			
Midterm Exam	10	15			25	15	10	25
Final exam		10	15	15	40	10	30	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Abraham Silberschatz, Peter B. Galvin, Greg Gagne, Operating System Concepts, 9th Edition, Wiley, 2014, ISBN: 9788126554270

Reference Books/ Other Supplementary Readings

William Stallings, Operating Systems: Internals and Design Principles, 9th Edition, Pearson, 2015, ISBN: 978-0134670959.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 233 Operating System & System Programming Lab

Course Code: SE 233		CIE Marks: 60
ISCED: 0613-233		LF Marks: 40
Course Title: Operating System & System Programming Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)	Contact Hours: 1 hour 30 minutes (14 weeks)	
Prerequisite: SE 222		
Course Type: Lab		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

This course introduces the tools used to develop modern operating systems (OS). The focus of this course is UNIX-based operating systems, though alternative operating systems, including MS Windows are introduced. This course

is divided into two parts. The first part is about the GNU/Linux command line interface to operate basic operations. The second parts are about the GNU/Linux application programming interface. In this part the creation and control of processes and threads are practiced. The cooperative process resource share management system and some advanced topics in the field of process scheduling, concurrency (synchronization, mutual exclusion, deadlock, and starvation), memory (both primary and secondary) management and input/output file organization are practiced.

Rationale of the Course

This course explores contemporary operating systems with a strong focus on UNIX-based platforms like GNU/Linux. Students first master the command line interface for efficient system navigation and management before shifting to the application programming interface. The curriculum features hands-on exercises in process and thread creation, control, and scheduling. Key operational dynamics covered include memory management, input/output file organization, and concurrency.

Course Objectives

The goal of this course is to introduce the students to the concept of command based UNIX operating system in practical terms. The main objectives of this course are,

- To operate the UNIX command line to explore open-source software under GNU/Linux Operating System.
- To write basic code on Shell Script and controlling processes and threads in Modern OS. Write code for handling process concurrency mechanisms such as Mutual Exclusion, Synchronization, Deadlock and Starvation using semaphore (GNU/Linux POSIX API).
- To implement Algorithm implementation using shell scripting programming.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcome s (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Interpret the GNU/Linux GUI and command line interface.	PLO5	C3	Fundamental Domain
CLO2	Demonstrate the file Handling in GNU/Linux, Programming with Shell Scripting.	PLO3	C3, P3	Fundamental Domain
CLO3	Implement technical aspects of modern operating systems.	PLO3	C4, P2	Fundamental Domain
CLO4	Write an effective report on the technical solutions of modern operating systems.	PLO10	C3, A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1					√							

CLO2			√									
CLO3			√									
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Discussion on OBE of Operating System & System Program Lab & Introduction to Linux Operating system	Syllabus Walkthrough, Environment Setup Lab	CLO1	Lab Performance
Week-2 [1.5 Hours]	Installation: Linux Installation using Virtual Machine Platform	Guided Installation, Environment Verification	CLO1	Lab Performance
Week-5 [1.5 Hours]	Windows Vs Linux: Type of files, directories, users	Comparative System Review, File Hierarchy Analysis	CLO1, CLO4	Lab Performance
Week-4 [1.5 Hours]	Commands Terminal V/s GUI File Commands (ls, cd, pwd, mkdir, cp, mv, rm, rmdir, cat, more, less, touch, head, tail, ls)	Hands-on Command execution, Terminal Syntax Practice	CLO1, CLO4	Lab Performance, Viva, Lab Final
Week-5 [1.5 Hours]	File Permissions (chmod), File Redirection (>, >>), vi Editor, nano editor and their functionalities	File Permission Masking, Text Editor Practice	CLO1, CLO4	Lab Performance, Viva, Lab Final
Week-6 [1.5 Hours]	Searching (grep, locate, find) System Information (date, cal, w, whoami, finger, uname, cat /proc/cpuinfo)	Pattern Matching Lab, System Information Querying	CLO1, CLO4	Lab Performance, Viva, Lab Final
Week-7 [1.5 Hours]	System Information (cat proc/meminfo, man, df, du, free, whereis App, which app)	Resource Allocation Tracing, Diagnostic Utilities Lab	CLO1, CLO4	Lab Performance, Viva, Lab Final
Week-8 [1.5 Hours]	Unix Administration (Creating a User, Deleting, disabling account, Adding users to the user groups, Finger, Linux/Unix User Management Commands)	Privilege Control Exercise, User Administration Practice	CLO1, CLO4	Viva, Lab Final
Week-9 [1.5 Hours]	Basic of Shell Scripting	Scripting Syntax Walkthrough	CLO2, CLO3	Lab Performance
Week-10 [1.5 Hours]	Implement Conditional statement in shell scripting.	Control Flow Scripting, Code Execution Task	CLO2, CLO3, CLO4	Lab Performance, Viva, Lab Final
Week-11 [1.5 Hours]	Implement Loop in shell scripting and Implementation of the scheduling algorithm.	Automation Scripting Lab, Process Algorithm Simulation	CLO2, CLO3, CLO4	Lab Performance, Viva, Lab Final

Week-12 [1.5 Hours]	Implementation of the scheduling algorithm and Deadlock avoidance algorithm	Automation Scripting Lab, Process Algorithm Simulation	CLO2, CLO3, CLO4	Lab Performance
Week-13 [1.5 Hours]	Industry Session.	Industry Expert Talk, Production Pipeline Review	CLO1, CLO2, CLO3, CLO4	Lab Performance, Presentation
Week-14 [1.5 Hours]	Lab Final	Independent Practical Exam, Practical Scripting Task	CLO1, CLO2, CLO3, CLO4	Viva, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO3	PLO3
Lab Final	40	CLO1	PLO5
		CLO2	PLO3
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Richard Blum, Christine Bresnahan, Linux Command Line and Shell Scripting Bible, 4th Edition, Wiley, 2021, ISBN: 978-1118983843.

Reference Books/ Other Supplementary Readings

1. William Stallings, Operating Systems: Internals and Design Principles, 9th Edition, Pearson, 2015, ISBN: 978-0134670959.

2. Lab Manual

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

VirtualBox, VMWare Workstation, Linux, Ubuntu, Online Platform

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

GE 235 Principles of Accounting, Business and Economics

Course Code: GE 235	CIE Marks: 60
ISCED: 0613-235	SEE Marks: 40
Course Title: Principles of Accounting, Business and Economics	Total Marks: 100
Semester:	
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)
Prerequisite: N/A	
Course Type: GED	

CIE: Continuous Internal Evaluation
SEE: Semester End Examination

Course Content (from syllabus)

Clear Concept of Accounting in action, Accounting Equation and effects of Transaction Analysis and preparing the Financial Statement, Recording Process (Journal, Ledger and Trial Balance), Adjusting the accounts, Completing the accounting cycle with worksheet, case study, study on Financial report of a company, Foundation of Business, Barter Theory, Factors and Resources of Business, Proprietorship, Partnership and Corporate Business, Basic concept of Economics, Micro and Macro Economics, Production Possibility Frontier, Factors of Production, Statements, Demand and Supply Curve with the schedule, Factor affecting on Demand and Supply, Law of Demand, Movement of the curve, Market Equilibrium.

Rationale of the Course

This course integrates fundamental accounting, business, and economic principles to build essential financial management and decision-making skills. In accounting, students master transaction analysis, financial statements, and the complete accounting cycle. The curriculum also introduces core business structures alongside microeconomic and macroeconomic concepts, including production possibility frontiers and the laws of supply

and demand. By bridging financial tracking with market theory, this course provides a holistic understanding of business operations and economic systems required for diverse professional environments.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of artificial intelligence. More specifically,

- To learn the basic concept of the accounting cycle, analysis, and recording of transactions.
- To learn about reporting financial information in accordance with Generally Accepted Accounting Principles (GAAP)
- To learn basic concepts of Business activities and different forms of Business.
- To learn fundamental concepts of economics and the theory of demand & supply are also included in this course. Students will be exposed to real-life scenarios by analyzing the financial statements of different companies in Bangladesh

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Interpret the concept of role and principles of financial accounting and reporting, Transaction to Record, handle and summarize accounting data.	PLO2	C2	Fundamental Domain
CLO2	Deploy the adjusting accounts to complete an accounting cycle using financial statement.	PLO1	C3	Fundamental Domain
CLO3	Discuss the core concept of Business.	PLO2	C2	Fundamental Domain
CLO4	Evaluate the role of economics and the theory of supply, demand and Equilibrium	PLO2	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2	√											
CLO3		√										
CLO4		√										

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
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Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: OBE Discussion, Introduction to Course.	Syllabus Walkthrough, Interactive Q&A	N/A	CLO1
	Lesson 2: Overview of Principles of Financial Accounting with Equation	Equation Breakdown, Algebraic Modeling Practice	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion about GAAP, Ethics and Accounts	Case Study Review, Ethical Dilemma Discussion	Class Test, Midterm exam	CLO1
	Lesson 2: Overview of Financial Transactions and their Analysis and Classwork on Transaction Analysis Practices	Double-Entry Tracing, Transaction Impact Analysis	Class Test, Presentation, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Types of Financial Statements and Preparing Financial Statements and class work	Statement Structure Review, Hands-on Statement Construction	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Overview of Recording Process	Flowchart Analysis, Sequential Step Review	Class Test, Assignment, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Making Journal Entries	Journalizing Practice, Account Classification Drill	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Class Work on Journal Practices and basic concept of Ledger Accounts Class Test -1	Whiteboard Demonstration, T- Account Mapping	Class Test, Assignment, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Preparing Ledger Accounts Procedure and Class Work on Ledger Accounts practices and making Trial Balances from the Ledger.	Ledger Posting Practice, Trial Balance Balancing Drill	Class Test, Midterm exam, Final Exam	CLO1
	Lesson 2: Gathering Knowledge about Adjusting Accounts with Principles and Assumptions with Classwork on Adjusting Entries	Accrual/Deferral Analysis, Adjusting Entry Practice	Class Test, Midterm exam, Final Exam	CLO1
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of Completing Accounting cycle and Class Work on practices of Worksheet	Step-by-step Process Walkthrough, Worksheet Column Mapping	Midterm exam, Final Exam	CLO1, CLO2

	Lesson 2: Practices on Classified Balance sheet and Closing Entries Class Test -2	Account Temporary-to-Permanent Closing, Ledger Reset Practice	Midterm exam, Final Exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Math Practices with case study based on Week 1- Week 6	Integrative Accounting Cycle Practice, Comprehensive Review	Class Test, Final Exam	CLO1, CLO3
	Lesson 2: Core concepts of Business	Visual Presentation, Business Ecosystem Mapping	Class Test, Final Exam	CLO1, CLO3
Week8 Lesson 1 & 2 [3 Hours]	Lesson 1: Different Types of Business and their Systems	Comparative Matrix Analysis, Entity Structures Discussion	Class Test, Final exam	CLO1, CLO3
	Lesson 2: Overview of basic Economics, Scarcity, Want and Demand.	Real-world Analogy Discussion, Resource Limit Modeling	Class Test, Final Exam	CLO1, CLO4
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Branches of Economics- Microeconomics and Macroeconomics concepts	Comparative Lecture, Boundary Concept Mapping	Class Test, Final Exam	CLO1, CLO4
	Lesson 2: Production Possibility Frontier	Graph Plotting Workshop, Trade-off Analysis	Class Test, Final Exam	CLO1, CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Factors of Production with statements	Matrix Classification, Resource Input Discussion	Final Exam	CLO1, CLO4
	Lesson 2: Theory of Demand and Supply with their factors	Determinants Mapping, Logic Puzzles	Final Exam	CLO1, CLO4
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Demand and Supply Curve, Laws, and Movement with the schedule.	Whiteboard Curve Plotting, Schedule-to-Graph Translation	Final Exam	CLO1, CLO4
	Lesson 2: Equilibrium with math practices. Class Test -3	Algebraic Intersection Solving, Market Clearing Analysis	Final Exam	CLO1, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Basic concept of Macroeconomics and Economic systems	Framework Modeling, Macro Variable Overview	Final Exam	CLO1, CLO4
	Lesson 2: Inflation, GDP, GNP, Investment	Formula Calculation Walkthrough, Macro Indicator Tracing	Final Exam	CLO1, CLO4
Week-13	Lesson 1: Financial Capital, Fiscal Policy and Monetary Policy.	Policy Matrix Breakdown, Instrument Analysis	Presentation, Final Exam	CLO1, CLO4

Lesson 1 & 2 [3 Hours]	Lesson 2: Real-life case scenario Analysis	Macro-Policy Case Analysis, Impact Evaluation Simulation	Presentation, Final Exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on the final examination	Integrative Review, Board Problem Solving	Final Exam	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's		Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	
Attendance	--	--	--	--	7			
Class Test 1	5(15)				Average 15			
Class Test 2		5(15)						
Class Test 3			5(15)					
Assignment				5	5			5
Presentation					8			
Midterm Exam	15	10			25	10	15	25
Final exam	14	12	5	9	40	12	28	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

J.J. Weygandt, D.E. Kieso, and Paul D. Kimmel, Accounting Principles, 12th Edition, Wiley, 2016, ISBN: 9788126561216

Reference Books/ Other Supplementary Readings

1. Companies Create Value for People, Latest Edition, Tata McGraw-Hill, 2007, ISBN: 978-0-07-352456-6
2. Parkin, M. Microeconomics, 8th Edition, Pearson, 2008, ISBN: 9780321493644.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 532 Introduction to Robotics

Course Code: SE 532		CIE Marks: 60
ISCED: 0612-532		SEE Marks: 40
Course Title: Introduction to Robotics		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE 213		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

This course includes definition of Robot, Types of Robots (manipulator, legged robot, wheeled robot, autonomous underwater vehicles), Use of Robots, Asimov's laws of Robotics, History of Robotics, Key components of Robot, Sensors: Introduction, working principles and use of sensors (vision, force, LDR, temperature, smoke, accelerometer gyroscope, laser, tilt, compass, PIR, Infrared, etc.), Actuators and different actuators (DC motor, servo motor, stepper motor, etc.) working principles and usage, Robot programming with AD conversion and interfacing different hardware, sensors, etc, Control theory of robotics; Obstacle avoidance, object tracking, and motion control, etc; Advance Robotic control and operations.

Rationale of the Course

This course addresses the rising industry demand for IoT, robotics, and intelligent cyber-physical systems. Covering applications from home automation to factory robotic arms, the curriculum teaches students to design standalone solutions using microcontrollers (MCUs), microprocessors (MPUs), systems-on-chip (SoCs), and single-board computers (SBCs). By integrating hardware concepts with modern software engineering, the course prepares learners to solve real-world automation challenges. Additionally, it serves as a foundational prerequisite for students aiming to specialize in advanced Robotics and Embedded Systems major programs.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of Computer Architecture, More specifically,

- . To learn the basics of Robotics and Robots.
- . Demonstrate critical thinking and problem-solving using mechatronics.
- . Develop Robotics programming and programming using MCUs.
- . Design robotics solutions through complex programming and assembling of modules.
- . Apply special tools, techniques and simulations in Robotics.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Interpret concept of Robotics and Robots.	PLO5	C2	Fundamental Domain
CLO2	Implement the knowledge of Electronics and Computer Programming for MCUs.	PLO1	C3	Fundamental Domain
CLO3	Develop robots using complex circuits, modules, and programming	PLO3	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1					√							
CLO2	√											
CLO3			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to OBE of Introduction to Robotics	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Overview of Instruction Set Architecture, Evolution of Computer.	Interactive Lecture, Whiteboard Diagramming	Class Test Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Power Electronics	Visual Presentation, Component Demonstration	Class Test, Midterm exam	CLO1
	Lesson 2: Converters, inverters and Buck-Boost Modules	Circuit Topology Analysis, Guided Discussion	Class Test, Midterm exam	CLO1
Week-3	Lesson 1: Introduction to Laboratory Tools, meters, soldering station	Tool Demonstration, Safety Walkthrough	Class Test, Midterm exam	CLO1

Lesson 1 & 2 [3 Hours]	Lesson 2: Measurements using different meters, oscilloscopes	Step-by-step Waveform Measurement, Lab Practice	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to MCUs and MPUs.	Visual Lecture, Architectural Diagramming	Class Test, Midterm exam	CLO1
	Lesson 2: Classification of MCUs and MPUs with features and grades (Industry, Hobby etc.) Class Test -1	Comparative Matrix Analysis, Group Discussion	Class Test, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Coding in MCUs, Programming Language.	Syntax Walkthrough, Logic Flowchart Drawing	Class Test, Midterm exam	CLO2
	Lesson 2: Installation of IDEs and coding through web editors.	Guided Installation, Environment Verification	Class Test, Midterm exam	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Programming in MCU (Arduino, Node MCU) with Basic I/Os (LED, Sonar etc.)	Live Coding Demonstration, Firmware Benchmarking	Class Test, Midterm exam, Final exam	CLO2
	Lesson 2: Programming through TinkerCad (Simulator). Class Test -2	Hands-on Simulation Practice, Virtual Circuit Tracing	Class Test, Presentation Midterm exam, Final exam	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Serial Communication, communication speed, Protocols, Observing the serial monitor using simulator (TinkerCad)	Timing Diagram Analysis, Serial Packet Decoding	Class Test, Presentation Midterm exam, Final exam	CLO2
	Lesson 2: Serial Communication with basic speed and	Live Trace Verification, Peer Problem-Solving	Class Test, Presentation Midterm exam, Final exam	CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Introducing Sensors, Classification and Characteristics of sensors. Actuators, Basic concepts of actuator, Electrical and Mechanical actuators. Pulse Width Modulation to control output.	Signal Waveform Walkthrough, Real-world Analogy Discussion	Class Test, Assignment, Final exam	CLO2
	Lesson 2: Programming of output to control speed of motor, brightness of LEDs	Duty-Cycle Scripting Lab, Hands-on Code Tuning	Class Test, Assignment, Final exam	CLO2
Week-9	Lesson 1: Connection of a display to the MCU (SPI, I2C, SERIAL etc.)	Bus Protocol Mapping, Pinout Analysis	Class Test, Assignment, Final exam	CLO2

Lesson 1 & 2 [3 Hours]	Lesson 2: Connection and programming for different output modules.	Modular Driver Scripting, Hardware Hookup Walkthrough	Class Test, Assignment, Final exam	CLO2
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Sensing the input for robots.	Data Stream Analysis, Signal Conditioning Review	Class Test, Assignment, Final exam	CLO2
	Lesson 2: ADC, DAC.	Step-by-step Resolution Mapping, Logic Puzzles	Class Test, Assignment, Final exam	CLO2
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Branch instructions, Skip instructions, Procedure call instruction.	Control Flow Charting, Assembly Trace Execution	Class Test, Assignment, Final exam	CLO2
	Lesson 2: Programming of digital input modules (3rd party) Class Test -3	Hands-on Register Configuration, Firmware Debugging	Class Test, Assignment, Final exam	CLO2
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to communication modules (Bluetooth, wifi, NRF, NFC etc.) and Connection, Programming of Bluetooth Module.	Wireless Packet Tracing, Live Command Testing	Final exam	CLO2
	Lesson 2: Kinematics: Basic Concepts, Concepts about Links, Joints, Degrees of Freedom	3D Spatial Modeling, Coordinate System Analysis	Final exam	CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Design of a complete project (Project-1) with basic I/Os, Modules	Collaborative Project Blueprinting, Visual Presentation	Final exam	CLO3
	Lesson 2: Developing of complete robotics projects with complex level of programming and simulation	Full System Logic Simulation, Peer Code Debugging	Final exam	CLO3
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Presentation, Assignment	CLO1, CLO2, CLO3
	Lesson 2: Review class on Final examination syllabus topic	Integrative Syllabus Review, Board Problem Solving	Presentation, Final exam	CLO2, CLO3

Overall Assessment Scheme

Assessment Task	CLO's			Mark (Total=100)	PLO's			Mark (Total =78)
	CLO1	CLO2	CLO3		PLO5	PLO1	PLO3	
Attendance	--	--	--	7				

Class Test 1	5(15)			Avg 15				
Class Test 2		5(15)						
Class Test 3			5(15)					
Assignment			5	5		5		5
Presentation				8		8		8
Midterm Exam	10	15		25	10	15		25
Final exam		25	15	40		25	15	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

John J. Craig, Introduction to Robotics: Mechanics and Control, 3rd Edition, Pearson, 2018, ISBN: 978-0132615622.

Reference Books/ Other Supplementary Readings

Robert J. Schilling, Fundamentals of robotics : analysis and control, 5th Edition, Pearson Education, 2003, ISBN: 0133444333

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Arduino, required hardware, Tinkercad

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 225 Data Communication and Computer Networking

Course Code: SE 225		CIE Marks: 60
ISCED: 0612-225		SEE Marks: 40
Course Title: Data Communication and Computer Networking		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)		Contact Hours: 3 (Total weeks: 14)
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation

Course Content (from syllabus)

Overview of Data Communication, Components of Data communication, Data representations, Data flow, Physical Structures of data communication, Topologies: Star, Ring, Bus, Mesh, And Hybrid Categories of Network- LAN, MAN, WAN, Internet Protocols and Standards, Data and Signals, Periodic Analog Signal, Digital Signal, Analog Data, Digital Data, Frequency, Bandwidth, Bit Rate, Transmission Impairments: Attenuation, Distortion, and Noise, Performance Measure: Throughput, Bandwidth, latency. Layered Task, Internet Model: Peer-to-Peer Process, Organization of the layer, Layers in the OSI Model. TCP/IP model, OSI Vs TCP/IP model. Physical Layer, Transmission Media, Guided media (Coaxial Cable, Twisted pair cable, Optical fiber). Unguided Media, Advantages and Disadvantages of various media, Framing, Fixed-Size Framing, Variable-Size Framing, Flow Control, Error Control Protocols, Types of Errors, Redundancy, Detection Versus Correction, Forward Error Correction Versus Retransmission. Hamming Distance, Minimum Hamming Distance, Linear Block Codes, One dimensional parity check, Two Dimensional parity checks, Cyclic Redundancy Check, Checksum, Basic concepts of Network layer, IP Address, Decimal-Binary data conversion, IP Address Classes, Network mask, Classful Address, Classless Address (CIDR), Basics of FLSM, VLSM, Sub netting concepts, VLSM for Classful Addressing, FLSM for Classless Addressing, Routing Table, Routing Protocol, Autonomous System, RIP, Transport layer Process-To-Process, Delivery, Client/Server, SCTP, TCP, UDP, Application Layer Overview, HTTP, DNS.

Rationale of the Course

This course provides a comprehensive foundation in data transmission principles and network architecture. Students explore signal characteristics, transmission mediums like copper and fiber optics, and network topologies spanning LANs and WANs. The curriculum breaks down the OSI and TCP/IP reference models layer by layer, focusing on error detection and flow control at the Data Link layer alongside routing mechanisms like RIP. Additionally, learners master the core protocols that power the internet, including TCP, UDP, DNS, and HTTP.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of data communications. More specifically,

- To explore the concept of data communication, its requirements, effectiveness, data Representation, data Flow, data transmission, transmission impairments, performance measurement.
- To understand the concept of concept of topology, its Advantages & Disadvantages, topology's working principles for Mesh, Star, Bus, Tree, Ring, Hybrid
- To perceive the concept of digital transmission and line coding scheme works.
- To gain the basic concepts of Network Layer, gain basic knowledge of IP address, Network mask, IP classes, solve problems on subletting (FLSM) for Classful addressing.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domain s (C, P, A)	BNQF Skill
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CLO1	Demonstrate the concepts of data communication and computer networking along with its requirements, effectiveness, and data transmission overview.	PLO2	C4	Fundamental Domain
CLO2	Compare the approaches of OSI & TCP/IP layering and different network models.	PLO2	C2	Fundamental Domain
CLO3	Apply the concept for error detection mechanisms	PLO1	C3	Fundamental Domain
CLO4	Analyze the basic concepts of Classful and Classless Addressing considering experimental design to provide valid conclusions.	PLO4	C4	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3	√											
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE	Syllabus Walkthrough, Interactive Q&A	None	CLO1
	Lesson 2: Course Outline Discussion, Overview of Data Communication	Interactive Lecture, Conceptual Mapping	Class Test	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of Data Communication, Characteristics, Components of Data communication, Data representations, Data flow, Physical Structures of data communication.	Communication Model Mapping, Flow Direction Analysis	Class Test, Midterm exam	CLO1
	Lesson 2: Topologies: Star, Ring, Bus, Mesh, And Hybrid. Categories of Network- LAN, MAN, WAN. Internet Protocols and Standards.	Network Topology Diagramming, Comparative Analysis	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Categories of Network- LAN, MAN, WAN. Internet Protocols and Standards.	Geographic Scale Analysis, Standards Matrix Walkthrough	Class Test, Midterm exam	CLO1
	Lesson 2: Data and Signals, Periodic Analog Signal, Digital Signal, Analog Data, Digital Data, Frequency, Bandwidth, Bit Rate.	Waveform Graph Analysis, Signal Parameter Math	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Transmission Impairments: Attenuation, Distortion, and Noise.	Signal Degradation Analysis, Decibel Math Exercise	Class Test, Midterm exam	CLO1
	Lesson 2: Performance Measure: Throughput, Bandwidth, latency. Class Test -1	Metric Calculation Practice, Network Delay Modeling	Class Test, Midterm exam	CLO1

Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Layered Task, Internet Model: Peer-to-Peer Process, Organization of the layer, Layers in the OSI Model.	Layered Stack Modeling, Protocol Data Unit (PDU) Analysis	Class Test, Midterm exam	CLO1, CLO4
	Lesson 2: TCP/IP model, OSI Vs TCP/IP model.	Architectural Cross-Mapping, Comparative Analysis	Class Test, Midterm exam	CLO1, CLO4
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Physical Layer, Transmission Media, Guided media (Coaxial Cable, Twisted pair cable, Optical fiber)	Physical Cable Demonstration, Structural Analysis	Class Test, Midterm exam	CLO1, CLO4
	Lesson 2: Unguided Media, Advantages and Disadvantages of various media. Class Test -2	Spectrum Allocation Review, Wireless Scenario Analysis	Class Test, Midterm exam	CLO1, CLO4
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: FRAMING, Fixed-Size Framing, Variable-Size Framing, Flow Control, Error Control Protocols	Byte-Stuffing Workshop, Frame Structure Dissection	Class Test, Midterm exam	CLO1, CLO4
	Lesson 2: Types of Errors, Redundancy, Detection Versus Correction, Forward Error Correction Versus Retransmission.	Error Scenario Analysis, Strategy Matrix Design	Class Test, Midterm exam	CLO1, CLO4
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Hamming Distance, Minimum Hamming Distance, Linear Block Codes	Codeword Distance Calculation, Vector Math Practice	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO4
	Lesson 2: One dimensional parity check, Two Dimensional parity checks	Bit-Matrix Parity Drawing, Error Detection Practice	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO4

Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Cyclic Redundancy Check, Checksum	Polynomial Long Division, Binary Checksum Computation	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO4
	Lesson 2: Basic concepts of Network layer, IP Address, Decimal-Binary data conversion	Binary Radix Conversion, Address Format Walkthrough	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: IP Address Classes, Network mask, Classful Address	Class Boundary Dissection, Default Mask Mapping	Class Test, Assignment, Final exam	CLO1, CLO4
	Lesson 2: Classless Address (CIDR), Basics of FLSM VLSM	Slash Notation Analysis, Address Space Block Modeling	Class Test, Assignment, Final exam	CLO1, CLO4
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Subnetting concepts, VLSM for Classful Addressing Class Test -3	Host Requirement Allocation, Step-by-step Subnet Splitting	Class Test, Assignment, Final exam	CLO1, CLO4
	Lesson 2: VLSM for Classless Addressing, FLSM for Classless Addressing	Custom Subnet Mask Derivation, Range Boundary Math	Assignment, Final exam	CLO1, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Routing Table, Routing Protocol	Next-Hop Lookup Exercises, Routing Entry Analysis	Assignment, Final exam	CLO1, CLO4
	Lesson 2: Autonomous System, RIP	Distance Vector Tracking, Bellman-Ford Metric Updates	Assignment, Final exam	CLO1, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Transport layer Process-To-Process Delivery, Client/Server Paradigm, Multiplexing and Demultiplexing	Socket Port Mapping, Segment Header Dissection	Assignment, Final exam	CLO1, CLO4
	Lesson 2: SCTP, TCP, UDP	Three-Way Handshake Tracing, Protocol Feature Comparison	Assignment, Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Application Layer Overview, HTTP, DNS	Request-Response Cycle Mapping, Name Resolution Tracing	Assignment, Final exam	CLO1, CLO4
	Lesson2: Industry Session	Guest Lecture / Industry Expert Talk, Infrastructure Blueprint Mapping	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =78)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO4	
Attendance	--	--	--		7				
Class Test 1	5(15)				Avg 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment			5		5	5			5
Presentation				8	8			8	8
Midtem Exam	10	10	5		25	5	20		25
Final exam			10	30	40	10		30	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, McGraw-Hill Education, 2013, ISBN: 9781259064753

Reference Books/ Other Supplementary Readings

James F. Kurose, Keith W. Ross, Computer Networking: A Top-Down Approach, 8th Edition, Pearson, 2021, ISBN: 978-0133594140.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 226 Data Communication and Networking Lab

Course Code: SE 226		CIE Marks: 60
ISCED: 0612-226		LF Marks: 40
Course Title: Data Communication and Networking Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)	Contact Hours: 1 hour 30 minutes (14 weeks)	
Prerequisite: 225		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

Overview of Networks and layered communications, understanding of Network equipment, wiring in detail, CAT6 UTP EIA/TIA 568A/B straight and cross-over wiring, testing, Overview of IP Addressing and sub-netting, static IP setting on Linux machine (Ubuntu) / Windows and testing, IP address and Packet Tracer, Creation of a LAN and connectivity test in the LAN, creation of VLAN and VLAN trunking, Basic concepts of Router Configuration -Static Routing Implementation, Implementation of Dynamic Routing (RIP, OSPF, BGP), Router Configuration using CLI, Firewall Implementation, Router Access Control List (ACL), Packet capture and header analysis by Wireshark (TCP, UDP, IP), Basic Frame Relay Implementation with PVC, DNS, Web, DHCP, FTP server configuration

Rationale of the Course

This course helps students with hands-on training regarding the design, troubleshooting, modeling, and evaluation of computer networks. In this course, students are going to experiment in a real test-bed networking environment and learn about network design and troubleshooting topics and tools such as Network address, Address Resolution Protocol (ARP), basic troubleshooting tools (e.g. ping, ICMP), IP routing (e.g. RIP), route discovery (e.g. Traceroute), TCP and UDP, IP fragmentation and many others. Students will also be introduced to network modeling and simulation, and they will have the opportunity to build some simple networking models using the tool and perform simulations that will help them evaluate their design approaches and expected network performance.

Course Objectives

The goal of this course is to provide the students with introductory concepts and technologies in networking. The main objectives of this course are,

- To familiarize students with some of the standard networking tools, devices, software, and operating systems used to build modern-day network infrastructures.
- To be able to explain, configure, verify, and troubleshoot complex computer network problems.
- To configure network routers and switches so that both LAN and WAN traffic successfully traverses the network.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Perform a wide range of data manipulation activities using the Packet Tracer Tool and command line interface.	PLO1	C3	Fundamental Domain
CLO2	Apply appropriate techniques for various protocols such as FTP, HTTP, Telnet, DNS.	PLO5	C3, P2	Fundamental Domain
CLO3	Conduct investigations of routing and congestion in network layer with routing algorithms and classify IPV4 addressing scheme.	PLO4	C3, P3	Fundamental Domain
CLO4	Write effective reports and design documentation for the computer networking-oriented applications.	PLO10	A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2					√							
CLO3				√								
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
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Week-1 [1.5 Hours]	Introduction class, and Explain course outline, CLO, PLO, objectives, assessment plan	Syllabus Walkthrough, Lab Safety Orientation	CLO1, CLO2,CLO3	Lab Performance
Week-2 [1.5 Hours]	Overview of Networks and layered communications, understanding of Network equipment	Hardware Device Identification, Layered Stack Mapping	CLO1, CLO2, CLO3	Lab Performance, Viva, Lab Final, Lab report
Week-3 [1.5 Hours]	UTP, STP straight and crossover wiring details	Physical Cable Crimping, Cable Tester Verification	CLO1, CLO2, CLO3	Lab Performance, Viva, Lab Final, Lab report
Week-4 [1.5 Hours]	Overview of IP Addressing and sub- netting, static IP setting	Host Interface Configuration, Subnet Allocation Practice	CLO1, CLO2, CLO3	Lab Performance, Viva, Lab Final, Lab report
Week-5 [1.5 Hours]	IP address and Packet Tracer: Experimental work using Cisco Packet tracer, Ping config on cmd,	Simulator Workspace Familiarization, ICMP Connectivity Testing	CLO2	Lab Performance, Viva, Lab Final, Lab report
Week-6 [1.5 Hours]	P2P connection, LAN connection using Hub, Introduction of Switch and established a connection with switch	Topology Building Exercise, Broadcast Domain Analysis	CLO2, CLO3	Lab Performance, Viva, Lab Final, Lab report
Week-7 [1.5 Hours]	Basic concepts of Router Configuration, single router configuration, setting name of the router.	Cisco IOS CLI Navigation, Basic Hostname Scripting	CLO1, CLO3, CLO4	Lab Performance, Viva, Lab Final, Lab report
Week-8 [1.5 Hours]	Basic concepts of Router Configuration, multiple router configuration, setup connection between multiple router: Experimental work using Cisco Packet tracer,	Interface IP Assignment, Serial Link Wiring	CLO1, CLO3, CLO4	Lab Performance, Viva, Lab Final, Lab report
Week-9 [1.5 Hours]	Static Routing Implementation: Experimental work using Cisco Packet tracer,	Route Table Configuration, Next-Hop Path Mapping	CLO1, CLO3, CLO4	Lab Performance, Viva, Lab Final, Lab report

Week-10 [1.5 Hours]	Dynamic Routing Implementation (RIP, OSPF): Experimental work using Cisco Packet tracer	Routing Protocol Convergence, Area Neighbor Dissection	CLO1, CLO3, CLO4	Lab Performance, Viva, Lab Final, Lab report
Week-11 [1.5 Hours]	DNS, DHCP, FTP server configuration: Experimental work using Cisco Packet tracer	Server Daemon Configuration, Client IP Pool Testing	CLO1, CLO4	Lab Performance, Viva, Lab Final, Lab report
Week-12 [1.5 Hours]	Telnet server configuration: Experimental work using Cisco Packet tracer	VTY Line Configuration, Remote Terminal Access	CLO1, CLO4	Lab Performance, Viva, Lab Final, Lab report
Week-13 [1.5 Hours]	Packet capture by traceroute (TCP,UDP,IP): Experimental work using Cisco Packet tracer	Simulation Mode Packet Sniffing, Protocol Header Analysis	CLO1, CLO4	Lab Performance, Viva, Lab Final, Lab report
Week-14 [1.5 Hours]	Final project documentation check and group presentation	Peer Review Evaluation, Live Topology Demonstration	CLO1, CLO2, CLO4	Lab Performance, Viva, Lab Final, Lab report

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO1
Lab Final	40	CLO2	PLO5
		CLO3	PLO4
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Allan Johnson, CCNA 200-301 Official Cert Guide, 1st Edition, Cisco Press, March 24, 2020, ISBN 978-0135792735

Reference Books/ Other Supplementary Readings

1. Chris Bryant, Brian McGahan, CCNA 200-301 Hands-on Mastery with Packet Tracer, 1st Edition, Cisco Press, 2024, ISBN 978-0135313091.

2. Lab Manual.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Packet Tracer

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)

Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 231 System Analysis & Design Capstone Project

Course Code: SE 231		CIE Marks: 60
ISCED: 0613-231		SEE Marks: 40
Course Title: System Analysis & Design Capstone Project		Total Marks: 100
Semester:		
Credit Value: 3 (Lab)	Contact Hours: 4.5hr (Total weeks: 14)	
Prerequisite: SE133, SE212, SE221		
Course Type: Lab-Project		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

The "System Analysis and Design" course provides students with a comprehensive exploration of fundamental concepts in system analysis and design, emphasizing practical applications. Beginning with foundational topics such as planning, proposal development, and feature listing, the course progresses to delve into use case design, UI principles, and prototyping. Students then navigate the intricacies of databases, including RDBMS, design principles, and normalization. Subsequent weeks involve visualizing system processes through Activity and Sequence Diagrams, with a focus on implementing the MVC architecture. The course culminates in a semester-long group project, where students apply their knowledge to analyze, design, and implement a system, fostering a hands-on understanding.

Rationale of the Course

The "System Analysis and Design" course is designed to provide students with a robust understanding of essential principles in system analysis and design and implement a system, with a strong emphasis on practical application. Beginning with foundational concepts such as planning, proposal development, and feature listing, students' progress to more advanced topics including use case design, user interface principles, and prototyping. As the course advances, students delve into database management systems, exploring relational database management, design principles, and normalization techniques. Subsequent weeks focus on visualizing system processes through Activity and Sequence Diagrams, with an emphasis on implementing the Model-View-Controller (MVC) architecture.

Course Objectives

This course aims to equip students with skills in project planning, user requirement elicitation and specification, graphical modeling of objects, data, and processes, and the design of UML diagrams, user interfaces, and relational databases. The primary objectives are as follows:

- Students will be able to identify, gather, and document system requirements using various analysis techniques to effectively understand and define real-world information systems.
- Students will learn to create detailed system design diagrams including UML diagrams.
- Students will be able to choose, justify, and effectively utilize appropriate software tools for developing, representing, and verifying analysis and design models, and for prototyping user interfaces.

- Students will gain proficiency in writing comprehensive system analysis and design documentation and reports, and effectively presenting their project outcomes collaboratively as part of a team.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Understand the components of information systems, and phases of System Development Life Cycle.	PLO1	C3	Fundamental Domain
CLO2	Apply various system analysis techniques to document system requirements, and user needs for developing real-life information systems.	PLO1	C4	Fundamental Domain
CLO3	Create system design diagrams, data models, and interface prototypes that guide the development process	PLO3	C4	Fundamental Domain
CLO4	Choose and justify appropriate tools for representing analysis and design models, and user interfaces	PLO5	P1, C4	Fundamental Domain
CLO5	Write effective reports and design documentation of the outcome of the capstone project.	PLO10	P2	Social, Thinking, Personal Domain
CLO6	Deliver the project documentation created collaboratively for intricate real-life applications as a team presentation.	PLO9	A1	Social, Thinking, Personal Domain

Knowledge Profile, Complex Engineering Problem, Complex Engineering Activities

Knowledge Profile	KP3 and KP5
Complex Engineering Problem	CEP1, CEP2, CEP3
Complex Engineering Activities	CEA1, CEA2

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2	√											
CLO3			√									
CLO4					√							
CLO5										√		
CLO6									√			

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Introduction class	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Explain course outline, CLO, PLO, objectives, assessment plan	Course Blueprint Mapping, Interactive Q&A	None	None
Week-2 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Identify real life project and its scope with key notes, Define project scenario and possible SDLC	Case Study Analysis, Brainstorming Sessions	CLO1, CLO2, CLO5	Viva, Class Performance, Final Report
	Lesson 2: Identify users and design user profile, State feasibility study and define elicitation techniques	Persona Design Workshop, Elicitation Modeling	CLO1, CLO2, CLO5	Viva, Class Performance, Final Report
Week-3 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Design proposed block diagram and tentative schedule plan, Prepare complete SRS documentation with description and stakeholders	System Block Diagramming, Requirements Tracing	CLO3, CLO4, CLO5, CLO6	Design & Implementation, Final Report, Group Presentation
	Lesson 2: Design use case diagram with defining primary and secondary actor, include, extend, generalization, prepare case description for each case considering project use case	Use Case Modeling, Scenario Mapping	CLO3, CLO4, CLO5, CLO6	Design & Implementation, Final Report, Group Presentation

Week-4 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Take proposal presentation for target case	Presentation Review, Peer Critique		
	Lesson 2: Take proposal presentation for target case	Presentation Review, Peer Critique		
Week-5 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Introduction to Data Flow Diagrams, Rules and Guidelines, Symbols and Notations, Levels of Data Flow Diagrams, Integrating DFDs with use cases to understand system behavior	Diagram Notation Walkthrough, Data Flow Analysis	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Viva, Class Performance, Design & Implementation, Final Report, Group Presentation
	Lesson 2: Creating Data Flow Diagrams, analyzing real-life scenarios and mapping them onto DFDs, Step-by-step process for creating DFDs	Step-by-step DFD Decomposition, Hands-on Modeling	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Viva, Class Performance, Design & Implementation, Final Report, Group Presentation
Week-6 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Define system modeling diagrams; introduce Activity Diagrams, their purpose, basic concepts, and symbols (states, transitions, decision points, start/end nodes). Cover action/control nodes and parallel activities using fork/join.	Control Flow Diagramming, Parallel Process Modeling	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Viva, Class Performance, Design & Implementation, Final Report, Group Presentation
	Lesson 2: Explore decision points and conditional branching in Activity Diagrams. Introduce State Diagrams: definition, purpose, fundamental elements (states, transitions, events, actions), and depicting states.	State Machine Modeling, Branching Logic Walkthrough	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Viva, Class Performance, Design & Implementation, Final Report, Group Presentation
Week-7 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Analyze composite and concurrent states in State Diagrams. Integrate both Activity and State Diagrams with use cases to model system behavior, demonstrating their real-world application and assigning projects for complex system modeling.	Behavioral Integration Modeling, Scenario Analysis	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Viva, Class Performance, Design & Implementation, Final Report, Group Presentation
	Lesson 2: Conduct class activities based on previous work	Collaborative Problem-Solving, Peer Review	CLO2	Class Performance
Week-8	Lesson 1: Definition and purpose of Sequence Diagrams in system modeling, Understanding the fundamental elements,	Interaction Sequence		Viva, Class Performance, Design & Implementation,

Lesson 1 & 2 [3 + 1.5 Hours]	including lifelines, messages, activations, and objects, illustrating how lifelines represent the existence of an object over time, Exploring different types of messages	Mapping, Object Timeline Analysis	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Final Report, Group Presentation
	Lesson 2: Representing decision points and iterations in interactions. Demonstrating how sequence diagrams are employed in real-world projects for system analysis and design	Message Flow Tracing, Combined Fragment Workshop	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Viva, Class Performance, Design & Implementation, Final Report, Group Presentation
Week-9 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Definition and purpose of Class Diagrams in system modeling, exploring different types of relationships, such as associations, aggregations, and compositions, Illustrating the concepts of inheritance and generalization	Structural Architecture Mapping, Relationship Matrix Analysis	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Viva, Class Performance, Design & Implementation, Final Report, Group Presentation
	Lesson 2: Integrating class diagrams with use cases to model system behavior, analyzing real-life scenarios and mapping them onto class diagrams,	Use-Case-to-Class Realization, Domain Modeling	CLO3, CLO4, CLO5, CLO6	Design & Implementation, Final Report, Group Presentation
Week-10 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Conduct class activities based on previous work	Code/Model Tracing Clinic, Logic Puzzles	CLO2	Class Performance
	Lesson 2: Define ERDs and their purpose in database design, emphasizing their role in visualizing relationships between entities. Understand entities as real-world objects and explore relationships (one-to-one, one-to-many, many-to-many). Learn common ERD notations, specifically the Crow's Foot method, and integrate ERDs with practical use cases to model data requirements, concluding with best practices for effective and maintainable ERD design.	Crow's Foot Notation Practice, Schema Translation	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Viva, Class Performance, Design & Implementation, Final Report, Group Presentation
Week-11 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Understand prototyping's role and significance in the SDLC for early visualization and testing. Differentiate between throwaway/rapid and evolutionary methods, covering tools for interactive UI creation and the concept, simplicity, and application (including	UI Prototyping Lab, Paper Wireframing Workshop	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Viva, Class Performance, Design & Implementation, Final Report, Group Presentation

	user scenarios and case studies) of paper prototypes.			
	Lesson 2: Explain the comprehensive project documentation process and assign projects requiring students to create complete and professional documentation for a system.	Documentation Blueprinting, Technical Writing Review	CLO5	Final Report
Week-12 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Review class	Integrative Diagram Review, Concept Mapping	None	None
	Lesson 2: Review class	Traceability Matrix Audit, Board Problem Solving	None	None
Week-13 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Industry Session	Industry Expert Talk, Production Blueprint Walkthrough		
	Lesson 2: Final project documentation check and Review	Milestone Code/Model Check, Technical Consultation		
Week-14 Lesson 1 & 2 [3 + 1.5 Hours]	Lesson 1: Final project documentation check and group presentation	Student Project Presentations, Peer Feedback		Group Presentation
	Lesson 2: Final project documentation check and group presentation	Student Project Presentations, Instructor Feedback		Group Presentation

Overall Assessment Scheme

Criteria	Assigned Marks	Sub Criteria	Sub Criteria Marks	CLO	PLO
Attendance	10				

Lab Report	25	Final Report	25		CLO5	PLO10
Lab Performance	25	Class Performance	15		CLO2	PLO1
		Viva	10		CLO1	PLO1
Lab Final	40	Design & Implementation	20	15	CLO3	PLO3
				5	CLO4	PLO5
		Group Presentation	20		CLO6	PLO9
Total	100		90			

Marks Distribution

Attendance	10
Lab Report	25
Lab Performance	25
Lab Final	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Alan Dennis, Barbara Wixom, Roberta Roth, Systems Analysis and Design, 8th Edition, Wiley, 2021, ISBN: 978-1119803782.

Reference Books/ Other Supplementary Readings

Jeffrey L Whitten; Lonnie D Bentley; Kevin C Dittman, Systems analysis and design methods, 7th Edition, Tata McGraw-Hill Edition 2003, ISBN: 9780070634176

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Microsoft Visio, Lucid chart, Canva

Exam Dates

According to the Examination Schedule



**Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)**

SE 234 Theory of Computing

Course Code: SE 234		CIE Marks: 60
ISCED: 0613-234		SEE Marks: 40
Course Title: Theory of Computing		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation,

SEE: Semester End Examination

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 10:00 AM	10:00 AM- 11:30 AM	11:30 AM- 01:00 PM	01:00 PM- 02:30 PM	2:30 PM- 4:00 PM	4:00 PM- 5:30 PM
Saturday						
Sunday						
Monday						
Tuesday						
Wednesday						
Thursday						

Course Content (from syllabus)

Basics of finite automata, Finite automata Identification and Computation, Deterministic Finite Automata (DFA), Non-deterministic finite automata (NFA), NFA to DFA conversion (subset construction method), Regular expressions (RE) Basics, RE to Regular language and Regular language to RE, RE to Finite Automata, DFA to RE, Context free grammar (CFG) Basics, CFG construction, Turing machine (TM).

Rationale of the Course

This course explores the theoretical foundations of computer science, mapping out the fundamental models of computation. Students begin with finite automata, mastering deterministic and non-deterministic systems, pattern recognition, and the subset construction method. The curriculum then covers regular expressions and their architectural conversions before moving to context-free grammars to analyze complex language syntax. Finally, learners examine Turing machines to define the absolute boundaries of computability. Emphasizing rigorous proofs and formal logic, this course builds the analytical precision required to evaluate computational models and solve complex algorithmic problems.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of data Structure, More specifically,

- To learn the basic concepts of Finite Automata.
- To understand different types of finite automata and their design approaches.
- To learn regular expressions and its conversion to Regular Language and finite automata.
- To learn context free grammar and its construction techniques.
- To learn Turing machine and its basics

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Design deterministic finite automata with explaining the terms of Finite Automata.	PLO3	C4	Fundamental Domain
CLO2	Design non-deterministic Finite Automata along with the comparison of DFA and NFA.	PLO3	C4	Fundamental Domain
CLO3	Apply conversion techniques of regular expression and Construct context free grammar by explaining derivation with parse tree and ambiguity and then construct Push down automata to accept context free language.	PLO1	C3	Fundamental Domain
CLO4	Illustrate Turing machines and its functionality.	PLO2	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1			√									
CLO2			√									
CLO3	√											
CLO4		√										

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
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Lesson (hour)				
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: OBE Discussion, Course outline brief discussion	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Introduction to mathematical terminologies, Introduction and mathematical terminologies [Automata Theory]	Interactive Lecture, Conceptual Mapping	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Use of mathematical notation, Introduction to Finite Automata [DFA,NFA Identification and computation]	Transition Diagram Walkthrough, Visual Presentation	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Practice formal definition, finite automata identification and Computation	Tuple Definition Practice, State Machine Analysis	Class Test, Midterm exam	CLO1, CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: DFA Construction [Basic length related problems]	Whiteboard Demonstration, Rule-based Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Exercise practice on DFA Construction problem [Specific symbol in mention position related problem]	Guided Problem-Solving, Peer Verification	Class Test, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: DFA Construction [Advanced Problem for single symbol condition]	Constraint Logic Puzzles	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Exercise practice on DFA Construction advanced problem [Inverse Problems] Class Test – 1	Complement Language Design, Whiteboard Practice	Class Test, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: NFA Construction [Basic problem and differentiate with DFA]	Comparative Analysis, Nondeterminism Mapping	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Exercise Practice on NFA Construction [Basic to advance]	Peer Problem-Solving	Class Test, Midterm exam	CLO1, CLO2

Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Conversion of NFA to DFA [Using Subset Construction Method to convert to equivalent DFA along with detailed method]	Step-by-step Matrix Construction, Live Tracking	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Class work on NFA to DFA conversion [Basic to advance practice problem solving] Class Test -2	Subset Power-Set Computation, Board Practice	Class Test, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Basics of Regular Expression (RE),	Syntax Walkthrough, Visual Presentation	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
	Lesson 2: Regular expression formal definition and applications of regular expression	Algebraic Modeling	Class Test, Assignment, Midterm exam	CLO1, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Conversion of RE to Regular Language [Length, position, number of symbol related problems]	Set Theory Conversion, Pattern Matching Drill	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
	Lesson 2: Exercise practice on Regular expression to Regular Language [Starts, ends conditions, complex problems, connecting multiple condition and/or]	Expression Dissection, Structural Mapping	Class Test, Midterm exam, Final exam	CLO1, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Regular Expression to Finite Automata conversion by using lemma method	Thompson's Construction, Induction Graphing	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Exercise practice on Regular expression to finite automata on different notations of regular expression	Transition Diagram Drawing, Peer Review	Class Test, Final exam	CLO1, CLO3
Week-10 Lesson 1 & 2 [3 Hours]		State Rip-Out Graphing, Step-by-step Reduction	Class Test, Assignment, Presentation,	CLO1, CLO3

	Lesson 1: DFA to Regular expression conversion with details of state elimination method		Final exam	
	Lesson 2: Classwork on DFA to RE for basic to advance conversion of DFA to RE	Systematic Elimination Practice, Algebraic Verification	Class Test, Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Basics of Context free grammar and derivation, parse tree Class Test -3	Derivation Tree Drawing, Syntax Walkthrough	Class Test, Final exam	CLO1, CLO3
	Lesson 2: Class work on derivation and parse tree drawing to generate a string, Ambiguity identification from CFG	Leftmost/Rightmost Comparison, Ambiguity Identification	Assignment, Final exam	CLO1, CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Class work on practicing exercises to identify ambiguity from context free grammar, Context free grammar construction and testing with derivation	Structural Parse Testing, Rule Elimination Lab	Assignment, Presentation, Final exam	CLO1, CLO3
	Lesson 2: Introduction to Push down Automata (PDA)		Final exam	CLO1, CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Implementation of Push down Automata (PDA)	Visual Presentation, Infinite Tape Analogy	Presentation, Final exam	CLO1, CLO3
	Lesson 2: Introduction to Turing Machine, Turing Machine and its Functionalities	Tape Head Mechanics Walkthrough, State Design	Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Class work on illustration turing machine basic problems	Delta-Function Tracing, Instantaneous Description Lab	Final exam	CLO1, CLO4
	Lesson 2: Industry Session	Guest Lecture / Industry Expert Talk, Compiler/Lexer Case Study	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO3	
Attendance	--	--	--		7				
Class Test 1	5(15)				Avg 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment		5			5		5		5
Presentation			8		8				
Midterm Exam	10	10	05		25	5		20	25
Final exam			35	05	40	35	05		40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Michael Sipser, Introduction to the Theory of Computation, 3rd Edition, Course Technology Inc, 2021, ISBN: 978-1133187790.

Reference Books/ Other Supplementary Readings

Harry R. Lewis, Christos H. Papadimitriou, Elements of the Theory of Computation, 2nd Edition, Pearson Education, 1998, ISBN: 9780132624787.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



**Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)**

SE 311 Design Pattern

Course Code: SE 311	CIE Marks: 60
ISCED: 0613-311	SEE Marks: 40
Course Title: Design Pattern	Total Marks: 100

Semester:		
Credit Value: 3 (Theory)		Contact Hours: 3 (Total weeks: 14)
Prerequisite: SE 221		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

The four pillars of object-oriented programming are abstraction, Encapsulation, Inheritance, and Polymorphism. Abstract class and interface. Definition of Refactoring and Code Smells. Different types of code smells and refactoring techniques. Use of different refactoring techniques for different purposes. Introduction to Design Patterns. Definition and classification of design patterns. Definition, Use Case, Code examples of Creational Patterns – Singleton, Factory Method, Abstract Factory, Prototype, Builder. Definition, Use Case, Code examples of Creational Patterns – Adapter Pattern, Bridge Pattern, Composite Pattern, Decorator Pattern, Facade Pattern, Flyweight Pattern, Proxy Pattern. Definition, Use Case, Code examples of Creational Patterns – Chain of Responsibility Pattern, Command Pattern, Interpreter Pattern, Iterator Pattern, Mediator Pattern, Memento Pattern, Observer Pattern, State Pattern, Strategy Pattern, Template Pattern, Visitor Pattern. Design principles (SOLID)- Single Responsibility Principle, Open Close Principle, Liskov Substitution Principle, Interface Segregation Principle, Dependency Inversion Principle.

Rationale of the Course

This course focuses on building adaptable, maintainable software systems by combining advanced object-oriented programming (OOP) principles with architectural design patterns. Students reinforce their understanding of the four OOP pillars, abstract classes, and interfaces before learning to eliminate code smells through strategic refactoring techniques. The core curriculum explores various creational, structural, and behavioral design patterns, providing hands-on experience with reusable solutions to common software engineering challenges. Additionally, the course integrates SOLID principles to instill industry-standard best practices, empowering learners to design highly flexible, testable, and scalable codebases.

Course Objectives

The goal of this course is to introduce the students to the concept of modern operating system principles. The main objectives of this course are,

- To Recognize the inherent and accidental complexities involved with developing object-oriented software.
- To Understand how pattern-oriented software architecture techniques can and cannot help to alleviate this complexity.
- To Apply key pattern-oriented software architecture techniques to develop reusable object-oriented software infrastructure and apps.
- To Utilize Java programming language features and libraries to object-oriented software.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Explain the OOP design norms, code smells, and refactoring techniques.	PLO2	C3	Fundamental Domain
CLO2	Apply the concepts of creational design patterns.	PLO3	C3	Fundamental Domain
CLO3	Solve software design problems using the concept of structural design patterns.	PLO4	C3	Fundamental Domain
CLO4	Construct classes according to the concept of behavioral design patterns and SOLID principles.	PLO4	C6	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2			√									
CLO3				√								
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Design Pattern	z	None	None
	Lesson 2: Class & Object, Encapsulation, Inheritance, Polymorphism.	Object Demonstration, Structural Diagramming	Class Test, Midterm exam	CLO1
Week-2	Lesson 1: Keywords, Exception Handling, Constructor, Collection Framework, Interface, Abstract Class	Live Coding Demonstration,	Class Test, Midterm exam	CLO1

Lesson 1 & 2 [3 Hours]		Framework Modeling		
	Lesson 2: Code Smells & Refactoring, Types of Code Smells & Refactoring	Anti-pattern Case Study, Code Tracing	Class Test, Assignment, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Inappropriate Naming, Commenting, Long Class, Long Method, Lazy Class	Code Smell Diagnostics, Interactive Review	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Refused Bequest, Feature Envy, Black Sheep, Speculative Generality, Long Parameter List Class Test -1	Metrics Matrix Analysis, Peer Problem-Solving	Class Test, Assignment, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Extraction method, Naming Convention, Duplicate Code, Types of duplication, Switch Statements	Refactoring Workshop, Whiteboard Restructuring	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Introduction to Design Pattern, Design Pattern in Smalltalk MVC, Describing Design Patterns, The Catalog of Design Patterns	Architectural Diagram Analysis, Visual Presentation	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Organizing the Catalog, Solve Design Pattern Problems, Selection of Design Pattern, Usage of Design Pattern.	Catalog Mapping, Problem-Solving Session	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Creational Patterns, Types of Creation Patterns, Usage of Creation Pattern Class Test -2	Creative Instantiation Analysis, Concept Mapping	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Singleton Pattern & its principle, Single object, Why it is necessary	Instance Restriction Analysis, Real-world Analogy	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Implementation of Singleton Pattern, Real life Problem, and Its Necessity	Live Coding Demonstration, Thread-Safety Tracing	Class Test, Assignment, Midterm exam	CLO1, CLO2

Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Factory Pattern, Principle of Factory Pattern, Necessity of Factory Pattern	Decoupling Analysis, Class Diagram Walkthrough	Class Test, Assignment, Midterm exam, Final Exam	CLO1, CLO2
	Lesson 2: Implementation of Factory Pattern, Real Life Problem, and Its Necessity	Object Coupling Lab, Hands-on Coding Practice	Class Test, Assignment, Midterm exam, Final Exam	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Prototype Pattern, Principle & Usage of Prototype Pattern, Implementation of Prototype Pattern, Real Life Problem, and Its Necessity	Clone Mechanics Walkthrough, Performance Benchmarking	Class Test, Assignment, Final exam	CLO1, CLO2
	Lesson 2: Structural design patterns, Types of structural patterns, Usage of Structural Pattern	Wrapper Framework Discussion, Visual Presentation	Class Test, Assignment, Final exam	CLO1, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Adapter Design Pattern & its principle, Implementation of Adapter Pattern, Real life Problem, and Its Necessity	Interface Interface Translation Mapping, Live Coding	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Decorator Design Pattern & its Principle, Usage of Decorator Pattern & its Necessity, Implementation of Decorator Pattern, Real life Problem, and Its Necessity Class Test -3	Dynamic Behavior Stacking Workshop, Code Tracing	Class Test, Assignment, Final exam	CLO1, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Facade Design Pattern & its Principle, Usage of Facade Pattern & its Necessity, Implementation of Facade Pattern, Real life Problem, and Its Necessity	Subsystem Simplification Mapping, Structural Review	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Behavioral Design Patterns, Types of Behavioral Patterns, Usage of Behavioral Patterns	Interaction Protocol Analysis, Concept Mapping	Class Test, Assignment, Final exam	CLO1, CLO4
Week-11	Lesson 1: Observer Design Pattern & its Principle, Usage of Observer Pattern & its Necessity, Implementation of Observer Pattern, Real-Life Problem, and Its Necessity	Publisher-Subscriber Event Flowcharting, Live Demo	Class Test, Assignment, Final exam	CLO1, CLO4

Lesson 1 & 2 [3 Hours]	Lesson 2: Chain of Responsibility Pattern & its principle, Usage of Chain of Responsibility Pattern & its Necessity, Implementation of Chain of Responsibility Pattern, Real Life Problem, and Its Necessity	Request Decoupling Simulation, Logic Puzzles	Class Test, Assignment, Final exam	CLO1, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: State Pattern & its Principle, Usage of State Pattern, Necessity of State Pattern	State Machine Diagramming, Context Flow Tracking	Assignment, Final exam	CLO1, CLO4
	Lesson 2: Implementation of State Pattern, Real life Problem, and Its Necessity	Polymorphic State Transition Lab, Code Tracing	Assignment, Final exam	CLO1, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Template Pattern & its principle, interface, abstract class, Real-life implementation, and its necessity	Skeleton Algorithm Tracing, Hook Method Analysis	Assignment, Final exam	CLO1, CLO4
	Lesson 2: Design Principles (SOLID), Design Patterns Principles real-life uses	Architecture Evaluation Matrix, Core Rule Checklist	Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Blueprint Review	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review Class on final exam syllabus.	Integrative Review, Board Problem Solving	Presentation	CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO2	PLO3	PLO4	
Attendance	--	--	--	--	7				
Class Test 1	5(15)				Avg 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment			5		5				
Presentation				8	8				

Midterm Exam	15	10			25	15	10		25
Final exam		10	15	15	40		15	25	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Eric Freeman, Elisabeth Robson, Head First Design Patterns: Building Extensible and Maintainable Object-Oriented Software, 2nd Edition, O'Reilly Media, 2021, ISBN-10 : 149207800X.

Reference Books/ Other Supplementary Readings

Hilary Campbell, Designing patterns : a fresh approach to pattern cutting, Stanley Thornes, 1980, ISBN: 0859504042

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 312 Software Quality Assurance & Testing

Course Code: SE 312		CIE Marks: 60
ISCED: 0613-312		SEE Marks: 40
Course Title: Software Quality Assurance & Testing		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 3)	
Prerequisite: SE 212, SE 221		
Course Type: Core		

CIE: Continuous Internal Evaluation,

SEE: Semester End Examination

Course Content (from syllabus)

Fundamentals of Testing, testing terminology, necessities of testing, Verification, validation, Testing Principles, The Psychology of Software Testing, Testing throughout Software Life Cycle, Software Development Model, V Model, Test Levels, Test Types, Black Box Testing, Basics on black box testing, types of black box testing, Boundary Value

Analysis, Equivalence Partitioning, Decision Table based Testing, State Transition based Testing, Use Case Testing , Test Planning & Documentation, Analyze the product, Design the Test Strategy, Define the Test Objectives, Define Test Criteria, Resource Planning, Plan Test Environment, Schedule & Estimation, Determine Test Deliverables, Test Case design, Test report, White box testing, Basic Path Testing, Cyclomatic complexity, Statement coverage, Branch coverage, Condition Coverage, Path Coverage, Bug Life Cycle, Capability Maturity Model, Capability Maturity Model Integration, Quality Metrics, Mutation Testing, Decision mutations, Mutation Testing, value mutations, Statement mutations, Test tools, Types of Test Tools, Selection of test tools.

Rationale of the Course

This comprehensive course explores Software Quality Assurance (SQA) and testing methodologies integrated across the software development life cycle. Students master core verification and validation principles and apply various testing levels within frameworks such as the V-Model. The curriculum covers functional black-box techniques—such as boundary value analysis and state transition testing—alongside structural white-box strategies, including cyclomatic complexity and code coverage metrics. Additionally, learners handle test planning, test case documentation, and bug life cycle tracking while exploring quality frameworks like CMMI.

Course Objectives

To provide a solid conceptual understanding of the Software Quality Assurance & Testing, more specifically,

- To learn the basic concepts of Software Quality Assurance & Testing.
- To learn different lifecycle models and explain the role of testing in the entire life cycle of a software system.
- To learn many techniques of Black Box Testing and White box testing.
- To learn about test management.
- To learn the concepts and techniques in error detection

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcome s (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Demonstrate the fundamental of testing and necessity of Software Quality Assurance & Testing in software life cycle	PLO2	C2	Fundamental Domain
CLO2	Analyze the approaches of black box testing and white box testing	PLO2	C4	Fundamental Domain
CLO3	Design test cases for different scenarios	PLO3	C3	Fundamental Domain

CLO4	Apply the right strategy and tools for testing	PLO4	C3, P1	Fundamental Domain
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Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3			√									
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1	Lesson 1: Overview of OBE (CLO, PLO, Knowledge level)	Syllabus Walkthrough, Interactive Q&A	None	None
Lesson 1 & 2 [3 Hours]	Lesson 2: Fundamentals of Testing- Testing terminology, necessities of Testing	Interactive Lecture, Conceptual Mapping	Class Test, Midterm exam	CLO1
Week-2	Lesson 1: Fundamentals of Testing- Testing terminology, necessities of Testing	Case Study Analysis, Brainstorming Sessions	Class Test, Midterm exam	CLO1
Lesson 1 & 2 [3 Hours]	Lesson 2: Fundamentals of Testing- Verification, and validation	Comparative Analysis, Process Matrix Walkthrough	Class Test, Midterm exam	CLO1
Week-3	Lesson 1: Fundamentals of Testing- Testing Principles	Real-world Analogy Discussion, Core Rules Review	Class Test, Midterm exam	CLO1
Lesson 1 & 2 [3 Hours]	Lesson 2: Fundamentals of Testing- The Psychology of Software Testing	Role-play Simulation, Collaborative Reflection	Class Test, Midterm exam	CLO1

Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Testing throughout Software Life Cycle- Software Development Model	Software Lifecycle Mapping, Visual Presentation	Class Test, Midterm exam	CLO1
	Lesson 2: Testing throughout Software Life Cycle - V Model Class Test -1	Dual-V Verification Walkthrough, Diagram Analysis	Class Test, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Testing throughout Software Life Cycle – Test Levels, Test Types	Architectural Layer Mapping, Visual Presentation	Class Test, Midterm exam	CLO1
	Lesson 2: Black Box Testing –Basics on black box testing, types of black box testing	Behavioral Framework Analysis, Interactive Lecture	Class Test, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Black Box Testing - Boundary Value Analysis , Equivalence Partitioning	Boundary Value Analysis Drill, Partition Mapping	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Black Box Testing -Decision Table based Testing Class Test -2	Truth Table Modeling, Combinatorial Logic Puzzles	Class Test, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Black Box Testing - State Transition based Testing	State Machine Diagramming, Matrix Review	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Black Box Testing - Use Case Testing	User Story Dissection, Scenario Mapping	Class Test, Midterm exam	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Test Planning & Documentation - Analyze the product, Design the Test Strategy, Define the Test Objectives, Define Test Criteria	Strategic Blueprinting, Product Breakdown Workshop	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Test Planning & Documentation – Resource Planning, Plan Test Environment, Schedule & Estimation, Test Report	Estimation Math Exercises, Timeline Scheduling	Class Test, Assignment, Final exam	CLO2, CLO3

Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: White box testing- Basic Path Testing, Cyclomatic complexity	Control Flow Graphing, Complexity Calculation, Execution Path Tracking	Class Test, Midterm exam	CLO1, CLO3
	Lesson 2: White box testing - Statement coverage, Branch coverage, Condition Coverage	Metric Report Drafting, Diagnostic Analysis	Class Test, Midterm exam	CLO2, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Test Case Design, Capability Maturity Model, Capability Maturity Model	Diagram Design, Process Level Auditing	Class Test, Final exam	CLO2, CLO3
	Lesson 2: Bug Life Cycle, Quality Metrics,	Lifecycle State Diagramming, Case Tracking Review, Quality Matrix Analysis	Class Test, Final exam	CLO2, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Generative AI & the AI Spectrum	Boolean Truth Mapping, Code Review Clinic	Class Test, Final exam	CLO3, CLO3
	Lesson 2: LLM Architecture, Embeddings & Multimodal Models for Testing Class Test -3	Technical walkthrough, case-study discussion of multimodal testing use cases	Class Test, Final exam	CLO3, CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Software Security Testing & Security Fundamentals — Security Testing concepts, objectives, CIA Triad, security requirements, threats, vulnerabilities, risks, authentication and authorization	Lecture, guided classroom discussion, real-world security breach case studies, instructor demonstration	Class Test, Assignment, Final exam	CLO3, CLO4
	Lesson 2: Security Test Management, Reporting & Risk-Based Security Testing — Security test planning, risk assessment, vulnerability severity, defect reporting, retesting, regression testing, security testing in CI/CD	Lecture, security test-report workshop, case-study discussion, mini-project presentation	Class Test, Final exam	CLO3, CLO4
Week-13	Lesson 1: Mutation Testing- Decision mutations, value mutations, Statement mutations	Fault Injection Analysis, Code Variant Tracing	Class Test, Final exam	CLO3, CLO4

Lesson 1 & 2 [3 Hours]	Lesson 2: Test tools- Types of Test Tools	Automation Tools Walkthrough, Comparative Matrix	Class Test, Presentation Final exam	CLO3, CLO4
Week-14	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
Lesson 1 & 2 [3 Hours]	Lesson 2: Review class	Integrative Syllabus Review, Board Problem Solving	Final exam	CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total=70)
	CLO1	CLO2	CLO3	CLO4		PLO2	PLO3	PLO4	
Attendance	--	--	--	--	7				
Class Test 1	5(15)				15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment			5		5		5		5
Presentation				8	8				
Midterm Exam	10	15			25	25			25
Final exam			20	20	40		20	20	40

Marks Distribution

Class attendance	7
Assignment	5

Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Graham, D., Black, R., & van Veenendaal, E., Foundations of Software Testing: ISTQB Certification ,4th ed.,Cengage Learning, 2019, ISBN: 978-1473764228

Reference Books/ Other Supplementary Readings

Srinivasan Desikan; Gopalaswamy Ramesh, Software testing : principles and practices , Pearson, c2006., ISBN: 9788177581218

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 313 Software Quality Assurance & Testing Lab

Course Code: SE 313		CIE Marks: 60
ISCED: 0613-313		LF Marks: 40
Course Title: Software Quality Assurance & Testing Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

Fundamentals of Testing, Test Levels, Test Types, Black Box Testing, Basics on black box testing, types of black box testing, Boundary Value Analysis, Equivalence Partitioning, Decision Table based Testing, State Transition based Testing, Use Case Testing, Test Planning & Documentation, Analyze the product, Design the Test Strategy, Define the Test Objectives, Define Test Criteria, Resource Planning, Plan Test Environment, Schedule & Estimation, Determine Test Deliverables, Test Case design, Test report, White box testing, Basic Path Testing, Cyclomatic complexity, Statement coverage, Branch coverage, Condition Coverage, Path Coverage, Bug Life Cycle, Capability Maturity Model,

Capability Maturity Model Integration, QualityMetrics, Mutation Testing, Decision mutations, Mutation Testing, value mutations, Statement mutations, Test tools, Types of Test Tools, Selection of test tools

Rationale of the Course

This course is a core course of Software Engineering. Software testing is particularly important in the Software Development Life Cycle (SDLC). The testing process can be broadly planned into two activities – Verification and Validation (V&V). A systematic realization of these activities can enhance software quality to a great extent. The aim of this course is to illustrate these aspects of software testing as a methodical process to produce quality software. It is expected that students, at the end of this course, will have a clear understanding of the foundations, methodologies and tools in the area of software testing

Course Objectives

The goal of this course is to introduce the students about the concept of software testing. The main objectives of this course are,

- To learn various testing techniques
- To develop skills to apply appropriate testing knowledge in problem-solving

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Develop test plans, identify test conditions	PLO1	C3	Fundamental Domain
CLO2	Design test cases for white box testing and black box testing	PLO3	C3, P2	Fundamental Domain
CLO3	Apply a wide variety of testing techniques for solving a given problem.	PLO5	C3, P3	Fundamental Domain
CLO4	Write effective reports and design test case documentation for the real life project applying a wide variety of testing techniques at various testing levels	PLO10	C5, A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2			√									
CLO3					√							
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Fundamentals of Testing, Test Levels, Test Types, Black Box Testing, Basics on black box testing	Syllabus Walkthrough, Lab Environment Setup	CLO1	Lab Performance
Week-2 [1.5 Hours]	Basics on black box testing, types of black box testing, Boundary Value Analysis	Boundary Value Scripting, Baseline Testing Exercise, Programming Language	CLO1	Lab Performance
Week-3 [1.5 Hours]	Three boundary value analysis techniques, boundary value check, robust testing, worst case testing	Extreme Input Testing, Robustness Debugging Lab, Programming Language	CLO1, CLO2, CLO3	Lab Performance, Lab Final
Week-4 [1.5 Hours]	Introduction to Equivalence Partitioning, different equivalence partitioning techniques	Dataset Partitioning Lab, Range Verification Coding, Programming Language	CLO1	Lab Performance
Week-5 [1.5 Hours]	Introduction to Decision Table based Testing, case design using decision table based testing	Combinatorial Logic Coding, Matrix-based Testing	CLO2, CLO3	Viva, Lab Final
Week-6 [1.5 Hours]	State Transition based Testing, draw state transition diagram	State Machine Drawing, Dynamic Execution Tracking, Programming Language	CLO1	Viva, Lab Final
Week-7 [1.5 Hours]	Use Case Testing, Test Planning & Documentation	User Scenario Scripting, Requirements Mapping, Using Software Tools	CLO3, CLO4	Viva, Lab Final
Week-8 [1.5 Hours]	Test Case design, Test report writing	Test Suite Composition, Bug Metric Logging, Coding	CLO2, CLO3	Viva, Lab Final
Week-9 [1.5 Hours]	White box testing, Basic Path Testing, Quality Metrics.	Source Code Inspection, Control Flow Mapping	CLO1	Lab Performance

Week-10 [1.5 Hours]	Cyclomatic complexity, Statement coverage, Branch coverage, Condition Coverage	Complexity Metric Profiling, Logical Path Coverage Lab	CLO1, CLO2, CLO3	Lab Performance, Viva, Lab Final
Week-11 [1.5 Hours]	Project based implementation	Documentation, Report Generation	CLO4	Lab Report
Week-12 [1.5 Hours]	Test tool: Debugging, Junit	Unit Testing Automation, Breakpoint Assertions Lab	CLO2, CLO3, CLO4	Lab Performance, Viva, Lab Final
Week-13 [1.5 Hours]	Test tool: Selenium	Browser Automation Scripting, Web UI Element Testing	CLO2, CLO4	Lab Performance
Week-14 [1.5 Hours]	Lab Final	Independent Practical Exam, Practical Scripting Task	CLO2, CLO3	Viva, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO1
Lab Final	40	CLO2	PLO3
		CLO3	PLO5
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Dosaj, C. R., The Self-Taught Software Tester: A Step By Step Guide to Learn Software Testing Using Real-Life Project. Independently published, 2020 ISBN: 979-8646813491.

Reference Books/ Other Supplementary Readings

1. Srinivasan Desikan; Gopalaswamy Ramesh, Software testing : principles and practices , Pearson, c2006., ISBN: 9788177581218

2. Lab Manual.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Junit, Selenium IDE, Jira, Java, C, Python

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

GE 324 Business Analysis and Communication

Course Code: GE 324		CIE Marks: 60
ISCED: 0417-324		SEE Marks: 40
Course Title: Business Analysis and Communication		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: GED		

CIE: Continuous Internal Evaluation
SEE: Semester End Examination

Course Content (from syllabus)

Introduction to Business Analysis (BA), BA Practices and uses, Business analysis Vs Business Analytics, DIKW model, Sources of data , Understanding problem and symptom, Business analysis techniques, Requirement, SMART Technique for writing Requirements, Requirement management, Requirement Implementation, Stakeholders, Managing Stakeholders, Communication & Business Communication, Written Communication, Nonverbal communication, Office memos, face to face communication, types of report, Types of Letters, CV.

Rationale of the Course

The program is designed specifically for professionals who wish to enhance their business analysis capabilities, as well as those who are seeking new opportunities in the field. Business Communication will provide students with communication principles, concepts, and techniques which are essential components for effective organizational behavior in oral and written communication situations.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of Business Analysis and Communication More specifically,

- To describe the basic concepts of Business Analysis and Communication
- To apply the components of Business Analysis and Communication
- To criticize Business Analysis and Communication components and case
- To compose Business Analysis and Communication tools for use.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcome s (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Contextualize the concepts of Business Analysis and Communication by examining their application and effects in different organizational environments.	PLO2	C4	Fundamental Domain
CLO2	Analyze the significance of Business Communication in real-life scenarios to understand its impact on organizational success and efficiency.	PLO2	C4	Fundamental Domain
CLO3	Justify the use of different types of business communications by evaluating their effectiveness in various organizational contexts.	PLO7	C5	Fundamental Domain
CLO4	Incorporate Business Analysis methods to develop a targeted business proposal addressing market needs.	PLO8	C6	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PL O7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3							√					
CLO4								√				

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to OBE	Syllabus Walkthrough, Interactive Q&A	None	CLO1
	Lesson 2: Business Analysis and Communication, Overview and Highlights, Meaning of Communication & Business Communication, Scope, Purposes, Process- Principles	Conceptual Mapping, Communication Flowchart Analysis	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Functions, Importance, Models of Communication, Nonverbal Communication, Symbols, Gestures, Body Language, Visual Communication	Behavioral Role-Play, Communication Model Walkthrough	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Interpersonal and Intrapersonal Communication, Downward, Upward-Horizontal, Mass Communication	Organizational Role-Play, Collaborative Discussion	Class Test, Midterm exam	CLO1 CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: The Principles of Collaborative Negotiations Planning for Impact	Negotiation Case Study, Strategy Matrix Review	Presentation Midterm exam	CLO1 CLO2
	Lesson 2: Negative News and Crisis Communication	Crisis Scenario Simulation, Interactive Review	Class Test, Midterm exam	CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: SMART Technique for writing Requirements. BABOK Guide V3 by IIBA, Requirement Management.	Requirements Dissection Workshop, Framework Modeling	Class Test, Midterm exam	CLO1 CLO 2
	Lesson 2: Crafting your Structure, Writing Clear and Engaging Sentences	Sentence Drafting Exercises, Peer Editing Clinic	Class Test Presentation	CLO1 CLO2

	Class Test 01			
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Agenda, Notice, Minutes of different meeting, Motion, Proxy, Quorum, and Resolution	Meeting Document Template Analysis, Live Drafting	Class Test, Midterm	CLO1 CLO 2
	Lesson 2: Types of Letters, Circular Letter, Letter of Inquiry, Letter of Complaints	Correspondence Format Drill, Writing Practice	Class Test, Midterm Exam	CLO1 CLO 2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Types of report, Characteristics and Importance of different types, Purposes, Scope	Typology Case Analysis, Structural Review	Class Test, Midterm Exam	CLO1
	Lesson 2: Techniques and Steps of Analytical and Marketing Report	Analytical Report Blueprinting, Writing Draft	Midterm exam, Assignment, Presentation	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Business Correspondence, Business Specific language Phrases, E-mail Etiquette, Professional Etiquette, Organizational Behavior, Cross Cultural Awareness	Cross-Cultural Role Play, Etiquette Case Review	Presentation Assignment, Midterm exam	CLO2
	Lesson 2: Speed Networking -The Elevator Pitch Being Diplomatic, Agreeing and Disagreeing	Elevator Pitch Workshop, Dialogue Simulation	Class Test, Assignment, Midterm exam	CLO2
	Class Test 02			
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Business Analysis, Techniques, Brainstorming, PESTLE Analysis, SWOT Analysis	Strategic Framework Matrix, Case Analysis	Class Test, Assignment, Final Exam.	CLO3
	Lesson 2: ERRC, Six Thinking Hats, The 5 Whys, MOST (Mission, Objectives, Strategies, and Tactics) Analysis	Cognitive Perspective Puzzles, Root-Cause Analysis	Class Test, Assignment, Final exam	CLO3 CLO 4

Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: MoSCoW (Must or Should, Could or Would), CATWOE, Non-Functional Requirement Analysis, Business Process Modeling (BPM)	Requirement Prioritization Workshop, Process Mapping	Class Test, Assignment, Final exam	CLO3 CLO 4
	Lesson 2: Structure of Business Proposal, Introduction of Stakeholders, Stakeholder Engagement Model	Proposal Blueprinting, Matrix Classification	Assignment, Presentation Final exam	CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Interacting and Dealing with Different types of Stakeholders, Developing and Delivering , Effective Presentation	Stakeholder Persona Management, Speech Practice	Class Test, Assignment, Final exam	CLO4
	Lesson 2: Quality Assurance, Continuous Improvement, Responding to Question, Preparing CV, Types of CV	Resume Structure Analysis, Diagnostic Layout Workshop	Class Test, Assignment, Final exam	CLO 3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Attire Etiquette Dinning Manner, Preparing for interview, Formal vs informal, In person and online	Visual Guide Presentation, Behavioral Modeling	Class Test, Assignment, Presentation Final exam	CLO4
	Lesson 2: Preparing for interview, Formal vs informal, In person and online Class Test 03	Interview Matrix Analysis, Peer Q&A Practice	Class Test, Assignment, Final exam	CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Preparing Video Resume	Video Recording Scripting, Script Tracing Lab	Presentation Assignment, Final exam	CLO4
	Lesson 2: Demonstration of Interview	Mock Interview Role-Play, Real-Time Evaluation	Presentation Assignment, Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Generate a Business Proposal and Present to the Audience	Capstone Project Presentation, Peer Q&A	Presentation Assignment, Final exam	CLO4
	Lesson 2: Generate a Business Proposal and Present to the Audience	Capstone Project Presentation, Instructor Feedback	Presentation, Assignment, Final exam	CLO4

Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Interactive Q&A	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class	Integrative Review, Board Problem Solving	Presentation, Open discussion	CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO2	PLO7	PLO8	
Attendance	--	--	--		7				
Class Test 1	15				Avg 15	5			
Class Test 2		15				5			
Class Test 3			15				5		
Assignment				5	5			5	5
Presentation				8	8			8	25
Midterm Exam	12	13			25	25			40
Final exam			15	25	40		15	25	70

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Joseph Grenny, Kerry Patterson, Crucial Conversations: Tools for Talking When Stakes are High, 3rd Edition, McGraw-Hill, 2021, ISBN: 978-1260474183.

Mary Ellen Guffey, Dana Loewy, Essentials of Business Communication, 12th Edition, Cengage Learning, 2022, ISBN: 0357714970.

Reference Books/ Other Supplementary Readings

Kathryn Rentz, Marie Flatley, Paula Lentz, Lesikar's Business Communication: Connecting in a Digital World, 12th Edition, McGraw-Hill, 2010, ISBN: 9780073377797.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Microsoft Office

Exam Dates

According to the Examination Schedule



**Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)**

SE 321 Software Engineering Web Application

Course Code: SE 321		CIE Marks: 60
ISCED: 0613-321		SEE Marks: 40
Course Title: Software Engineering Web Application		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE 121, SE 122		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Software Engineering Web Application; History of Internet and Web; Protocols; Domain Name and URL; Attributes & Categories of Web Application; Anatomy of Web Page and URL; Overview of HTTP; Persistent and Non-Persistent HTTP; Response Time Modeling; HTTP GET & POST Request; HTTP Methods, Response Code; User-Server Interaction; Basic structure of HTML; Examples and usage of various HTML Elements; HTML Forms; Types of HTML Elements; Introduction to Bootstrap; Responsive Web Design; Rendering Mode: Flexbox, Position; CSS Variables, Web Fonts; Mobile Web; Getting started with Bootstrap; Basic Grid and Columns in Bootstrap; Using Common Components in Bootstrap; Introduction to JavaScript; Basic JavaScript Syntax; JavaScript DOM; Introduction to Server-side rendering using PHP; Basic PHP Syntax; Introduction to Node.js & Express.js; Database Connection and CRUD operations using EXPRESS.JS; Introduction to MVC Framework; CRUD operations using Node.js & MySQL Library; Using Firebase as a Database Server; Connecting and Operating with Firebase using .

Rationale of the Course

This course tracks the evolution of web development from basic protocols to dynamic full-stack engineering. Students master frontend fundamentals including responsive HTML, CSS, and interactive JavaScript for DOM manipulation before advancing to server-side development. The curriculum emphasizes object-oriented PHP and clean system architecture using the Model-View-Controller (MVC) pattern within the Laravel framework. Through hands-on projects, learners build core CRUD operations and integrate NoSQL databases like Firebase.

Course Objectives

To introduce different technologies that are used for building complex web applications for solving real world problems. The main objectives of this course are:

- To understand the working principles of rendering a website from the internet.
- To be able to design web applications using different frontend technologies based on the use requirements.
- To grow expertise in creating interactive components in web pages for better user experience.
- To learn techniques and technologies for developing dynamic web applications that meet the requirement specifications.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcome s (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Analyze the concepts and principles of web applications including the procedures and protocols that are related with the rendering of a webpage.	PLO2	C4	Fundamental Domain
CLO2	Design different web components using frontend technologies that comprise a full web application	PLO3	C3	Fundamental Domain
CLO3	Construct interactive components to improve user experience and dynamic functionality using client-side programming.	PLO4	C6 A2	Fundamental Domain
CLO4	Develop dynamic web applications using server-side programming that can provide information based on user requests.	PLO4	C6	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO 1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2			√									
CLO3				√								
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of OBE (CLO, PLO, Knowledge level)	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Software Engineering Web Application, Use of Software Engineering in Web Applications, Web Vs. Internet, History of	Architectural Layer Mapping, Visual Presentation	Class Test,	CLO1

	Internet and Web, Protocols, Domain Name and URL, Attributes & Categories of Web Application		Assignment Midterm exam	
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Anatomy of Web Page and URL, Overview of HTTP, Types of HTTP, Persistent and Non-Persistent HTTP, Response Time Modeling, HTTP GET & POST Request, HTTP, Methods, Response, Code, User-Server Interaction, Introduction to HTML, Basic structure of an HTML Document, Tag and Attribute in HTML, Basic HTML Elements - Headings, Paragraph, Image, Anchor, List, Table	Request-Response Cycle Mapping, Document Tree Analysis	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2
	Lesson 2: Class Practice on HTML Table - Rowspan and Colspan, Introduction to HTML Forms - Form Attributes, Input Elements, Input Attributes, Types of HTML Elements - Block vs Inline Elements	Form Structure Blueprinting, Tag Dissection Practice	Class Test, Midterm exam, Final exam	CLO1, CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to MVC Architecture. Meaning of Model, View, and Controller; separation of concerns; MVC in traditional applications; MVC in an Express.js backend; relationship among the frontend view, controller logic, and database model.	MVC Flow Diagramming, Guided Discussion	Class Test, Assignment, Midterm Exam	CLO1
	Lesson 2: Modern Full-Stack Web Application Architecture. React frontend, Express backend, REST API, and MySQL database; why React does not directly connect to MySQL; role of the API layer; local development workflow; project folder structure.	System Architecture Mapping, Project Discussion	Class Test, Assignment, Midterm Exam	CLO1, CLO4
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: React.js Core Concepts: Props and Reusability. Props; reusable components; component composition; difference between ordinary HTML pages and React component design.	Reusable Component Design, Props Mapping	Class Test - 1, Assignment, Midterm Exam	CLO2
	Lesson 2: Bootstrap for Responsive Frontend Design. Bootstrap grid system; container, row, and column; navbar; buttons; cards; forms; tables; responsive design; mobile-first layout; using Bootstrap classes inside React components.	Responsive Grid Sketching, Bootstrap Practice	Assignment, Midterm Exam	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Designing Web Pages and Layout Components. Home, Products, Add Product, Login, Register, and Dashboard pages; header, footer, sidebar, content area; UI consistency; visual hierarchy; basic accessibility awareness.	Page and Component Mapping, UI Workshop	Class Test, Assignment, Midterm Exam	CLO2
	Lesson 2: Client-Side Programming: State and Events. React state concept; event handling; user interaction; state-driven interface updates; how dynamic UI improves user experience.	State Transition Tracing, Event Demonstration	Class Test, Assignment, Midterm Exam	CLO3

Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Controlled Forms and Frontend Validation. Controlled form inputs; form submission; storing input values in state; frontend validation; preparing form data before sending it to the backend.	Controlled Form Tracing, Validation Exercise	Assignment, Midterm Exam	CLO3
	Lesson 2: Client-Side Data Handling and API Communication. JSON format; Fetch API or Axios concept; asynchronous programming; promises and async-await; loading and error states; displaying API data in cards or tables.	API Data-Flow Tracing, Async Code Walkthrough	Class Test, Assignment, Midterm Exam	CLO3, CLO4
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Frontend Routing and Multi-Page Application Concept. Single Page Application concept; page navigation; route idea; frontend route versus backend route; protected page concept; 404 page; navigation flow for a complete web application.	Navigation Flow Design, Routing Demonstration	Class Test - 2, Midterm Exam	CLO2, CLO3
	Lesson 2: Backend Development with Node.js. Node.js as a server-side JavaScript runtime; npm; backend project structure; package.json; backend dependencies; server.js; environment variables; frontend runtime versus backend runtime.	Backend Structure Analysis, Node.js Demonstration	Midterm Exam, Final Exam	CLO4
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Express.js Fundamentals: Server and Routes. Express.js server; routes; request and response objects; route parameters; handling basic client requests and responses.	Request Handling Walkthrough, Route Coding	Class Test, Assignment, Final Exam	CLO4
	Lesson 2: Express.js Fundamentals: Middleware and Controllers. Middleware concept; CORS; dotenv; nodemon; organizing backend code into routes and controllers; request-processing flow.	Middleware Pipeline Mapping, Controller Exercise	Class Test, Assignment, Final Exam	CLO4
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: REST API Design Principles. REST API concept; resource-based URL design; HTTP methods; endpoint naming; request and response bodies; status codes; API table; relationship between REST APIs and frontend communication.	API Endpoint Table Design, REST Scenario Analysis	Class Test, Assignment, Final Exam	CLO1, CLO4
	Lesson 2: Backend Error Handling, Validation, and API Testing. Backend input validation; error-response format; try-catch concept; common API errors; Postman or Thunder Client testing; debugging route errors; frontend-friendly API responses.	API Error Analysis, API Testing Practice	Assignment, Final Exam	CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to MySQL Database for Web Applications. Database role in dynamic web applications; relational database concept; table, row, column; primary key; foreign key; data types; CRUD from the database perspective; suitability of MySQL for structured applications.	Database Table Identification, Concept Mapping	Class Test, Assignment, Final Exam	CLO1, CLO4
	Lesson 2: Database Design and Relationships. Creating a database and tables; products and users tables; primary and foreign keys; simple	ER Diagram Design, Schema Planning	Assignment, Final Exam	CLO4

	relationships; basic ER diagram; mapping project requirements to database tables.			
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: SQL Basics and CRUD Queries. INSERT, SELECT, UPDATE, and DELETE statements; WHERE condition; creating and manipulating records in products and users tables.	SQL Query Construction, CRUD Practice	Class Test - 3, Assignment, Final Exam	CLO4
	Lesson 2: Connecting Express.js with MySQL. MySQL connection from the backend; database configuration file; environment variables for credentials; query execution; promise-based operations; separation of database and route logic; connection error handling.	Connection Flow Tracing, Configuration Walkthrough	Assignment, Final Exam	CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: CRUD Operation Flow in Full-Stack Applications. Full CRUD workflow; React form to REST API; Express controller; MySQL query; JSON response; React display; create, read, update, and delete; frontend and backend responsibilities.	End-to-End CRUD Mapping, Integration Exercise	Assignment, Final Exam	CLO3, CLO4
	Lesson 2: Search, Filter, and Dynamic Data Presentation. Searching product data; category filtering; sorting; displaying records in cards and tables; frontend versus backend filtering; loading messages; empty-data messages; usability improvement.	Filtering Comparison, Data Presentation Exercise	Class Test, Assignment, Final Exam	CLO3, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Authentication: Registration and Login System. Need for authentication; users table; registration and login flows; password hashing concept; session versus token idea; JWT concept; storing login state; risks of plain-text passwords.	Authentication Workflow Diagramming, Security Discussion	Assignment, Final Exam	CLO4
	Lesson 2: Testing, Debugging, GitHub, and Documentation. Frontend, backend API, and database testing; common bugs; browser and backend consoles; Git and GitHub workflow; commit messages; README writing; screenshots; API and database schema documentation..	Role-Based Access Design, Dashboard Sketching	Assignment, Presentation, Final Exam	CLO3, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Deployment, Final Integration, and Course Review. Preparing the project for deployment; Vercel deployment; hosted MySQL requirement; environment variables; live-link testing; final presentation guidelines; review of frontend, backend, API, database, GitHub, and deployment.	Submission Checklist, GitHub Documentation Practice	Assignment, Presentation, Final Exam	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on Final Exam Syllabus topic	Deployment Plan Presentation, Integrative Review	Project Presentation, Final Exam	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =78)
	CLO1	CLO2	CLO3	CLO4		PLO2	PLO3	PLO4	
Attendance	--	--	--	—	7				
Class Test 1	5	10			Avg 15				
Class Test 2		5	10						
Class Test 3			5	10					
Assignment		2	3		5		2	3	5
Presentation			4	4	8			8	8
Midterm Exam		10	10	5	25		10	15	25
Final exam		5	10	25	40		5	35	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Gerti Kappel, Web engineering : the discipline of systematic development of web applications, John Wiley & Sons, c2006., ISBN: 9780470015544

Reference Books/ Other Supplementary Readings

Nixon, Robin. Learning PHP, MySQL & JavaScript with jQuery, CSS & HTML5, 4th Edition, O'Reilly Media, 2015. ISBN: 9789352130153.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Visual Studio Code, PHP, Laravel, Git

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 322 Software Engineering Web Application Lab

Course Code: SE 322		CIE Marks: 60
ISCED: 0613-322		LF Marks: 40
Course Title: Software Engineering Web Application Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)	Contact Hours: 1 hour 30 minutes (14 weeks)	
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

Introduction to Local Web Servers (Node.js environment); Setting up the development environment; Google Developer Tools; Create a web page with Basic HTML Elements; HTML audio, video and YouTube; Develop HTML Forms; Applying different text CSS properties; Border properties and converting link to button; CSS Font properties;

CSS Background property; Responsive layouts using Flexbox, media queries; JavaScript DOM manipulation, events, form validation, modal components and interactive UI; React.js fundamentals, components, JSX, props, state, events, forms, conditional rendering, list rendering and hooks; Building responsive single-page front-end applications with React; Node.js and Express.js fundamentals; Routing, Middleware, Controllers and REST APIs; Request-Response handling, req.params, req.query and req.body; MySQL integration and CRUD operations using Express; Building full-stack web applications with React, Node.js and Express.

Rationale of the Course

This comprehensive course covers the essential skills and technologies required for modern web and software development. This lab course introduces students with the tools and technologies required to build modern web applications using HTML, CSS, JavaScript and PHP. Students will have hands-on experience with developing complex websites that solve real-world problems. The course will start with designing responsive websites and expands to develop back-end logics to process complications of the world.

Course Objectives

The goal of this course is to introduce the students to the concept of principles in practice. The main objectives of this course are,

- . To create aesthetic and responsive websites that meet modern web trends.
- . To produce interactive web components for improving user experience.
- . To build applications that can be updated in real-time.
- . To develop web applications that will be able to solve real-world problems and perform complex operations.
- . To grow skills for designing and developing successful web applications by maintaining software engineering steps for ensuring accomplishment of user's requirements.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Implementation of HTML and CSS for creating aesthetic web applications.	PLO3	C6	Fundamental Domain
CLO2	Construction of interactive web components using JavaScript for betterment of user experience	PLO5	C3	Fundamental Domain
CLO3	Development of dynamic web applications using PHP and Laravel that can handle user requests and communicate with databases for data management.	PLO3	C6	Fundamental Domain
CLO4	Apply engineering and management principles to develop web applications for project management,	PLO11	C3	Fundamental Domain

	functioning effectively as a team member or leader in multidisciplinary environments			
CLO5	Apply Software Engineering skills to design and develop web applications according to problem specifications for sustainable development.	PLO 7	C6, A2	Fundamental Domain, Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1			√									
CLO2					√							
CLO3			√									
CLO4											√	
CLO5							√					

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning, and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Discussion on OBE & Introduction Local Web Servers (XAMPP); Online Free Web Servers	Syllabus Walkthrough, Environment Setup Lab	N/A	N/A
Week-2 [1.5 Hours]	Setting Up the Development Environment and HTML Audio, Video and YouTube; Develop an HTML Forms	Media Embedding Exercise, Form Structure Coding	CLO1	Lab Performance
Week-3 [1.5 Hours]	Applying Different Text CSS Properties; Border Properties and converting link to button; CSS Font Properties; CSS Background Property	Style Sheet Dissection, Inline-to-Block Overriding	CLO1	Lab Performance

Week-4 [1.5 Hours]	Creating a web layout using float, overflow and box-sizing properties.	Box-Model Structural Walkthrough, Layout Mapping	CLO1	Lab Performance
Week-5 [1.5 Hours]	Use flexbox to create an responsive web layout; Create responsive navbar using flexbox & media queries, Creating a web section using Bootstrap Framework	Flexbox Axis Alignment, Breakpoint Tracing Lab	CLO1	Viva, Lab Final
Week-6 [1.5 Hours]	Create a solution to block a user if he reloads a page more than 10 times within 1 minute	Rate Limiting Logic, Stateful Session Scripting	CLO2	Viva, Lab Final
Week-7 [1.5 Hours]	Creating Modal Elements using vanilla JavaScript; Build a Digital Clock using Vanilla JavaScript.	DOM Object Manipulation, Async Timer Control Lab	CLO2	Viva, Lab Final
Week-8 [1.5 Hours]	JavaScript form validation for client-side validation, Review Front-end Design using HTML, CSS and javascript	Regex Pattern Matching, Interface Integration Practice	CLO2	Lab Performance, Viva, Lab Final
Week-9 [1.5 Hours]	Problem Solving using PHP, Session and Cookies, Handling Submitted Form using PHP with proper validation and file upload mechanism	Server-Side Input Sanitization, Superglobal Stream Processing	CLO3, CLO5	Lab Performance, Viva, Lab Final
Week-10 [1.5 Hours]	Practice Object Oriented PHP on real-life scenarios.	Class Instantiation Modeling, Object-Coupling Lab	CLO3, CLO5	Lab Performance, Viva, Lab Final
Week-11 [1.5 Hours]	Create Registration System using PHP Form and Database CRUD Operation; Developing Model Class for Database, CRUD Operation using PHP	SQL-to-PHP Statement Binding, Data Access Layer Coding	CLO3, CLO5	Lab Performance, Viva, Lab Final
Week-12 [1.5 Hours]	Introduction to ORM in Laravel; Authentication and Authorization in Laravel; User management using Firebase and Laravel	Eloquent Model Mapping, Driver Wiring Practice	CLO3, CLO5	Lab Performance, Viva, Lab Final
Week-13 [1.5 Hours]	Auth, Multi User Dashboard in Laravel, API Development Using Passport	Route Middleware Guarding, Token Bearer Generation	CLO1, CLO2, CLO3, CLO4	Lab Performance

Week-14 [1.5 Hours]	Lab Final	Independent Practical Exam, Practical Task Execution	CLO1, CLO2, CLO3, CLO4	Viva, Lab Final
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Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO11
Lab Performance	25	CLO1, CLO3	PLO3
Lab Final	40	CLO2	PLO5
		CLO5	PLO7
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Gerti Kappel, Web engineering : the discipline of systematic development of web applications, John Wiley & Sons, c2006., ISBN: 9780470015544

Reference Books/ Other Supplementary Readings

Nixon, Robin. Learning PHP, MySQL & JavaScript with jQuery, CSS & HTML5, 4th Edition, O'Reilly Media, 2015. ISBN: 9789352130153.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Visual Studio Code, PHP, Laravel, Git

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 323 Software Architecture & Design

Course Code: SE 323/GED 323		CIE Marks: 60
ISCED: 0613-323		SEE Marks: 40
Course Title: Software Architecture & Design		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE 212, SE 221, SE223, SE311		
Course Type: Core/GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Software Architecture Introduction, Architecture Overview and Process, Architectural Structures and Views, The 4+1 View Model of Architecture, Layered Architecture, Broker Architecture, Monolithic Architecture, Microservices Architecture, Architecture Evaluation, S.O.L.I.D. Principles, Architecture Trade-off Analysis Method (ATAM) Understanding Quality Attributes, Availability, Performance, Usability, Modifiability.

Rationale of the Course

This course covers the architectural frameworks and design principles required to build robust, scalable software systems. Students map system structures using the 4+1 View Model and evaluate diverse structural paradigms, including Monolithic, Layered, Broker, and Microservices architectures. The curriculum focuses on calculated decision-making using SOLID principles and the Architecture Trade-off Analysis Method (ATAM) to balance complex design trade-offs. Additionally, learners analyze methods for optimizing critical software quality attributes like availability, performance, and modifiability.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of data communications. More specifically, to learn

- To learn the basic concepts of software architecture..
- To learn different architecture models and explain their importance.

- To learn many approaches to gathering requirements from users and designing with proper documents.
- To learn how to design an architecture for a software.
- To learn the quality attributes and tradeoff analysis of a software architecture.
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Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Explain the concept and principles of software architecture.	PLO2	C2	Fundamental Domain
CLO2	Analyze the concept of recognizing software architectural styles, design patterns, and frameworks within existing software systems	PLO2	C3	Fundamental Domain
CLO3	Identify the concept of the current trends and technologies in the field by designing model architectures.	PLO3	C4	Fundamental Domain
CLO4	Evaluate different quality attributes of software.	PLO4	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√										
CLO3			√									
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
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Week-1 Lesson 1 & 2 [3 Hours]	Lesson1: Introductory lecture on OBE and course contents	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Introduction to software architecture Terminology, Enterprise architecture	Interactive Lecture, Conceptual Mapping	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: System architecture, Software architecture	Comparative Analysis, Structural View Identification	Class Test, Midterm exam	CLO1
	Lesson 2: Architecture Overview and Process, Software <u>architecture</u> definition, <u>Factors need to be considered in Software architecture</u>	Architectural Process Walkthrough, Visual Presentation	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Why we need a Software architecture, Advantages of Software architecture, Stakeholders of software architecture, Architectural Structure and Views	Stakeholder Persona Mapping, Interactive Lecture	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Different types of Architectural Structure	Structural Taxonomy Modeling, Visual Presentation	Class Test, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: The 4+1 View Model of Architecture	View Model Deconstruction, Visual Walkthrough	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Different layers of the 4+1 View Model of Architecture Class Test -1	View Layer Mapping, Case-based Allocation	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Layered Architecture	Architectural Blueprint Mapping, Tier Analysis	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
	Lesson 2: Application of Layered Architecture	Real-world Case Study, Component Isolation Drill	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
Week-6	Lesson 1: Introduction to Broker Architecture, Broker Architecture Components	Messaging Stream Tracing, Component Mapping	Class Test, Assignment,	CLO1, CLO3

Lesson 1 & 2 [3 Hours]			Midterm exam, Final exam	
	Lesson 2: Monolithic Architecture Monolithic Architecture, Advantages and Disadvantages of Architecture Class Test -2	Structural Complexity Analysis, Anti-pattern Review	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Load balancing, Examples of Monolithic Architecture, Why are Microservices used? Microservices Architecture	Traffic Simulation Routing, Distribution Mapping	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
	Lesson 2: Monolithic Vs Microservices Architecture Advantages and Disadvantages of Microservices Architecture Examples of Microservices Architecture	Comparative Matrix Analysis, Migration Path Review	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Architecture Evaluation. Why, When, Benefits of Architecture Evaluation, Evaluation Techniques	Review Method Walkthrough, Metric Checklist Analysis	Class Test, Final exam	CLO2, CLO3
	Lesson 2: Architecture Trade-off Analysis Method (ATAM)	ATAM Scenario Generation, Collaborative Trade-off Analysis	Class Test, Final exam	CLO2, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: What is S.O.L.I.D principles? What does not support S.O.L.I.D?	Design Principle Violations Analysis, Code Review	Class Test, Final exam	CLO2, CLO4
	Lesson 2: Understanding Quality Attributes Architecture and Requirements, Functionality Quality Attribute Considerations	Non-Functional Requirements Dissection, Matrix Design	Class Test, Final exam	CLO2, CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Performance General Scenario, Tactics for Performance	Latency Math Exercises, Resource Management Tactic Drill	Class Test, Final exam	CLO2, CLO4
	Lesson 2: Usability General Scenario, Tactics for Usability	User Interaction Tracing, System State Feedback Review	Class Test, Final exam	CLO2, CLO4
Week-11	Lesson 1: Availability General Scenario	Fault Tree Diagramming,	Class Test, Final exam	CLO2, CLO4

Lesson 1 & 2 [3 Hours]		MTBF Math Analysis		
	Lesson 2: Tactics for Availability Class Test -3	Redundancy Strategy Design, Failover Flowcharting	Class Test, Final exam	CLO2, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Modifiability General Scenario	Dependency Coupling Audit, Change Impact Analysis	Class Test, Final exam	CLO2, CLO4
	Lesson 2: Tactics for modifiability	Interface Encapsulation Workshop, Refactoring Strategy	Class Test, Assignment, Final exam	CLO2, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Balancing multiple quality attributes, Incorporating trade-off concept	Conflict Matrix Modeling, Multi- Attribute Optimization	Class Test, Final exam	CLO2, CLO4
	Lesson 2: Overcoming common challenges during software development	Architectural Anti- pattern Recovery, Diagnostic Analysis	Class Test, Final exam	CLO2, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Architecture Review	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on topic discussed for final exam	Integrative Syllabus Review, Board Problem Solving	Presentation, Final exam	CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =78)
	CLO1	CLO2	CLO3	CLO4		PLO2	PLO3	PLO4	
Attendance	--	--	--	--	7				
Class Test 1	10	5			Avg 15				
Class Test 2		10	5						
Class Test 3			5	10					

Assignment			5		5		5		5
Presentation				8	8			8	8
Midterm Exam	11	9	5		25	20	5		25
Final exam		10	23	7	40	20	23	7	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Len Bass, Paul Clements, Rick Kazman, Software Architecture in Practice, 4th Edition, Addison-Wesley, 2021, ISBN: 978-0321815736.

Reference Books/ Other Supplementary Readings

Ian Sommerville, Software Engineering, 9th edition, Pearson, 2021, ISBN: 9780133943030

Carlo Ghezzi, Mehdi Jazayeri, Dino Mandrioli, Fundamentals of software engineering, 2nd Edition, Pearson Education, c2003., ISBN: 8129700468

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 332 Information System Security

Course Code: SE 332		CIE Marks: 60
ISCED: 0613-332		SEE Marks: 40
Course Title: Information System Security		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: NA		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 10:00 AM	10:00 AM- 11:30 AM	11:30 AM- 01:00 PM	01:00 PM- 02:30 PM	2:30 PM- 4:00 PM	4:00 PM- 5:30 PM
Saturday						
Sunday						
Monday						
Tuesday						
Wednesday						
Thursday						

Course Content (from syllabus)

Information security concept, secured software development life cycle (secSDLC), secured web application design solution, threats and defense mechanisms on current web application, the classifications of malware and its prevention. Classic Cryptographic algorithm that includes Caesar Cipher, One Time Pad, Transposition, Playfair, and hill cipher. Feistel Network, Data Encryption Standard, Advanced Encryption Standard, Public Key Crypto System, RSA, Key management- Diffie-Hellman, Elliptic Curve, etc..Digital Signature, authentication protocol.

Rationale of the Course

This course covers the fundamental principles and defensive strategies required to safeguard modern infrastructure against evolving cyber threats. Students explore secure software development and web application design alongside threat classification and malware mitigation. The curriculum establishes a deep foundation in cryptography, transitioning from classical mechanisms to advanced encryption standards like AES and RSA. Additionally, learners examine secure key management through Diffie-Hellman and Elliptic Curve cryptography, while mastering digital signatures and authentication protocols to guarantee data integrity and authenticity.

Course Objectives

- To provide a solid conceptual understanding of the fundamentals of data communications. More specifically,
- Demonstrate the basic security concept in information systems and explain the measures of securing the software development life cycle.
 - Illustrate the 10 common vulnerabilities in web applications due to bad design and understand how such weaknesses can be mitigated using current best practices.
 - Demonstrate the classifications of malware and explain the countermeasures to hinder them.
 - Apply the principles of classic and modern cryptography and use it to secure the system.
 - Analyze the key management, message authentication and hash function.
 - Construct the digital signature and authentication protocol.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Contextualize the security design and ethical concepts of the information system.	PLO8	C3	Fundamental Domain
CLO2	Apply the principles of Classical Cryptographic Algorithms to evaluate secure encryption and decryption processes.	PLO1	C3	Fundamental Domain
CLO3	Formulate the methods of modern cryptographic algorithms to ensure its effectiveness.	PLO12	C5	Fundamental, Social, Thinking, Personal Domain
CLO4	Analyze the construction process of digital signatures and authentication protocols.	PLO3	C4	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1								√				
CLO2	√											
CLO3												√
CLO4			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction and Icebreaking session.	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Overview of the terminologies of information security. [Asset, Threat, Risk, Attack, Vulnerability, Exploitation, Security, etc.]	Threat Modeling Framework, Conceptual Mapping	Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of the terminologies of information security. [Confidentiality, Integrity, Availability, Non- Repudiation, Authenticity, doxing, Zero-day, etc.]	CIA Triad Analysis, Real-world Incident Review	Midterm exam	CLO1
	Lesson 2: Explain the measures of securing the software development life cycle (secSDLC). [Investigation, Analysis, and Logical Design]	Secure SDLC Mapping, Architectural Review	Midterm exam	CLO1
Week-3	Lesson 1: Explain the measures of securing the software development life cycle (secSDLC). [Physical	Threat Surface Analysis, Code Review Clinic	Class Test, Midterm exam	CLO1

Lesson 1 & 2 [3 Hours]	Design, Implementation, and Maintenance & Changes].			
	Lesson 2: Malware [Trojan horse, Backdoor, Rootkit, Ransomware, Adware, Virus, etc.]	Malware Taxonomy Mapping, Indicator of Compromise (IoC) Review	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Malware [Worms, Spyware, Botnet, Crypter, Logic bomb, Trapdoor. Etc.]	Botnet Command-and-Control Tracing, Attack Tree Analysis	Midterm exam	CLO1
	Lesson 2: Designing Secure Web Application Design [Input Validation, Authentication] Class Test -1	Injection Prevention Workshop, Live Authentication Tracing	Presentation, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Designing Secure Web Application Design [Authorization, Configuration Management, and Sensitive Data]	Access Control Matrix Design, Privilege Least-Principle Review	Presentation Midterm exam	CLO1
	Lesson 2: Designing Secure Web Application Design [Input Validation, Authentication, Authorization]	Defensive Coding Architecture, Threat Scenario Discussion	Presentation Midterm exam	CLO1
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Designing Secure Web Application Design [Configuration Management, and Sensitive Data]	Data-at-Rest Protection Mapping, Hardening Checklist	Presentation Midterm exam	CLO1
	Lesson 2: Designing Secure Web Application Design [Parameter Manipulation, Exception Management, Auditing and Logging] Class Test -2	Secure Log Architecture Design, Error Handling Walkthrough	Presentation Midterm exam	CLO1
Week-7	Lesson 1: Overview of Cryptography. Classic Cryptographic Algorithm	Modular Arithmetic Exercises, Cryptanalysis Drill	Class Test, Assignment, Midterm exam	CLO1, CLO2

Lesson 1 & 2 [3 Hours]	[Caesar Cipher].			
	Lesson 2: Classic Cryptographic Algorithm [One Time Pad]	Perfect Secrecy Proof, Key Stream Logic Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson1: Classic Cryptographic Algorithm [Transposition, Play Fair]	Matrix Permutation Mapping, Bigram Calculation Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Classic Cryptographic Algorithm [Hill Cipher]	Linear Algebra Matrix Multiplication, Key Inversion Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Concept of Modern Cryptographic Algorithm, Confusion, Diffusion, Classic Feistel Network.	Feistel Block Diagram Analysis, Bit Substitution Tracing	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: General Depiction of DES Encryption and Decryption Algorithm.	DES Pipeline Walkthrough, S-Box Permutation Tracing	Class Test, Assignment, Final exam,	CLO1, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Single Round of DES Algorithm, Overview of Advanced Encryption Standard.	Round-Function Bitwise Processing, Comparative Analysis	Class Test, Final exam	CLO1, CLO3
	Lesson 2: AES Parameters, AES Encryption	AES State Matrix Transformation, Visual Presentation	Class Test, Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: AES Decryption, AES Key Expansion Process	Inverse Transformation Mapping, Cipher Math Practices	Class Test, Final exam	CLO1, CLO3

	Lesson 2: AES Key Expansion Process Class Test -3	RotWord/SubWord Schedule Generation, Board Practice	Class Test, Final exam	CLO1, CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of Public key cryptosystem	Asymmetric Key Exchange Paradigm, Protocol Mapping	Class Test, Final exam	CLO1, CLO4
	Lesson 2: RSA Algorithm, Overview of the Digital Signature	Prime Factorization Math, Key Generation Calculations	Class Test, Final exam	CLO1, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Authentication Protocols, RSA based Digital Signature Standards	Trust Chain Verification, Signature Hash Tracing	Class Test, Final exam	CLO1, CLO4
	Lesson 1: DSA/DSS based Digital Signature	Mathematical Verification Drill, Parameter Analysis	Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Infrastructure Blueprint Mapping	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on topic discussed for final exam	Brainstorming sessions, Voice over PPT, Lecture note, Open discussion	Final exam	CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =78)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO3	PLO8	PLO12	
Attendance	--	--	--	--	7					
Class Test 1	5(15)				Avg 15					
Class Test 2		5(15)								
Class Test 3			5(15)							
Assignment		5			5		5			5

Presentation	8				8	8				8
Midterm Exam	16	9			25	9		16		25
Final exam			30	10	40		10		30	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

William Stallings, Cryptography and Network Security: Principles and Practice, 8th Edition, Pearson, 2020, ISBN: 978-1292403245.

Reference Books/ Other Supplementary Readings

Buchanan, W. Cryptography, CRC Press, 2022. ISBN: 978-0367466030.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 342 Compiler Design

Course Code: SE 342		CIE Marks: 60
ISCED: 0613-342		SEE Marks: 40
Course Title: Compiler Design		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation,
SEE: Semester End Examination

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 10:00 AM	10:00 AM- 11:30 AM	11:30 AM- 01:00 PM	01:00 PM- 02:30 PM	2:30 PM- 4:00 PM	4:00 PM- 5:30 PM
Saturday						
Sunday						
Monday						
Tuesday						
Wednesday						
Thursday						

Course Content (from syllabus)

Compiler Basics, Compiler History, Language Processing System, Lexical Analysis, Syntax Analysis, Semantic Analysis, Intermediate Code Generation, Code optimization, Phases of Compiler, Grammar Constituents, Unambiguous Grammar, Left Recursion, Deterministic Grammar, Syntax Directed Translation, Parsing Techniques, LL1 Parsing, LR

Parsing, Three Address Code, Operator Precedence Parsing, Code Optimization Techniques, Syntax Directed Translation Scheme.

Rationale of the Course

This course provides a comprehensive exploration of compiler architecture and the evolution of translating high-level programming languages into machine-readable code. Students analyze the core phases of compilation, including lexical, syntax, and semantic analysis, alongside intermediate code generation and optimization. The curriculum covers grammar clarity, recursion handling, and parsing techniques like LL1 and LR to ensure error-free compilation. By balancing language theory with translation mechanics, this course equips learners with the technical skills required to design, analyze, and optimize efficient compilers for various programming environments.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of data communications. More specifically,

- To learn the major phases of compilation and how they work manually.
- To learn the analysis, parsing, code generation, and optimization phases of compilation.
- To learn regular expressions and their conversion to Regular Language and finite automata.
- To learn the design basics of compilers
- To learn the identification and selection of suitable parsing strategies for a compiler for various cases.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Illustrate the major phases of compilation and the knowledge of how they work manually.	PLO2	C2	Fundamental Domain
CLO2	Apply the automata theory in the phases of a compiler.	PLO1	C3	Fundamental Domain
CLO3	Translate a given source code to intermediate code generation, assembly code generation, and optimization.	PLO2	C2	Fundamental Domain
CLO4	Construct parsers and explore various parser designs to demonstrate proficiency in parsing techniques.	PLO4	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2	√											
CLO3		√										
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction of OBE	Syllabus Walkthrough, Interactive Q&A	None	None
	Lesson 2: Compiler Basics, Compiler Definition, Features of Compiler, Types of Compiler	Comparative Analysis, Architecture Walkthrough	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Task of a Compiler, History of Compiler, Language Processing System, Compiler History and Application, Steps of LPS, Use of the compiler, Application of Compiler	Pipeline Modeling, Conceptual Mapping	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Phases of Compiler, Work with examples to understand the overview of phases	Multi-stage Translation Walkthrough, Visual Presentation	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Lexical Analysis and Tokenization, Lexical Analysis, Lexical Error Identifying	Token Stream Dissection, Error Scanning Workshop	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Symbol Table, Tokenization	Hash Table Structure Mapping, Active Tracing	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Finite Automata Finite Automata and its roles in Lexical analysis	State Machine Modeling, Pattern Matching Review	Class Test, Assignment, Midterm exam	CLO1, CLO2

	Lesson 2: Exercise practice Finite Automata (DFA/NFA Construction from Regular language) Class Test-1	Transition Diagram Drawing, Board Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Regular expression, Regular expression and how it works with lexical analyzer	Token Pattern Mapping, Interactive Lecture	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Exercise practice on regular expression conversion from regular language	Syntax Pattern Drafting, Peer Verification	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2[3 Hours]	Lesson 1: Fundamentals of Syntax Analysis, Syntax Analysis, Syntax error Identifying, Syntax tree generation after lexical analysis	Structural Parse Tree Modeling, Code Inspection	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2
	Lesson 2: Context free Grammar (CFG), Role of CFG in Syntax analysis, Derivation and parse tree basics, Ambiguity of grammar Class Test-2	Leftmost/Rightmost Derivation Comparison, Ambiguity Resolution	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Exercise practice Context Free Grammar Construction, Semantic Analysis , Semantic error Identifying	Type Checking Logic Review, Rule Construction Practice	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Intermediate Code Generation, ICG 3-address coding, Assembly Language, Loader and Linker	Three-Address Code Linearization, Object Linkage Analysis	Class Test, Assignment, Final exam	CLO1, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Optimization, Code optimization, Work with ICG code to get the optimized one	Redundant Code Elimination, Basic Block Restructuring	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Overview of all phases with details for one specific line of high level programming language program to convert low level language	End-to-End Statement Compiling, Comprehensive Review	Class Test, Assignment, Final exam	CLO1, CLO3
Week-9	Lesson 1: Syntax Directed Translation, Simple SDT, Synthesized Attributes, Postfix Notation	Attribute Evaluation Tracing, Grammar Annotation Practice	Class Test, Assignment, Final exam	CLO1, CLO3

Lesson 1 & 2 [3 Hours]	Lesson 2: Tree traversals, Translation schemas, Techniques of Adding syntax rules with CFG	Tree Walk Semantic Simulation, Annotation Rules Mapping	Class Test, Assignment, Final exam	CLO1, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Context free Grammar Ambiguity: Prepare CFG for Parsing (Elimination of Ambiguity), Unambiguous vs Ambiguous Grammar, Conversion- Ambiguous - Unambiguous	Precedence Tier Restructuring, Structural Grammar Redesign	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Recursive Grammar : Prepare CFG for Parsing (Elimination of Left recursion), Left Recursion, Left Recursion Elimination	Rule Rewriting Drill, Algorithmic Grammar Modification	Class Test, Assignment, Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Determinism of Grammar : Prepare CFG for Parsing (Elimination of non determinism), Deterministic and no deterministic grammar, Non-determinism elimination	Left Factoring Workshop, Non-determinism Parsing Analysis	Class Test, Assignment, Final exam	CLO1, CLO4
	Lesson 2: Basics of parsing and different types of parsing techniques, Parsing, Classification of parsing with details Class Test-3	Parser Taxonomy Mapping, Pushdown Mechanics Walkthrough	Class Test, Assignment, Final exam	CLO1, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Top-down and bottom parsing, LL1 parsing, FIRST () and FOLLOW(), LL1 Grammar	First/Follow Set Math, Prediction Analysis	Class Test, Assignment, Final exam	CLO1, CLO4
	Lesson 2: LL1 Parsing table generation from FIRST () and FOLLOW(), Common Prefix Problem	Table Construction Matrix, Branching Conflict Tracking	Class Test, Assignment, Final exam	CLO1, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: LL1 Exercise including First () and follow(), LL1 Parsing Table Construction work with practice problems	Error Recovery Mapping, Systematic Table Construction	Class Test, Assignment, Final exam	CLO1, CLO4
	Lesson 2: LR (0) Parsing, LR (0) items, Parsing Table, SLR (1), Examples of LR (0) and SLR (1)	Closure and Goto Graphing, Canonical Collection Tracking	Class Test, Assignment, Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 2: Industry Session	Guest Lecture / Industry Expert Talk, Static Code Optimization Case Study	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on topic discussed in Week-15- Week-18	Integrative Review, Board Problem Solving	Final exam	None

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO4	
Attendance	--	--	--		7				
Class Test 1	5(15)				Avg 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment		5			5		5		5
Presentation			8		8				
Midterm Exam	18	7			25	7	18		25
Final exam		5	8	27	40	5	8	27	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Keith D. Cooper, Linda Torczon, Engineering a Compiler, 3rd Edition, Morgan Kaufmann, 2022, ISBN: 978-0120884780.

Reference Books/ Other Supplementary Readings

Pradeep K. Sinha, Priti Sinha, Computer Fundamentals: Concepts, Systems & Applications, 8th Edition, BPB Publications, 2021. ISBN: 9789389949328

Course Materials and Slides

Alfred V. Aho, Jeffrey D. Ullman, & Rajesh Sunderraman Sethi, Compiler Principles, Techniques, and Tools, 2nd edition, Addison-Wesley, 2006, ISBN: 978-0321486813.

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 441 Software Engineering Professional Ethics

Course Code: SE 441/GEDS 218		CIE Marks: 60
ISCED: 0613-441		SEE Marks: 40
Course Title: Software Engineering Professional Ethics		Total Marks: 100
Semester:		
Credits: 3 (Theory)	Contact Hours: 3 (Total week: 14)	
Prerequisite: (If any)		
Course Type: GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Ethics and professional ethics, ethical Frameworks: Deontological ethics (Duty-based ethics), Utilitarian ethics (Outcome-based ethics), Aristotelian ethics (Virtue ethics). Professional Ethics and responsibilities: ICT/IT/Computer ethics, some ethical guidelines for computer professionals, Examine and discuss professional codes of ethics, conduct, and practice (IEEE, ACM, SE, AITP, and so forth). Privacy and personal information: privacy, effect of ICT on privacy, Ethical and legal basis for privacy protection; Technological strategies for privacy protection. Freedom of Expression: freedom of expression and speech, hate speech, censorship, pornography, promoting an understanding of the balance between free expression and responsible communication. Intellectual Property: Intellectual property issues including copyrights, patents, and trade secrets, along with engaging in debates on software piracy, software patents, fair use, and the principles of free and open-source software. Employment and Health: impact of ICT on employment, work environments, employee monitoring, individual health and well-being, fostering an understanding of the ethical implications of technological advancements in the workplace. Cyber Laws: Information Communication Technology Act of 2006 and its amendment in 2013, Pornography Control Act 2012, Digital Security Act 2018; Copyright Act of 2000, Data Protection Act, Computer Misuse Act,

Rationale of the Course

This course provides a comprehensive exploration of ethical considerations and cyber laws within the software engineering industry. Students examine general and professional ethics through philosophical frameworks like consequentialism and deontologism, alongside professional codes from organizations like IEEE and ACM. The curriculum covers critical issues surrounding digital privacy, intellectual property rights, freedom of speech, and the societal impacts of technology on employment and health. Additionally, it details the legal landscape by analyzing key cyber regulations, including the Information Communication Technology Act and Data Protection Act.

Course Objectives

The purpose of this course is to introduce the ideas of professional ethics in software engineering to the students. The specific objectives of this course are given below:

- To define and differentiate between general ethics and professional ethics within the context of software engineering.
- To discuss real-world case studies where software engineers faced ethical dilemmas and analyze the decisions made through the lens of different professional codes.
- To explore the ethical and legal aspects of privacy protection in the digital age.
- To analyze intellectual property laws, including copyrights, patents, and trade secrets, and their implications for software engineers regarding ownership, collaboration, and innovation.
- To explore the ethical considerations surrounding the impact of computer technology on employment, the work environment, and individual health and well-being.
- To provide an overview of key cyber laws.
- To instill a sense of professional responsibility in students, highlighting their role in developing ethical and socially responsible software solutions.
- To cultivate a commitment to lifelong learning and staying informed about emerging ethical challenges and legal developments in the software industry.

Course Learning Outcomes (CLOs) with Mappings PLOs

At the end of the course, students will be able to:

CLO's	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C,P,A)	BNQF Skill
CLO1	Interpret the concept of ethics and ICT-related ethics and ethical frameworks as an ICT professional.	PLO1	C2	Fundamental Domain
CLO2	Analyze and apply professional codes of ethics (IEEE, ACM, Software Engineering Code of Ethics, and AITP) to resolve ethical issues and support responsible engineering practice.	PLO8	C4	Fundamental, Social, Thinking, Personal Domain
CLO3	Assess the societal, legal, cultural, and professional impacts of ICT on individuals and organizations to support responsible decision-making in engineering practice.	PLO6	C5	Fundamental, Social, Thinking, Personal Domain
CLO4	Evaluate cyber laws, regulations, and organizational policies to recommend ethical and effective governance and management practices for ICT systems.	PLO11	C5	Fundamental, Social, Thinking, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12

CLO1	√											
CLO2								√				
CLO3						√						
CLO4											√	

Course Delivery Plan/Lesson Delivery Plan:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on course outcomes (CO) and program outcomes (PO) related to the course.	Syllabus Walkthrough, Outcome Mapping Q&A	CLO1	Class Test, Midterm
	Lesson 2: Introduction: what is Ethics and professional ethics? ethical Frameworks: Deontological ethics (Duty-based ethics)	Ethical Framework Analysis, Case Dilemma Discussion	CLO1	Class Test, Midterm
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction: Utilitarian ethics (Outcome-based ethics), Aristotelian ethics (Virtue ethics)	Comparative Philosophy Matrix, Scenario Evaluation	CLO1	Class Test, Midterm
	Lesson 1: Introduction: Definition of computer ethics, Scenarios of computer ethics	Case Study Analysis, Ethical Scenario Modeling	CLO1	Class Test, Midterm
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Social Context of Computing: Similarities of Computer Ethics to Other Ethics, Professional ethics & general ethics.	Boundary Concept Mapping, Interactive Lecture	CLO1	Class Test, Assignment, Midterm
	Lesson 2: Social Context of Computing: Technical & professional ethics, Introduction to the Social Implications of Computing	Impact Matrix Breakdown, Visual Presentation	CLO1	Class Test, Assignment, Midterm
Week-4	Lesson 1: Social Context of Computing: Social Implications of Networked Communications	Network Impact Analysis, Open Discussion	CLO2	Class Test, Assignment, Midterm

Lesson 1 & 2 [3 Hours]	Lesson 2 : Responsibility, Liability: What includes in Professional responsibility. Class Test 01	Legal Responsibility Mapping, Liability Case Review	CLO2	Class Test, Assignment, Midterm
Week-5	Lesson 1: Responsibility, Liability: How to behave with office colleagues, Professional responsibility and liability.	Workplace Role-Play, Corporate Ethics Case Study	CLO2	Class Test, Assignment, Midterm, Final Exam
Lesson 1 & 2 [3 Hours]	Lesson 2: Professional Ethics: What includes in Professional Ethics, Professional Ethics: Components of professional ethics, Principle of ethics.	Core Principles Walkthrough, Framework Modeling	CLO2	Class Test, Assignment, Midterm, Final Exam
Week-6	Lesson 1: Code of ethics: Code of ethics purpose, Versions of code of ethics, Do's and Don'ts for the ethics culture of the computer professionals, Ethics Guidelines, Ethics in business.	Industry Code Review (ACM/IEEE), Guideline Analysis	CLO2	Class Test, Assignment, Midterm, Final Exam
Lesson 1 & 2 [3 Hours]	Lesson 2: Code of ethics: Code of ethics purpose, Versions of code of ethics, Code of ethics: Do's and Don'ts for the ethics culture of the computer professionals, Ethics Guidelines, Ethics in business. Class Test 02	Compliance Checklist Exercise, Business Scenario Audit	CLO2, CLO3	Class Test, Assignment, Midterm, Final Exam
Week-7	Computer Crime, Viruses and Hacking: computer crime, Types of computer crime, Prevention of computer crime.	Attack Vector Mapping, Mitigation Strategy Analysis Regional Case Review, Counterfeiting Impact Analysis	CLO3, CLO4	Class Test, Assignment, Final Exam
Lesson 1 & 2 [3 Hours]	Lesson 2: Computer Crime, Viruses and Hacking: Software piracy, Computer crime in Bangladesh.	Licensing Matrix Analysis, OSS Ecosystem Walkthrough	CLO3, CLO4	Class Test, Assignment, Final Exam
Week-8	Open Source Movement : Application areas of open source, List of open-source languages, List of open-source database, List of Open-Source Operating	Policy Impact Evaluation, Structural Trend Debate	CLO3, CLO4	Class Test, Assignment, Final Exam
Lesson 1 & 2 [3 Hours]				

	Systems, List of Open-Source Servers, Open-source challenges			
	Lesson 1: Discussion about open-source applications and combined with real life example: Censorship and the Internet;	Legal Privacy Assessment, Regulatory Flowcharting	CLO3, CLO4	Class Test, Assignment, Final Exam
Week-9	Lesson 1: Data Protection and Information Privacy Issues: censorship	Comparative Data Laws, Threat Boundary Discussion	CLO3, CLO4	Class Test, Assignment, Final Exam
Lesson 1 & 2 [3 Hours]	Lesson 2: Censorship and the Internet; Data Protection and Information Privacy Issues	Credential Theft Profiling, Defense Mechanism Mapping	CLO3, CLO4	Class Test, Assignment, Final Exam
Week-10	Lesson 1: privacy, identity theft, Data protect mechanism, critical Information Infrastructure, Investigation of offense and trail.	Peer Synthesis Clinic, Legal Challenge Review	CLO3, CLO4	Class Test, Assignment, Final Exam
Lesson 1 & 2 [3 Hours]	Lesson 2: Group discussion	Statute Framework Mapping, Legal Code Dissection	CLO3, CLO4	Class Test, Assignment, Final Exam
Week-11	Lesson 1: Digital Act 2018: Overview, cyber law, cyber-crime	Infrastructure Hardening Review, Judicial Process Tracing	CLO3, CLO4	Class Test, Assignment, Final Exam
Lesson 1 & 2 [3 Hours]	Lesson 2: Digital Act 2018: Preliminary of Digital Act 2018, Digital Security Agency, Critical Information Infrastructure, Investigation of offense and trail. Class Test 03	Statutory Crime Matrix, Interactive Diagnostics	CLO3, CLO4	Class Test, Assignment, Final Exam
Week-12	Lesson 1: Class work on types of crime and law	Infringement Case Drafting, Peer Audit Workshop	CLO3, CLO4	Class Test, Assignment, Final Exam

Lesson 1 & 2 [3 Hours]	Lesson 2: Class work on Software piracy and list on computer crimes	Data Flow Privacy Auditing, Vulnerability Assessment	CLO3, CLO4	Class Test, Assignment, Final Exam
Week-13	Lesson 1: Discuss about data and information privacy with proper example	Career Practice Blueprinting, Course Reflection	CLO4	Class Test, Assignment, Final Exam
Lesson 1 & 2 [3 Hours]	Lesson 2: How to learn from this course and where it uses.	Guest Lecture / Industry Expert Talk, Compliance Roadmap Review	CLO4	Class Test, Assignment, Final Exam
Week-14	Industry Week Session	Integrative Syllabus Review, Case Study Analysis	CLO4	Presentation
Lesson 1 & 2 [3 Hours]	Lesson 2: Review class on topic discussed in Week-11- Week-15	Syllabus Walkthrough, Outcome Mapping Q&A	CLO4	Final

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO8	PLO6	PLO11	Total (65)
Attendance	--	--	--	--	7					
Class Test 1	5 (15)				Avg 15					
Class Test 2		5 (15)								
Class Test 3				5 (15)						
Assignment		2	3		5					
Presentation					8					
Midterm	10	8	7		25	10	8	7		25
Final	0	10	15	15	40		10	15	15	40
Total					100	Total				65

Marks Distribution

Class attendance	7
Assignment	6
Presentation (Mandatory)	7
3 Quizzes	15
Midterm Test	25

Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

George W. Reynolds, Ethics in Information Technology, 6th edition, Cengage Learning, 2019, ISBN 978-0357127162.

Reference Books/ Other Supplementary Readings

Deborah G. Johnson, Computer Ethics, 4th Edition, Pearson, 2009, ISBN: 978-0131112414.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software

Engineering (SWE)

Faculty of Science and Information Technology (FSIT)

Daffodil International University (DIU)

(Version 3)

SE 411 Software Project Management & Documentation

Course Code: SE 411/GEDS 415		CIE Marks: 60
ISCED: 0613-411		SEE Marks: 40
Course Title: Software Project Management & Documentation		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE312		
Course Type: GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Introduction to Software Project Management, understanding of the profession of project manager. Project Scope, Schedule and Budget Management. Quality issues in project, Resources, Risk and Communication Management, Network Diagram, Gantt Chart, Project Budget, PERT analysis, Quality tools-Six Sigma, Quality Plan & Quality documents, Procurement management Process. DevOps, Documentation Management Process, Documentation Communication, Risk Identification, Risk Communication, Finally project Communication, Stakeholder and integration of a project.

Rationale of the Course

This course explores the core principles and practices of project management within the software development domain. Students learn to plan, execute, and control critical project parameters, including scope, scheduling, and budget management. The curriculum covers quality assurance methodologies to mitigate issues across the project lifecycle, alongside strategies for resource management, risk mitigation, and professional communication. Special emphasis is placed on stakeholder engagement and project integration to align development with organizational goals.

Course Objectives

The goal of this Project Management and Documentation course is to provide the concept of modern information systems for different levels of management in business organizations. The main objectives of this course are,

- To make students understand the requirements and the contents of project management.
- To make students understand the role of the manager in each phase of the software development life cycle.
- To make students capable of writing a plan for a project according to an established standard.
- To make students capable of using current Software Project Management (SPM) tools and develop new SPM ideas.
- To grow capability among students to work in a team environment and be aware of different modes of communications.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
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CLO1	Demonstrate different terms, concepts and models related to software project management and documentation.	PLO2	C3	Fundamental, Social, Thinking, Personal Domain
CLO2	Utilizing management tools and techniques to meet the project objectives from the perspective of a project manager.	PLO11	C4	Social, Thinking, Personal Domain
CLO3	Develop a project management plan and document it with all relevant aspects in context of scientific and business application.	PLO7	C6	Fundamental, Social, Thinking, Personal Domain
CLO4	Determine appropriate functions for quality maintenance, motivate teams in managing big projects.	PLO9	C5	Social, Thinking, Personal Domain
CLO5	Prescribe appropriate support in a changed environment for successfully complete.	PLO12	C5	Social, Thinking, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2											√	
CLO3							√					
CLO4									√			
CLO5												√

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction and Icebreaking session. Discussion of OBE & Blooms Taxonomy. Introduction to Project Management – Understand what a project is with its characteristics.	Syllabus Walkthrough, Taxonomy Mapping Q&A	Class Test, Midterm exam	CLO1
	Lesson 2: The Project Management and Information Technology Context.	Enterprise Systems Review, Context Mapping	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: The Project Management Process Groups: Case Study Analysis – Understand the processes. Related documents' introductions.	Lifecycle Case Analysis, Lifecycle Mapping	Class Test, Midterm exam	CLO1
	Lesson 2: Challenges of Project Management Role of Project Manager	Matrix Role Evaluation, Open Discussion	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Project Success Factors, Stakeholders Role. Related Documents	Success Criteria Blueprinting, Stakeholder Matrix Review	Class Test, Midterm exam	CLO1
	Lesson 2: Skills for Project Management, Project Manager's Role	Core Competency Review, Visual Presentation	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Scope Management Processes – the boundary of a project.	Scope Boundary Mapping, Framework Analysis	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Project Charter and Requirement Traceability Matrix	Charter Drafting Exercises, Requirements Traceability Trace Table	Class Test, Assignment, Midterm exam	CLO1

	Class Test -1			
Week-5	Lesson 1: Project Scope Management - Work Breakdown Structure (WBS)	Hierarchical Tree Decomposition, WBS Dictionary Drafting	Class Test, Assignment, Presentation, Midterm exam	CLO1, CLO5
Lesson 1 & 2 [3 Hours]	Lesson 2: Project Time Management Processes – Related issues	Sequence Dependency Modeling, Timeline Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-6	Lesson 1: Network Diagram or AOA, Gantt Chart drawing \Class work on Project Scheduling.	Critical Path Calculation, Gantt Chart Drawing Workshop	Class Test, Assignment, Midterm exam	CLO1, CLO2
Lesson 1 & 2 [3 Hours]	Lesson 2: Project Budget Management Processes. Class Test -2	Cost Estimation Modeling, Budget Baseline Review	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-7	Lesson 1: Understanding of Special Terms Related to Budget	Financial Acronym Matrix, Analytical Review	Class Test, Assignment, Midterm exam	CLO1, CLO2
Lesson 1 & 2 [3 Hours]	Lesson 2: Developing Budget of a Project, Cash flow analysis and problem solution.	Cash Flow Cumulative Calculation, S-Curve Plotting	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-8	Lesson 1: PERT Analysis and Planning Budget	Three-Point Probability Math, Variance Calculation Practice	Class Test, Assignment, Midterm exam	CLO1, CLO2
Lesson 1 & 2 [3 Hours]	Lesson 2: Project Quality Management Processes and Quality Issues.	Standard Audit Mapping, Quality Metrics Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-9	Lesson 1: Quality Tools - Ishikawa	Root-Cause Brainstorming, Cause-and-Effect Diagramming	Class Test, Assignment, Midterm exam	CLO1, CLO3

Lesson 1 & 2 [3 Hours]	Lesson 2:Quality Tools – six sigma, Introducing DevOps With all features	Defect Rate Math, Pipeline Infrastructure Review	Class Test, Assignment, Midterm exam	CLO1, CLO3
Week-10	Lesson 1: Quality plan and quality documents.	Control Plan Blueprinting, Inspection Checklist Design	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
Lesson 1 & 2 [3 Hours]	Lesson 2: Procurement Management processes, Procurement Tools & Problem solving,	Make-or-Buy Calculation, Vendor Evaluation Matrix	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
Week-11	Lesson 1: DevOps and project management	Agile-DevOps Integration, Automation Pipeline Analysis	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
Lesson 1 & 2 [3 Hours]	Lesson 2: Project Communication Management and Process. Documenting Communication, Communication channels and Problems solving Class Test -3	Channel Math Calculations, Communication Plan Design	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO3
Week-12	Lesson 1: Project Risk Management Processes, Risk Identification techniques, Risks evaluation techniques, Positive risks issues	Probability-Impact Matrix, Qualitative Risk Ranking	Assignment, Presentation, Midterm exam, Final exam	CLO1, CLO3
Lesson 1 & 2 [3 Hours]	Lesson 2: Risk Management Responses. Documenting Risks. Risk Management Planning and Control	Response Strategy Selection, Risk Register Maintenance	Class Test, Assignment, Presentation, Midterm exam, Final exam	CLO1, CLO3
Week-13	Lesson 1:			
Lesson 1 & 2 [3 Hours]	Stakeholders Management issues. Documenting Stake management, Roles of Stakeholders.	Power-Interest Grid Mapping, Engagement Strategy Design	Class Test, Assignment, Presentation, Midterm exam, Final exam	CLO1, CLO4, CLO

	Lesson 1: Project Integration Management Process, Documenting Integration, Integrated Documentation in a project.	Change Control Workflow, Project Plan Consolidation	Assignment, Presentation, Midterm exam, Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Integration management planning. & Documentation Related documentation, Project Controlling according to plan.	Performance Variance Tracing, Earned Value Audit	Class Test, Assignment, Presentation Final exam	CLO1, CLO4, CLO 5
	Lesson 2: Industry Session	Guest Lecture / Industry Expert Talk, Production Blueprint Review	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assess ment Task	CLO's					Mark (Total=100)	PLO's					Mark (Total =65)
	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5		PLO 2	PLO1 1	PLO7	PLO9	PLO1 2	
Attend ance	--	--	--		–	7						
Class Test 1	5(15)					Avg 15						
Class Test 2		5(15)										
Class Test 3			5(15)									
Assign ment				5		5						
Present ation					8	8					8	
Midter m exam	6	19				25	6	19				25
Final exam			30	10		40			30	10		40

Marks Distribution

Class attendance	7
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Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Kathy Schwalbe, Information Technology Project Management, 9th edition, Cengage Learning, Inc., 2019, ISBN: 978-1305502046.

Reference Books/ Other Supplementary Readings

Scott Berkun. Making Things Happen: Mastering Project Management, Revised Edition, O'Reilly Media, 2020. ISBN: 978-0596517699.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 333 Artificial Intelligence

Course Code: SE 333		CIE Marks: 60
ISCED: 0613-333		SEE Marks: 40
Course Title: Artificial Intelligence		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE 214, SE 234		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Introduction to AI, Components of AI, Agents, Environments, and Fundamental Challenges of AI, Limitation of Electronic Systems, The Concept of Human Complex, Human Perception to Logical Sentence Conversion, Terminologies related to AI, and Forming the based idea of Search Algorithms, Working Principle for Informed Search Algorithms: BFS, and DFS. Working Principle of Uninformed Search Algorithms: GBFs, A* Search. Introduction to Adversarial Search Algorithm & Designing Simple AI Game Logic, Forming Logical Sentences, Compound Statements using Logical Connectives, and Statement Modeling. Inference Algorithm, and Knowledge Engineering. Inference through Resolution, Bayesian Networks. Approximate Inference. Markov Assumption, Markov Chain, Markov Hidden Models, and Sensor Model. Concept of Optimization, and Formation of the State Space Landscape, AI Optimization Algorithm Design. Different Existing Optimization Algorithms, and Their Applications. Mathematical Background of Machine Learning. Learning Algorithm Formation, and Optimization. Single Layer Neural Network, Deep Neural Network, Activation Functions, Backpropagation Algorithm, Handling Overfitting, and Underfitting, Different Optimization Algorithms. Convolutional Neural Networks. Syntax &

Semantics, Formal Grammar to Context Free Grammar, Tokenization, Markov Model. Text Categorization, Naïve Bayes, Smoothing (Additive, Laplace), Topic Modeling. Word Vector Formation, Training Neural Network for NLP, Evaluating the Performance of the Trained Model, Exploring the Limitations of the NLP Project.

Rationale of the Course

This course provides a comprehensive foundation in artificial intelligence, bridging classical symbolic AI with modern machine learning paradigms. Student progress from AI agents and environments to logical sentence formulation, knowledge engineering, and search algorithms, including informed, uninformed, and adversarial methods. The curriculum covers probabilistic reasoning through Bayesian networks alongside optimization techniques across state space landscapes. Furthermore, learners explore the mathematical foundations of machine learning, neural networks, and convolutional neural networks, before concluding with natural language processing concepts like tokenization and syntax analysis.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of artificial intelligence. More specifically,

- Understand the foundational principles of Artificial Intelligence (AI) and its components.
- Explore search algorithms, both informed and uninformed, and gain proficiency in adversarial search techniques.
- Master logical sentence formation, compound statements, and inference algorithms, including resolution and Bayesian networks.
- Acquire knowledge in optimization algorithms, machine learning, and neural networks, with a focus on practical applications.
- Develop expertise in natural language processing, covering syntax, semantics, and text categorization, while evaluating the performance and addressing limitations of NLP projects.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Observe AI's mathematical background and concepts regarding data structure and other abstractions.	PLO2	C2	Fundamental Domain
CLO2	Implement AI algorithms using Python programming language.	PLO5	C3, P1	Fundamental Domain
CLO3	Analyze AI-based solutions to solve real-world problems.	PLO2	C4	Fundamental Domain
CLO4	Assess reusable AI projects for academic and business purposes.	PLO3	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2					√							
CLO3		√										
CLO4			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: OBE Discussion, Introduction to AI, Components of AI, Agents, Environments, and Fundamental Challenges of AI.	Syllabus Walkthrough, Interactive Q&A	Class Test, Midterm exam	CLO1
	Lesson 2: Intelligence, limitation of electronic systems, the concept of human complex human perception to logical sentence conversion	Propositional Logic Mapping, Conceptual Discussion	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Terminologies related to AI and forming the based idea of search algorithms.	State-Space Graph Modeling, Visual Presentation	Class Test, Midterm exam	CLO1
	Lesson 2: Working principle for informed search algorithms: BFS and DFS.	Tree Traversal Simulation, Queue/Stack Data Tracing	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Working principle of uninformed search algorithms: GBFS, A* Search.	Heuristic Function Calculation, Path Cost Optimization	Class Test, Midterm exam	CLO1, CLO2

	Lesson 2: Introduction to Adversarial Search Algorithm & designing simple AI game logic.	Game Tree Minimax Modeling, Alpha-Beta Pruning Practice	Class Test, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Forming logical sentences, compound statements using logical connectives and statement modeling.	Boolean Truth Table Drawing, Syntax Sentence Parsing	Extempore Q&A, Midterm exam	CLO1
	Lesson 2: Inference Algorithm and Knowledge Engineering. Inference through Resolution. Class Test -1	Conjunctive Normal Form (CNF) Conversion, Resolution Proofs	Extempore Q&A, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Bayesian Networks, Approximate inference. Markov assumption	Conditional Probability Tables (CPT) Mapping, Graph Modeling	Extempore Q&A, Midterm exam	CLO1, CLO2
	Lesson 2: Markov chain, Markov hidden models, and sensor model.	State Transition Matrix Math, Hidden State Sequence Tracing	Extempore Q&A, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Concept of optimization and formation of the state space landscape.	Local Search Topography Evaluation, Global Maxima Tracking	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Overview of Cryptography, Classic Cryptographic Algorithm [Caesar Cipher], AI optimization algorithm design. Class Test -2	Genetic Algorithm Hyperparameter Tuning, Substitution Matrix Drill	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Mathematical background of machine learning.	Linear Algebra Vector Multiplication, Calculus Partial Derivatives	Extempore Q&A, Final exam	CLO1, CLO3
	Lesson 2: Learning algorithm formation and optimization.	Cost Function Minimization, Gradient Descent Optimization	Extempore Q&A, Final exam	CLO1, CLO3
Week-8	Lesson 1: Different types of machine learning algorithms.	Paradigm Classification Framework, Comparative Mapping	Extempore Q&A, Final exam	CLO1, CLO3

Lesson 1 & 2 [3 Hours]	Lesson 2: Implementation of machine learning algorithms.	Model Training Demonstration, Code Trace Walkthrough	Extempore Q&A, Final exam	CLO1, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Single Layer Neural Network, Deep Neural Network	Perceptron Forward Pass Math, Layer Architecture Blueprinting	Class Test, Group Discussion, Final exam	CLO1, CLO3
	Lesson 2: Activation functions, Different Learning Algorithm	Non-linear Function Mapping (Sigmoid/ReLU), Vector Weight Updates	Class Test, Group Discussion, Final exam	CLO1, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Backpropagation Algorithm.	Matrix Chain Rule Calculus, Error Derivative Tracing	Class Test, Group Discussion, Final exam	CLO1, CLO3
	Lesson 2: Handling Overfitting, and Under fitting	Regularization Technique Analysis, Loss Curve Evaluation	Class Test, Group Discussion, Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Different optimization algorithms.	Optimizer Variance Review (SGD vs Adam), Convergence Tracking	Class Test, Group Discussion, Final exam	CLO1, CLO4
	Lesson 2: Convolutional Neural Networks. Class Test -1	Kernel Feature Matrix Multiplying, Pooling Layer Reduction	Class Test, Group Discussion, Final exam	CLO1, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Syntax & semantics, formal grammar to context free grammar, tokenization, Markov model.	Text String Tokenization, Production Rule Parsing	Class Work, Group Discussion, Final exam	CLO1, CLO4
	Lesson 2: Text Categorization, Naïve Bayes, Smoothing (Additive, Laplace), Topic Modeling, Word vector formation	Probability Likelihood Estimation, Word Embedding Matrix Mapping	Class Work, Group Discussion, Final exam	CLO1, CLO4

Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Training neural network for NLP, Evaluating the performance of the trained model	Loss Convergence Benchmarking, Confusion Matrix Audit	Class Work, Group Discussion, Final exam	CLO1, CLO4
	Lesson 2: Exploring the limitations of the NLP project	Out-of-Vocabulary Analysis, Bias Diagnostic Review	Class Work, Group Discussion, Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Pipeline Case Study	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class before the final examination	Integrative Syllabus Review, Board Problem Solving	None	CLO1, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =78)
	CLO 1	CLO2	CLO3	CLO4			PLO2	PLO3	PLO5	
Attendance	--	--	--	--	7					
Class Test 1	5(15)				Avg 15					
Class Test 2		5(15)								
Class Test 3			5(15)							
Assignment			5		5		5			5
Presentation			8		8		8			8
Midterm Exam	15	10			25		15		10	25
Final exam			20	20	40		20	20		40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Stuart Russell, & Peter Norvig, Artificial Intelligence: A Modern Approach, 3rd edition, Pearson, 2010, ISBN: 9789332518698.

Reference Books/ Other Supplementary Readings

Padgham, Lin, and Michael Winikoff. Developing Intelligent Agent Systems: A Practical Guide, 1st Edition, Wiley, 2004. ISBN: 978-0470861219.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 334 Artificial Intelligence Lab

SE 334 Artificial Intelligence Lab		
Course Code: SE 334		CIE Marks: 60
ISCED: 0613-334		LF Marks: 40
Course Title: Artificial Intelligence Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)	Contact Hours: 1 hour 30 minutes (14 weeks)	
Prerequisite: N/A		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

The "Artificial Intelligence Lab" course provides students with a comprehensive exploration of Basic Python Programming, Implementation of Search Algorithms, Knowledge Representation using Propositional Statements, Dealing with Uncertainty, Optimization Algorithms Implementation, Machine Learning Coding, Implementation of Neural Network, and Natural Language Processing.

Rationale of the Course

It is essential to develop the skill set of implementing the Artificial Intelligence (AI) algorithms to develop intelligent systems. We live in an AI-enabled world where almost every technological aspect of our lives is dependent on AI.

Course Objectives

The goal of this course is to guide the students to develop the skill sets necessary to implement AI algorithms. The main objectives of this course are:

- . Be able to apply the concept of basic object-oriented programming using Python.
- . Developing the skillset to think algorithmically and plan solutions using AI algorithms.

- . Gaining the capability of implementing AI algorithms using python.
- . Understanding the application domain of AI to solve real-world problems.
- . Be able to design, implement and present AI-based solutions.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Perform environment setup by installing the necessary Integrated Development Environment (IDE), Python interpreter, necessary libraries, and modules for AI programming by following instructions	PLO1	C3	Fundamental Domain
CLO2	Apply Python programming language, syntax, and coding norms to solve different types of problems	PLO5	C3, P2	Fundamental Domain
CLO3	Conduct AI algorithm implementation experiments	PLO4	C3, P3	Fundamental Domain
CLO4	Write effective reports and design documentation for the AI project implementation.	PLO10	C5, A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2					√							
CLO3				√								
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Correspond ing CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Environment setup to work with Python programming language	Guided Installation, Environment Verification	CLO1	Environment inspection and explanation
Week-2 [1.5 Hours]	Structured programming basics in Python	Live Coding Demonstration, Console Syntax Practice	CLO2	Evaluation of the solutions
Week-3 [1.5 Hours]	Object Oriented Programming (OOP) in Python	Class Instantiation Modeling, Object-Coupling Lab	CLO2	Evaluation of the solutions
Week-4 [1.5 Hours]	Problem Solving using Python Programming Language	Independent Coding Practice, Script Debugging	CLO2	Evaluation of the solutions
Week-5 [1.5 Hours]	Solving Maze problem using DFS and BFS algorithms	Graph Traversal Simulation, Queue/Stack Data Tracing	CLO3	Assessing the output and depth of understanding
Week-6 [1.5 Hours]	Min-Max Algorithm - game tree construction, maximizing and minimizing players, terminal state evaluation, and recursive search on a two-player zero-sum game	Game Tree Minimax Modeling, Terminal State Evaluation	CLO3	Output evaluation and explanation of game tree decisions
Week-7 [1.5 Hours]	Alpha-Beta Pruning - pruning condition intuition, alpha and beta bounds, tracing pruned branches on the same game tree, and comparative analysis of nodes evaluated with and without pruning	Pruning Condition Tracing, Boundary Value Optimization	CLO3	Evaluation of pruning implementation and depth of understanding of efficiency gains
Week-8 [1.5 Hours]	Implementation of the Hill-Climbing algorithm and analysis of its random restart variants.	Local Search Landscape Evaluation, Global Maxima Tracking	CLO3	Output evaluation and explanation
Week-9 [1.5 Hours]	Implementation of Informed Search: A* Algorithm (path-	Heuristic Function Calculation, Path Cost Optimization	CLO3	Output evaluation and explanation of heuristic design
Week-10 [1.5 Hours]	Implementation of Greedy Best-First Search and comparative analysis with A* on the same problem	Heuristic Performance Comparison, Comparative Matrix Analysis	CLO3	Evaluation of solutions and comparison of heuristic performance

Week-11 [1.5 Hours]	Applying the Genetic Algorithm to an optimization problem (e.g., Traveling Salesman / function optimization)	Fitness Evaluation Scripting, Chromosome Selection Simulation	CLO3	Performance evaluation of the GA-based solution
Week-12 [1.5 Hours]	Basic Data Analysis using Python Libraries (NumPy, Pandas, and Matplotlib)	Matrix Manipulation Lab, Vector Plotting Workshop	CLO2	Evaluation of implemented data analysis scripts and plots
Week-13 [1.5 Hours]	Project implementation applying Informed Search and/or Genetic Algorithm to a real-world optimization problem	Full System Logic Simulation, Capstone Project Demonstration	CLO3	
Week-14 [1.5 Hours]	Lab Final	Independent Practical Exam, Practical Task Execution	CLO2, CLO3	Lab final exam performance assessment

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO1
Lab Final	40	CLO2	PLO5
		CLO3	PLO4
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Stuart Russell, & Peter Norvig, Artificial Intelligence: A Modern Approach, 3rd edition, Pearson, 2010, ISBN: 9789332518698.

Reference Books/ Other Supplementary Readings

Padgham, Lin, and Michael Winikoff. Developing Intelligent Agent Systems: A Practical Guide, 1st Edition, Wiley, 2004. ISBN: 978-0470861219.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom.

(Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Python, Visual Studio Code, Jupyter Notebook, Google Colab, GitHub

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 544 Introduction to Machine Learning

Course Code: SE 544		CIE Marks: 60
ISCED: 0613-544		SEE Marks: 40
Course Title: Introduction to Machine Learning		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: MAT 101, MAT 102, STA 101, SE 121		
Course Type: Guided Elective II		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 10:00 AM	10:00 AM- 11:30 AM	11:30 AM- 01:00 PM	01:00 PM- 02:30 PM	2:30 PM- 4:00 PM	4:00 PM- 5:30 PM
Saturday						
Sunday						
Monday						
Tuesday						
Wednesday						
Thursday						

Course Content (from syllabus)

Definitions, Types, K-Nearest Neighbor, Euclidean Distance, Manhattan Distance, Differentiation, Partial Differentiation, Gradient Descent, Hypothesis Function, Loss Function, Learning Rate, Learning Rate Scheduling, Early Stopping, Stochastic Gradient Descent, Batch Gradient Descent, Mini-Batch Gradient Descent, Simple Linear Regression, Multiple Linear Regression, Logistic Regression, Binary Log-Loss Function, Polynomial Regression, Overfitting And Underfitting, Bias-Variance Trade Offs, Regularization Methods, Ridge Regression, Lasso Regression, Elastic Net, Ensemble Learning: Voting, Bagging, Pasting, Random Patches, Random Subspaces, Random Forest, Boosting, and Stacking, Model Selection: Hyperparameter Tuning, Grid Search, Random Search and Cross-Validation, Model Evaluation: Bootstrapping, Mean Absolute Error, Mean Square Error, Root Mean Square Error, Confusion Matrix, Type I Error, Type II Error, Accuracy, Precision, Recall, Sensitivity, Specificity, True Positive Rate, False Positive Rate, AUC-ROC, Precision-Recall Curve, Feature Engineering: Normalization, Standardization, and Augmentation, K-Means Clustering, Agglomerative Clustering, Single Linkage, Complete Linkage, Average Linkage, DBSCAN, Dimensionality Reduction: PCA and t-SNE.

Rationale of the Course

It is essential to develop the skill set of implementing the Artificial Intelligence (AI) algorithms to develop intelligent systems. We live in an AI-enabled world where almost every technological aspect of our lives is dependent on AI.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of Machine Learning. More specifically,

- To be taught about the basic concepts of machine learning techniques such as its types, fundamental challenges, and applications.
- To be taught about the theories of supervised and unsupervised machine learning algorithms so that they can apply them in real world scenarios.
- To evaluate and interpret machine learning models properly to the stakeholders.
- To discover any potential biases, misinterpretations, and ethical or fairness concerns of machine learning models.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Explain different types of machine learning models and their scopes and fundamental challenges.	PLO2	C2	Fundamental Domain
CLO2	Apply appropriate machine learning models to solve real-world problems.	PLO1	C3	Fundamental Domain
CLO3	Analyze machine learning models to verify their design, analysis, results, conclusion, and interpretation.	PLO4	C4	Fundamental Domain

CLO4	Evaluate models to discover potential bias, fairness issues, or societal, cultural, or ethical concerns.	PLO6	C5	Fundamental Domain
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Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2	√											
CLO3				√								
CLO4						√						

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to course syllabus, assessment methods, learning outcomes, deadlines, consultation hour, code of conduct, and other administrative procedures regarding the course.	Syllabus Walkthrough, Outcome Mapping Q&A	None	None
	Lesson 2: Definition of Machine Learning, Advantages of machine learning based software against traditional software, Applications of Machine Learning,	Architectural Paradigm Comparison, Conceptual Mapping	Class Test, Assignment, Presentation, Midterm exam, Final exam	CLO1, CLO4
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to different types of machine learning models such as supervised, unsupervised, self-supervised, reinforcement learning, model-based and instance-based learning, batch, and online learning, etc.	Taxonomy Matrix Analysis, Visual Presentation	Class Test, Assignment, Presentation, Midterm exam, Final exam	CLO1, CLO4
	Lesson 2: Fundamental Challenges of machine learning, the necessity and way of testing, and validating machine learning algorithms.	Validation Strategy Design, Threat Modeling Review	Class Test, Assignment, Presentation, Midterm exam, Final exam	CLO1, CLO4

Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Regression with K-Nearest Neighbor, Euclidean distance, Manhattan distance, and other distance metrics, their advantages, and disadvantages.	Distance Vector Calculations, Visual Geometry Mapping	Class Test, Midterm exam, Final exam	CLO2, CLO3
	Lesson 2: The big picture of Machine Learning, visualizing data, preparing data for machine learning, selecting, training, and fine-tuning machine learning models and hyper parameters.	Pipeline Architecture Walkthrough, Exploratory Analysis	Class Test, Midterm exam, Final exam	CLO2, CLO3
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Classification with K-Nearest Neighbor and different evaluation metrics of classification models, Error analysis of classification models, Multilabel Vs. Multi-output classification algorithms.	Confusion Matrix Analysis, Multi-label Mapping	Class Test, Midterm exam, Final exam	CLO2, CLO3
	Lesson 2: Linear Regression, The Normal Equation, Computational Complexity of normal equation, Gradient descent, learning rates and learning rate scheduling, early stopping, batch vs. mini batch vs. stochastic gradient descent. Class Test -1	Cost Function Optimization, Step-size Optimization Tracing	Class Test, Midterm exam, Final exam	CLO2, CLO3
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Polynomial regression, detecting underfitting and overfitting, understanding bias-variance trade-offs.	Variance Fitting Breakdown, Error Curve Graphing	Class Test, Midterm exam, Final exam	CLO2, CLO3
	Lesson 2: Regularization, Ridge Regression, Lasso Regression, and ElasticNet.	Penalty Constraint Mapping, Coefficient Shrinkage Drill	Class Test, Midterm exam, Final exam	CLO2, CLO3
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Binary and multiclass classification using Logistic Regression, Understanding sigmoid function, and SoftMax function.	Probability Curve Mapping, Log-Odds Equation Tracing	Class Test, Midterm exam, Final exam	CLO2, CLO3
	Lesson 2: Understanding ROC-AUC curve, Precision-Recall curve, and techniques of determining the best thresholds. Class Test -2	Threshold Optimization Plotting, Curve Coordinate Analysis	Class Test, Midterm exam, Final exam	CLO2, CLO3
Week-7	Lesson 1: Classification using Decision Tree, computational complexity, computing node	Recursive Partitioning Modeling, Impurity Calculus Practice	Class Test, Midterm exam, Final exam	CLO2, CLO3

Lesson 1 & 2 [3 Hours]	impurity using Gini/Entropy, and determining best split using Information Gain, CART algorithm for building decision tree.			
	Lesson 2: Regression using Decision Tree, Regularization, hyper parameter tuning, and Random Forest, Identifying potential source of biasness, fairness issues, or any other societal, cultural, or ethical concerns in unsupervised machine learning models.	Ensemble Strategy Review, Bias Assessment Framework	Class Test, Assignment, Presentation, Midterm exam, Final Exam	CLO2, CLO3, CLO4
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: The curse of dimensionality, Main approaches of Dimensionality Reduction, their advantages, and disadvantages.	Vector Space Sparsity Analysis, Geometric Metaphor Drill	Class Test, Final exam	CLO2, CLO3
	Lesson 2: Principal Component Analysis (PCA), Kernel PCA	Eigenvector Covariance Projection, Coordinate Rotation Mapping	Class Test, Final exam	CLO2, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Locally Linear Embedding (LLE), and other dimensionality reduction techniques, K- Means Clustering, Euclidean, Manhattan, and other distance metrics, their advantages, and disadvantages.	Topology Manifold Graphing, Metric Space Comparison	Class Test, Final exam	CLO1, CLO3, CLO4
	Lesson 2: Advantages and Disadvantages of K-Means clustering and different ways to solve them.	Initialization Trait Analysis, Cluster Hardening Lab	Class Test, Final exam	CLO2, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Hierarchical / Agglomerative clustering algorithm and identifying best clusters using Dendrograms, Identifying best clusters using inter-cluster distance and intra-cluster distance (inertia).	Tree Linkage Diagramming, Variance Bound Calculation	Class Test, Final exam	CLO2, CLO3
	Lesson 2: Identifying potential source of biasness, fairness issues, or any other societal, cultural, or ethical concerns in unsupervised machine learning models.	Ethical Impact Analysis, Dataset Equity Review	Assignment, Presentation, Final exam	CLO1, CLO4
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Machine learning model training best practices: Data Leakage, Data Sufficiency, trying different range of models, trying appropriate models, optimizing hyperparameters, and feature selection.	Pipeline Integrity Verification, Data Volume Analysis	Class Test, Assignment, Presentation, Final exam	CLO3, CLO4

	Lesson 2: Machine learning model training best practices: Data Leakage, Data Sufficiency, trying different range of models, trying appropriate models, optimizing hyperparameters, and feature selection.	Feature Space Selection, Hyperparameter Search Matrix	Class Test, Assignment, Presentation, Final exam	CLO3, CLO4
	Class Test -3			
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Machine learning model evaluation best practices: Train-Test-Validation sets and their purpose, Stratified Splitting Vs. Random Splitting	Population Balance Auditing, Partition Space Modeling	Class Test, Assignment, Final exam	CLO3, CLO4
	Lesson 2: Machine learning model evaluation best practices: Cross-validation, and Bootstrapping, Hyper parameter Tuning, managing imbalanced datasets.	K-Fold Split Sequencing, Resampling Matrix Design	Class Test, Assignment, Final exam	CLO3, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Best practices of comparing multiple models: use of statistical tests in model comparison, multiple comparisons, combining models, and evaluating community benchmarks.	Statistical Significance Testing, Error Interval Mapping	Class Test, Assignment, Presentation, Final exam	CLO3, CLO4
	Lesson 2: Best practices of comparing multiple models: use of statistical tests in model comparison, multiple comparisons, combining models, and evaluating community benchmarks.	Benchmark Matrix Dissection, Meta-Learning Analysis	Class Test, Assignment, Presentation, Final exam	CLO3, CLO4
Week-14 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Blueprint Review	Assignment, Presentation,	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Best practices of reporting results of machine learning models: transparency, generalizing beyond data, interpreting statistical significance, ethical considerations during reporting, and so on.	Transparency Metric Checklist, Interpretability Review	Class Test, Assignment, Presentation, Final exam	CLO4, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total=78)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO4	PLO6	

Attendance	--	--	--	–	7					
Class Test 1	5(15)				Avg 15					
Class Test 2		5(15)								
Class Test 3			5(15)							
Assignment				5	5				5	5
Presentation				8	8				8	8
Midterm Exam	5	10	5	5	25	10	5	5	5	25
Final exam	5	10	20	5	40	10	5	20	5	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd edition, O'Reilly Media, 2022, ISBN: 978-1492032649.

Reference Books/ Other Supplementary Readings

Ian Goodfellow, Yoshua Bengio, & Aaron Courville, Deep Learning, MIT Press, 2016, ISBN: 978-0262035613.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 331 Software Engineering Design Capstone Project

Course Code: SE 331		CIE Marks: 60
ISCED: 0613-331		SEE Marks: 40
Course Title: Software Engineering Design Capstone Project		Total Marks: 100
Semester:		
Credit Value: 3 (Project)	Contact Hours: 4.5 (Total week: As per BNQF Recruitment)	
Prerequisite: SE321, SE322		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

The “Software Engineering Design Capstone Project” course provides hands-on experience and practical application of web and software development concepts using web technologies. Students will be guided through a systematic approach to develop a project, emphasizing real life problem-solving skills and logical thinking. The course covers a range of topics from problem formulation to the evaluation of solutions, ensuring that students acquire a comprehensive understanding of the software and web development process. The course encompasses front end web technologies, framework and backend web technologies and frameworks. These core concepts form the backbone of the curriculum, providing students with a comprehensive understanding of essential web application principles.

Rationale of the Course

The Software Engineering Design Capstone Project is an essential culmination of theoretical knowledge and practical application in software development. This course equips students with advanced skills in project management, problem-solving, and collaboration within a simulated professional environment. Through real-world scenarios and industry-standard methodologies, students tackle complex software design challenges, fostering creativity and innovation. By integrating diverse perspectives and leveraging cutting-edge technologies, the course ensures graduates are prepared to excel in dynamic software engineering roles, bridging the gap between academia and industry seamlessly.

Course Objectives

The goal of this course is to introduce the students to the development process of a project using structured programming language. The main objectives of this course are-

- Enable students to apply theoretical knowledge gained throughout their academic journey to real-world software engineering projects, fostering practical skills development.
- Facilitate effective teamwork, communication, and collaboration among students, mirroring the dynamics of professional software development environments.
- Cultivate critical thinking and problem-solving abilities by challenging students with complex software design problems and guiding them through the process of finding innovative solutions.
- Develop project management skills by requiring students to plan, organize, execute, and monitor the progress of their software projects, adhering to deadlines and budgets.
- Promote ethical conduct and professionalism by emphasizing the importance of integrity, accountability, and responsibility in all aspects of software engineering design and development.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	Knowledge Profile	Complex Engineering Problem/engineering activities	BNQF Skill
CLO1	Analyze real life problems to find effective solutions	PLO 2	A3,P1	K3	CEP2, CEP3	Fundamental Domain
CLO2	Interpret the suitable frameworks to reach and justify the solutions	PLO 5	A1,P1,P7	K6	CEP3, CEP4	Fundamental Domain
CLO3	Design possible solutions for projects that meet valid conclusions.	PLO 4	A3	K8	CEP4, CEP5	Fundamental Domain
CLO4	Apply engineering and management principles to plan and coordinate project tasks effectively as a team member or leader in a multidisciplinary setting.	PLO11	A4,P2	K7	CEP2, CEP3, CEP6	Social, Thinking, Personal Domain
CLO5	Deliver the project documentation created collaboratively for intricate real-life applications as a team presentation	PLO9	A2,P4	K5	CEA4, CEA5	Social, Thinking, Personal Domain

CLO6	Write effective reports and make an effective presentation of the outcome of the capstone project.	PLO 10	A5,P1	K4	CEP6, CEP7, CEP8	Social, Thinking, Personal Domain
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Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2					√							
CLO3				√								
CLO4							√					
CLO5									√			
CLO6										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Introduction class & Explain course outline, CLO, PLO	Syllabus Walkthrough, Blueprint Orientation	None	None
	Lesson 2: Introduction To Laravel	Framework Lifecycle Review, Environment Setup	Submission of lab task of given problem	CLO2
	Lesson 3: Mastering Concept in laravel	Architecture Pattern Analysis, Service Container Lab	Submission of lab task of given problem	CLO3
Week-2 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Introduction To Livewire	Reactive Directive Walkthrough, Live Coding	Submission of lab task of given problem	CLO2
	Lesson 2: Single Page Application Development with livewire	Dynamic Component Injection, Dom Diffing Lab	Submission of lab task of given problem	CLO3

	Lesson 3: Complete project buildup using livewire	Modular State Synchronization, Flowchart Tracking	Submission of lab task of given problem	CLO3
Week-3 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Introduction to Model Controller View Route in laravel, Resource Controller & Normal Controller	MVC Schema Mapping, Route Pipeline Dissection	Submission of lab task of given problem	CLO3
	Lesson 2: CRUD operation, Auth in Laravel	Database Statement Binding, Auth Guard Configuration	Submission of lab task of given problem	CLO1
	Lesson 3: Multi User Dashboard, Role Management Using Spatie	RBAC Matrix Modeling, Middleware Filter Wiring	Submission of lab task of given problem	CLO1
Week-4 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Eloquent ORM, Query Builder	ORM Schema Mapping, Query Profiling Laboratory	Submission of lab task of given problem	CLO2
	Lesson 2: API development using passport, Mailing In Laravel	Bearer Token Tracing, SMTP Stream Debugging	Submission of lab task of given problem	CLO2
	Lesson 3: Yajra Datatable, Laravel Live notification system using firebase	Server-side Pagination Grid, Socket Event Broadcast	Submission of lab task of given problem	CLO1, CLO2
Week-5 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Introduction to Vue.js	Virtual DOM Walkthrough, Reactive Data Binding	Submission of lab task of given problem	CLO2
	Lesson 2: Vue.js Components	Component State Flowcharting, Props/Events Wiring	Submission of lab task of given problem	CLO2
	Lesson 3: Vue Router	Single Page Router Guarding, Lazy Loading Lab	Submission of lab task of given problem	CLO2
Week-6 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Vuex, Forms and Validation	Centralized Flux Mapping, Client-Side Input Validation	Submission of lab task of given problem	CLO1
	Lesson 2: HTTP Requests with Vue.js, Vue.js Directives	Axios Interceptor Tracing, Custom Directive Building	Submission of lab task of given problem	CLO1, CLO2

	Lesson 3: Vue.js Plugins, Vue.js Animations and Transitions	Plugin Prototype Injection, DOM Transition Hooking	Submission of lab task of given problem	CLO2, CLO3
Week-7 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Server-Side Rendering (SSR) with Vue.js	Hydration Cycle Analysis, Universal Build Setup	Submission of lab task of given problem	CLO3
	Lesson 2: Testing Vue.js Applications	Component Assertion Scripting, Mocking API Wrappers	Submission of lab task of given problem	CLO3
	Lesson 3: Vue.js Best Practices and Optimization	Bundle Chunk Optimization, Performance Profiling Lab	Submission of lab task of given problem	CLO2
Week-8 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Installing VUE In Laravel	Build-Engine Asset Compilation, Vite Config Hardening	Submission of lab task of given problem	CLO3
	Lesson 2: Building API with passport api	Secure Payload Structure, Header Verification Lab	Submission of lab task of given problem	CLO3
	Lesson 3: API Consumption from vue using fetch and axios	Cross-Origin Data Streaming, Async State Resolution	Submission of lab task of given problem	CLO3
Week-9 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Introduction To Github, Introduction To Tortoise Git	Git Branching Simulation, Merge Conflict Resolution	Submission of lab task of given problem	CLO2
	Lesson 2: Introduction To cpanel and project deployment in cpanel using GitHub	Webhook Automation Scripting, Symlink Target Tuning	Submission of lab task of given problem	CLO5
	Lesson 3: Introduction To AWS, EC2 & installing Ubuntu in EC2, Deployment Of Project In EC2	Linux Server Configuration, Reverse Proxy Setup	Submission of lab task of given problem	CLO3, CLO4, CLO5
Week-10 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Engaging the students in group work to design and develop the project step by step	Capstone System Architecture Designing, SRS Drafting	Submission of SRS	CLO5
	Lesson 2: Engaging the students in group work to design and develop the project step by step	Database Schema Normalization, UML Structure Analysis	Submission of documentation of ER, Use Case, Sequence, Activity and class diagram	CLO5

	Lesson 3: Engaging the students in group work to design and develop the project step by step	Socio-Technical System Discussion, Project Prototyping	Project idea presentations and possible impact of the project on society	CLO6
Week-11 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Team Wise Work Progress showing the mastering vue and laravel integration	Code-Integration Code Clinic, Milestone Code Verification	Team Meeting and progress showing of teams	CLO2
	Lesson 2: Team Wise Work Progress showing the mastering vue and laravel integration	Interface Connection Troubleshooting, Technical Review	Team Meeting and progress showing of teams	CLO2
	Lesson 3: Team Wise Work Progress showing the mastering vue and laravel integration	Integration Flow Optimization, Peer Review Consultation	Team Meeting and progress showing of teams	CLO2
Week-12 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Project Development meeting with teams and working on things and developing quality of the project for solving real world problem	Product Hardening Checklist, Vulnerability Verification	Final Documentation Preparation	CLO4
	Lesson2: Project Development meeting with teams and working on things and developing quality of the project for solving real world problem	Final Compilation Profiling, Package Bundling Analysis	Final Documentation Submission	CLO4
	Lesson 3: Project Development meeting with teams and working on things and developing quality of the project for solving real world problem	System Diagnostics Verification, Group Oral Review	Viva on Final Report	CLO4
Week-13 Lesson 1 & 2 [4.5 Hours]	Lesson 1: Final project demonstration showing and group presentation	System Architecture Showcasing, Production Verification	Project Demonstration and Final Presentation	CLO 4, CLO 5, CLO 6
	Lesson 2: Final project demonstration showing and group presentation	System Architecture Showcasing, Production Verification	Project Demonstration and Final Presentation	CLO 4, CLO 5, CLO 6
	Lesson 3: Industry Session	Industry Expert Talk, Production Blueprint Review	Presentation	CLO 4, CLO 5, CLO 6
Week-14	Lesson 1: Final project demonstration showing and group presentation	System Architecture Showcasing, Production Verification	Project Demonstration and Final Presentation	CLO 4, CLO 5, CLO 6

Lesson 1 & 2 [4.5 Hours]	Lesson 2: Final project demonstration showing and group presentation	System Architecture Showcasing, Production Verification	Project Demonstration and Final Presentation	CLO 4, CLO 5, CLO 6
	Lesson 3: Final project demonstration showing and group presentation	System Architecture Showcasing, Production Verification	Project Demonstration and Final Presentation	CLO 4, CLO 5, CLO 6

Overall Assessment Scheme

Criteria	Assigned Marks	Sub Criteria	Sub Criteria Marks	CLO	PLO
Attendance	10				
Lab Report	25	SRS Submission And Project Plan Documentation	25	CLO1 CLO5 CLO6	PLO2 PLO7 PLO9
Lab Performance	25	Continuous assessment	15	CLO1,CLO2,CLO3	PLO2,PLO4,PLO5
		Group Activities	10	CLO5	PLO9
Lab Final	40	Project Demonstration	25	CLO1, CLO2, CLO3	PLO1, PLO2, PLO3
		Final presentation	15	CLO4, CLO5, CLO6	PLO7, PLO9, PLO10
Total	100		90		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Matt Stauffer, Laravel: Up & Running: A Framework for Building Modern PHP Apps, 3rd edition, O'Reilly Media, 2023, ISBN: 978-1492041184

Reference Books/ Other Supplementary Readings

Jon Duckett, PHP & MySQL: Server-side Web Development, 1st Edition, Wiley, 2022, ISBN: 978-1119149224

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Figma, Git, HTML, CSS, JavaScript

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

EMP 101 Employability 360

Course Code: EMP 101		CIE Marks: 60
ISCED: 0613-13		SEE Marks: 40
Course Title: Employability 360		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: GED		

CIE: Continuous Internal Evaluation,

SEE: Semester End Examination

Course Content (from syllabus)

Overview - Employability 360, self-assessment, career mapping and planning, transferable soft skills, administrative tools, business communication tools, collaboration and teamwork, interview skills, finding cutting edge engineering technology and market research, engineering tech skills showcasing for Software/Computer Engineering students, Foster adaptability & lifelong learning in today's rapidly changing tech-job market.

Rationale of the Course

This "Employability 360" course, it helps individuals develop skills that are highly valued by employers by focusing on foundational skills, technical skills, personal attributes, career management skills, and professional development which can help individuals become more competitive in the job market and more successful in their careers. The course focuses on developing a broad range of skills that can help individuals become more employable, as they will be better equipped to meet the demands of today's as well as the future job market.

Course Objectives

The objective of Employability 360 for computer/software engineering students is to equip them with a range of skills, knowledge, and experience that are essential for success in their chosen field. Specifically, the program aims:

- To provide an overview of Employability 360 with Self-Assessment, Career Mapping, and Planning along with soft skills and general job recruitment process.
- To enhance communication skills which is essential for any software engineer, as they need to collaborate with team members, stakeholders, and clients.
- To make students understand of problem-solving and project management skills developing. These skills include proficiency in various development tools, solve complex technical problems of information management systems. Computer/software engineering students are often involved in managing projects, including planning, scheduling, and tracking progress. Also by providing opportunities to lead projects and manage risks.
- To foster adaptability. The field of software engineering is constantly evolving, and software engineers need to be adaptable and willing to learn new skills and technologies. Employability 360 aims to foster adaptability in computer/software engineering students by providing opportunities to learn new skills and technologies quickly.
- To provide practical experience: Employability 360 aims to provide computer/software engineering students with practical experience in their chosen field through internships, work placements, or other work-related experiences.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Knowing about different terms, concepts and models related to Employability 360, self-assessment, career mapping and planning, transferable tech & soft skills, administrative tools, business communication tools, collaboration and teamwork, interview skills, personal branding and market research.	PLO2	C2	Fundamental Domain
CLO2	Applying the learned concepts and models to find and relate to cutting edge tech market, and skills on Artificial Intelligence & Machine Learning, Blockchain, Cybersecurity, Cloud Computing, DevOps, IoT, Data Science etc.	PLO3	C3	Fundamental Domain
CLO3	Creating portfolio to demonstrate to potential employers including descriptions of projects, the technologies and methodologies used, and the results achieved by collaborating with team members, stakeholders and clients of a project from the perspective of a project manager including planning, scheduling, tracking progress and risk management.	PLO5	C5	Fundamental Domain

CLO4	Creating Online Presence like in GitHub, Stack Overflow, GitLab, Bitbucket, Jira, Docker, AWS, Google Cloud Platform, LinkedIn etc. Participate in Hackathons and Competitions, Attend Conferences and Networking Events, Publish Research Papers and Articles.	PLO12	C6	Social, Thinking, Personal Domain
CLO5	Demonstrating the skills of Employability 360 along with knowledge & experience of their chosen fields.	PLO10	C6	Social, Thinking, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2			√									
CLO3					√							
CLO4												√
CLO5										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: OBE Discussion, Course outline brief discussion, Overview Employability360, Job Market.	Syllabus Walkthrough, Market Overview Discussion	None	CLO1
	Lesson 02: Understanding the employers' need, Understand what is self-assessment, and how to assess self. SWOT.	SWOT Matrix Analysis, Self-Reflective Mapping	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 01: Value proposition, how to add value to self by skills development, Career mapping and planning.	Value Proposition Design, Career Path Charting	Class Test, Midterm exam	CLO1
	Lesson 02: Communication skills (oral and written), Knowing pros & cons, writing negative letters.	Correspondence Case Study, Written Drafting Drill	Class Test, Midterm exam	CLO1

Week-3 Lesson 1 & 2 [3 Hours]	Lesson 01: Self-management (time management, self-motivation, etc.)	Priority Matrix Workshop, Time-Blocking Practice	Class Test, Midterm exam	CLO1
	Lesson 02: Personal Attributes: Adaptability and flexibility, Resilience and persistence	Scenario Role-Play, Behavioral Case Review	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 01: Personal Attributes: Initiative and creativity, Cultural competence, Emotional intelligence	EQ Self-Assessment, Cross-Cultural Case Study	Class Test, Midterm exam	CLO1
	Lesson 02 : Resume and cover letter writing. Class Test-1	Resume Layout Dissection, Formatting Workshop	Class Test, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 01: Networking and relationship building, Job search strategies	Cold Outreach Simulation, Pipeline Mapping	Class Test, Presentation, Midterm exam	CLO1, CLO2
	Lesson 02: Research Company & mapping the required skills	Target Account Research, Skills Gap Analysis	Class Test, Presentation, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 01: Finding cutting edge engineering technology like Artificial Intelligence & Machine Learning, Blockchain.	Tech Stack Landscape Review, Visual Presentation	Class Test, Presentation, Midterm exam	CLO1, CLO2
	Lesson 2: Skills on Cybersecurity, Cloud Computing, DevOps, IoT, Data Science etc. Class Test-2	Infrastructure Roles Mapping, Industry Roadmap Review	Class Test, Presentation, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lecture 01: Project management skills on cutting edge tech application.	Lifecycle Phase Analysis, Agile Framework Overview	Class Test, Presentation, Midterm exam	CLO1, CLO2
	Lesson 02: Scheduling, budgeting and risk managing in projects, Soft skills and practice of soft skills.	Gantt Chart Scheduling, Risk Register Blueprinting	Class Test, Presentation, Midterm exam	CLO1, CLO2
Week-8	Lesson 01: Conflict management and negotiation skills, Interview skills for both interviewee and Interviewer.	Negotiation Role-Play, Mock Interview Simulation	Class Test, Assignment, Midterm exam	CLO1, CLO2

Lesson 1 & 2 [3 Hours]	Lesson 2: Review Session before Mid Term.	Integrative Concept Review, Q&A Clinic	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 01: Understand cutting edge tech market.	Market Demand Analysis, Ecosystem Walkthrough	Class Test, Assignment, Presentation, Final exam	CLO1, CLO3
	Lesson 2: Create portfolio with appropriate skills set.	Portfolio Architecture Blueprinting, Hosting Setup	Class Test, Assignment, Presentation, Final exam	CLO1, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 01: Professional certifications and licensure	Certification Roadmap Planning, Vendor Matrix Analysis	Class Test, Assignment, Presentation, Final exam	CLO1, CLO3
	Lesson 2: Understand mentoring and coaching Volunteerism and community involvement.	Mentorship Framework Discussion, Open Discussion	Class Test, Assignment, Presentation, Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 01: Showcasing of skills in portfolio.	Asset Deployment Workshop, Project Presentation	Class Test, Assignment, Presentation, Final exam	CLO1, CLO3
	Lesson 02: Showcasing of Problem Solving skills. Class Test-3	Code Repository Walkthrough, Readme Optimization	Class Test, Assignment, Presentation, Final exam	CLO1, CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 01: Networking and relationship building. Attend workshops and seminars.	Event Preparation Matrix, Pitch Deck Practice	Assignment, Presentation, Final exam	CLO1, CLO3
	Lesson 02: Attend job fairs and defending own projects, Use of data from online resources and demonstration of various skills.	Live Project Defense, Field Interaction Practice	Assignment, Presentation, Final exam	CLO1, CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 01: Participate in Hackathons and Competitions, enrich portfolio, Attend Conferences and Networking	Competition Strategy Workshop, Writing Review	Assignment, Presentation,	CLO1, CLO3, CLO4

	Events, Publish Research Papers and Articles.		Final exam	
	Lesson 02: Positive risks issues, Risk Management Responses. What can go wrong and how can we overcome our shortcomings?	Crisis Scenario Diagnostics, Resiliency Blueprinting	Assignment, Presentation, Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 2: Industry Session	Strategic Exam Preparation, Speed Interview Clinic	Assignment, Presentation	CLO1-CLO5
	Lesson 1: Briefing before Final exam, Preparatory mock interview	Guest Lecture / Industry Expert Talk, Interactive Q&A	Assignment, Presentation, Final exam	CLO1, CLO5

Overall Assessment Scheme

Assessment Task	CLO's					Mark (Total=100)	PLO's					Mark (Total=65)
	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5		PLO 2	PLO3	PLO5	PLO1 0	PLO1 2	
Attendance	--	--	--			7						
Class Test 1						Avg 15						
Class Test 2												
Class Test 3												
Assignment						5						
Presentation						8						
Midterm Exam	10	15				25	10	15			25	25
Final exam			22	12	6	40			22	6	12	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8

3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Md. Sabur Khan, Handbook of entrepreneurship development, DIU Press, 2013

Reference Books/ Other Supplementary Readings

Harvard Business Review Press. Emotional Intelligence Series. HBR Press, 2020.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software

**Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)**

Engineering (SWE)

SE 444 Data Warehouse and Data Mining

Course Code: SE 444		CIE Marks: 60
ISCED: 0613-444		SEE Marks: 40
Course Title: Data Warehouse and Data Mining		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Open Elective I		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Transactional Processing Vs. Decision Support Systems, Data Warehousing Fundamentals, Characteristics of Data Warehouse, Data Warehouse Architectures, Maturity Models, Potential Applications, OLTP, OLAP, Data Cubes, Fact Tables, Dimensions, Relational Models, Integrity Rules, Entity Relationship Diagrams, Modification Anomalies, Database Normalizations, Star Schema, Snowflake Schema, Constellation Schema, Time Representation, Types of Change, ETL, Naïve Bayes, Train-Test-Validation Set, Cross-Validation, Bootstrapping, Mean Square Error, Mean Absolute Error, Root Mean Square Error, Confusion Matrix, Accuracy, Precision, Recall, F1 Score, Association Rule Mining, Apriori Algorithm, Linear Regression, Logistic Regression, Decision Tree, K-Means Clustering, Agglomerative Clustering, Graph Theory, Graph Clustering.. Community Detection, Girvan Algorithm, Louvain Algorithm.

Rationale of the Course

This text outlines the distinct roles and synergies of data warehousing and data mining in business intelligence. Data warehouses consolidate data from multiple sources to ensure accuracy and consistency, optimizing online analytical processing (OLAP) by separating historical analysis from transactional databases (OLTP). By leveraging statistics, artificial intelligence, and machine learning, data mining enables organizations to predict future trends and efficiently resolve complex business queries.

Course Objectives

The goal of this course is to introduce the students about the concept of Data warehouse and Data Mining principles. The main objectives of this course are:

- Students will be taught about the necessity and the appropriate scenarios where warehouse and data mining are relevant and beneficial.
- Students will be taught about different components and architectures of data warehouse.
- Students will be taught to apply different data mining algorithms.
- Students will be taught to evaluate and interpret the results of data mining algorithms.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
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CLO1	Design data warehouse architectures to solve real-world business problems.	PLO3	C6	Fundamental Domain
CLO2	Apply appropriate data mining algorithms to answer different business queries.	PLO2	C3	Fundamental Domain
CLO3	Communicate the findings of different data mining algorithms properly to the stakeholders.	PLO10	C5, A3, P3	Social, Thinking, Personal Domain
CLO4	Evaluate different components of a data warehouse – data mining solution for the correctness of design, analysis, and interpretation.	PLO4	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1			√									
CLO2		√										
CLO3										√		
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Introduction to Software Engineering, Software, Types of software, Characteristics of a quality Software, Basics of Software Engineering and terminologies.	Syllabus Walkthrough, Quality Attribute Mapping	Class Test, Midterm exam	CLO1
	Lesson 2: Importance of Software Engineering, challenges of Software Engineering. The History of Software Engineering, software process, software process models, software development practice, activities and task, software methodology, motivations behind the use of software development Methodologies	Process History Timeline, Methodology Case Review	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2	Lesson 1: Different phases of software development life cycle. Differentiating Phases, Practice, Activities and Tasks in SDLC	SDLC Phase Dissection, Task Allocation Exercise	Class Test, Midterm exam	CLO1, CLO2

[3 Hours]	Lesson 2: Relationship among Software process model, Software development Life cycle, and Software development methodology	Lifecycle Parameter Venn-Diagramming, Open Debate	Class Test, Midterm exam	CLO1, CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Waterfall classical model and Waterfall V model. Usages, advantages, and disadvantages of the Models.	Dual-V Verification Walkthrough, Rigid Design Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Prototype Model, Iterative model, Incremental model. Usages, advantages, and disadvantages of the Models.	Prototype Feedback Mapping, Evolutionary Lifecycle Review	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Spiral Model. Usages, advantages, and disadvantages of the Models.	Risk Spiral Graphing, Visual Paradigm Presentation	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Agile methodologies, Value and principles of Agile methodology. Class Test -1	Agile Manifesto Deconstruction, Core Rules Checklist	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: SCRUM Model. Usages, advantages & disadvantages of SCRUM model.	Sprint Routine Simulation, Burn-down Chart Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: XP Model. Usages, advantages & disadvantages of the model.	Pair-Programming Simulation, Refactoring Workflow Review	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hs]	Lesson 1: Kanban Model. Usages, advantages & disadvantages of the model, Comparative view among Scrum, XP and Kanban Class Test -2	Work-In-Progress (WIP) Limit Board Modeling, Matrix Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Concepts of user and system requirements and why these requirements should be written in different ways; functional and non-functional software requirements, Requirements engineering activities of elicitation, analysis, and validation, and the relationships between these activities.	Requirement Dissection Blueprinting, Traceability Mapping	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Design process, characteristics of a good design, different design concepts such as abstraction, architecture, Patterns, separation of concerns, modularity etc., Design model, design modeling	System Modular Tree Sketching, Architectural Review	Class Test, Assignment, Final exam	CLO1, CLO2

	principles, data design elements, architectural design elements			
	Lesson 2: Interface design elements, component-level design elements, deployment-level design elements, to visualize software structure and behavior	Wireframe Flowcharting, Structural Node Mapping	Class Test, Assignment, Final exam	CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Verification and validation, software testing, important of testing, different types of testing with definition,	V&V Process Matrix, Test Topology Mapping	Class Test, Assignment, Final exam	CLO1, CLO2
	Lesson 2: Test-case design: requirements and use cases, traceability, white box testing: basis path testing with cyclomatic complexity, black box testing: interface testing, equivalence partitioning, boundary value analysis,	Control Flow Graphing, Boundary Logic Math Drill	Class Test, Assignment, Final exam	CLO1, CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: object-oriented software testing: class testing, behavioral testing, Halstead's complexity, Decision table	Halstead Complexity Equation practice, Truth Table Modeling	Class Test, Final exam	CLO1, CLO2
	Lesson 2: Software quality, quality factors, qualitative quality assessment, quantitative quality assessment, software quality dilemma, achieving software quality	Quality Metric Checklist, Cost-of-Quality (CoQ) Analysis	Class Test, Final exam	CLO1, CLO2
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Dependability, principal dependability properties, Attributes of dependable processes	Process Attribute Auditing, Failure Mode Evaluation	Final exam	CLO1, CLO2
	Lesson 2: Reliability Engineering (availability and reliability, reliability requirements, fault-tolerant architectures, programming for reliability, reliability measurement)	MTBF Reliability Formula Practice, Redundancy Analysis	Final exam	CLO1, CLO2
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Security and dependability, Security terminology, Security risk assessment, security requirements	Threat Surface Matrix, Secure Requirement Mapping	Final exam	CLO1, CLO2
	Lesson 2: Software maintenance, steps in a maintenance process, strategies of software maintenance, software maintenance, reverse engineering	Legacy Code Diagnostics, Restructuring Flowchart Review	Final exam	CLO1, CLO2
Class Test -3				

Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Software project management, Project management in Waterfall model-based projects, Project management in agile model-based projects, team management of software projects	Governance Model Comparison, Matrix Role Evaluation	Assignment, Final exam	CLO1, CLO3
	Lesson 2: Decomposition and Estimation techniques: Software Sizing, FP-Based Estimation, related mathematics	Function Point Factor Math, Sizing Table Practice	Assignment, Final exam	CLO1, CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: COCOMO models (Basic and Intermediate) and related mathematics, Software project scheduling, basic principles of scheduling, task set for project planning	Nonlinear Effort Calculation, Work Breakdown Structuring	Assignment, Final exam	CLO1, CLO3
	Lesson2: Scheduling: Time-Line Charts, scheduling related mathematics, Software risks, seven principles of risk management, risk identification, assessing overall project risk, risk estimation, Risk mitigation, risk monitoring, and risk management	Critical Path Evaluation, Risk Register Blueprinting	Assignment, Final exam	CLO1,CLO2, CLO3
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Blueprint Review	Assignment, Presentation	CLO1, CLO2, CLO3
	Lesson 2: Review class on final exam syllabus. Presentation on a given topic	Capstone Concept Mapping, Board Problem Solving	Presentation, Open discussion	CLO1, CLO2

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO3	PLO4	PLO10	
Attendance	--	--	--	--	7					
Class Test 1	15				Avg 15					
Class Test 2		15								
Class Test 3				15						
Assignment				5	5					
Presentation			8		8					
Midterm Exam	10	10		5	25	10	10	5		25

Final exam	10	10	10	10	40	10	10	10	10	40
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Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Michael Mannino, Database Design, Application Development, and Administration, 2nd edition, McGraw-Hill/Irwin, c2004., ISBN: 007282350X

Reference Books/ Other Supplementary Readings

Mark A. Hall, Ian H. Witten, Eibe Frank, Christopher Pal, Data Mining: Practical Machine Learning Tools and Techniques, 3rd Edition, Morgan Kaufmann, 2011, ISBN: 9780123748560

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Python, Google Colab

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 447 Human Computer Interaction

Course Code: SE 447		CIE Marks: 60
ISCED: 0613-447		SEE Marks: 40
Course Title: Human Computer Interaction		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Open Elective II		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Foundations of Human Computer Interaction (Human Capabilities, Computer usability Interaction Paradigms), The Design Process (Interaction Design Basics, HCI in the Software Process, Design Rules, Universal Design), Implementation Support (Implementation Tools), Device Evaluation and User Support (Evaluation ,User Support), User-Centered Design (Participant Observation and User Personas, Low-Fidelity Prototyping, Usability Testing Computer Prototyping), Users Models (Cognitive Models, Socio-organizational Issues and Stakeholder Requirements), Task Models and Dialogs (Analyzing Tasks, Dialog Notations and Design), Groupware, Ubiquitous Computing, Virtual and Augmented Reality, Hypertext and Multimedia (Groupware and Computer-supported Collaborative Work, Ubiquitous Computing, Virtual Reality and Augmented Reality, Hypertext, Multimedia and the World Wide Web), Challenges and technologies (Perception and Visualization, Design and evaluation challenges), Interactions (Wearable and Mobile Interaction, UbiComp, Haptics), Physiological & Social Computing (Tangible and Gestural interfaces, Physiological Computing, Social Computing, Digital addiction), Behavioral Computing (Human persistency and actuator compliance).

Rationale of the Course

This course provides a comprehensive foundation in human-computer interaction, focusing on the principles and methodologies required to design intuitive, user-centered digital interfaces. Students explore human cognitive capabilities, usability standards, and diverse interaction paradigms. The curriculum emphasizes the end-to-end design

lifecycle, covering participant observation, user personas, prototyping, and iterative evaluation based on user feedback.

Course Objectives

The goal of this course is to introduce the students about human-computer interaction and the design and evaluation of user interfaces. The main objectives of this course:

- To apply an interactive design process and universal design principles to designing HCI systems.
- To describe and use HCI design principles, standards and guidelines.
- To analyze and identify user models, user support, socio-organizational issues, and stakeholder requirements of HCI systems.
- To illustrate typical human-computer interaction (HCI) models and styles, as well as various historic HCI paradigms
- To analyze and discuss HCI issues in groupware, ubiquitous computing, virtual reality, multimedia, and Word Wide Web-related environments.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Describe user-centered design methods to conduct the formative and summative evaluations.	PLO1	C1	Fundamental Domain
CLO2	Develop effective and usable prototyping based on behavioral analysis.	PLO3	C3	Fundamental Domain
CLO3	Demonstrate low-fidelity prototypes based on scenarios for better user experience.	PLO2	C3	Fundamental Domain
CLO4	Evaluate the user behaviors, preferences, and the proliferation of connected devices, for personalized digital experiences.	PLO4	C4	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2			√									

CLO3		√										
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1	Lesson 1: Introduction to course content	Syllabus Walkthrough, Interactive Q&A	N/A	N/A
Lesson 1 & 2 [3 Hours]	Lesson 2: Objective, history, and components of HCI	Interactive Lecture, Timeline Mapping	Class Test, Midterm	CLO1
Week-2	Lesson 1: Human Capabilities, Computer usability Interaction Paradigms in HCI, HCI design processes	Cognitive Limit Analysis, Process Flowcharting	Class Test, Assignment, Midterm	CLO1
Lesson 1 & 2 [3 Hours]	Lesson 2: Interaction Design Basics, HCI in the Software Process, Design Rules, Universal Design	Design Heuristics Audit, Scenario Mapping	Class Test, Assignment, Midterm	CLO1
Week-3	Lesson 1: Ergonomics, Interaction Styles, Device Evaluation and User Support (Evaluation, User Support)	Interface Evaluation Workshop, Fitts' Law Drills	Class Test, Assignment, Midterm	CLO1
Lesson 1 & 2 [3 Hours]	Lesson 2: User-Centered Design, Participant Observation and User Personas	Persona Canvas Design, Observation Analysis	Class Test, Assignment, Midterm	CLO1
Week-4	Lesson 1: Low-Fidelity Prototyping, Usability Testing Computer Prototyping with Figma, Adobe XD	Paper Wireframing Lab, Interactive Prototyping	Assignment, Midterm, Final Exam	CLO2
Lesson 1 & 2 [3 Hours]	Lesson 2: Users Models	GOMS Model Calculation, Task Trace Table	Assignment, Midterm, Final Exam	CLO2
	Class Test - 1			

Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Cognitive Models, Socio-organizational Issues and Stakeholder Requirements	Mental Model Graphing, Stakeholder Matrix Analysis	Assignment, Midterm, Final Exam	CLO2
	Lesson 2: Task Models and Dialogs	Hierarchical Task Analysis (HTA), State Diagramming	Assignment, Midterm, Final Exam	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Analyzing Tasks, Dialog Notations and Design basics to intermediate level	STN Notation Construction, Dialog Flow Mapping	Assignment, Midterm, Final Exam	CLO3
	Lesson 2: Groupware, Ubiquitous Computing, Virtual and Augmented Reality, Hypertext and Multimedia. Class Test - 2	Immersive Tech Case Study, Framework Analysis	Assignment, Midterm, Final Exam	CLO3
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Groupware and Computer-supported Collaborative Work.	Synchronous Matrix Modeling, CSCW Core Review	Assignment, Final Exam	CLO3
	Lesson 2: Ubiquitous Computing, Virtual Reality and Augmented Reality, Hypertext, Multimedia and the World Wide Web.	Context-Aware Architecture Mapping, Visual Presentation	Assignment, Final Exam	CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Challenges and technologies, Perception and Visualization	Visual Hierarchy Auditing, Perception Puzzles	Assignment, Final Exam	CLO3
	Lesson 2: Design and evaluation challenges, Interactions with digital device	Heuristic Evaluation Clinic, Defect Logging	Assignment, Final Exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Wearable and Mobile Interaction, UbiComp, Haptics.	Mobile Gesture Mapping, Haptic Feedback Review	Class Test, Presentation, Assignment, Final Exam	CLO4
	Lesson 2: Physiological & Social Computing	Biosensor Input Analysis, Sociogram Mapping	Class Test, Presentation, Assignment, Final Exam	CLO4
Week-10	Lesson 1: <u>Tangible</u> and Gestural interfaces.	Spatial Mapping Analysis, Physical Token Design	Class Test, Presentation,	CLO4

Lesson 1 & 2 [3 Hour]			Assignment, Final Exam	
	Lesson 2: Physiological Computing.	Signal Processing Mapping, Affective State Analysis	Class Test, Presentation, Assignment, Final Exam	CLO4
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Social Computing, Digital addiction, Behavioral Computing	Dark Pattern Diagnostics, Habit Loop Modeling	Presentation, Assignment, Final Exam	CLO4
	Lesson 2: Human persistency and Device accumulation, Actuator compliance Class Test - 3	Hardware Interface Review, Ergonomic Math Practice	Presentation, Assignment, Final Exam	CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: User-centered design (UCD), device evaluation	UCD Lifecycle Tracking, Empirical Testing Review	Presentation, Assignment, Final Exam	CLO4
	Lesson2: HCI interaction lifecycle	Execution-Evaluation Gulf Tracing, System State Review	Presentation, Assignment, Final Exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Handwriting recognition, gesture recognition, computer vision	CV Feature Matrix Analysis, Input Vector Tracking	Presentation, Final Exam	CLO4
	Lesson 2: CSCW Issues and Theory (face-to-face communication, conversation, Multisensory systems)	Multi-channel Grounding Analysis, Dialogue Review	Presentation, Final Exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, UX Case Study Review	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review Class on Final Syllabus	Integrative Review, Board Problem Solving	Final Exam	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =65)
	CLO1	CLO2	CLO3				PLO2	PLO3	PLO4	

				CLO4						
Attendance	--	--	--		7					
Class Test 1	15				Avg 15					
Class Test 2		15								
Class Test 3				15						
Assignment			5		5					
Presentation				8	8					
Midterm Exam	10	15			25		10	15		25
Final exam			25	15	40		25		15	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Jennifer Preece, Yvonne Rogers, Helen Sharp, Interaction Design: Beyond Human-Computer Interaction, 5th Edition, Wiley, 2019, ISBN: 978-1119547259.

Reference Books/ Other Supplementary Readings

1. Andrew Dix, Janet Finlay, Gregory Abowd, & Russell Beale, Human-Computer Interaction, 4th edition, Pearson, 2004, ISBN: 978-0130461094

2. Julie A. Jacko (Editor), Human Computer Interaction Handbook: Fundamentals, Evolving Technologies, and Emerging Applications, 3rd Edition, CRC Press (Taylor & Francis), 2012, ISBN: 978-1439829431.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Figma

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)

Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 599 Research Methodology & Scientific Writing

Course Code: SE 599/ GED 347		CIE Marks: 60
ISCED: 0613-599		SEE Marks: 40
Course Title: Research Methodology & Scientific Writing		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: GED		

CIE: Continuous Internal Evaluation,

SEE: Semester End Examination

Course Content (from syllabus)

The aim of this course is to develop students' knowledge and understanding of the role and conduct of quantitative and qualitative research methods in planning [and urban design]. Intellectual and methodological debates will be discussed in order to assist students to develop informed opinions and a critical appreciation for other's research. The imperative for ethical research practice will be presented. The course equips students with the skills to review and conduct methodologically sound research as a part of their professional work. Students develop the skills to recognise and reflect on the strengths and limitations of different research methodologies, understand the links between theory and practice, critically assess research, and address ethical and practical issues. The course takes a step-by-step approach to the design and implementation of quantitative and qualitative techniques including case study and precedent studies, surveys, interviews, focus groups, participant observation, textual and media analysis. How to manage and analyse data (including computer assisted), and how to write up and present findings are core components of this course. Students will be equipped with the knowledge and ability to undertake original research projects and develop a set of transferable workplace skills.

Rationale of the Course

This course establishes a robust foundation in quantitative and qualitative research methodologies essential for professional planning and urban design. Students develop critical thinking and ethical research practices while mastering practical techniques such as surveys, interviews, and case studies. The curriculum highlights data management and computer-assisted analysis, bridging theoretical concepts with hands-on practice. Through original research projects, report writing, and presentations, learners build the skills necessary to evaluate research limitations, communicate insights effectively, and succeed in professional industry environments.

Course Objectives

- To provide a solid conceptual understanding of the fundamentals of Research Methodology, More specifically,
- To locate, analyze and synthesise information about the diversity of research approaches.
- Develop an ability to apply effective, creative and innovative solutions to research problems.
- Critically evaluate the efficacy of virtual means of delivering or developing research strategies.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Develop skills that will enhance the fulfilment of ongoing and continuous learning and intellectual curiosity via independent learning exercises	PLO1	C3	Fundamental Domain
CLO2	Demonstrate , by use of role model examples, how research can help support leadership and better decision making within the community and in the public and private sectors	PLO1	C5	Fundamental Domain
CLO3	Develop an understanding of cross cultural contexts and the nuances/implications of cross cultural research.	PLO1	C2	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2	√											
CLO3	√											

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [2.5 Hours]	Lesson 1: OBE Discussion, Course outline brief discussion	Syllabus Walkthrough, Outcome Mapping Q&A	None	None
	Lesson 2: Introduction to research	Scientific Method Walkthrough, Concept Mapping	None	None

Week-2 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Literature Reviews	Critical Appraisal Exercise, Taxonomy Matrix Analysis	Class Test, Midterm exam	CLO1
	Lesson 2: Database Searches	Boolean Search Boolean Logic Drills, Query Optimization	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Communicating Research Outcomes	Abstract Structure Breakdown, Visual Presentation	Class Test, Midterm exam	CLO2
	Lesson 2: Communicating Research Outcomes Class Test - 1	Peer Review Simulation, Manuscript Formatting Drill	Class Test, Midterm exam	CLO2
Week-4 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Communicating Research Outcome	Citation Management Review, Plagiarism Verification	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Communicating Research Outcomes	Editorial Guideline Audit, Interactive Open Discussion	Class Test, Midterm exam	CLO2
Week-5 Lesson 1 & 2 [2.5 Hours]	Lesson 1: The Role of Theory in Research Design and Delivery	Theoretical Framework Mapping, Concept Boundary Design	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: The Role of Theory in Research Design and Delivery	Hypothesis Formulation Practice, Structural Variables Review	Class Test, Midterm exam	CLO2
Week-6 Lesson 1 & 2 [2.5 Hours]	Lesson 1: The Role of Theory in Research Design and Delivery	Model Validity Analysis, Predictive Flowcharting	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Research Ethics and Engaging Cultures	Institutional Review Board (IRB) Protocols, Case Debate	Class Test, Assignment, Midterm exam	CLO2
Week-7 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Research Ethics and Engaging Cultures	Informed Consent Drafting, Privacy Constraint Review	Class Test, Assignment, Midterm exam, Final exam	CLO2
	Lesson 2: Research Ethics and Engaging Cultures	Cross-Cultural Vulnerability	Class Test, Midterm exam, Final exam	CLO2

		Evaluation, Bias Diagnostics		
	Class Test -2			
Week-8 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Research Ethics and Engaging Cultures	Ethical Dilemma Analysis, Regulatory Code Review	Class Test, Mid exam	CLO2
	Lesson 2: Research Ethics and Engaging Cultures	Socio-Technical Impact Assessment, Open Discussion	Class Test, Mid exam	CLO2
Week-9 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Qualitative Methods I	Coding Frame Matrix, Theme Identification Workshop	Class Test, Assignment, Final exam	CLO2
	Lesson 2: Qualitative Methods I	Semi-Structured Interview Design, Dialogue Analysis	Class Test, Final exam	CLO2
Week-10 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Concept of Database integration (SQL, NoSQL)	Schema Translation Mapping, Relational vs Document Audit	Class Test, Final exam	CLO2
	Lesson 2: Qualitative Methods I	Transcript Content Interrogation, Triangulation Mapping	Assignment, Final exam	CLO2
Week-11 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Qualitative Methods II	Grounded Theory Modeling, Constant Comparative Analysis	Assignment, Final exam	CLO1
	Lesson 2: Qualitative Methods II	Ethnographic Field Blueprinting, Case Study Profiling	Assignment, Final exam	CLO1
Week-12 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Research Analysis	Analytical Metric Graphing, Variable Variance Tracking	Presentation, Final exam	CLO1
	Lesson 2: Research Analysis	Empirical Output Interrogation, Error Margin Calculation	Final exam	CLO1
Week-13 Lesson 1 & 2 [2.5 Hours]	Lesson 1: How to Write up a Research Project	Technical Document Blueprinting, Structural Layout Review	Final exam	CLO1
	Lesson 2: How to Write up a Research Project	Data Synthesis Drafting, Peer Editing Workshop	Final exam	CLO1

Week-14 Lesson 1 & 2 [2.5 Hours]	Lesson 1: How to Write up a Research Project Proposal	Proposal Budget Matrix, Timeline Gantt Charting	Final exam	CLO1
	Lesson 2: How to Write up a Research Project Proposal	Capstone Pitch Formulation, Defensive Argument Construction	Final exam	CLO1

Overall Assessment Scheme

Assessment Task	CLO's			Mark (Total=100)	PLO's	Mark (Total =70)
	CLO1	CLO2	CLO3		PLO1	
Attendance	--	--	--	7		
Class Test 1	5(15)			Avg 15		
Class Test 2		5(15)				
Class Test 3			5(15)			
Assignment		5		5		5
Presentation			8	8		
Midterm Exam	5	20		25	25	25
Final exam	15	10	15	40	40	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

Different research paper from google scholar

Reference Books/ Other Supplementary Readings

N/A.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 442 Management Information System

Course Code: SE 442/GEDS 347	CIE Marks: 60
ISCED: 0613-442	SEE Marks: 40
Course Title: Management Information System	Total Marks: 100
Semester:	

Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)
Prerequisite: N/A	
Course Type: GED	

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Management of Information Systems addresses the management of the information resources from a senior management viewpoint. This course provides students with an overview of the utilization of business application software and problem-solving using that software. The course provides instruction so students can use information technology to create collaborative environments for employees, customers, and suppliers; securely manage information assets including data and Internet-based technologies; manage system applications and data to best support business decisions operational excellence through ERP, SCM and CRM; and understand the fundamentals of how to build and manage local and global information systems. Further, this course will provide hands-on experience with data visualization, predictive analytics, prototype development and SQL data management.

Rationale of the Course

This course takes a senior management perspective on the strategic utilization and secure management of information resources within modern organizations. Students explore how to build collaborative environments and achieve operational excellence through enterprise architectures like ERP, SCM, and CRM. The curriculum covers the fundamentals of designing both local and global information systems while safeguarding critical digital assets. Through hands-on components, learners gain practical skills in data visualization, predictive analytics, SQL database management, and prototype development.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of data communications. More specifically,

- To develop knowledge on management information system
- To develop critical thinking, and problem-solving skills using IT

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Apply fundamental concepts and tools of technology and information management in work tasks and organizations.	PLO1	C3	Fundamental Domain

CLO2	Analyze business-related issues in the context of information systems in global business today.	PLO2	C4	Fundamental Domain
CLO3	Utilize technology and information management to attain strategic business objectives.	PLO5	C3,P1	Fundamental Domain
CLO4	Develop skills in technology and information management for personal development and managerial decision-making.	PLO9	A4	Social Thinking, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3					√							
CLO4									√			

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of OBE (CLO, PLO, Knowledge level)	Syllabus Walkthrough, Outcome Mapping Q&A	N/A	N/A
	Lesson 2: Overview of MIS, Business Model Canvas, MIS in Industry, TPS, MIS, DSS and ESS	Systems Tier Mapping, Canvas Canvas Dissection Workshop	Class Test, Midterm, Assignment	CLO1, CLO2, CLO3, CLO4
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: ERP, SCM and CRM	Enterprise Value Chain Analysis, Concept Mapping	Class Test, Midterm, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: ERP, Types, cost model	Lifecycle TCO Calculation, Implementation Case Study	Class Test, Midterm, Assignment	CLO1, CLO2, CLO3, CLO4
Week-3	Lesson 1: E- Commerce, Types of E-Commerce, E-Commerce Revenue Model, Payment gateways	Gateway Transaction Flowcharting, Revenue Matrix Review	Class Test, Midterm, Assignment	CLO1, CLO2, CLO3, CLO4

Lesson 1 & 2 [3 Hours]	Class Test -1			
	Lesson 2: Selling and marketing in the web	Funnel Conversion Analysis, Marketing Strategy Debate	Class Test, Midterm	CLO1, CLO2, CLO3, CLO4
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Usability, Tools and techniques. SUS, SUMI, QUIS and Super-Q	Metric Questionnaire Auditing, Usability Benchmarking	Class Test, Midterm	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Project Management, project management tools, WBS, Kanban, SCRUM	Agile Sprint Backlog Simulation, WBS Tree Decomposition	Class Test, Midterm	CLO1, CLO2, CLO3, CLO4
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Software Project maintenance, maintenance techniques	Legacy Code Diagnostics, Regression Risk Analysis	Class Test, Midterm	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Design user profile, identify implied requirements, identify user pattern	Persona Cohort Modeling, Interaction Trace Review	Class Test, Midterm	CLO1, CLO2, CLO3, CLO4
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Data visualization Concept. Hands-on Data visualization-I	Chart Typology Selection, Visual Design Mapping	Class Test, Midterm Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Hands-on Data visualization-II Class Test -2	Interactive Dashboard Building, Chart Presentation Practice	Class Test, Midterm, Assignment	CLO1, CLO2, CLO3, CLO4
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Predictive analytics fundamentals and concept	Regression Line Math, Feature Parameter Review	Class Test, Midterm, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Hands- on predictive analytics-II	Forecast Validation Lab, Error Metric Auditing	Class Test, Midterm, Assignment	CLO1, CLO2, CLO3, CLO4
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of Social media marketing techniques	Acquisition Metric Dissection, Campaign Blueprinting	Class Test, Final Assignment , Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Hands- On with social media analytics and chatbot development	Dialog Flow Tree Automation, Sentiment Stream Profiling	Class Test, Final Assignment , Presentation	CLO1, CLO2, CLO3, CLO4
Week-9	Lesson 1: Database software fundamentals, and how to use the	Relational Statement Binding, Query Construction Lab	Class Test, Final Assignment , Presentation	CLO1, CLO2, CLO3, CLO4

Lesson 1 & 2 [3 Hours]	Structured Query Language (SQL) to work with database			
	Lesson 2: Data visualization from MySQL database, Statistics for Industry research, ANN	DB-to-Chart Stream Configuration, Neural Pass Simulation	Class Test, Final, Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Hands- on NON-ML based statistics tools, t-test, ANOVA, Web analytics for industry, importance of web analysis.	Variance Math Calculations, Hypothesis Testing Lab	Final, Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Hands- on with web analytics like google analytics, google trends and other web analytics tools	Traffic Segment Isolation, Conversion Goal Tracking	Final, Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: E- learning fundamentals, learning strategies.	Instructional Scaffolding Mapping, Paradigm Analysis	Final, Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Learning tools and techniques. Like: Moodle, wake let, Mendeley, kahoot Class Test -3	Platform Engagement Configuration, Gamification Flow Design	Final, Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: CRM systems, customer segmentations, customer analytics, memory model	RFM Clustering Framework, Customer Lifetime Value Calculation	Final, Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Decision matrix, Prioritization, Traceability Matrix	MoSCoW Prioritization Matrix, Requirements Trace Table	Final, Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Over of Prototyping, Prototyping Development, Low fidelity prototype using PowerPoint, Paper Prototyping	Paper Interactive Blueprinting, Click-through Mapping	Final, Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Project distribution, Hands-on with video presentation	Technical Project Showcasing, Production Delivery Scripting	Final, Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Architecture Review	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on the final exam topic of the syllabus	Integrative Review, Board Problem Solving	Final	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =65)
	CLO 1	CLO 2	CLO 3	CLO4		PLO 1	PLO 2	PLO5	PLO9	
Attendance	--	--	--	--	7					
Class Test 1					15					
Class Test 2										
Class Test 3										
Assignment			5		5			5		5
Presentation				8	8			8		8
Midterm Exam	10	15			25	10	15			25
Final exam	15	25			40	15	25			40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Kenneth C. Laudon, & Jane P. Laudon, Management Information Systems: Managing the Digital Firm, 14th edition, Pearson, 2016, ISBN: 9789332548909.

Reference Books/ Other Supplementary Readings

Gary Schneider (2015). E-Business. 11th Ed. Cengage Learning. ISBN: 978-1285425436.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

RE 331 Embedded Programming

Course Code: RE331		CIE Marks: 60
ISCED: 0714/331		SEE Marks: 40
Course Title: Embedded Programming		Total Marks: 100
Semester:		
Credits: 2(Theory)	Contact Hours: 3 (Total weeks: 12)	
Prerequisite: SE 213, SE 121, SE 532		
Course Type: Major		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content

Introduction to Embedded Programming, Low-Level Programming for Embedded Systems, Real-Time Operating Systems (RTOS), Device Drivers and Hardware Interfacing, Debugging and Testing Techniques for Embedded Systems, Power Optimization Techniques in Embedded Programming, Communication Protocols for Embedded Systems, Embedded System Security, ROS, Important ROS Concepts, ROS Commands, Middleware and Simulation, Advanced ROS Concepts, Project Development, Compilers for Embedded systems, Dynamic voltage scaling, Dynamic power management, Interrupts, timers, and advanced features in microcontroller programming, Advanced motor control techniques, Overview of robotics middleware (e.g., Robot Middleware (RTM), DDS), Understanding control systems in robotics PID controllers and their applications, Trajectory planning and motion control, Hands-on projects on robot control systems, Sensors and actuators for robotics applications, Circuit design and analysis, prototyping circuits on breadboards and PCB's, Overview of Arduino platform, Arduino IDE and basic programming, GPIO programming with Arduino

Rationale of the Course

This course provides a comprehensive understanding of embedded programming, focusing on the development of software for embedded systems. Students will learn low-level programming techniques, real-time operating systems, device drivers, and debugging strategies. The Embedded System Design & Development course is a comprehensive program designed to equip participants with the knowledge and skills necessary to design, implement, and optimize embedded systems. Embedded systems are at the core of modern technology, powering a wide range of devices and applications, from consumer electronics to industrial machinery. The course aims to equip students with the skills necessary to design and develop efficient and reliable software for embedded systems.

Course Objectives

The objective of this course is to provide a solid conceptual understanding of the fundamentals of embedded programming. More specifically,

- To Introduce students to the fundamentals of embedded programming
- To familiarize students with low-level programming techniques for microcontrollers
- To provide an understanding of real-time operating systems and their application in embedded systems
- To develop skills in device driver development and hardware interfacing
- To equip students with debugging and testing strategies for embedded systems
- To explore power optimization techniques in embedded programming
- To introduce various communication protocols used in embedded systems
- To create awareness of security considerations in embedded programming
- To enhance practical skills through a project development component.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Identify the fundamentals of different advanced robotics components and its role in system development.	PLO4	C1	Fundamental Domain

CLO2	Apply robot kinematics to solve mathematical problems in relation to social perspective.	PLO6	C3	Fundamental Domain, Social Thinking, Personal Domain
CLO3	Implement robot software architecture using Robot Operating system including task scheduling and inter-task communication.	PLO3	C6	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1				√								
CLO2						√						
CLO3			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Teaching Learning Strategy	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to course syllabus, assessment methods, learning outcomes, deadlines, consultation hour, code of conduct, and other administrative procedures regarding the course.	Syllabus Walkthrough, Administrative Q&A	None	None
	Lesson 2: Role of embedded programming in system development. Overview of the embedded programming workflow	Hardware Lifecycle Review, Process Mapping	Class Test, Midterm	CLO1, CLO2
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Embedded Programming, Introduction to Microcontrollers and Microprocessors, Basic components and architecture of embedded systems.	System Architecture Breakdown, Comparative Review	Class Test, Midterm	CLO1, CLO2

	Lesson 2: Overview of programming languages for embedded systems, Overview of C Programming for Embedded Systems. Basics of C programming language, Data types, variables, and operators.	Cross-Language Paradigm Analysis, Syntax Drill	Class Test, Midterm	CLO1, CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Control flow and loops, Functions and pointers, Debugging and troubleshooting in C, Microcontroller Architecture and Programming (e.g. Arduino), Introduction to popular microcontrollers (e.g., AVR, ARM-based)	Memory Trace Tables, Pointer Logic Breakdown	Class Test, Midterm	CLO1, CLO2
	Lesson 2: Introduction to popular microcontrollers (e.g., AVR, ARM-based), Architecture and features of microcontrollers. Class Test-1	Register Map Deconstruction, Board Feature Analysis	Class Test, Midterm	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Microcontroller Architecture and Programming (e.g., Arduino), GPIO programming, Interrupts and timers.	Vector Interrupt Tracing, Timer Configuration Math	Class Test, Midterm	CLO2
	Lesson 2: GPIO programming, Interrupts and timers, Analog-to-digital conversion (ADC). Advanced Programming for Embedded Systems, Memory management in embedded systems, Advanced data structures	Bitwise Operations Mapping, ADC Resolution Math	Class Test, Midterm	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: File handling in embedded systems, Multi-file projects and modular programming.	Modular Source Decoupling, Header Structuring Workshop	Class Test, Midterm	CLO2
	Lesson 2: Real-Time Operating Systems (RTOS) basic, Introduction to RTOS, Tasks, scheduling, and priorities	Context-Switching Flowcharting, Preemption Priority Mapping	Class Test, Midterm	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Semaphore and Mutex concepts, RTOS for microcontrollers	Deadlock Scenario Analysis, Resource Sharing Matrix	Class Test, Midterm	CLO1, CLO2
	Lesson 2: Review class Class Test-2	Integrative Review, Board Problem Solving	Class Test, Midterm	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Microcontroller Communication Protocols, Serial communication (UART, SPI, I2C), Introduction to wireless communication (Bluetooth, Wi-Fi), Communication protocols for sensors and actuators	Bus Waveform Dissection, Protocol Frame Comparison	Class Test, Assignment, Final Exam, Presentation	CLO2, CLO3
	Lesson 2: Introduction to Robot Operating System (ROS), Overview of ROS and its architecture.	Distributed Publish-Subscribe Graphing, Graph Walkthrough	Class Test, Assignment, Final Exam, Presentation	CLO2, CLO3

Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of ROS and its architecture, ROS nodes, topics, and messages. Introduction to Robot Operating System (ROS) ROS services and actions.	Async/Sync Protocol Comparison, Pipeline Blueprinting	Class Test, Assignment, Final Exam, Presentation	CLO2, CLO3
	Lesson 2: Introduction to ROS programming in Python and C++. Important Concept of ROS, Coordinate Transformation (TF), File system (Configuration, Installation and workspace)	Transform Tree Analysis, Spatial Vector Matrix Mapping	Class Test, Assignment, Final Exam, Presentation	CLO2, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Creating Package, modifying (package.xml), build configuration (CMakeLists.txt) , Running the node. ROS Commands Shell Commands and Execution commands (roscd, rosls, roscore, rosrn etc)	Manifest Dependency Validation, CLI Tool Drills	Class Test, Assignment, Final Exam, Presentation	CLO3
	Lesson 2: Information, Catkin and Package Commands (roscd, rosls, roscore etc), ROS Middleware and Simulation, Building and launching ROS packages Class Test-3	Launch File Structure Analysis, Runtime Parameter Tuning	Assignment, Final Exam, Presentation	CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Integration of ROS with microcontrollers. ROS Middleware and Simulation Building and launching ROS packages.	Serial Bridge Stream Configuration, Node Telemetry Tuning	Assignment, Final Exam, Presentation	CLO3
	Lesson 2: Advanced ROS Concepts, ROS navigation stack, GMapping	SLAM Costmap Decomposition, Kinematic Flowcharting	Assignment, Final Exam	CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Robot perception and sensing	Sensor Point-Cloud Analysis, Laser Scan Transform Tracking	Assignment, Final Exam	CLO2, CLO3
	Lesson 2: Advanced ROS Concepts, ROS navigation stack, GMapping	Path-finding Algorithm Optimization, Localization Audit	Assignment, Final Exam	CLO2, CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson - 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Blueprint Review	Assignment, Presentation	CLO2, CLO3, CLO4
	Lesson 2: Review class	Comprehensive Blueprint Mapping, Core Problem Troubleshooting	Assignment, Final Exam	CLO2, CLO3

Overall Assessment Scheme

Assessment Task	CLO's	Mark		PLO's	Mark
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	CLO1	CLO2	CLO3	(Total=100)	PLO4	PLO6	PLO3	(Total =78)
Attendance	--	--	--	7				
Class Test 1	5(15)			Avg 15				
Class Test 2		5(15)						
Class Test 3			5(15)					
Assignment			5	5			5	5
Presentation			8	8			8	8
Midterm Exam	15	10		25	10	15		25
Final exam		15	25	40		15	25	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Yifeng Zhu, Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C, 1st Edition, E-man Press LLC, 2023, ISBN:978-0982692677

Reference Books/ Other Supplementary Readings

Michael Barr, Embedded C coding standard, BARR-C:2018 edition, Barr Group, 2018, ISBN 978-172112798

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

RE 332 Embedded Programming Lab

Course Code: RE 332		CIE Marks: 60
ISCED: 0613-332		SEE Marks: 40
Course Title: Embedded Programming Lab		Total Marks: 100
Semester:		
Credit Value: 1	Contact Hours: 15 (Total 14 weeks)	
Prerequisite: SE 213, SE 121, SE 532		
Course Type: Major		

CIE: Continuous Internal Evaluation

Course Content (from syllabus)

Introduction to Arduino and ESP Microcontrollers, Programming with Arduino and ESP, Real-time Operating Systems (RTOS) Implementation, Hardware Description Languages (HDLs) and FPGA Programming, IoT Concepts and Applications with Arduino and ESP, Wireless Communication Protocols in Embedded Systems, Building a Wireless Sensor Network, ROS Introduction, ROS Setup, ROS setup and run, Integrating Arduino and ESP with ROS, Project Development.

Rationale of the Course

This course provides a comprehensive introduction to embedded programming, focusing on the essential concepts of microcontroller and microprocessor programming, as well as the integration of the Robot Operating System (ROS) for advanced robotics applications. Students will gain practical skills in programming, interfacing with hardware, and designing embedded systems for real-world scenarios.

Course Objectives

The Objective of this Course is to provide a solid conceptual understanding of the fundamentals of embedded programming. More specifically,

- To Introduce students to the fundamentals of embedded programming
- To familiarize students with low-level programming techniques for microcontrollers
- To provide an understanding of real-time operating systems and their application in embedded systems
- To develop skills in device driver development and hardware interfacing
- To equip students with debugging and testing strategies for embedded systems
- To explore power optimization techniques in embedded programming
- To Introduce various communication protocols used in embedded systems
- To create awareness of security considerations in embedded programming
- To enhance practical skills through a project development component

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Understand the fundamentals of embedded programming and its role in system development.	PLO2	C2	Fundamental Domain
CLO2	Apply low-level programming techniques for microcontrollers, including register manipulation and interrupt handling.	PLO4	C3,P2	Fundamental Domain

CLO3	Write effective reports and design documentation for the advanced robotics solutions.	PLO10	A3	Social Domain, Thinking Domain, Personal Domain
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Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2				√								
CLO3										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 [1.5 Hours]	Discussion on OBE of Embedded Lab	Syllabus Walkthrough, Lab Safety Orientation	--	--
Week-2 [1.5 Hours]	Introduction to Arduino and ESP Microcontrollers. Programming with Arduino and ESP, Writing programs to interface with DHT11 and DHT22 sensors	Toolchain Installation, Host- to-Board Firmware Flashing	Lab Performance	CLO1, CLO2
Week-3 [1.5 Hours]	Programming with Arduino and ESP. Writing programs to interface with DHT11 and DHT22 sensors	Serial Monitoring Debugging, Logic Signal Scrutiny Analog Threshold Ingestion, Gas Calibration Math Drill	Viva, Lab Performance, Lab Final	CLO2

Week-4 [1.5 Hours]	Programming with Arduino and ESP. Writing programs to interface with MQ and Grove-Dust sensors	FreeRTOS Task Spawning, Preemptive Context-Switching Lab Verilog Register-Transfer-Level (RTL) Compilation, Gate Bitstream Synthesis	Viva, Lab Performance, Lab Final	CLO1, CLO2
Week-5 [1.5 Hours]	Real-time Operating Systems (RTOS) Implementation. Task scheduling and management using RTOS, Implementing a real-time task scheduler in Arduino and ESP.	Edge-Node Topography Layout, Dynamic Client-Server Mapping Broker Session Establishment, Publish-Subscribe Topic Wiring	Lab Performance, Viva, Lab Final	CLO1, CLO2
Week-6 [1.5 Hours]	Hardware Description Languages (HDLs) and FPGA Programming. Basics of FPGA programming, Designing and programming a simple FPGA-based system.	RF Beacon Scanning, Socket Stream Configuration Access Point (AP) Handshaking, OTA Firmware Scripting	Lab Performance, Viva, Lab Final	CLO1, CLO2
Week-7 [1.5 Hours]	IoT Concepts and Applications with Arduino and ESP. Understanding the basics of IoT and its applications	Multi-hop Mesh Tree Mapping, Time-Series Cloud Ingestion Linux Environment Validation, Catkin Workspace Sourcing	Lab Performance	CLO2
Week-8 [1.5 Hours]	IoT Concepts and Applications with Arduino and ESP. Implementing IoT communication protocols (MQTT, HTTP) Building a basic IoT-enabled embedded system.	Rosserial Hardware Bridge Wiring, String/Integer Array Binding	Lab Performance, Viva	CLO3

Week-9 [1.5 Hours]	Wireless Communication Protocols in Embedded Systems. Exploring wireless communication protocols (Bluetooth, Wi-Fi).	Syllabus Walkthrough, Lab Safety Orientation Toolchain Installation, Host-to-Board Firmware Flashing	Lab Performance, Viva	CLO3
Week-10 [1.5 Hours]	Wireless Communication Protocols in Embedded Systems. Implementing wireless communication in Arduino and ESP projects. Developing IoT applications with wireless connectivity.	Serial Monitoring Debugging, Logic Signal Scrutiny Analog Threshold Ingestion, Gas Calibration Math Drill	Lab Performance, Viva	CLO2, CLO3
Week-11 [1.5 Hours]	Building a Wireless Sensor Network. Extending wireless communication to create a sensor network. Analyzing and visualizing sensor data in real-time.	FreeRTOS Task Spawning, Preemptive Context-Switching Lab Verilog Register-Transfer-Level (RTL) Compilation, Gate Bitstream Synthesis	Lab Performance, Viva, Lab Final	CLO2, CLO3
Week-12 [1.5 Hours]	ROS Introduction. Overview of ROS and its components. ROS Setup, Setting up the ROS environment and configuring Arduino and ESP and understanding the ROS ecosystem	Edge-Node Topography Layout, Dynamic Client-Server Mapping Broker Session Establishment, Publish-Subscribe Topic Wiring	Lab Performance, Viva	CLO2, CLO3

Week-13 [1.5 Hours]	Interfacing Arduino and ESP with ROS nodes. Integrating Arduino and ESP with ROS. Publishing and subscribing to topics in ROS.	RF Beacon Scanning, Socket Stream Configuration Access Point (AP) Handshaking, OTA Firmware Scripting	Lab Performance, Viva, Lab Final	CLO2, CLO3
Week-14 [1.5 Hours]	Integrating Arduino and ESP with ROS. Debugging and testing ROS-controlled systems	Multi-hop Mesh Tree Mapping, Time-Series Cloud Ingestion	Lab Performance, Viva, Lab Final	CLO2, CLO3

Overall Assessment Scheme and Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10	--	--
Lab Report	25	CLO1	PLO2
Lab Performance	25	CLO2	PLO4
Lab Final	40	CLO2	PLO4
		CLO3	PLO10
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Yifeng Zhu, Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C, 1st Edition, E-man Press LLC, 2023, ISBN:978-0982692677

Reference Books/ Other Supplementary Readings

Michael Barr, Embedded C coding standard, BARR-C:2018 edition, Barr Group, 2018, ISBN 978-1721127986

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Sensors, Arduino

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

RE 411 Embedded System Design and Development

Course Code: RE411		CIE Marks: 60
ISCED: 0714/411		SEE Marks: 40
Course Title: Embedded System Design and Development		Total Marks: 100
Semester:		
Credit Value: 2 (Theory)	Contact Hours: 3 (Total weeks: 12)	
Prerequisite: RE331, RE332		
Course Type: Major		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content

Overview of Embedded Systems, Importance of Embedded Systems in various applications, Overview of robotics applications and industries, Introduction to Microcontrollers and Microprocessors, types, Understanding electronic components used in robotics, Sensors and actuators for robotics applications, Circuit design and analysis, prototyping circuits on breadboards and PCB's, Overview of Arduino platform, Arduino IDE and basic programming, GPIO programming with Arduino, Sensor interfacing with Arduino and ESP boards, analog electronics, digital electronics, tri-state outputs and logic gates, VLSI and Integrated circuit design, Electronic design Automation tools, Embedded Firmware, Understanding the kinematics of robotic systems, Task level concurrency management, High level optimizations (Loop tiling/blocking, loop splitting, Array folding), Compilers for Embedded systems, Dynamic voltage scaling, Dynamic power management, Interrupts, timers, and advanced features in microcontroller programming, Advanced motor control techniques, Overview of robotics middleware (e.g., Robot Middleware (RTM), DDS), Understanding control systems in robotics PID controllers and their applications, Trajectory planning and motion control, Hands-on projects on robot control systems.

Rationale of the Course

This course provides a comprehensive understanding of embedded programming, focusing on the development of software for embedded systems. Students will learn low-level programming techniques, real-time operating systems, device drivers, and debugging strategies. The Embedded System Design & Development course is a comprehensive program designed to equip participants with the knowledge and skills necessary to design, implement, and optimize embedded systems. Embedded systems are at the core of modern technology, powering a wide range of devices and applications, from consumer electronics to industrial machinery. The course aims to equip students with the skills necessary to design and develop efficient and reliable software for embedded systems.

Course Objectives

The Objective of the fundamentals of embedded programming is To provide a solid conceptual understanding. More specifically,

- To Introduce students to the fundamentals of embedded programming
- To familiarize students with low-level programming techniques for microcontrollers
- To provide an understanding of real-time operating systems and their application in embedded systems
- To develop skills in device driver development and hardware interfacing
- To equip students with debugging and testing strategies for embedded systems
- To explore power optimization techniques in embedded programming
- To Introduce various communication protocols used in embedded systems
- To create awareness of security considerations in embedded programming
- To enhance practical skills through a project development component

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Identify the fundamentals of embedded programming and its role in system development.	PLO2	C1	Fundamental Domain
CLO2	Apply low-level programming techniques for microcontrollers, including register manipulation and interrupt handling in relation to social perspective.	PLO7	C3	Fundamental, Social Thinking, Personal Domain
CLO3	Implement real-time operating systems for embedded systems, including task scheduling and inter-task communication.	PLO3	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2							√					
CLO3			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to the course syllabus, assessment methods, learning outcomes, deadlines, consultation hours, code of conduct, and other administrative procedures regarding the course.	Syllabus Walkthrough, Administrative Q&A	None	None

	Lesson 2: Overview of Embedded Systems, Importance of Embedded Systems in various applications. Overview of robotics applications and industries.	Enterprise Systems Review, Industry Taxonomy Mapping	Class Test, Midterm	CLO1, CLO2
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Embedded Systems, Basic components and architecture of embedded systems. Introduction to Embedded Systems, Basic components and architecture of embedded systems.	System Architecture Breakdown, Comparative Review	Class Test, Midterm	CLO2
	Lesson 2: Basics of Electronics and Microcontrollers Introduction to Microcontrollers and Microprocessors, types	CPU Architecture Dissection, Block Diagram Mapping	Class Test, Midterm	CLO1,CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Basics of Electronics and Microcontrollers, Understanding electronic components used in robotics, Sensors and actuators for robotics applications	Transducer Paradigm Classification, Signal Flow Analysis	Class Test, Midterm	CLO1,CLO2
	Lesson 2: Basics of Electronics and Microcontrollers Circuit design and analysis, prototyping circuits on breadboards and PCB's. Class Test-1	Schematic CAD Walkthrough, Board Routing Layout Analysis	Class Test, Midterm	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Embedded System Development boards (Ex. Arduino and ESP boards) Overview of Arduino platform, Arduino IDE and basic programming, GPIO programming with Arduino	IDE Compiler Toolchain Review, Live Syntax Tracing	Class Test, Midterm	CLO2
	Lesson 2: Analog and Digital Electronic Components Analog electronics, digital electronics, tri-state outputs and logic gates.	Boolean Gate Truth Mapping, Propagation Delay Dissection	Class Test, Midterm	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Analog and Digital Electronic Components, VLSI and Integrated circuit design, Electronic design Automation tools, Embedded Firmware, Raspberry Pi Basics Introduction to Raspberry Pi, Operating system installation and configuration	Linux Kernel Sourcing Analysis, VLSI Tool Pipeline Walkthrough	Class Test, Midterm	CLO2
	Lesson 2: Raspberry Pi Basics GPIO programming with Python, Basic projects with Raspberry Pi.	Python Hardware Library Analysis, Threading Script Review	Class Test, Midterm	CLO1, CLO2

Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Robotics Platforms. Overview of popular robotics platforms and kits. Understanding the kinematics of robotic systems.	Matrix Transformation Algebra, DH Parameter Modeling	Class Test, Midterm	CLO1,CLO2
	Lesson 2: Embedded Software General requirements, Real-time OS. Embedded Software, Virtual machines, Resource access protocols. Class Test-2	Priority Inversion Simulation, Context Switching Analysis	Class Test, Midterm	CLO1,CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Application Mapping Problem definition, scheduling in real-time systems, scheduling algorithms.	Rate-Monotonic Math Exercises, Timeline Scheduling Bounds	Class Test, Assignment, Final Exam	CLO2, CLO3
	Lesson 2: Application Mapping Hardware and software partitioning, Mapping to heterogeneous multi-processors.	Co-design Trade-off Optimization, Hardware/Software Boundary Allocation	Class Test, Assignment, Final Exam, Presentation	CLO2, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Optimization, Task level concurrency management, High level optimizations (Loop tiling/blocking, loop splitting, Array folding)	Memory Cache Locality Analysis, Loop Transformation Math	Class Test, Assignment, Final Exam, Presentation	CLO2, CLO3
	Lesson 2: Advanced Microcontroller Programming for Embedded Systems	Bare-Metal Pipeline Diagnostics, Direct Memory Access Review	Class Test, Assignment, Final Exam, Presentation	CLO2, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Advanced Microcontroller Programming for Embedded Systems Interrupts, timers, and advanced features in microcontroller programming Communication protocols for robotics applications (e.g., I2C, SPI)	Vector Table Routing Analysis, Bus Frame Protocol Dissection	Class Test, Assignment, Final Exam, Presentation	CLO3
	Lesson 2: Advanced Microcontroller Programming for Embedded Systems: Advanced motor control techniques. Developing custom drivers for robotic sensors and actuators. Class Test-3	Register Bitwise Script Review, H-Bridge Logic Gating	Class Test, Assignment, Final Exam, Presentation	CLO2,CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Advanced ROS Programming Advanced concepts in ROS, including services and actions Navigation stack in ROS for robotic movement	Sync/Async Callback Tracing, Costmap Inflation Layout	Class Test, Assignment, Final Exam	CLO3
	Lesson 2: Advanced ROS Programming, Integrating sensors and perception in ROS, Real-world robotics applications with ROS	Sensor Point-Cloud Transformation, Node Telemetry Graphing	Class Test, Assignment, Final Exam	CLO3
Week-11	Lesson 1: Robotics Middleware and Simulation, Overview of robotics middleware (e.g., Robot Middleware	Pub-Sub Wire QoS Analysis, Middleware Taxonomy Review	Class Test, Assignment, Final Exam	CLO3

Lesson 1 & 2 [3 Hours]	(RTM), DDS), Simulation tools for testing and debugging robotic systems			
	Lesson 2: Robotics Middleware and Simulation: Building and simulating robotic scenarios, Integrating simulation with physical robots.	Physics Engine Constraint Mapping, Kinematic Joint Auditing	Class Test, Assignment, Final Exam	CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Week Session	Guest Lecture / Industry Expert Talk, Compliance Roadmap Review	Assignment, Presentation	CLO2, CLO3, CLO4
	Lesson 2: Robot Control Systems: Understanding control systems in robotics, PID controllers and their applications, Trajectory planning and motion control, Hands-on projects on robot control systems.	Laplace Domain Stability Analysis, Error Derivative Calculus	Class Test, Assignment, Final Exam	CLO2, CLO3

Overall Assessment Scheme

Assessment Task	CLO's			Mark (Total=100)	PLO2	PLO's		Mark (Total =78)
	CLO1	CLO2	CLO3			PLO7	PLO3	
Attendance	--	--	--	7				
Class Test 1	15			Avg 15				
Class Test 2		15						
Class Test 3			15					
Assignment			5	5			5	5
Presentation			8	8			8	8
Midterm Exam	15	10		25	10	15		25
Final exam		15	25	40		15	25	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Jagdeep Singh, & Srinivasa Raghavendra, A Textbook on Embedded System Design for Engineering Students, self-published, 2023.

Reference Books/ Other Supplementary Readings

Thomas Bräunl, Embedded Robotics: From Mobile Robots to Autonomous Vehicles with Raspberry Pi and Arduino, 4th edition, Springer, 2022, ISBN 978-9811608032

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

RE 412 Embedded System Design and Development Lab

Course Code: RE412		CIE Marks: 60
ISCED: 0613-412		SEE Marks: 40
Course Title: Embedded System Design and Development Lab		Total Marks: 100
Semester:		
Credit Value: 1	Contact Hours: 1.5 (Total 14 weeks)	
Prerequisite: RE 331, RE332		
Course Type: Major		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Microcontroller Architecture and I/O Interfacing, Advanced Programming with Arduino and ESP, Embedded Programming Techniques, Actuator Control and Closed-Loop Systems, Implementing Navigation in ROS Projects, Advanced ROS Features and Applications, Security Measures in Embedded Systems, Optimizing Code for Performance, Project-Based Experiments and Project Development.

Rationale of the Course

This course provides a comprehensive introduction to embedded programming, focusing on the essential concepts of microcontroller and microprocessor programming, as well as the integration of Robot Operating System (ROS) for advanced robotics applications. Students will gain practical skills in programming, interfacing with hardware, and designing embedded systems for real-world scenarios.

Course Objectives

The objective of this course is to provide a solid conceptual understanding of the fundamentals of embedded programming. More specifically,

- To Introduce students to the fundamentals of embedded programming
- To familiarize students with low-level programming techniques for microcontrollers
- To provide an understanding of real-time operating systems and their application in embedded systems
- Develop skills in device driver development and hardware interfacing
- Equip students with debugging and testing strategies for embedded systems
- Explore power optimization techniques in embedded programming
- Introduce various communication protocols used in embedded systems
- Create awareness of security considerations in embedded programming
- Enhance practical skills through a project development component

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Understand the fundamentals of embedded programming and its role in system development.	PLO1	C1	Fundamental Domain
CLO2	Understand the fundamentals of embedded programming and its role in system development.	PLO4	C3, P2	Fundamental Domain
CLO3	Design real-time operating systems for embedded systems, including task scheduling and inter-task communication & Write effective reports and design documentation	PLO10	A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2				√								
CLO3										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 [1.5 Hours]	Discussion on OBE of Embedded System Design & Development Lab	Syllabus Walkthrough, Lab Safety Orientation General-Purpose Input/Output (GPIO) Pin Mapping, Logic Gating Implementation	--	--
Week-2 [1.5 Hours]	Microcontroller Architecture and I/O Interfacing. I/O interfacing with basic input and output devices	Sensor Hardware Setup, Serial Communication Troubleshooting Operational Amplifier Conditioning, Raw Value Ingestion Scripting	Lab Performance	CLO1
Week-3 [1.5 Hours]	Advanced Programming with Arduino and ESP. Selection and interfacing of various sensors (temperature, humidity, etc.)	Non-blocking Event Intercepting, Timer Counter Optimization Dynamic Allocation Auditing, Stack-Overflow Protection Lab	Lab Performance	CLO1,CLO2
Week-4 [1.5 Hours]	Advanced Programming with Arduino and ESP. Signal conditioning and processing techniques, Hands-on projects integrating sensors into embedded systems.	H-Bridge Driver Wiring, Encoder Interfacing Mechanics Proportional-Integral-Derivative (PID) Parameter Tuning, Error Correction Testing	Viva,Lab Performance, Lab Final	CLO1, CLO2
Week-5 [1.5 Hours]	Embedded Programming Techniques. Advanced programming techniques for embedded systems. Embedded Programming Techniques. Developing efficient code for time-sensitive applications.	Robot Kinematic Vector Mapping, Transformation Tree Configuration Costmap Configuration Tuning, Gazebo	Lab Performance, Viva, Lab Final	CLO1, CLO2

		Simulator Obstacle Traversal		
Week-6 [1.5 Hours]	Embedded Programming Techniques. Memory management and optimization strategies. Hands-on coding exercises for real-world scenarios.	Client-Server Message Tracing, Async Goal Handle Binding Multi-node Pipeline Orchestration, Graph Telemetry Verification	Lab Performance, Viva, Lab Final	CLO2
Week-7 [1.5 Hours]	Actuator Control and Closed-Loop Systems. Understanding different types of actuators (motors, servos). Closed-loop control systems and feedback mechanisms.	Cipher Stream Encryption, Secure Socket Layer Initialization	Lab Performance, Viva, Lab Final	CLO1, CLO2
Week-8 [1.5 Hours]	Actuator Control and Closed-Loop Systems. Implementing control algorithms for actuators. Developing a closed-loop system with real-time feedback.	Syllabus Walkthrough, Lab Safety Orientation General-Purpose Input/Output (GPIO) Pin Mapping, Logic Gating Implementation	Lab Performance, Viva, Lab Final	CLO2
Week-9 [1.5 Hours]	Implementing Navigation in ROS Projects. Understanding navigation concepts in ROS	Sensor Hardware Setup, Serial Communication Troubleshooting Operational Amplifier Conditioning, Raw Value Ingestion Scripting	Lab Performance, Viva, Lab Final	CLO2
Week-10 [1.5 Hours]	Implementing Navigation in ROS Projects. Implementing path planning and obstacle avoidance. Simulating and testing ROS-controlled navigation systems	Non-blocking Event Intercepting, Timer Counter Optimization Dynamic Allocation Auditing, Stack-Overflow Protection Lab	Lab Performance, Viva, Lab Viva	CLO2, CLO3
Week-11 [1.5 Hours]	Advanced ROS Features and Applications. Exploring advanced features of ROS (e.g., services, actions) Integrating additional sensors and actuators with ROS	H-Bridge Driver Wiring, Encoder Interfacing Mechanics Proportional-Integral-Derivative (PID) Parameter Tuning, Error Correction Testing	Lab Performance, Viva, Lab Final	CLO2, CLO3
Week-12 [1.5 Hours]	Advanced ROS Features and Applications Developing a more complex ROS-controlled system	Robot Kinematic Vector Mapping, Transformation Tree Configuration Costmap Configuration Tuning, Gazebo	Lab Performance, Viva, Lab Final	CLO3

		Simulator Obstacle Traversal		
Week-13 [1.5 Hours]	Security Measures in Embedded Systems. Implementing cryptographic techniques for secure communication	Client-Server Message Tracing, Async Goal Handle Binding Multi-node Pipeline Orchestration, Graph Telemetry Verification	Lab Performance, Viva, Lab Final	CLO2, CLO3
Week-14 [1.5 Hours]	Optimizing Code for Performance , Analyzing and optimizing code for better performance, and Implementing techniques to reduce memory usage. Profiling and benchmarking Arduino and ESP applications.	Cipher Stream Encryption, Secure Socket Layer Initialization	Lab Performance, Viva, Lab Final	CLO3

Overall Assessment Scheme and Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10	--	--
Lab Report	25	CLO1	PLO1
Lab Performance	25	CLO2	PLO4
Lab Final	40	CLO2	PLO4
		CLO3	PLO10
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Jagdeep Singh, & Srinivasa Raghavendra, A Textbook on Embedded System Design for Engineering Students, self-published, 2023.

Reference Books/ Other Supplementary Readings

Thomas Bräunl, Embedded Robotics: From Mobile Robots to Autonomous Vehicles with Raspberry Pi and Arduino, 4th edition, Springer, 2022, ISBN 978-9811608032

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Sensors, Microcontroller, required elements

Exam Dates

According to the Examination Schedule



Department of Software

Engineering (SWE)

Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

RE 421 Robotic Process Automation Design & Development

Course Code: RE 421	CIE Marks: 60
ISCED: 0714-421	SEE Marks: 40
Course Title: Robotic Process Automation Design & Development	Total Marks: 100

Semester:		
Credit Value: 2 (Theory)	Contact Hours: 3 (Total weeks: 12)	
Prerequisite: SE 442, SE 331, RE331, RE332, RE 411, RE 412		
Course Type: Major		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

This course includes history of Automation; Story of Work; Introduction to RPA; RPA vs Automation; RPA and AI; RPA and emerging ecosystem; Industries best suited for RPA; Processes that can be automated, UiPath and its Products; Robots and their Types; Studio Overview; Orchestrator; UiPath Studio Installation & Updating; The User Interface; Features of Studio; Building 'Hello World' Robot, Variables and their Types; Variables Panel; Scope of Variable; Arguments; Arguments Panel; Argument Directions; Arguments vs. Variables, UI interactions; Input Actions and Input Methods; Containers; Recording and its types; Selectors and their types; Anchors; Debugging Selectors, Sequences; Control Flow and Its Types; Decision Control; Loops; Other Control Flow Activities; Flowcharts; Error Handling, Data Manipulation and Its importance; String Manipulations ; Data Table Manipulations; Collection, Types and Manipulations, Extraction and Its Techniques; Automation Techniques; Orchestrator Overview; Publishing a Robot to Orchestrator; Orchestrator Functionalities.

Rationale of the Course

The Robotic Process Automation (RPA) Design & Development course offers comprehensive knowledge and professional-level skills focused on developing and deploying software robots. The course assumes no prior knowledge of RPA. This course aims at providing an understanding of basic concepts of Robotic Process Automation to University students. It introduces vital RPA Design and Development strategies and methodologies specified in the context of UiPath products. The student undergoing the course shall develop the competence to design and develop a robot for a defined process. The course is developed for Instructor-led delivery.

Course Objectives

- To prepare students to be Junior RPA Developers.
- To learn the basic concepts of Robotic Process Automation.
- To develop familiarity and deep understanding of UiPath tools.
- To develop the ability to independently design and create robots for business processes.
- To develop skills required to pass the UiPath RPA Associate v1.0 Exam successfully.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
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CLO1	Identify Robotics Process Automation (RPA) Fundamentals to formulate efficient RPA Programming to reach sustained conclusion.	PLO 4	C5	Fundamental Domain
CLO2	Apply contextual knowledge and ethical reasoning in RPA development to address societal concerns.	PLO 6	C3	Fundamental Domain, Social Thinking, Personal Domain
CLO3	Design RPA Solutions and Deployment of RPA workflows using UiPath Attended and Unattended robots for efficient automation.	PLO 3	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1				✓								
CLO2						✓						
CLO3			✓									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1	Lesson 1: Introduction to OBE, Introduction to course syllabus, assessment methods, learning outcomes, deadlines, consultation hour, code of conduct, and other administrative procedures regarding the course.	Syllabus Walkthrough, Administrative Q&A	--	--

Lesson 1 & 2 [3 Hours]	Lesson 2: History of Automation: Explore the evolution of automation from its early beginnings to modern-day applications. Story of Work; Introduction to RPA: Understand the changing landscape of work and the role of RPA in revolutionizing business processes.	Business Evolution Timeline, Paradigm Analysis	Class Test, Midterm	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: RPA vs Automation: Differentiate between traditional automation methods and Robotics Process Automation. RPA and AI: Explore the intersection of RPA and Artificial Intelligence, and how they complement each other in automation.	API vs UI Driving Comparison, Cognitive Automation Mapping	Class Test,, Midterm	CLO1
	Lesson 2: RPA and emerging ecosystem: Discover the expanding ecosystem surrounding RPA technology and its implications for businesses. Industries best suited for RPA: Identify industries and sectors that can benefit most from implementing RPA solutions. Class Test-1	Corporate Bottleneck Analysis, Industry Matrix Scoring	Class Test, Midterm	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: UiPath and its Products: Learn about UiPath's suite of RPA products and their functionalities.	Product Stack Ecosystem Walkthrough, Platform Mapping	Class Test, Midterm	CLO1
	Lesson 2: Robots and their Types; Studio Overview; Orchestrator: Understand the different types of robots used in RPA, get an overview of UiPath Studio, and learn about UiPath Orchestrator.	Attended vs Unattended Deconstruction, Control Architecture Review	Class Test,, Midterm	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: UiPath Studio Installation & Updating; The User Interface: Install and update UiPath Studio, and familiarize yourself with its user interface.	Panel Navigation Guide, Installation Environment Audit	Class Test, Assignment, Midterm	CLO1
	Lesson 2: Features of Studio; Building 'Hello World' Robot: Explore the features of UiPath Studio and build your first 'Hello World' robot.	Activity Drag-and-Drop Walkthrough, Execution Trace Checking	Class Test, Midterm	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Variables and their Types; Variables Panel: Understand the concept of variables in RPA and learn how to work with them in UiPath Studio.	Data Type Assignment Drill, Panel Allocation Practice	Class Test, Midterm	CLO2
	Lesson 2: Scope of Variable: Explore the scope of variables and how they are used in different contexts within RPA workflows. Class Test-2	Container Visibility Mapping, Boundary Logic Review	Midterm	CLO2

Week-6 Lesson 1 & 2 [3 Hours]	Lesson1: Arguments; Arguments Panel: Learn about arguments in RPA and how they facilitate communication between different parts of a workflow. Argument Directions; Arguments vs. Variables: Understand the directionality of arguments and differentiate between arguments and variables.	In/Out Directional Flowcharting, Workflow Invocation Mapping	Class Test, Midterm	CLO2
	Lesson 2: UI interactions: Explore how RPA robots interact with user interfaces to perform tasks such as data entry and navigation.	Selector Target Mapping, Target Element Verification	Assignment, Midterm	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Input Actions and Input Methods; Containers: Learn about different input actions and methods used in RPA, and understand the concept of containers.	Hardware Driver vs API Simulation, Scope Window Attachment	Class Test, Final Exam	CLO2
	Lesson 2: Recording and its types: Understand the concept of recording in RPA and explore different types of recording methods.	Basic/Desktop Script Scraping, Sequence Generation Walkthrough	Class Test, Assignment, Final Exam	CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Selectors and their types: Learn about selectors in RPA and the different types used to identify and interact with UI elements.	XML Node Dissection, Dynamic Tag Customization	Class Test, Assignment, Final Exam	CLO2
	Lesson 2: Anchors: Explore the concept of anchors and their role in ensuring reliable automation in RPA workflows.	Visual Element Pairing, Interface Instability Modeling	Class Test, Assignment, Final Exam	CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Debugging Selectors: Learn techniques for debugging selectors to troubleshoot issues and improve the performance of RPA robots.	UI Explorer Property Patching, Live Validation Diagnostics	Class Test, Final Exam	CLO2
	Lesson 2: Sequences: Understand the concept of sequences in RPA workflow design and how they organize and structure automation tasks. Control Flow and Its Types: Explore various control flow techniques used in RPA, such as decision control and loops.	Linear Logic Sequencing, Branching Strategy Mapping	Class Test, Final Exam	CLO2
Week-10 Lesson 1 & 2 [3 Hours]	Lesson-1: Decision Control: Master decision control techniques in RPA and learn how to implement conditional logic in workflows. Loops; Other Control Flow Activities: Explore different types of loops and other control flow activities used for iterative automation tasks. Class Test-3	If/Switch Branch Auditing, Index Iteration Tracing	Class Test, Final Exam	CLO2

	Lesson 2: Flowcharts; Error Handling: Understand the role of flowcharts in RPA process design and learn error handling strategies for robust automation. Data Manipulation and Its importance, Explore techniques for manipulating data in RPA workflows and understand the importance of data management in automation.	Try-Catch Block Architecture, Data Type Cast Verification	Final Exam	CLO2
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: String Manipulations: Learn techniques for manipulating strings in RPA, such as concatenation, substring extraction, and string formatting. Data Table Manipulations: Explore methods for working with data tables in RPA, including sorting, filtering, and aggregating data.	Regex Pattern Extraction, Row-Filter Expression Drafting	Assignment, Final Exam	CLO2
	Lesson 2: Collection, Types and Manipulations: Understand the concept of collections in RPA and explore different types of collections and their manipulations. Extraction and Its Techniques: Learn techniques for extracting data from various sources in RPA workflows, such as web pages, documents, and databases. Automation Techniques: Dive into advanced automation techniques for complex RPA workflows, including exception handling and retry mechanisms.	List/Dictionary Parsing, Structured Web Scraping Wizard	Final Exam	CLO2
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Week Session	Guest Lecture / Industry Expert Talk, Production Blueprint Review	Assignment, Presentation, Final Exam	CLO2, CLO3, CLO4
	Lesson 2: Orchestrator Overview: Gain an overview of UiPath Orchestrator and its role in managing and monitoring RPA deployments. Publishing a Robot to Orchestrator: Learn how to publish a robot from UiPath Studio to UiPath Orchestrator for centralized management, Orchestrator Functionalities: Explore various functionalities offered by UiPath Orchestrator, such as scheduling, monitoring, and reporting.	Asset and Queue Blueprinting, Package Release Deployment	Final Exam	CLO3

Overall Assessment Scheme

Assessment Task	CLO's			Mark (Total=100)	PLO's			Mark (Total=78)
	CLO1	CLO2	CLO3		PLO2	PLO6	PLO12	
Attendance	--	--	--	7				
Class Test 1	15	--	--	Avg.				

Class Test 2	--	15	--	15				
Class Test 3	--	--	15					
Assignment	--	--	5	5			5	5
Presentation	--	--	8	8			8	8
Midterm Exam	10	15	--	25	10	15		25
Final exam	--	25	15	40		25	15	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Ankur M. Tripathi, Learning Robotic Process Automation, Packt Publishing, 2021, ISBN: 978-1801074857.

Reference Books/ Other Supplementary Readings

Richard Murdoch, Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant, 1st edition, self-published, 2018, ISBN 978-1983036835

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 2.0)

RE 422 Robotic Process Automation Design & Development Lab

Course Code: RE 422		CIE Marks: 60
ISCED: 0714-422		SEE Marks: 40
Course Title: Robotic Process Automation Design & Development Lab		Total Marks: 100
Semester: Spring 2025		
Credit Value: 1	Contact Hours: 1.5 (Total 14 weeks)	
Prerequisite: SE 442, SE 331, RE331, RE332, RE 411, RE 412		
Course Type: Major		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

Robotics Process Automation (RPA) Lab course introduces students to the fundamental principles and practical application of RPA using UiPath Studio. Beginning with an overview of automation history and the RPA landscape, students delve into UiPath Studio installation, user interface navigation, and the construction of foundational robots. Subsequent weeks cover essential topics such as UI interactions, control flow, error handling, data manipulation, and automation techniques. Advanced concepts, including the integration of RPA with AI, are explored, followed by an examination of industry-specific applications and case studies. The course culminates with a focus on Orchestrator functionality, exploration of emerging trends in RPA, and hands-on preparation and presentation of a final project, ensuring students acquire comprehensive skills in designing, implementing, and managing RPA solutions for real-world scenarios

Rationale of the Course

In today's rapidly evolving technological landscape, organizations are increasingly adopting Robotics Process Automation (RPA) to streamline business processes, enhance efficiency, and reduce operational costs. This lab course is designed to provide students with practical, hands-on experience in utilizing UiPath Studio, a leading RPA platform, to automate tasks and processes. By gaining proficiency in RPA, students will be well-equipped to contribute to the digital transformation efforts of businesses across various industries.

Course Objectives

- To prepare students to be Junior RPA Developers.
- To learn the basic concepts of Robotic Process Automation.
- To develop familiarity and deep understanding of UiPath tools.
- To develop the ability to independently design and create robots for business processes.
- To develop skills required to pass the UiPath RPA Associate v1.0 Exam successfully.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Apply the principles of mathematics, science, and engineering fundamentals to design and develop robotics process automation solutions using UiPath.	PLO1	C3	Fundamental Domain
CLO2	Conduct experiments, analyze data, and synthesize information to design and implement effective robotics process automation systems while considering public health, safety, and environmental concerns.	PLO4	C4, P3	Fundamental Domain
CLO3	Write effective reports and design documentation for the advanced robotics solutions.	PLO10	A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	✓											
CLO2				✓								
CLO3										✓		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Discussion on OBE of Robotics Process Automation, Introduction to RPA and UiPath - Install UiPath Studio on students' computers. - Explore UiPath Studio interface and features. - Create a simple "Hello World" automation workflow. - Navigate UiPath documentation and online resources.	IDE Toolchain Setup, Activity Drag-and-Drop Walkthrough	CLO1	Lab Performance

Week-2 [1.5 Hours]	Basic Concepts in UiPath - Build a basic automation workflow for data entry tasks. - Use data scraping techniques to extract information from a website. - Implement error handling mechanisms in automation workflows.	Dynamic UI Targeting, Try-Catch Block Architecture	CLO1, CLO3	Lab Performance
Week-3 [1.5 Hours]	Advanced UiPath Activities - Integrate UiPath with external systems such as databases or APIs. - Explore orchestrator features and deploy automation projects. - Develop a library of custom activities for automation.	API Client Invocation, Package Release Deployment	CLO1	Lab Performance
Week-4 [1.5 Hours]	Real-world Applications of RPA: Analyze case studies of RPA implementations across industries, design and implement RPA solutions for specific business processes, implement version control using Git for managing automation project files, debug automation workflows using UiPath tools, optimize performance and scalability of automation solutions.	Source Control Merging, Execution Breakpoint Inspection	CLO1, CLO3	Viva, Lab Final
Week-5 [1.5 Hours]	Integrating UiPath with External Systems - Automate email processing tasks using UiPath. - Integrate UiPath with Excel and CSV files for data manipulation. - Develop automation solutions for document processing tasks.	SMTP/IMAP Protocol Binding, Workbook Cell Parsing	CLO1, CLO3	Viva, Lab Final
Week-6 [1.5 Hours]	Advanced Topics in UiPath - Explore cognitive automation capabilities in UiPath. - Experiment with machine learning models for automation. - Design and implement attended and unattended automation workflows.	AI Fabric Endpoint Wiring, Hybrid Robot Provisioning	CLO1	Viva, Lab Final
Week-7 [1.5 Hours]	Project Development and Implementation: Form project teams and brainstorm project ideas, collaboratively design and develop an automation project, test and iterate on the project based on feedback, prepare a comprehensive presentation showcasing the automation project, present the project to the class, participate in peer evaluation sessions, reflect on the course journey, and discuss future opportunities.	Team Sprint Backlog Simulation, Mock Presentation Practice	CLO2, CLO3	Lab Performance, Viva, Lab Final
Week-8 [1.5 Hours]	Advanced Topics in RPA - Explore advanced concepts such as attended and unattended automation. - Learn about orchestrator queues and scheduling. - Experiment with AI and cognitive automation in UiPath.	Orchestrator Queue Linearization, Cron-Trigger Automation	CLO2, CLO3	Lab Performance, Viva, Lab Final

Week-9 [1.5 Hours]	Scalability and Performance Optimization: Implement strategies for scaling automation solutions, optimize automation workflows for performance, manage resources efficiently to ensure scalability, understand regulatory compliance requirements in RPA projects, implement governance policies and controls, ensure data privacy and security in automation workflows.	Execution Log Profiling, Asset Encrypted Credential Hardening	CLO2	Lab Performance, Viva, Lab Final
Week-10 [1.5 Hours]	Integration with AI and ML - Explore integration possibilities with AI and machine learning technologies. - Experiment with natural language processing (NLP) and image recognition in UiPath. - Develop AI-powered automation solutions.	Intelligent Document Processing (IDP), Pattern Matching Extraction	CLO2	Lab Performance, Viva, Lab Final
Week-11 [1.5 Hours]	Continuous Improvement and Maintenance - Implement continuous improvement processes in automation projects. - Monitor and maintain automation solutions for efficiency and effectiveness. - Handle exceptions and errors in automated processes.	Operational KPI Metric Tracking, Global Exception Handler Injection	CLO2, CLO3	Lab Performance, Viva, Lab Final
Week-12 [1.5 Hours]	Real-world Project Implementation - Work on real-world automation projects in teams. - Apply learned concepts and skills to solve practical challenges. - Implement end-to-end automation solutions.	End-to-End System Integration, Production Deployment Mapping	CLO2	Lab Performance
Week-13 [1.5 Hours]	Project Testing and Deployment - Test automation projects rigorously for functionality and reliability. - Deploy automation solutions to production environments. - Conduct user acceptance testing (UAT) and gather feedback.	UAT Test Scenario Execution, Production Switch-Over Audit	CLO1, CLO2, CLO3	Lab Performance
Week-14 [1.5 Hours]	Project Presentation and Reflection - Present completed automation projects to the class. - Reflect on the learning journey and share insights and experiences. - Discuss future opportunities and advancements in RPA.	Enterprise Architecture Showcasing, Technical Retrospective Defense	CLO1, CLO2, CLO3	Viva, Lab Final

Overall Assessment Scheme and Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10	--	--
Lab Report	25	CLO3	PLO10
Lab Performance	25	CLO1	PLO1
Lab Final	40	CLO1	PLO1

		CLO2	PLO4
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Ankur M. Tripathi, Learning Robotic Process Automation, Packt Publishing, 2021, ISBN: 978-1801074857.

Reference Books/ Other Supplementary Readings

Richard Murdoch, Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant, 1st edition, self-published, 2018, ISBN 978-1983036835

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

UiPath, Google Colab

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

CS 211 Cyber Security Fundamentals

Course Code: CS 211		CIE Marks: 60
ISCED: 0613-211		SEE Marks: 40
Course Title: Cyber Security Fundamentals		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE 332		
Course Type: Major Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Cybersecurity is one of the most significant challenges of the contemporary world, due to both the complexity of information systems and the business they support. Software running on current systems is exploited by attackers despite many deployed defense mechanisms and best practices for developing new software. In this course students will learn about overview of cybersecurity domains, global challenges related to cyber security, cyber security governance, concepts of digital trusts, cyber risk, common attack types and attack vectors, implementing cyber security controls, security architecture principles, different architect model like OSI, defense in depth, information flow control, isolation and segmentation, logging, monitoring and detection, encryption fundamentals, techniques and applications of encryption, security of networks, security of systems, security of applications, security of data, risk assessments, vulnerability management, penetration testing, network security, operating system security, application security, data security, security implications and adoption of evolving technology, like threat landscape, cloud and digital collaboration, block chain, zero trust, privilege access management (PAM), security devices, cyber security incident management, investigations, legal holds and preservation, forensics, disaster recovery plan, business continuity plans, etc. The students will work with real-world problems and technical challenges by implementing security solutions in web applications.

Rationale of the Course

This course explores the strategic management of information assets and the implementation of robust cybersecurity frameworks within systems design, development, and administration. Students examine the core domains of security—including application, data, communications, and networks—while addressing risk factors like unauthorized access, incident response, and contingency planning. The curriculum emphasizes proactive threat assessment, administrative auditing, and the evaluation of cost-benefit criteria to balance security complexities with organizational objectives. Ultimately, this course equips learners with the defensive and management skills necessary to safeguard digital infrastructure and maintain a competitive operational advantage.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of cyber security. More specifically,

- Interpret Key Cybersecurity Concepts: Gain an in-depth understanding of key cybersecurity principles using real-life examples to demonstrate comprehensive knowledge and application.
- Analyze and Design Cyber Security Architectures: Develop skills in analyzing diverse technology architectures, network models, and defense-in-depth concepts, and adeptly design and evaluate comprehensive cybersecurity architectures.
- Validate Cybersecurity Controls Implementation: Acquire the ability to validate the implementation of cybersecurity controls across networks, systems, applications, data, and emerging technologies.
- Measure and Manage Cyber Incidents: Learn to measure the impact of cyber incidents and demonstrate the ability to manage these incidents effectively, implementing appropriate controls.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLO's	CLO Descriptions	PLO's	Learning Domains (C,P,A)	BNQF Skill

CLO1	Interpret the key cybersecurity concepts using real-life examples to demonstrate a comprehensive understanding and application of these principles.	PLO1	C2	Fundamental Domain
CLO2	Analyze diverse technology architectures, network models, and defense-in-depth concepts to adeptly design and evaluate comprehensive cybersecurity architectures.	PLO2	C4	Fundamental Domain
CLO3	Validate the implementation of cyber security controls in Networks, Systems, Applications, Data and emerging technologies.	PLO2	C5	Fundamental Domain
CLO4	Measure the impact of cyber incident and demonstrate the ability to manage the cyber incidents and implement appropriate controls.	PLO3	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3		√										
CLO4			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview to cybersecurity, Difference between Information Security and Cybersecurity	Security Perimeter Comparison, Conceptual Mapping	Class Test, Mid-term	CLO1

	Lesson 2: Cybersecurity Objectives, Cybersecurity Governance	Risk Governance Framework, Strategy Discussion	Class Test, Mid-term	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Cybersecurity Domains, digital trust concepts, cyber risk	Attack Surface Deconstruction, Asset Threat Review	Class Test, Mid-term	CLO1
	Lesson 2: Case study discussion - 3 out of 10 top cyber-attacks in history	Attack Vector Dissection, Root-Cause Analysis	Class Test, Mid-term	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Case study discussion - 4 out of 10 top cyber-attacks in history	Incident Timeline Reconstruction, Exploit Vector Review	Class Test, Mid-term	CLO1
	Lesson 2: Case study discussion - 3 out of 10 top cyber-attacks in history	Mitigation Failure Assessment, Post-Mortem Analysis	Class Test, Mid-term	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Cyber threat actors, cyber-attack Types	Threat Actor Profiling, Attack Taxonomy Classification	Class Test, Mid-term	CLO1
	Lesson 2: Cyber-attack, cyber Policies, Cybersecurity Controls Class Test -1	Policy Framework Auditing, Controls Gap Analysis	Class Test, Mid-term	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of security architecture	Enterprise Zone Blueprinting, Defensive Boundary Analysis	Class Test, Mid-term	CLO2
	Lesson 2: OSI model, Defense in depth, Information flow control	Layered Packet Encap-Decap, Trust-Boundary Logic Review	Class Test, Mid-term	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Isolation and segmentation, logging, Monitoring, and detection	Network Vlan Partitioning, SIEM Log Signature Analysis	Class Test, Mid-term	CLO2
	Lesson 2: Encryption Basic, Symmetric Encryption	Shared-Key Cryptanalysis, Block Cipher Substitution Math	Class Test, Mid-term	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Asymmetric encryption	Public-Private Key Pair Math, Mathematical Factorization Review	Class Test, Mid-term	CLO2

	Lesson 2: Application of encryption Class Test -2	SSL/TLS Handshake Tracing, Digital Certificate Validation	Class Test, Mid term	CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Process Controls in Risk Assessments	Risk Likelihood Calculation, Qualitative Matrix Assessment	Class Test, Assignment, Presentation, Final exam	CLO3
	Lesson 2: Process Controls in Vulnerability Management	Vulnerability Scan Profiling, CVSS Severity Score Review	Class Test, Assignment, Presentation, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Process Controls in Penetration Testing, Network Security	Penetration Testing Scope Rule Mapping, Firewall Rule Auditing	Class Test, Assignment, Presentation, Final exam	CLO3
	Lesson 2: Operating System Security	OS Kernel Hardening, Privilege Least-Principle Checklist	Class Test, Assignment, Presentation, Final exam	CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Application Security	OWASP Top 10 Defect Analysis, Secure Coding Review	Class Test, Assignment, Presentation, Final exam	CLO3
	Lesson 2: Data Security	Data Flow Lifetime Classification, DLP Rules Mapping	Class Test, Assignment, Presentation, Final exam	CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Current Threat Landscape	Active Exploit Feed Interrogation, Zero-Day Trend Review	Class Test, Assignment, Presentation, Final exam	CLO3
	Lesson 2: Advanced Persistent Threats, Cloud and Digital Collaboration Class Test -3	APT Kill-Chain Phase Analysis, Cloud IAM Guardrail Review	Class Test, Assignment, Presentation, Final exam	CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Blockchain, Zero trust, Different Security devices	Identity Context Verification, Inline Appliance Layout Mapping	Class Test, Assignment, Presentation, Final exam	CLO3
	Lesson 2: Events vs incidents, Incident categories and types.	Alert Triage Classification, Threshold Impact Assessment	Class Test, Assignment, Presentation, Final exam	CLO3, CLO4

Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Blueprint Review	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Security event management, Incident response plans	IR Runbook Workflow Mapping, Playbook Verification	Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: legal requirements of investigation and evidence preservation, Digital forensic investigations	Chain-of-Custody Verification, Hash Integrity Analysis	Final	CLO4
	Lesson 2: Business continuity planning and disaster recovery	BIA RTO/RPO Metric Math, Failover Architecture Simulation	Final	CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO3	
Attendance			--	--	7				
Class Test 1	5				Avg 15				
Class Test 2		5							
Class Test 3			5						
Assignment			5		5		5		5
Presentation					8				
Midterm Exam	15	10			25	15	10		25
Final exam			20	20	40		20	20	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

ISACA, Cybersecurity Fundamentals Study Guide, 3rd edition, ISACA, 2021, ISBN: 978-1604207174.

Reference Books/ Other Supplementary Readings

Kutub Thakur, Al-Sakib Khan Pathan, Cybersecurity Fundamentals, 1st Edition, CRC Press, 2020, 978-0367476489

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

CS 418 Network & Communication Security

Course Code: CS 418		CIE Marks: 60
ISCED: 0612-418		SEE Marks: 40
Course Title: Network & Communication Security		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: (If any)		
Course Type: Major		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

In this course, students will learn about Basic Computer Networks, Introduction to Computer Security, Aspects of Computer Security, Cryptographic Processes, Network Security-issues, Cryptographic Hash Functions, Security Workshop, Firewall Configuration and Administration, Overview of VPN, Advanced Features of VPN, Advanced Routing, NAT & PAT, Network Monitoring and Tools and Network Security infrastructure.

Rationale of the Course

This course delivers a comprehensive understanding of the foundational concepts and strategic importance of cybersecurity within enterprise environments. Students examine the landscape of cyber-attacks, threat vectors, and risk assessment while exploring defensive controls and core architecture principles. The curriculum covers the systematic protection of networks, systems, applications, and data, as well as incident management, digital forensics, and the security implications of adopting emerging technologies

Course Objectives

To provide a solid conceptual understanding of the fundamentals of data communications. More specifically,

- To know about network and cyber security concepts
- To know about cyber-attack, vectors, risk, controls
- To know about Security architecture principles
- To know about cybersecurity of network, systems, applications and data
- To know about cyber incident Management and Monitoring

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skills
CLO1	Articulate the fundamental principles, process and protocols of Network and Communication Security.	PLO1	C4	Fundamental Domain
CLO2	Simulate effective security solutions to protect network infrastructure.	PLO3	C4	Fundamental Domain
CLO3	Evaluate and address various network security vulnerabilities and threats using relevant tools and techniques.	PLO5	C5	Fundamental Domain
CLO4	Develop and configure network devices to implement security measures in accordance with industry standards.	PLO6	C6	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2			√									
CLO3					√							
CLO3						√						

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Introduction to Software Engineering Basic Computer Network - OSI Reference Model - Network Infrastructure	Syllabus Walkthrough, OSI Layer Mapping	Class Test, Midterm exam	CLO1
	Lesson 2: Basic Computer Network, A brief discussion on device IOS, Functions of IOS, Console Access Method, Navigating between IOS Modes, Securing Device Access	CLI Command Shell Navigation, Privilege Guard Setup	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Computer Network Security, Securing Networks, Network Threats, Mitigating Threats Summary	Threat Vector Mapping, Perimeter Vulnerability Analysis	Class Test, Midterm exam	CLO1
	Lesson 2: IP Addressing Explain the structure of IP addressing, Classify by type and describe how it is used in the network, Explain how addresses are assigned to networks by ISPs and within networks by administrators IPv4 addressing design criteria, calculate the appropriate addressing components, Use common testing utilities to verify and test network connectivity	Octet Bit Reconstruction, ICMP Utility Troubleshooting	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: IP Subnetting, Knowledge on Subnet Mask, Why Subnetting is required? How to calculate Subnetting? How to calculate Subnets and Hosts?	Variable Length Subnet Masking (VLSM) Math, Host Bounds Allocation	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Introduction to Cryptography, Classification of the field of Cryptology, Purpose of Cryptography, Symmetric	Cipher Block Deconstruction, Encryption Pipeline Analysis	Class Test, Midterm exam	CLO1, CLO2

	Cryptography Types of Cryptography Process of Cryptography			
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Dynamic Routing Protocols • Introduction to Dynamic and Static Routing Protocols • Interior Gateway Protocols (IGP) • Exterior Gateway Protocols (EGP) • Comparison between Dynamic Routing Protocols and Static Routing Protocols • Functions of Dynamic Routing Protocols • The purpose of a dynamic routing protocol • Classifying Routing Protocols Comparison of Distance Vector & Link State Routing Protocols • Classful and Classless routing protocols - Identifying the Administrative Distance (AD) Class Test-1	Convergence Metric Calculation, Routing Table Topology Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Access Control List (ACL) • To learn standard and extended ACLs • To learn the Placement of ACLs • To create and apply named ACLs • To learn the Function of firewalls Use ACLs to restrict virtual terminal access	Wildcard Masking Calculation, Packet Filtering Matrix Design	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Network Security- issues.	Perimeter Vulnerability Auditing, Breach Impact Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Cryptographic Hash Functions	Message Digest Collision Analysis, Bit Hashing Drill	Class Test, Assignment, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hs]	Lesson 1: Security Workshop, Security Workshop	Attack Scenario Role-Play, Incident Defense Lab	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Firewall Configuration and Administration Class Test-2	Access Policy Rules Mapping, Inspection Engine Review	Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Advanced Routing, NAT & PAT	Inside/Outside Address Translation Mapping, Port Multiplexing Tracing	Class Test, Assignment, Final exam	CLO1, CLO2

	Lesson 2: Overview of VPN, Advanced Feature of VPN	Tunneling Overhead Analysis, IPSec Security Association Mapping	Class Test, Assignment, Final exam	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Network Monitoring and Tools - What is the importance of keeping networks secured and protected? - What are the different tools or devices used for network protection? Discuss about the importance of port mirroring technique.	Traffic Capture Interrogation, Switched Port Analyzer (SPAN) Configuration	Class Test, Assignment, Final exam	CLO1, CLO2
	Lesson 2: Understanding Defense - Explain the concept of Defense-in-Depth. - Explain the ways used by cyber security analysts to identify the assets, vulnerabilities and threats.	Layered Security Ring Modeling, Asset Threat Matrix Analysis	Class Test, Assignment, Final exam	CLO1, CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Network Security Infrastructure - Familiarize the learners with the important terms in network topology. - Demonstrate a basic topology with some of the key icons. - Reinforce the differences between the physical and logical topologies. - In detail discuss on Security Operations Center (SOC) Explain the three-layer network design model	Hierarchical Core-Distribution Layout, Logic Topology Mapping	Class Test, Final exam	CLO1, CLO2
	Lesson 2: Attackers and their Tools - Discuss the importance of cybersecurity in an organization. - Discuss the term threat, vulnerability, risk. Briefly explain four ways to manage risk.	Risk Formulation Calculation, Risk Mitigation Strategy Selection	Class Test, Final exam	CLO1, CLO2
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Endpoint Protection Describe the three different approaches to detecting viruses.	Signature vs Heuristic Pattern Analysis, Malicious Heuristic Evaluation	Final exam	CLO1, CLO2
	Lesson 2: Endpoint Protection Explain the devices and techniques used to provide endpoint protection.	Host IPS Policy Mapping, Endpoint Firewall Hardening	Final exam	CLO1, CLO2
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Endpoint Vulnerability Assessment	Vulnerability CVE Database Cross-Referencing, Patch Integrity Audit	Final exam	CLO1, CLO2
	Lesson 2: Secure Device Management, Technologies and Protocols Class Test-3	Cryptographic SSH Configuration, Secure Device Access Management	Assignment, Final exam	CLO1, CLO2, CLO3

Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Network Security Data, Explain how Syslog and NTP are important in network monitoring.	Syslog Severity Parsing, Time-Stamp Sync Auditing	Assignment, Final exam	CLO1, CLO3, CLO4
	Lesson 2: Ask the class to share their understanding on the types of security data.	Log Telemetry Segment Isolation, Telemetry Analytics Review	Assignment, Final exam	CLO1, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Digital Forensics and Incident Analysis and Response - Explain the digital forensic process with a real-life example. - Define types of evidences. Later, provide examples to the learners and ask them to identify.	Chain-of-Custody Integrity Analysis, Evidence Hash Verification	Assignment, Final exam	CLO1, CLO3, CLO4
	Lesson 2: Real-time Lab experiment on Network Security	Live Exploit Capture Lab, Wire-Level Packet Dissection	Assignment, Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Infrastructure Deployment Case Study	Assignment, Presentation	CLO4
	Lesson 2: Review	Integrative Syllabus Review, Board Problem Solving	Presentation, Open discussion	CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total=70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO3	
Attendance			--	--	7				
Class Test 1	5				Avg 15				
Class Test 2		5							
Class Test 3			5						
Assignment			5		5		5		5
Presentation					8				
Midterm Exam	15	10			25	15	10		25
Final exam			20	20	40		20	20	40

Marks Distribution

Class attendance	7
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Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Tugrul Ozcelebi, & Johan I. den Hartog, Computer Networks and Security (2IC60), version 0.38, self-published, 2019.

Reference Books/ Other Supplementary Readings

William Stallings, Cryptography and network security: principles and practice, 5th Edition, Pearson/Prentice Hall, c2011. 2013, ISBN: 9788131761663

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

CS 422 Digital Forensics

Course Code: CS 422		CIE Marks: 60
ISCED: 0612-422		SEE Marks: 40
Course Title: Digital Forensics		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Major		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

The necessity of Digital Forensic and Computer Forensics Investigation Process, Understanding the Hard Disks, Understanding the File Systems, Data Acquisition and Duplication, Defeating Anti-Forensics Techniques, Fundamentals of Operating Systems Forensics, Windows Forensic and Linux Forensic, Fundamentals of Network Security Architecture, Different Network Attack Methodologies, Network Attack Forensics, Fundamentals of Web Application Security Posture, Web Application Attack Methodologies, Investigating Web Attacks, Database Forensics, Cloud Forensics, Investigating Email Crimes, Malware Forensics, Mobile Forensics, IoT Forensics, Fundamentals of Web

Application Security Posture, Web Application Attack Methodologies, Investigating Web Attacks, Database Forensics, Cloud Forensics, Investigating Email Crimes, Mobile Forensics, Malware Forensics, Importance of different types, Purposes, Scope of Forensic.

Rationale of the Course

This course addresses the growing demand for specialized professionals equipped to investigate cybercrimes and analyze digital evidence. Students learn foundational digital forensic principles, methodologies, and investigative techniques necessary to counter sophisticated cyber threats. The curriculum emphasizes the full lifecycle of an investigation, from analyzing digital evidence to properly presenting findings in legal proceedings.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of data communications. More specifically,

- To understand the fundamental concepts of digital forensics, including data acquisition, preservation, and analysis.
- To develop practical skills in forensic investigation techniques, tools, and methodologies used in cybercrime investigations.
- To learn how to identify, collect, and analyze digital evidence from various sources, such as computers, networks, and mobile devices.
- To gain knowledge of legal and ethical considerations in digital forensics, including chain of custody, privacy laws, and expert witness testimony.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Articulate proficiency in digital forensic investigation methodologies.	PLO4	C4	Fundamental Domain
CLO2	Apply digital forensic tools and techniques to collect, preserve, and analyze digital evidence.	PLO5	C3	Fundamental Domain
CLO3	Validate the digital evidence to identify security breaches, cybercrimes, and unauthorized activities.	PLO2	C5	Fundamental Domain
CLO4	Formulate effective strategies for presenting digital forensic findings in legal proceedings and other contexts.	PLO7	C6	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1				√								
CLO2					√							
CLO3		√										
CLO4							√					

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Overview of Course OBE	Voice Over PPT, Open Discussion	N/A	N/A
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Necessity of Digital Forensic and Computer Forensics Investigation Process	Ice breaker activity, Open discussion. Oral lecture.	Class Test	CLO1
	Lesson 2: Understanding the Hard Disks, Understanding the File Systems	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note.	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Data Acquisition and Duplication, Defeating Anti-Forensics Techniques	Brainstorming sessions, Lecture video, Lecture note, Open discussion.	Class Test, Midterm exam	CLO1
	Lesson 2: Fundamentals of Operating Systems Forensics	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video.	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Windows Forensic and Linux Forensic Class Test-1	Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	Class Test, Midterm exam	CLO2
	Lesson 2: Fundamentals of Network Security Architecture, Different Network Attack Methodologies	Review, Student activities, Lecture video.	Class Test, Midterm exam	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Network Attack Forensics, Fundamentals of Web Application Security Posture	Brainstorming sessions, Voice over PPT, Lecture note, Open discussion.	Class Test, Midterm exam	CLO2
	Lesson 2: Web Application Attack Methodologies	Lecture video, Lecture note, Open discussion.	Midterm exam	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Investigating Web Attacks, Database Forensics	Brainstorming sessions, Voice over PPT, Open discussion.	Class test, Midterm exam	CLO3

	Lesson 2: Cloud Forensics, Investigating Email Crimes Class Test-2	Voice over PPT, Lecture video, Lecture note.	Class Test, Assignment, Final exam	CLO3
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Malware Forensics	Classroom discussion, Lecture note, Open discussion.	Class Test, Assignment, Final exam	CLO4
	Lesson 2: Mobile Forensics	Brainstorming sessions, Lecture video, Lecture note, Open discussion.	Class Test, Assignment, Final exam	CLO4
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: IoT Forensics	Classroom discussion, Voice over PPT, Open discussion.	Class Test, Presentation, Final exam	CLO3
	Lesson 2: Fundamentals of Web Application Security Posture	Brainstorming sessions, Lecture video, Open discussion.	Class Test, Presentation, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Web Application Attack Methodologies	Brainstorming sessions, Open discussion, Lecture video.	Class Test, Presentation, Final exam	CLO4
	Lesson 2: Investigating Web Attacks	Voice over PPT, Open discussion.	Class Test, Presentation, Final exam	CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Database Forensics, Cloud Forensics	Brainstorming sessions, Lecture video, Lecture note, Open discussion.	Class Test, Final exam	CLO4
	Lesson 2: Investigating Email Crimes, Investigating Email Crimes Methods	Brainstorming sessions, Lecture video, Lecture note, Open discussion.	Class Test, Final exam	CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Mobile Forensics Class Test-3	Brainstorming sessions, Classroom discussion, Voice over PPT.	Final exam	CLO3
	Lesson 2: Mobile Forensics Analysis	Brainstorming sessions, Classroom discussion, Voice over PPT.	Final exam	CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Malware Forensics	Voice over PPT, Lecture video, Lecture note, Open discussion.	Final exam	CLO4
	Lesson 1: Malware Forensics Analysis	Voice over PPT, Lecture video, Lecture note, Open discussion.	Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Importance of different types of Forensic	Lecture video, Lecture note, Open discussion.	Final exam	CLO4
	Lesson 1: Importance of different purpose, scope of Forensic	Brainstorming sessions, Classroom discussion, Voice over PPT.	Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Interactive Lectures and Presentations, Expert Talks, Hands-On Workshops, Site Visits and Virtual Tours	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4

	Lesson 2: Review class	Brainstorming sessions, Lecture note, Open discussion.	Final exam	CLO4
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Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO4	
Attendance	--	--	--		7				
Class Test 1	5				Avg 15				
Class Test 2		5							
Class Test 3			5						
Assignment			5		5		5		5
Presentation				8	8				
Midterm Exam	10	10	5		25	18	7		25
Final exam			10	30	40	10	30		40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Mangesh M. Ghonge, Sabyasachi Pramanik, Ramchandra Mangrulkar , Cyber Security and Digital Forensics, 2nd Edition, USA : Wiley, 2022

Reference Books/ Other Supplementary Readings

Gerard Johansen, Digital Forensics and Incident response : incident response tools and techniques for effective cyber threat response, 3rd Edition, New York ; Packt publisher : 2022, ISBN: 9781803258678

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

DS 331 Introduction to Data Science and Data Management & Analysis

Course Code: DS 331		CIE Marks: 60
ISCED: 0613-331		SEE Marks: 40
Course Title: Introduction to Data Science and Data Management & Analysis		Total Marks: 100
Semester:		
Credit Value: 2 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: SE 121, STAT 101		
Course Type: Major		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Intro to Data Science, Learn about Data Scientist, Data Analyst, and Data Engineer, Old Problems, New Problems, Data Science Solutions, Applications of Data Science, Data Science Real-Life Problems and Solutions, Data Sets: Powering Data Science, Types and Sources of Data, Data Quality and Data Collection, Data Understanding, Data Preparation: Concepts, Data Management for Data Science, ETL Process and Architecture, Introduction to Data Analysis, Introduction to Data Visualization, Data Visualization Tools and Techniques, Data Ethics, Applications of Machine Learning and Machine Learning Workflow, Introduction to Statistics, Time Series Forecasting, Introduction to Deep Learning, Basic Neural Network Concepts, Deep Learning for Text, Image, and Sequence Data, Introduction to Large Language Models, Transformer, Attention, and Embeddings, Introduction to RAG Architecture, Business Understanding in Data Science, Model Deployment and Business Value.

Rationale of the Course

This course provides a comprehensive foundation in data science and data management, highlighting the distinct roles and responsibilities of data science engineers and data analysts. By mapping theoretical modeling to real-world case studies and deployment strategies, this course equips beginners and aspiring professionals with the cross-disciplinary skills needed to solve complex, data-driven problems.

Course Objectives

The goal of this course is to introduce the students about the concept of Data Science. The main objectives of this course are-

- To Learn the introduction of data science.
- To Learn and understand the data engineering tools.
- To learn and demonstrate data visualization and algorithm.
- To learn and demonstrate Artificial Intelligence techniques and approaches.
- To learn and apply data science methodology and understand the ETL.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Explain data science, its tools and Technique	PLO1	C2	Fundamental Domain
CLO2	Evaluate different data visualization techniques and algorithms.	PLO2	C5	Fundamental Domain
CLO3	Integrate different data science methodology	PLO4	C4	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of OBE (CLO, PLO, Knowledge level, Learning Domains, Bloom's Taxonomy)	Syllabus Walkthrough, Taxonomy Mapping Q&A	None	
	Lesson 2: Intro To Data Science	Data Pipeline Overview, Core Paradigm Comparison	None	
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Learn about Data Scientist, Data Analyst, and Data Engineer	Roles Matrix Analysis, Responsibility Mapping	Class Test, Midterm exam	CLO1
	Lesson 2: Old problems, new problems, Data Science solutions	Case Scenario Assessment, Open Discussion	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Applications of Data Science	Domain Impact Analysis, Industry Trend Review	Class Test, Midterm exam	CLO1
	Lesson 2: Data Science Real-Life Problems and Solutions Class Test -1	Problem Statement Formulation, Solution Architecture Review	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Data Sets: Powering Data Science	Feature Matrix Dissection, Sample Distribution Analysis	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Types and Sources of Data, Data Quality & Data Collection	Storage Ingestion Profiling, Schema Categorization	Class Test, Midterm exam	CLO1, CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Data Understanding	Exploratory Data Profiling, Statistical Attribute Scanning	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Data Preparation: Concepts	Missing Value Imputation, Outlier Boundary Identification	Class Test, Assignment, Midterm exam	CLO1, CLO2

Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Data Management for Data Science	Schema Design Verification, Storage Partition Assessment	Class Test, Assignment, Midterm exam	CLO2, CLO3
	Lesson 2: ETL Process and Architecture	Pipeline Ingestion Flowcharting, Data Stream Transformation	Class Test, Assignment, Midterm exam	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Data Analysis Class Test-2	Bivariate Correlation Drilling, Hypothesis Testing Review	Midterm exam, Final exam	CLO2
	Lesson 2: Introduction to Data Visualization	Chart Typology Selection, Visual Encoding Mechanics	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Data Ethics	Fairness Metric Auditing, Privacy Constraint Review	Class Test, Assignment, Midterm exam, Final exam	CLO2,CLO3
	Lesson 2: Applications of Machine Learning & Machine Learning Workflow	Pipeline State Modeling, Inference Cycle Analysis	Class Test, Assignment, Midterm exam, Final exam	CLO2,CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Statistics	Central Tendency Calculation, Probability Distribution Mapping	Class Test, Assignment, Mid Term, Final exam	CLO2,CLO3
	Lesson 2: Time Series Forecasting	Trend Decomposition Analysis, Autocorrelation Analysis	Class Test, Assignment, Mid Term, Final exam	CLO2, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Deep Learning	Layer Hierarchy Modeling, Feature Abstraction Review	Class Test, Assignment, Midterm exam, Final exam	CLO2, CLO3
	Lesson 2: Basic Neural Network Concepts	Forward Pass Calculation, Weight Delta Optimization	Class Test, Assignment, Midterm exam, Final exam	CLO2,CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Deep Learning for Text, Image, and Sequence Data	Convolution Window Striding, Recurrent Sequence Tracking	Class Test, Assignment, Final exam	CLO1, CLO2,CLO3
	Lesson 2: Introduction to Large Language Models	Autoregressive Logic Dissection, Token Context Analysis	Class Test, Assignment, Final exam	CLO1, CLO2.CLO3

Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Transformer, Attention, and Embeddings	Attention Matrix Calculation, Vector Space Alignment	Class Test, Assignment, Final exam	CLO1, CLO2,CLO3
	Lesson 2: Introduction to RAG Architecture Class Test-3	Vector DB Vector Search Indexing, Prompt Context Formulation	Class Test, Assignment, Final exam	CLO2, CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Business Understanding in Data Science	KPI Alignment Matrix, Objective Scope Framing	Assignment, Final exam	CLO2,CLO3
	Lesson 2: Model Deployment and Business Value	REST API Endpoint Verification, Production Value Ingestion	Assignment, Final exam	CLO2,CLO3
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Industry Expert Talk, Production Blueprint Review	Assignment, Final exam	CLO2,CLO3
	Lesson 2: Review class on topic discussed in Week-10, Week-11, Week-12 & Week-13	Integrative Review, Comprehensive Problem Solving	Presentation, Assignment, Final exam	CLO2, CLO3

Overall Assessment Scheme

Assessment Task	CLO's			Mark (Total=100)	PLO's			Mark (Total =78)
	CLO1	CLO2	CLO3		PLO1	PLO2	PLO4	
Attendance	--	--	--	7				
Class Test 1	5(15)			Avg 15				
Class Test 2		5(15)						
Class Test 3			5(15)					
Assignment				5				
Presentation				8				
Midterm Exam	15	10		25	15	10		25
Final exam		25	15	40		25	15	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Chirag Shah, A Hands-On Introduction to Data Science, 1st edition, MIT Press, 2020, ISBN: 978-0262043786.

Reference Books/ Other Supplementary Readings

1. Introduction to Machine Learning with Python: A Guide for Data Scientists [Book by Andreas C. Müller and Sarah Guido]
2. Hands-On Large Language Models: Language Understanding and Generation [Book by Jay Alammar & Maarten Grootendorst]
3. <https://www.youtube.com/watch?v=N6BghzuFLIg>

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Jupyter Notebook, Python

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

DS 332 Introduction to Data Science and Data Management & Analysis Lab

Course Code: DS 332		CIE Marks: 60
ISCED: 0613-332		LF Marks: 40
Course Title: Introduction to Data Science and Data Management & Analysis Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: STAT 101, SE 121		
Course Type: DS Major		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

This course provides a broad introduction to Data Science: (i) Learn about data science engineer and data analyst, Careers in Data Science, Application of machine learning (ii) Introduction about data science tools (iii) and data set collection and data visualization (iv) data set analysis and model applyStatistical (v) statistical basics (vi) Time series data analysis and ETL.

Rationale of the Course

The "Introduction to Data Science and Data Management & Analysis Lab" course is designed for Data Science is a combination of multiple disciplines that uses statistics, data analysis, and machine learning to analyze data and to extract knowledge and insights from it. The course will Describe what data science and machine learning are, their applications & use cases, and various types of tasks performed by data scientists. Students will Gain hands-on familiarity with common data science tools including JupyterLab, R Studio. The course will develop the mindset to work like a data scientist, and follow a methodology to tackle different types of data science problems etc.

Course Objectives

The goal of this course is to introduce the students to the concept of Data Science in practice. The main objectives of this course are:

- To Learn the introduction of data science.
- To Learn and understand the data engineering tools.
- To learn and demonstrate data visualization and algorithms..
- To learn and apply data science methodology and understand the ETL

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Demonstrate the data science and analytics technique.	PLO2	C6, P5	Fundamental Domain
CLO2	Appraise Data Visualization tools and techniques.	PLO5	C4, P2	Fundamental Domain
CLO3	Illustrate Databases and SQL for Data Science & ETL.	PLO3	C3, P2	Fundamental Domain
CLO4	Write effective reports and design documentation for the data science oriented applications.	PLO10	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2					√							
CLO3			√									
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning, and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
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Week-1 [1.5 Hours]	Learn course outline and OBE Discussion	Syllabus Walkthrough, Environment Preparation	CLO1	None
Week-2 [1.5 Hours]	Data Science Application, problem and implementation	Notebook Session Initialization, Workspace Dependency Configuration	CLO1	Lab Performance, Viva, Lab Final
Week-3 [1.5 Hours]	Python Libraries for Data Science and Introduction to Jupyter Notebook	Array Matrix Manipulation, Dataframe Series Ingestion	CLO1	Lab Performance, Viva, Lab Final
Week-4 [1.5 Hours]	Python Visualization Libraries Matplotlib, Seaborn	Static Chart Plotting, Statistical Distribution Mapping	CLO2	Lab Performance
Week-5 [1.5 Hours]	Python Visualization Libraries Bokeh, Geoplotlib	Interactive Graphic Rendering, Geospatial Coordinate Mapping	CLO2	Viva, Lab Final
Week-6 [1.5 Hours]	Open-Source Tools for Data Science git ,GitHub, bit bucket	Repository Branching Simulation, Commit Conflict Resolution	CLO1	Viva, Lab Final
Week-7 [1.5 Hours]	Data analytic using Tableau /Power BI Visualization	BI Dashboard Component Dragging, Business KPI Aggregation	CLO2	Lab Performance, Viva, Lab Final
Week-8 [1.5 Hours]	SQL, Relational Database, Table, Tuple	Relational Statement Binding, Join Aggregation Queries	CLO3	Lab Performance, Viva, Lab Final
Week-9 [1.5 Hours]	No-SQL Non-relational, Schema-free, Simple API, Distributed	JSON Document Store Instantiation, BSON Field Ingestion	CLO3	Lab Performance, Viva, Lab Final
Week-10 [1.5 Hours]	ETL architecture and Data Warehouse	Pipeline Stream Extraction, Warehouse Dimension Schema Modeling	CLO3	Lab Performance, Viva, Lab Final
Week-11 [1.5 Hours]	ETL tools Introduction named – Hadoop	MapReduce Execution Profiling, HDFS File Appending	CLO3	Lab Performance, Viva, Lab Final
Week-12 [1.5 Hours]	ETL tools for data warehouse and big Data – Scala	Functional Stream Processing, Dataframe Schema Mapping	CLO3	Lab Performance, Viva, Lab Final

Week-13 [1.5 Hours]	Implement Time series Model development and Implement model development through API in data analysis and visualization	Rolling Window Forecasting, REST Endpoint Integration	CLO4	Lab Performance
Week-14 [1.5 Hours]	Lab Final	Independent Practical Exam, Production Pipeline Execution	CLO1, CLO2, CLO3, CLO4	Viva, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO2
Lab Final	40	CLO2	PLO5
		CLO3	PLO3
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Chirag Shah, A Hands-On Introduction to Data Science, 1st edition, MIT Press, 2020, ISBN: 978-0262043786.

Reference Books/ Other Supplementary Readings

Andreas C. Müller, & Sarah Guido, Introduction to Machine Learning with Python: A Guide for Data Scientists, 1st edition, O'Reilly Media, 2017, ISBN: 9781449369415

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Google Colab, Jupyter Notebook, MySQL

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

DS 411 Statistical Data Analysis

Course Code: DS 411		CIE Marks: 60
ISCED: 0613-411		SEE Marks: 40
Course Title: Statistical Data Analysis		Total Marks: 100
Semester:		
Credit Value: 2 (Theory)	Contact Hours: 2 (Total weeks: 14)	
Prerequisite: MAT 101, STA 101		
Course Type: Major		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Data Types, Data Quality, Mean, Median, Mode, Percentiles, Quantiles, Variance, Standard Deviation, Interquartile Range, Range, Correlation, Pearson Correlation, Spearman's Rank Correlation, Histogram, Density Plot, Bar Chart, Grouped Bar Chart, Stacked Bar Chart, Heatmap, Scatter Plot, Line Plot, Sampling Methods, Probability Sampling, Simple Random Sampling, Systematic Sampling, Cluster Sampling, Stratified Sampling, Non-Probability Sampling, Snowball Sampling, Quota Sampling, Convenience Sampling, Probability Theory, Counting Theory, Permutation, Combinations, Random Variables, Discrete Random Variable, Continuous Random Variable, Probability Mass Function, Probability Density Function, Cumulative Density Function, Probability Distributions, Bernoulli Distribution, Binomial Distributions, Poisson Distribution, Hypergeometric Distribution, Normal Distribution, Uniform Distribution, Sampling Distribution, Central Limit Theorem, Confidence Intervals, Hypothesis Testing, P-Value, Type-I Error, Type-II Error, Statistical Power, Z-Test, T-Test, Two Sample T-Test, Paired T-Test, Analysis Of Variance, Chi-Square Test, F-Statistic, Simple Linear Regression, Multiple Linear Regression.

Rationale of the Course

This course provides a comprehensive foundation in statistical data analysis, bridging mathematical theory with practical applications. Student progress from descriptive statistics and probability theory to core inferential

techniques, including sampling distributions, confidence intervals, and hypothesis testing. The curriculum covers advanced analytical methods such as ANOVA, Chi-Square tests, and linear regression for predictive modeling. Beyond technical execution, the course emphasizes critical analysis, teaching learners to identify statistical bias, design flaws, and ethical or sociological implications in research.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of statistics. More specifically,

- Students will be taught to apply appropriate statistical tests to answer research queries that will solve real-world problems.
- Students will be taught to critique existing statistical experiments so that they can identify any issues with experiment design, analysis, and interpretation.
- Students will be taught to evaluate any societal, cultural, or ethical issues in statistical experiments.
- Students will be taught to develop their self-learning capabilities so that they can keep themselves updated with the dynamic field of data science.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Adopt suitable statistical techniques to resolve real-world issues.	PLO2	C3	Fundamental Domain
CLO2	Analyze statistical experiments to verify their design, analysis, conclusion, and interpretation.	PLO4	C4	Fundamental Domain
CLO3	Evaluate statistical information to discover societal, cultural, or ethical concerns.	PLO6	C5	Fundamental Domain, Social Thinking, Personal Domain
CLO4	Analyze the latest research literature to keep themselves updated.	PLO12	C4	Social Thinking, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2				√								

CLO3						√						
CLO4												√

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to course syllabus, assessment methods, learning outcomes, deadlines, consultation hour, code of conduct, and other administrative procedures regarding the course.	Syllabus Walkthrough, Administrative Q&A	N/A	N/A
	Lesson 2: Analyzing categorical data, identifying individuals and variables, reading pictographs, bar graphs, pie graphs, constructing and reading frequency tables, and so on.	Frequency Table Calculation, Categorical Proportion Mapping	Class Test, Midterm exam, Final exam	CLO1, CLO2
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Representing data, dot plots, histograms, stem-and leaf plots, shapes of distributions, clusters, gaps, peaks, and outliers, comparison of distributions.	Distribution Shape Graphing, Outlier Geometry Mapping	Class Test, Midterm exam, Final exam	CLO1, CLO2
	Lesson 2: Summarizing quantitative data, measure of central tendencies like mean, median, mode, and percentiles, and choosing the best metric, comparing means of distributions, impact of outliers on mean and median, missing values, etc.	Central Tendency Calculation, Metric Sensitivity Review	Class Test, Midterm exam, Final exam	CLO1, CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Measure of spread such as variance, standard deviation, IQR, range, difference between sample and population statistics, bias and unbiased statistics, boxplot, mean absolute deviation, etc.	Dispersion Variance Math, Boxplot Boundary Modeling	Class Test, Presentation, Midterm, Final	CLO1, CLO2, CLO4
	Lesson 2: Bivariate relationships, scatter plots, positive and negative association, outliers and clusters in scatter plots, correlation coefficients, etc. Class Test-1	Scatter Plotting Co-variance, Correlation Delta Tracing	Class Test, Presentation, Midterm, Final	CLO1, CLO2, CLO4
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Linear Regression, fitting line to data, estimating the best fitted line, residuals, interpreting slope and intercept of regression lines, R-squared coefficient, impact of outliers on regression, etc.	Ordinary Least Squares Math, Residual Vector Graphing	Class Test, Assignment, Midterm, Final	CLO1, CLO2, CLO3
	Lesson 2: Understanding how different designs, analyses, interpretations, and data collection methods create social or ethical concerns.	Bias Distortion Detection, Ethical Impact Analysis	Class Test, Assignment, Midterm,	CLO1, CLO2, CLO3

			Final	
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Study design, statistical vs. non-statistical questions, identifying sample and population, bias in surveys, correlation vs. causality, sampling methods, etc.	Sampling Method Mapping, Confounding Variable Isolation	Class Test, Assignment, Midterm, Final	CLO1, CLO2, CLO3
	Lesson 2: Different types of statistical studies such as experiment, observational studies, sample studies, etc., their advantages and disadvantages, ethical or societal issues.	Experimental Protocol Deconstruction, Open Discussion	Class Test, Assignment, Midterm, Final	CLO1, CLO2, CLO3
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Probability theory, set theory, experimental vs. theoretical probability, etc.	Sample Space Set Mapping, Combinatorial Ratio Math	Class Test, Midterm, Final	CLO1, CLO2
	Lesson 2: Addition and multiplication rules probability, Venn diagrams, independent events, dependent events and multiplication rule, and conditional probability.	Bayes Theorem Formula Practice, Venn Space Decomposition	Class Test, Midterm, Final	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Counting outcomes using tree diagrams, permutations, combinations, computing probabilities using counting principles.	Permutation Combinatoric Multiplication, Tree Path Tracing	Class Test, Midterm, Final	CLO1, CLO2
	Lesson 2: Random variables, types, constructing probability distribution, mean, variance, and standard deviations of random variables, scaling and shifting random variables, combining random variables, etc. Class Test-2	Expectation Operator Math, Linear Combination Scaling	Class Test, Midterm, Final	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Binomial, geometric, and Poisson random variables, calculating expected values and variance of the random variables.	PMF Function Equation Practice, Discrete Metric Tracking	Class Test, Midterm, Final	CLO1, CLO2
	Lesson 2: The normal and the standard normal distribution, probability density functions, etc.	Z-Score Integration Calculus, Normal Curve Integration	Class Test, Midterm, Final	CLO1, CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Sampling distributions, biased vs. unbiased sample statistics, sampling distribution of sample proportions and means, the central limit theorem, etc.	Central Limit Convergence Graphing, Standard Error Reduction	Class Test, Midterm, Final	CLO1, CLO2
	Lesson 2: : Confidence intervals, margin of error, interpreting confidence intervals and confidence levels, conditions for valid confidence intervals for both mean and proportion, t-statistic, confidence interval for mean with paired data, calculating sample size for given margin, etc.	Interval Margin Estimation, Critical T-Value Mapping	Class Test, Midterm, Final	CLO1, CLO2

Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Hypothesis testing, null and alternate hypothesis, estimating, Interpreting P-values to draw conclusions, type-I and type-II errors, statistical power, etc.	Type Error Boundary Analysis, Power Function Calculation	Class Test, Final	CLO1, CLO2
	Lesson 2: Tests about population proportions, conditions for a z test about proportions, calculating z statistics about a proportion	Z-Test Test-Statistic Practice, Tail Probability Analysis	Class Test, Final	CLO1, CLO2
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Calculating p-value from z-statistic and t-statistic, drawing conclusions.	Tail Area Probability Calculation, Decision Rule Analysis	Class Test, Final	CLO1, CLO2
	Lesson 2: Hypothesis testing about population mean, conditions for t-test about a mean, z-test vs. t-test Class Test-3	Degrees-of-Freedom Calculations, Student-T Bound Evaluation	Class Test, Final	CLO1, CLO2
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Comparing population proportions, statistical significance of experiment, distribution of difference of sample means	Pooled Proportion Calculations, Difference Mean Modeling	Final	CLO1, CLO2
	Lesson 2: Chi-square test for homogeneity, Chi-square test for association, contingency table Chi-square test, Chi-square distribution, Pearson's Chi-square test, Chi-square statistic for hypothesis test, Chi-square goodness of fit.	Observed-Expected Matrix Analysis, Goodness-of-Fit Testing	Final	CLO1, CLO2
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Chi-square test for homogeneity and association, contingency table Chi-square test, etc. Analysis of variance (ANOVA), calculating total sum of squared with and between groups, hypothesis test with F-statistic, etc.	Sum-of-Squares Table Calculus, F-Ratio Balance Tracking	Final	CLO1, CLO2
	Lesson 2: Evaluating statistical results and interpretations for any potential source of bias, misinterpretation, and societal, cultural, or ethical concerns. Inference about slope in linear regression, conditions for inference on slope, confidence intervals for the slope of a regression line, hypothesis testing about the slope.	Regression Parameter Estimation, Slope Confidence Interval Calculation	Final	CLO1, CLO2
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Architecture Review	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Evaluating linear regression results and interpretations for any potential source of bias, misinterpretation, and societal, cultural, or ethical concerns. Review Class	Final Syllabus Cross-Mapping, Core Formula Breakdown	Assignment, Final	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO2	PLO4	PLO6	PLO12	
Attendance	--	--	--	--	7	--	--	--	--	
Class Test 1	15				AVG 15					
Class Test 2		15								
Class Test 3			15							
Assignment			5		5					
Presentation				8	8					
Midterm Exam	10	10	5		25	10	10	5		25
Final exam	10	10	10	10	40	10	10	10	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Larry Wasserman, All of Statistics: A Concise Course in Statistical Inference, Springer, 2022, ISBN: 978-0387402727.

Reference Books/ Other Supplementary Readings

1. Andy Field, Jeremy Miles, & Zoë Field, Discovering Statistics Using R, 1st edition, Sage Publications, 2012, ISBN: 978-1446200469.
2. David Griffiths, Head First Statistics, 1st edition, O'Reilly Media, 2009, ISBN: 978-0596527590.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Google Colab, Jupyter Notebook

Exam Dates

According to the Examination Schedule



Department of Software

Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

Engineering (SWE)

DS 412 Statistical Data Analysis Lab

Course Code: DS 412		CIE Marks: 60
ISCED: 0613-412		LF Marks: 40
Course Title: Statistical Data Analysis Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)	Contact Hours: 1 hour 30 minutes (14 weeks)	
Prerequisite: STA 101		
Course Type: Data Science Major		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

Data Types, Variables, Data Structures, Functions, Conditions, Loops, Objects, Exploratory Data Analysis, Histograms, Density Plots, Boxplots, Scatter Plots, Bar Charts, Heatmaps, Linear Regression, Logistic Regression, Independent T-Test, Dependent T-Test, ANOVA, Factorial ANOVA, Repeated ANOVA, Mixed ANOVA, ANCOVA, MANOVA, Chi-Square Test, A/B Test, Wilcoxon, Kruskal-Wallis, Friedman's Test, Multilevel Linear Model, Sample Size Calculation, Power Calculation, Intraclass Correlation, Effect Size, Report Writing.

Rationale of the Course

Statistical analysis is a scientific method for collecting and analyzing vast quantities of data to uncover common patterns and trends and transform them into meaningful information. Simply said, statistical analysis is a tool for analyzing data that aids in drawing meaningful conclusions from unstructured and unprocessed data. This course will teach students how to use modern programming languages such as R, Python, etc., and different statistical software. Students will learn how to use these tools and its libraries to do exploratory analysis, build and test models, and run statistical tests.

Course Objectives

The goal of this course is to introduce the students to the concept of data science principles in practice. The main objectives of this course are,

- To write advanced level scripts using the R programming language
- To load, wrangle, and visualize data..
- To learn to perform different statistical tests..
- To learn the best practices of a reproducible research workflow

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Code fluently in any modern programming language such as R, Python, etc.	PLO5	C3	Fundamental Domain
CLO2	Carry out statistical data analysis using any modern programming language such as R, Python, etc.	PLO5	C3	Fundamental Domain
CLO3	Code collaboratively with peers to develop statistical solutions and software.	PLO9	C3	Fundamental Domain
CLO4	Communicate findings properly to the stakeholders through presentations, reports, etc.	PLO10	C5, P4	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1					√							
CLO2					√							
CLO3									√			
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning, and assessment strategy mapped with CLOs:

Week-1 [1.5 Hours]	Discussion on OBE and Setting up a programming environment, learning to install packages and setting up workspace,	Virtual Environment Initialization, Package Dependency Pip-Flashing	CLO1	None
Week-2 [1.5 Hours]	Writing conditional logics, loops, functions, and working with classes and objects, Working with different data formats and developing data preprocessing pipelines for analysis.	Object-Oriented Instance Structuring, CSV/JSON File Parsing	CLO1	Lab Performance
Week-3 [1.5 Hours]	Introduction to different tools for data collection, available data repositories and APIs, and writing scripts to connect them.	HTTP Request Verification, REST API Payload Ingestion	CLO2	Lab Performance
Week-4 [1.5 Hours]	Generating different types of graphs for exploratory data analysis and exploring the best practices of data visualization.	Subplot Grid Matrix Plotting, Visual Density Mapping	CLO2, CLO4	Viva, Lab Final

Week-5 [1.5 Hours]	Calculating summary statistics such as measure of central tendency, measure of spread, and measure of association.	Descriptive Matrix Aggregation, Covariance Coefficient Calculation	CLO2, CLO4	Viva, Lab Final
Week-6 [1.5 Hours]	Designing, training, and evaluating simple linear regression, multiple linear regression models, and logistic regression models	Least-Squares Cost Optimization, Confusion Matrix Profiling	CLO2, CLO4	Viva, Lab Final
Week-7 [1.5 Hours]	Implementation of different types of operators in SQL Arithmetic Operators, Logical Operators, Comparison Operator, Special Operator, Set Operation.	Relational Operator Statement Binding, Set Union/Intersect Queries	CLO2, CLO4	Viva, Lab Final
Week-8 [1.5 Hours]	Performing the ANOVA and its variants, calculating effect size, reporting and interpreting the results of ANOVA.	Variance F-Ratio Calculation, Eta-Squared Factor Evaluation	CLO2, CLO4	Lab Performance, Viva, Lab Final
Week-9 [1.5 Hours]	Performing Chi-Square tests, measuring goodness of fits and independence, and performing A/B testing.	Contingency Table Observed-Expected Analysis, Variant Split Traffic Tracking	CLO2, CLO4	Lab Performance
Week-10 [1.5 Hours]	Performing Wilcoxon rank-sum test, Wilcoxon signed-rank test, Kruskal-Wallis's test, etc.	Non-Parametric Rank Transmutation, Median Variance Testing	CLO2, CLO4	Lab Performance, Viva, Lab Final
Week-11 [1.5 Hours]	Developing custom packages, testing codes, and writing documentations.	Module Namespace Structuring, PyTest Unit-Assertion Sweeping	CLO3	Lab Performance, Viva, Lab Final
Week-12 [1.5 Hours]	Creating dashboards for data visualization and analytics: connecting data from various databases and APIs.	BI Connector Stream Wiring, Real-Time Pipeline Mapping	CLO3	Lab Performance, Viva, Lab Final
Week-13 [1.5 Hours]	Creating dashboards for data visualization and analytics: data preprocessing, data visualization and analysis.	Multi-Panel Widget Event Binding, Interactive Callback Configuration	CLO3	Lab Performance, Viva, Lab Final
Week-14 [1.5 Hours]	Best practices of presenting statistical findings and writing statistical reports.	Metric Telemetry Sheet Compilation, Analytical Asset Showcasing	CLO4	Lab Performance

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10

Lab Performance	25	CLO1	PLO5
Lab Final	40	CLO2	PLO5
		CLO3	PLO9
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Peter Bruce, Andrew Bruce, & Peter Gedeck, Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python, O'Reilly Media, 2020, ISBN: 978-1491952961.

Reference Books/ Other Supplementary Readings

- 1.Wes McKinney, Python for Data Analysis, 3rd edition, O'Reilly Media, 2022, ISBN: 978-1491957660
- 2.Jeremy Grus, Data Science from Scratch: First Principles with Python, 2nd edition, O'Reilly Media, 2019, ISBN: 978-1492041139.
- 3.Larry Wasserman, All of Statistics: A Concise Course in Statistical Inference, Springer, 2010, ISBN: 978-0387402727.
- 4.Andy Field, Jeremy Miles, & Zoë Field, Discovering Statistics Using R, 1st edition, Sage Publications, 2012, ISBN: 978-1446200469.
- 5.Lab Manual.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Google Colab, Jupyter Notebook, Git, Python

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

DS 421 Machine Learning Driven Data Analysis I

Course Code: DS 421		CIE Marks: 60
ISCED: 0613-421		SEE Marks: 40
Course Title: Machine Learning Driven Data Analysis I		Total Marks: 100
Semester:		
Credit Value: 2 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: DS 331, DS 332, DS 411, DS 412		
Course Type: Major		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Gradient Descent, Hypothesis Function, Loss Function, Learning Rate, Learning Rate Scheduling, Early Stopping, Stochastic Gradient Descent, Batch Gradient Descent, Mini-Batch Gradient Descent, Simple Linear Regression, Multiple Linear Regression, Vectorization of Gradient Descent, Logistic Regression, Binary Log-Loss Function, Polynomial Regression, Overfitting And Underfitting, Bias-Variance Tradeoffs, Regularization Methods, L1 Penalty, L2 Penalty, Ridge Regression, Lasso Regression, Elastic Net, Artificial Neural Networks, Activation Functions: ReLU, Hyperbolic Tangent Function, SoftMax, etc., Dropout Layer, Advanced Optimization Techniques: Momentum, RMSProp, and ADAM, Decision Tree, Gini, Entropy, CART Algorithm, Ensemble Learning: Voting, Bagging, Pasting, Random Patches, Random Subspaces, Random Forest, Boosting, and Stacking, XGBoost, Model Selection: Hyperparameter Tuning, Grid Search, Random Search and Cross-Validation, Model Evaluation: Bootstrapping, Mean Absolute Error, Mean Square Error, Root Mean Square Error, Confusion Matrix, Type I Error, Type II Error, Accuracy, Precision, Recall, Sensitivity, Specificity, True Positive Rate, False Positive Rate, AUC-ROC, Precision-Recall Curve, Feature Engineering: Normalization, Standardization, and Augmentation, K-Means Clustering, Agglomerative Clustering, Single Linkage, Complete Linkage, Average Linkage, DBSCAN, Anomaly Detection, Recommendation Systems, Collaborative Filtering, Content-Based Filtering, Dimensionality Reduction, Principal Component Analysis, Locally Linear Embedding, t-SNE, Reinforcement Learning, Q-Learning.

Rationale of the Course

This course offers a comprehensive overview of contemporary machine learning, encompassing supervised learning techniques such as logistic regression, neural networks, and decision trees; unsupervised learning approaches like clustering, dimensionality reduction, and recommender systems; and industry-recognized best practices for machine learning and artificial intelligence innovation (including model tuning, data-centric performance enhancement, and more). Students will have mastered fundamental principles and acquired the practical skills necessary to implement machine learning techniques efficiently and effectively on complex real-world issues by the conclusion of this course.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of Machine Learning. More specifically,

- Students will be taught to apply appropriate machine learning models to automate real-world problems.
- Students will be taught to critique existing models so that they can identify any issues with experiment design, analysis, results, conclusions, and interpretation.
- Students will be taught to evaluate any societal, cultural, or ethical issues in existing machine learning.
- Students will be taught to develop their self-learning capabilities so that they can keep themselves updated with the dynamic field of machine learning.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Apply appropriate machine learning models to solve real-world problems.	PLO2	C3	Fundamental Domain
CLO2	Analyze machine learning models to verify their design, analysis, results, conclusion, and interpretation.	PLO4	C4	Fundamental Domain
CLO3	Evaluate models to discover potential bias, fairness issues, or societal, cultural, or ethical concerns.	PLO6	C5	Fundamental Domain, Social Thinking, Personal Domain
CLO4	Analyze the latest research literature to keep themselves updated which will help for life-long learning.	PLO12	C4	Social Thinking, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1			√									
CLO2				√								
CLO3						√						
CLO4												√

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to course syllabus, learning outcomes, assessment methods, deadlines, consultation hour, code of conduct, and other administrative procedures regarding the course.	Syllabus Walkthrough, Administrative Q&A	N/A	N/A
	Lesson 2: Linear Regression, cost function formula and intuition, visualization of cost function, gradient descent for linear regression, learning rates, etc.	Ordinary Least Squares Math, Cost Surface Graphing	Class Test, Midterm Final	CLO1,CLO2
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Simple Linear Regression, multiple linear regression, vectorization of gradient descent.	Matrix Vectorization Calculations, Linear Algebra Transformation	Class Test, Midterm Final	CLO1,CLO2
	Lesson 2: Feature Engineering, Feature scaling, checking gradient descent for convergence, choosing learning rate, and stopping criteria of gradient descent.	Feature Normalization Scaling, Convergence Monitoring Metric	Class Test, Midterm Final	CLO1,CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Polynomial regression, Classification with logistic regression, cost function of logistic regression and gradient descent implementation.	Log-Odds Equation Tracing, Sigmoid Probability Mapping	Class Test, Midterm Final	CLO1,CLO2
	Lesson 2: The problem of overfitting, regularization with Ridge, Lasso, and ElasticNet, Evaluate regression models for fairness and other societal, ethical, or cultural issues.	Penalty Constraint Mapping, Fairness Disparate Assessment	Class Test, Midterm Final	CLO1,CLO2, CLO3
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Neural Networks, Neural Network layers, activation functions, and forward propagation, matrix multiplications, tricks of implementing neural networks efficiently.	Layer Architecture Blueprinting, Matrix Multiplying Tracing	Class Test, Midterm Final	CLO1,CLO2
	Lesson 2: Training Neural Networks, choosing activation functions, SoftMax for multiclass classifications, and classification with multiple outputs. Class Test - 1	Backpropagation Chain Rule Calculus, Categorical Softmax Cross-Entropy	Class Test, Midterm Final	CLO1,CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Advanced optimization techniques such as ADAM, RMSProp, etc., and additional layer types.	Hyperparameter Learning Decay Tracking, Momentum Optimization	Class Test, Midterm Final	CLO1,CLO2
	Lesson 2: Model evaluation and model selection, training / cross-validation / and test	Cross-Validation Space Modeling, Variance Overfitting Diagnosis	Class Test, Midterm Final	CLO1,CLO2

	sets, diagnosing bias and variance, and regularization.			
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Establishing baseline performance, learning curves, error analysis, error metrics for skewed data, precision recall tradeoffs, fairness, bias, and ethics.	Confusion Matrix Analysis, PR Curve Threshold Optimization	Class Test, Assignment, Midterm, Final	CLO1,CLO2,CLO3
	Lesson 2: Evaluate artificial neural network models for fairness and other societal, ethical, or cultural issues, Decision Tree model, learning process, measuring impurity, and choosing a split using information gain. Class Test - 2	Recursive Partitioning Modeling, Entropy Impurity Calculus	Class Test, Assignment, Midterm, Final	CLO1,CLO2,CLO3
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: One-hot encoding for categorical features and regression trees., Using multiple decision trees, sampling with replacements, and Random Forest algorithm.	Categorical Sparse Representation Mapping, Bootstrap Aggregation Review	Class Test, Midterm Final	CLO1,CLO2
	Lesson 2: XGBoost algorithm and other ensemble learning methods.	Residual Loss Gradient Boosting, Sequential Tree Addition	Class Test, Midterm Final	CLO1,CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to clustering algorithms, K-Means clustering intuition and algorithm, and optimization objective.	Cluster Centroid Vector Calculation, Geometry Metric Optimization	Class Test, Final	CLO1,CLO2
	Lesson 2: K-Means initialization techniques, choosing the best number of clusters, advantages and disadvantages of K-Means and ways to overcome them such as multiple initialization and elbow method.	Variance Inertia Minimization, Elbow Plot Coordinate Analysis	Class Test, Final	CLO1,CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Hierarchical / Agglomerative clustering algorithm and choosing the best number of clusters from dendrograms.	Tree Linkage Agglomeration Parsing, Dendrogram Tree Slicing	Class Test, Final	CLO1,CLO2
	Lesson 2: Evaluating clustering algorithms using inertia / intra-cluster distance and inter-cluster distance, and other cluster evaluation metrics.	Silhouette Width Calculations, Variance Ratio Tracking	Class Test, Final	CLO1,CLO2
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Finding unusual events, Gaussian distributions, and anomaly detection algorithms.	Multivariate Probability Density Function Practice, Elliptical Contour Analysis	Class Test, Final	CLO1,CLO2
	Lesson 2: Developing and evaluating anomaly detection system, anomaly detection vs. supervised learning, and choosing appropriate features.	F1-Score Anomaly Threshold Adjusting, Feature Space Review	Class Test, Final	CLO1,CLO2
Week-11	Lesson 1: Recommendation systems, using per-item features, and collaborative filtering algorithms.	Latent Factor Dot Product Modeling,	Class Test, Assignment, Final	CLO1,CLO2,CLO3

Lesson 1 & 2 [3 Hours]		User Item Matrix Factorization		
	Lesson 2: Incorporating binary labels, mean normalization, and finding related items, and ethical concerns of collaborative filtering. Class Test - 3	Rating Bias Normalization Math, Cosine Similarity Vector Ranking	Class Test, Assignment, Final	CLO1,CLO2,CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Shortcomings of collaborative filtering algorithm, introduction to content-based filtering method.	Cold-Start Trait Vector Extraction, User Profile Feature Modeling	Assignment, Final	CLO1,CLO2,CLO3
	Lesson 2: Deep learning for content-based filtering, recommending large catalog, and ethical use of recommender systems.	Two-Tower Neural Network Embedding, Catalog Retrieval Pipeline	Assignment, Final	CLO1,CLO2,CLO3
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Curse of dimensionality, main approaches of dimension reduction, and reducing the number of features using Principal Component Analysis (PCA) algorithm, Other approaches of dimension reduction such as Kernel PCA, Locally Linear Embedding (LLE) and t-SNE.	Covariance Orthogonal Vector Rotation, Non-linear Manifold Embedding	Final	CLO1,CLO2
	Lesson 2: Reinforcement Learning, optimizing rewards, Policy search, Q-Learning, Evaluating actions, Markov decision process, Deep Q-Learning variants.	Bellman Equation Iteration Practice, Value Function Network Approximation	Final	CLO1,CLO2
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Industry Expert Talk, Production Blueprint Review	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review and discussion on latest machine learning literatures.	Peer Synthesis Review, Academic Manuscript Dissection	Final, Presentation	CLO1,CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total = 100)	PLO's				Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO3	PLO4	PLO6	PLO12	
Attendance	--	--	--	--	7					
Class Test 1	15				Avg 15					
Class Test 2		15								
Class Test 3			15							

Assignment			5		5					
Presentation				8	8					
Midterm Exam	10	10	5		25	10	10	5		25
Final exam	10	10	10	10	40	10	10	10	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd edition, O'Reilly Media, 2023, ISBN: 978-1492032649.

Reference Books/ Other Supplementary Readings

1. Mykel J. Kochenderfer, & Tim A. Wheeler Algorithms for Optimization, 1st Edition, The MIT Press, 2019, ISBN: 978-0262039420
2. Ian Goodfellow, Yoshua Bengio, & Aaron Courville (2016). Deep Learning. MIT Press. ISBN: 978-0262035613

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

DS 422 Machine Learning Driven Data Analysis Lab

Course Code: DS 422		CIE Marks: 60
ISCED: 0613-422		LF Marks: 40
Course Title: Machine Learning Driven Data Analysis Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: DS 331, DS 332, DS 411, DS 412		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

Python, its Variables and Built-in Data Structures, Object-Oriented Python, Scikit-Learn, Pandas, Matplotlib, NumPy, Supervised Machine Learning, Linear Regression, Logistic Regression, Lasso Regression, Ridge Regression, Elastic Net, Decision Tree, CART Algorithm, Support Vector Machine (SVM), Kernel Methods, Unsupervised Machine Learning, K-Means Clustering, DBSCAN, Hierarchical Clustering, Gaussian Mixture Models, Bayesian Gaussian Mixture Models, Dimensionality Reduction, Principal Component Analysis (PCA), Kernel PCA Locally Linear Embedding,

Rationale of the Course

Machine learning is a subfield of artificial intelligence (AI) and computer science that focuses on the use of data and algorithms to emulate how humans learn while gradually enhancing its accuracy. Machine learning is an integral part of the rapidly expanding discipline of data science. As big data continues to spread and flourish, there will be a greater demand for machine learning engineers. They will be expected to assist in identifying the most pertinent business queries and the corresponding data. This course examines the fundamentals of machine learning using the well-known and approachable programming language Python.

Course Objectives

The goal of this course is to introduce the students about the concept of Data Science. The main objectives of this course are:

- . Students will be taught to write advanced level scripts using the Python programming language.
- . Students will be taught to load, wrangle, and visualize data.
- . Students will be taught to develop different machine learning models.
- . Students will be taught to learn the best practices of a reproducible research workflow.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Code fluently in any modern programming language such as Python.	PLO5	C3	Fundamental Domain
CLO2	Build machine learning models using any modern programming language such as Python..	PLO6	C3	Fundamental Domain
CLO3	Code collaboratively with peers to develop machine learning solutions.	PLO9	C3,	Fundamental Domain
CLO4	Communicate findings properly to the stakeholders through presentations, reports, etc	PLO10	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1					√							
CLO2						√						
CLO3									√			

CLO4										√		
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Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Discussion on course outline, course learning outcomes, assessment methods, deadlines, and other administrative information.	Syllabus Walkthrough, Infrastructure Orientation	None	None
Week-2 [1.5 Hours]	Setup Python programming environment, learn to install packages and setup working directory, work with different data types of Pythons.	Virtual Environment Initialization, Package Dependency Pip-Flashing	CLO1	Lab Performance / Lab Final
Week-3 [1.5 Hours]	Familiarization with the machine learning ecosystem of Python, Review of Object-Oriented Python, Introduction to NumPy, Pandas, Matplotlib, Seaborn, and Sklearn.	Multi-dimensional Array Vectorization, DataFrame Ingestion Mechanics	CLO1	Lab Performance / Lab Final
Week-4 [1.5 Hours]	Implement Linear Regression to perform regression task and implement metrics for model evaluation.	Ordinary Least Squares Optimization, Residual Variance Evaluation	CLO2 CLO3 CLO4	Lab Project
Week-5 [1.5 Hours]	Implement Ridge, Lasso, and Elastic Net to perform regression task and implement metrics for model evaluation.	L1/L2 Regularization Penalty Tuning, Feature Weight Shrinkage Audit	CLO2 CLO3 CLO4	Lab Project
Week-6 [1.5 Hours]	Implement Logistic Regression to perform classification task and implement metrics for model evaluation.	Sigmoid Decision Boundary Mapping, Confusion Matrix Profiling	CLO2 CLO3 CLO4	Lab Project
Week-7 [1.5 Hours]	Implement linear Support Vector Machine to perform classification task and implement metrics for model evaluation.	Hard/Soft Margin Parameter Customization, Hyperplane Vector Spacing	CLO2 CLO3 CLO4	Lab Project
Week-8 [1.5 Hours]	Implement non-linear Support Vector Machine to perform classification task and implement metrics for model evaluation.	RBF/Polynomial Kernel Hyperparameter Configuration, Dual Space Transformation	CLO2 CLO3 CLO4	Lab Project
Week-9 [1.5 Hours]	Implement Support Vector Machine to perform regression task and implement metrics for model evaluation.	Epsilon-Tube Width Adjustment, Cost Function Margin Hardening	CLO2 CLO3 CLO4	Lab Project

Week-10 [1.5 Hours]	Implement Decision Tree algorithm to perform classification task and implement different metrics for model evaluation.	Gini/Entropy Node Binary Splitting, Tree Depth Pruning Hardening	CLO2 CLO3 CLO4	Lab Project
Week-11 [1.5 Hours]	Combine multiple models using voting, bagging, pasting, random patches, and random subspaces ensemble techniques.	Bootstrap Row-Column Subsampling, Soft/Hard Aggregate Decision Voting	CLO2 CLO3 CLO4	Lab Project
Week-12 [1.5 Hours]	Combine multiple models using Boosting, Stacking, and Random Forest ensemble techniques.	Sequential Error Weight Updating, Multi-tier Meta-Learner Layering	CLO2 CLO3 CLO4	Lab Performance, Lab Final
Week-13 [1.5 Hours]	Implement Principal Component Analysis (PCA) and Locally Linear Embeddings (LLE) algorithms to reduce data dimension.	Covariance Singular Value Decomposition (SVD), Neighborhood Manifold Unrolling	CLO2 CLO3 CLO4	Lab Performance, Lab Final
Week-14 [1.5 Hours]	Implement K-Means, DBSCAN, and Hierarchical clustering algorithms to segment images, perform image preprocessing tasks and semi-supervised tasks and Gaussian Mixture models and Bayesian Gaussian Mixture models to detect anomaly and novelty.	Density-Based Core Point Traversal, Expectation- Maximization Ellipsoid Boundary Analysis	CLO2 CLO3 CLO4	Lab Performance, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO5
		CLO3	PLO9
Lab Final	40	CLO1	PLO5
		CLO2	PLO5
		CLO3	PLO9
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd edition, O'Reilly Media, 2023, ISBN: 978-1492032649.

Reference Books/ Other Supplementary Readings

1. McKinney, W., Python for Data Analysis, 3rd edition, O'Reilly Media, 2022, ISBN: 978-1098104035.
2. Hastie, T., Tibshirani, R., & Friedman, J., The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd edition, Springer Series in Statistics ,2017, ISBN: 978-0387848570.
3. Bishop, C. M., Pattern Recognition and Machine Learning, 1st edition, Information Science and Statistics. Springer, 2016, ISBN: 978-0387310732.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Google Colab, NumPy, Pandas, Matplotlib, Seaborn, and Sklearn

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

SE 431 Numerical Analysis

Course Code: SE 341		CIE Marks: 60
ISCED: 0613-341		SEE Marks: 40
Course Title: Numerical Analysis		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Guided Elective V		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

In this course students will be introduced to the mathematical analysis of numerical methods by emphasizing on different algorithms that are encountered in many disciplines like physical sciences and engineering. By the end of this course,

students will be competent to solve complex mathematical problems using simple arithmetic operations such as bisection method, Newton Raphson's method, differentiation, matrix problems, finding roots of equations, overview of Gaussian elimination, partial pivoting, LU decomposition, boundary value.

Rationale of the Course

This course offers a comprehensive grounding in the mathematical analysis of numerical methods and algorithms applicable across engineering and the physical sciences. Students explore essential root-finding and calculus techniques, including the bisection method, Newton-Raphson method, and numerical differentiation. The curriculum details advanced linear algebra operations such as Gaussian elimination, partial pivoting, and LU decomposition, alongside boundary value analysis

Course Objectives

This course is an introduction to numerical analysis. The primary objective of the course is to develop the basic understanding of numerical algorithms and skills to implement algorithms to solve mathematical problems on the computer. The main objectives of this course are that students will be able to

- Demonstrate understanding of common numerical methods and how they are used to obtain approximate solutions to otherwise intractable mathematical problems.
- Apply numerical methods to obtain approximate solutions to mathematical problems.
- Derive numerical methods for various mathematical operations and tasks, such as interpolation, differentiation, integration, the solution of linear and nonlinear equations, and the solution of differential equations.
- Analyze and evaluate the accuracy of common numerical methods.

Course Learning Outcomes (CLOs) with Mappings PLOs

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Demonstrate the basics of numerical calculation	PLO1	C2	Fundamental Domain
CLO2	Classify the solution of non-linear equation and set of simultaneous linear equation	PLO2	C2,C4	Fundamental Domain
CLO3	Evaluate basic solutions of ordinary differential equations.	PLO2	C3, C5	Fundamental Domain
CLO4	Identify proper solutions for solving the linear and nonlinear equations	PLO4	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3		√										
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Introduction about OBE, CLO,PLO of course	Syllabus Walkthrough, Outcome Mapping Q&A	None	N/A
	Lesson 2: Introduction and fundamental concepts of numerical methods.	Continuous-to-Discrete Mapping, Algorithm Complexity Review	Class Test, Midterm	CLO1
Week-2 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Concept of Approximation, procession & Accuracy.	Sig-Fig Error Analysis, Measurement Variance Review	Class Test, Midterm	CLO1
	Lesson 2: Overview on different types of errors, Concept of floating point arithmetic.	IEEE-754 Mantissa Dissection, Underflow/Overflow Diagnostics	Class Test, Midterm	CLO1
Week-3 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Defining True error, approximate error, truncation error and Rounding error.	Taylor Series Remainder Estimation, Absolute/Relative Math Drills	Class Test, Midterm	CLO1
	Lesson 2: Basic Concepts of non- linear equations Class Test - 1	Interval Sign-Change Verification, Graph Bounds Exploration	Class Test, Midterm	CLO1
Week-4 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Define algebraic, polynomial, Transcendental equations.	Function Degree Classification, Root Multiplicity Review	Class Test, Midterm	CLO1
	Lesson 2: Overview on solutions of non-linear Equations.	Bracket vs Open-Interval Comparison, Solvability Analysis	Class Test, Midterm	CLO1
Week-5 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Finding roots of nonlinear equations. (direct , graphical, trial & error, iterative methods	Visual Intersection Plotting, Fixed-Point Iteration Mapping	Class Test, Midterm	CLO2

	Lesson 2: Overview on Bi-section method. (Algorithm, convergence, advantages and disadvantages)	Interval Halving Calculus, Linear Convergence Bound Check	Class Test, Midterm	CLO2
Week-6 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Mathematical Analysis of Bisection Method. Overview on False Position Method	Iteration Error Bound Math, Secant Chording Walkthrough	Class Test, Midterm, Final Examination	CLO2
	Lesson 2: Mathematical analysis of false position Method, Overview on Newton Raphson method	Regula-Falsi Proportionality Math, Tangent Line Derivative Mapping	Class Test, Midterm, Final Examination	CLO2
Week-7 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Mathematical analysis on Newton Raphson Method. Class Test - 2	Quadratic Convergence Calculus, Stationary Point Defect Review	Class Test, Midterm, Final Examination	CLO2
	Lesson 2: Overview on Secant method, Mathematical analysis on secant method	Finite Difference Derivative Approximation, Convergence Order Math	Class Test, Midterm, Final Examination	CLO2
Week-8 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Mathematical analysis on secant Method, Overview on differentiation.	Root Convergence Rate Auditing, Slopes Calculus Review	Class Test, Midterm, Final Examination	CLO3
	Lesson 2: Overview on Numerical differentiation. (differentiation, approximate error) Differentiating Continuous function. (forward and backward)	Forward/Backward Stencil Taylor Derivation, Step-Size Error Math	Class Test, Midterm, Final Examination	CLO3
Week-9 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Successive Differentiation. Overview on Gaussian Elimination	Central Difference Higher-Order Math, Linear System Matrix Setup	Class Test, Final Exam	CLO3
	Lesson 2: Forward Elimination Technique, Overview on Back Substitution method	Elementary Row Op Matrix Reduction, Upper-Triangular Pivot Setup	Class Test, Final Exam	CLO3

Week-10 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Mathematical analysis of Back Substitution method Class Test - 3	Triangular Kernel System Solving, Row-Wise Loop Execution	Class Test, Final Exam	CLO3
	Lesson 2: Overview on Partial Pivoting	Pivot Row Swapping Heuristics, Round-off Ill-Conditioning Diagnostics	Class Test, Final Exam	CLO3
Week-11 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Mathematical Analysis of Partial Pivoting.	Permutation Matrix Algebra, Numerical Stability Verification	Class Test, Final Exam	CLO4
	Lesson 2: Overview on LU Decomposition	Doolittle/Crout Matrix Factoring, Lower-Upper Matrix Mapping	Class Test, Final Exam	CLO4
Week-12 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Finding the matrix and inverse matrix	Identity Target Gauss-Jordan Row Math, Inverse Transform Calculation	Class Test, Assignment, Final Exam	CLO4
	Lesson 2: Using LU decomposition to solve SLE, Concept of Block LU Decomposition	Forward/Backward Matrix Substitution Cascade, Sub-matrix Partition Review	Class Test, Assignment, Final Exam	CLO4
Week-13 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Overview on Boundary value & Solution of Boundary Value using Shooting method.	IVP Target Parameter Bounded Search, Root-Finding Integration Cascade	Class Test, Assignment, Final Exam	CLO4
	Lesson 2: Industry Session	Industry Expert Guest Lecture, Production Computational Scaling Review	Class Test, Assignment, Final Exam	CLO1, CLO2, CLO3, CLO4
Week-14 Lesson 1 & 2 [2.5 Hours]	Lesson 1: Overview on Eigenvalue & Power method.	Dominant Eigenvector Power Iteration, Rayleigh Quotient Calculation	Presentation, Final Exam	CLO4
	Lesson 2: Review on final syllabus topic	Comprehensive Method Cross-Mapping, Error Bound Troubleshooting	Presentation, Final Exam	CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO4	
Attendance	--	--	--	--	7				
Class Test 1	15				Avg 15				
Class Test 2		15							

Class Test 3			15						
Assignment					5				
Presentation					8				
Midterm Exam	10	15			25	10	15		25
Final exam			30	10	40		30	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

45% to less than 50%	C	2.25	Below Average
40% to less than 45%	D	2.00	Pass
Less than 40%	F	0.00	Fail

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

1. Numerical Methods for Engineers, By Raymond Canale and Steven C. Chapra, 7th Edition.

Reference Books/ Other Supplementary Readings

1. Numerical Methods by Balagurusamy

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering(SWE)
Faculty of Science and Information Technology
Daffodil International University (DIU)
(Version 3)

RE 423 Advanced Robotics

Course Code: RE 423			CIE Marks: 60
ISCED: 0714-423			SEE Marks: 40
Course Title: Advanced Robotics			Total Marks: 100
Semester:			
Credit Value: 2		Contact Hours: 3 (Total weeks: 12)	
Prerequisite: SE532			
Course Type: Major			

CIE: Continuous Internal Evaluation

LF: Lab Final

Class Content (from syllabus)

The basic terminologies related to Advanced Robotics applications, advanced sensors, actuators and algorithms. Working principles of advanced robotics components like, Lidar sensor, IMU sensor, encoder sensor, GPS sensor, Depth Camera, encoder motor, hub motor, Brush less motor. In-depth knowledge about robot kinematics. Mathematical representation of forward kinematics and inverse kinematics. Introduction to Robot Operating System (ROS). Components of ROS. Brief discussion about Gazebo, RVIZ and Rqt graph. Introduction to robot navigation system. In-depth knowledge about robot localization, sensor fusion, mapping, and motion planning. Designing advanced robotics solution architecture.

Rationale of the Course

This course covers advanced programming and hardware concepts associated with industrial robots. In this course, several robots are programmed to work together and with other common automation systems to increase the efficiency and throughput of industrial automation processes. Robot safety procedures and standards will be emphasized throughout the course. Two main categories: robotic manipulator and mobile robotics. Robotic manipulator includes: classification of robotic systems; transformation of coordinates; kinematics and inverse kinematics; Jacobians and robot dynamics; trajectory generation; modeling; control. Mobile robotics may include any of the following: wheeled mobile robot, machine vision basics, space and underwater robots; robot plume tracing, mobile robot trajectory

generation; motion planning, robot navigation and other new robotic developments process.

Course Objective

To provide a solid understanding of different components and design of Advanced Robotics solutions. More specifically

- . To explore advanced robotics components.
- . To learn the advanced concepts of advanced robotics.
- . To develop a deep understanding of advanced robot control systems.
- . To develop the ability to solve social problems using advanced robotics.

Course Learning Outcome (CLO) with Mapping

At the end of the course, students will be able to:

CLO's	CLO Descriptions	PLO's	Learning Domains (C, P,A)	BNQF Skills
CLO1	Identify different advanced robotics components.	PLO1	C1	Fundamental Skills
CLO2	Solve the mathematical problems of robot kinematics to reach substantiated conclusions	PLO2	C3	Fundamental Skills
CLO3	Evaluate robotics solutions architecture to solve social problems.	PLO6	C5	Fundamental Skills

Mapping of CLOs with PLOs

Course Learning Outcome	PLO 1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3						√						

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1	Lesson 1: Discussion on OBE of Advanced robotics course. Introduction to Advanced	Syllabus Walkthrough, Field Taxonomy Mapping	Class Test, Midterm exam	CLO1

Lesson 1 & 2 [3 Hours]	robotics applications, scopes and challenges.			
	Lesson 2: Exploring different types of Advance sensors for advanced robotics applications.	Transducer Profile Auditing, Sensor Platform Review	Class Test, Midterm exam	CLO1, CLO3
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Working principle of Lidar Sensor, classifications, applications.	Time-of-Flight (ToF) Mathematical Drills, Point-Cloud Ingestion Analysis	Class Test, Midterm exam	CLO1
	Lesson 2: Background study and working principles of Wheel Encoder and encoder motor. Exercise some mathematical problems related wheel encoder.	Pulse-to-Distance Calculatons, Kinematic Trajectory Modeling	Class Test, Midterm exam	CLO1, CLO3
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: In-Depth discussion about basics IMU. Applications of IMU in robotics aspects.	Sensor Dead-Reckoning Tracing, Gyro/Accel Integration Review	Class Test, Midterm exam	CLO1, CLO3
	Lesson 2: Classification of IMU. Working principles of IMU. Class Test-1	Degree-of-Freedom (DoF) Matrix Dissection, Drift Error Modeling	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Robot Vision Systems. Working principles of the Depth camera.	Structured Light Geometry Analysis, RGB-D Frame Realignment	Class Test, Assignment Midterm exam	CLO1
	Lesson 2: Differential drive working principles, advantages and disadvantages.	Non-holonomic Instantaneous Center of Curvature (ICC) Math, Kinematic Bounds Tracking	Class Test, Assignment Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Ackermann drive working principles, advantages and disadvantages.	Steering Geometry Calculations, Turning Radius Curve Analysis	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Transformation to Kinematics, Vector Operation, 2D Coordinate Vector Transformation. Notation, Homogenous transformation,	Matrix Row-Column Multiplication, Pose Operator Transformation	Class Test, Assignment, Midterm exam	CLO1

Week-6 Lesson 1 & 2 [3 Hours]]	Lesson 1: Spatial Description and Transformation (Rotation around axis, Vector in Space, Examples	Direction Cosine Rotation Matrix Setup, Reference Frame Projections	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Solving mathematical problems of Spatial Description and Transformation Class Test -2	SE(3) Transform Compounding Drills, Vector Rotation Translations	Class Test, Assignment, Midterm exam	CLO1
Week-7 Lesson 1 & 2 [3 Hours]]	Lesson 1: Frame relative to a fixed reference frame, Rigid body, Pure translation, Pure Rotation, Combined Transformation.	Composite Transformation Chaining Calculus, Frame Tree Dependency Mapping	Class Test, Assignment, Final exam	CLO1, CLO2
	Lesson 2: Solving Mathematical problems of Inverse kinematics	Trigonometric Geometric Dissection, Joint Variable Extraction Math	Class Test, Assignment, Final exam	CLO2
Week-8 Lesson 1 & 2 [3 Hours]]	Lesson 1: Industrial Robot: Functional Architecture.	Controller-Manipulator Linkage Topology Analysis, Block Diagram Blueprinting	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Industrial Robot: classifications and applications.	Cartesian/Articulated Workspace Envelope Profiling, Pay-Load Capability Audit	Class Test, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]]	Lesson 1: Introduction to Robot Operating System (ROS).	Linux IPC Architecture Verification, Environment Setup	Class Test, Final exam	CLO1, CLO3
	Lesson 2: ROS, ROS Components, Nodes, Topics, Services, Messages, ROS Master, Mobile Robot, ROS Tools	Distributed Graph Topography Mapping, Pub-Sub Pipeline Tracing	Class Test, Assignment, Final exam	CLO1, CLO2
Week-10 Lesson 1 & 2 [3 Hours]]	Lesson 1: ROS node and message commutation over the network.	TCPROS Wire Protocol Dissection, Socket Packet Tracking	Class Test, Assignment, Final exam	CLO2
	Lesson 2: Robot Localization	Odometry-Laser Scan Alignment, Probability Pose Estimation	Class Test, Assignment, Final exam	CLO2

Week-11 Lesson 1 & 2 [3 Hours]]	Lesson 1: Sensor Fusion using ROS, Robot software architecture designing using ROS navigation stack. Class Test -3	Extended Kalman Filter (EKF) Setup, Costmap Layer Tuning	Class Test, Assignment, Final exam	CLO1, CLO3
	Lesson 2: Design robotics solutions to solve social problems.	Operational Pipeline Ideation, Scope Blueprinting	Class Test, Assignment Final exam	CLO1, CLO3
Week-12 Lesson 1 & 2 [3 Hours]]	Lesson 1: Industry Week Session	Guest Lecture / Industry Expert Talk, Production Deployment Architecture Review	Assignment, Presentation	CLO1, CLO3
	Lesson 2: Review class on final exam syllabus. Presentation on a given topic	Comprehensive Blueprint Cross-Mapping, Technical Defense Showcasing	Presentation, Open discussion	CLO1, CLO2, CLO3

Overall Assessment Scheme

Assessment Task	CLO's			Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3		PLO1	PLO2	PLO6	
Attendance	--	--	--	7				
Class Test 1	5(15)			AVG 15				
Class Test 2		5(15)						
Class Test 3			5(15)					
Assignment		3	2	5			5	5
Presentation				8				
Midterm Exam	15	10		25	15	10		25
Final exam	5	15	20	40	15	5	20	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

John J. Craig, Introduction to Robotics: Mechanics and Control, Global edition, 4th edition, Pearson, 2021, ISBN: 978-1292164953.

Reference Books/ Other Supplementary Readings

Lentin Joseph, Mastering ROS for Robotics Programming - Third Edition: Best practices and troubleshooting solutions when working with ROS, 3rd edition, Packt Publishing, 2021, ISBN 978-1801071024

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology
Daffodil International University (DIU)
(Version 3)

RE 424 Advanced Robotics Lab

Course Code: RE424		CIE Marks: 60
ISCED: 0714-424		LF Marks: 40
Course Title: Advanced Robotics Lab		Total Marks: 100
Semester:		
Credits: 1 (Lab)	Contact Hours: 1.5 (Total 15 weeks)	
Prerequisite: SE532		
Course Type: Major		

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

The Advanced Robotics Lab course encompasses a diverse range of topics aimed at providing students with comprehensive knowledge and practical skills in robotics. Beginning with the study and implementation of DC encoder motors and HUB motors, participants delve into sensor technologies such as IMUs and Lidar, exploring their integration with the Robot Operating System (ROS) environment. Concepts like PID control and fuzzy logic are covered for advanced motion control and decision-making. Students learn to install and utilize ROS on Ubuntu systems, write basic nodes, and implement navigation stacks for autonomous robot movement. Practical exercises involve sensor fusion, 2D map building with Lidar, and optimizing navigation algorithms. By the end of the course, participants gain proficiency in various robotic components, algorithms, and systems integration, preparing them for complex robotics applications in research and industry.

Rationale of the Course

This lab course provides a practical, hands-on foundation in robotics principles and Robot Operating System (ROS) applications. Students gain direct experience designing, implementing, and controlling robotic systems through core curriculum topics such as motor control, sensor integration, and software programming. The course emphasizes an interdisciplinary approach, blending insights from computer science, mechanical engineering, electrical engineering, and mathematics to build comprehensive technical solutions. By bridging the gap between theoretical frameworks and real-world deployment, this course prepares learners to excel in dynamic engineering, research, and product development roles within the robotics industry.

Course Objective

The goal of this course is to introduce the students about the concept of Advanced Robotics applications. The main objectives of this course are,

- To learn the implementation of advanced robotics components.

- . To practice advanced concepts of robotics.
- . To develop familiarity and deep understanding of advanced robotics tools.
- . To develop the ability to independently design and create robotics solutions for industrial purposes.
- .

Course Learning Outcome (CLO) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Perform data collection activities of advanced robotics sensors using C++ and Python language.	PLO5	C3	Fundamental Domain
CLO2	Develop ROS packages for advanced robot applications in societal and environmental contexts	PLO7	C3, P2	Fundamental Domain
CLO3	Write effective reports and design documentation for advanced robotics solutions.	PLO10	A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1					√							
CLO2							√					
CLO3										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Discussion on OBE of Advanced Robotics Lab, Introduction to advanced robotics applications.	Syllabus Walkthrough, Lab Safety Orientation	CLO1, CLO2, CLO3	Lab Performance

Week-2 [1.5 Hours]	Software Installation, Study and Implementation of DC encoder motor	Toolchain Installation, Host-to-Board Firmware Flashing	CLO1	Lab Performance
Week-3 [1.5 Hours]	Study and Implementation of HUB motor: Integrate HUB motor and do simple programming to run hub motor.	GPIO Pin Mapping, PWM Frequency Tuning	CLO1	Lab Performance
Week-4 [1.5 Hours]	Study and Implementation of HUB motor: Do complex programming for complex control of hub motor	Multi-channel Register Interrupt Configuration, Kinematic Bound Verification	CLO1	Lab Performance
Week-5 [1.5 Hours]	Study and Implementation of IMU sensor: Integrate IMU sensor and do programming to read IMU basic data.	I2C/SPI Bus Frame Interrogation, Register Bitwise Data Parsing	CLO1	Lab Performance
Week-6 [1.5 Hours]	Study and Implementation of PID. Develop PID control system with necessary programming	Closed-Loop Error Derivation, Gain Variable Fine-Tuning	CLO1	Lab Performance
Week-7 [1.5 Hours]	Implementation of Fuzzy control. Develop Fuzzy control system with necessary programming	Membership Function Mapping, Defuzzification Rule Base Composition	CLO1	Lab Performance
Week-8 [1.5 Hours]	Installation of ROS on Ubuntu system and basic configuration. Writing Basic Node on ROS system: Write basic subscriber and publisher node	Bash Environment Sourcing, Catkin Package Dependency Compiling	CLO2, CLO3	Lab Performance, Viva, Lab Final
Week-9 [1.5 Hours]	Implementation of Lidar sensor with ROS: Integrate LiDAR sensor with ROS and do programming to read Lidar data.	Serial Driver Interfacing, LaserScan Array Buffer Ingestion	CLO2, CLO3	Lab Performance, Viva, Lab Final
Week-10 [1.5 Hours]	Implementation of Lidar sensor application with ROS: 2D Map building using Lidar.	GMapping Node Configuration, Occupancy Grid SLAM Generation	CLO2, CLO3	Lab Performance , Viva, Lab Final
Week-11 [1.5 Hours]	Implementation of Sensor Fusion: IMU, Encoder, Lidar	Extended Kalman Filter (EKF) Covariance Weighting, Transformation Tree (TF) Alignment	CLO2, CLO3	Lab Performance , Viva, Lab Final
Week-12 [1.5 Hours]	Implementation of ROS navigation Stack: Localization, Path Planning and Motion planning.	Adaptive Monte Carlo Localization (AMCL) Initialization, Dijkstra	CLO2, CLO3	Lab Performance , Viva, Lab Final

		Global Costmap Planning		
Week-13 [1.5 Hours]	Implementation of ROS navigation Stack: Parameter tuning	Dynamic Reconfigure Trajectory Optimization, Inflation Radius Hardening	CLO2, CLO3	Lab Performance , Viva, Lab Final
Week-14 [1.5 Hours]	Review of previous classes and discussion on final exam syllabus.	Integrated Production Verification, System Troubleshooting Audit	CLO1, CLO2, CLO3	Lab Performance

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO3	PLO10
Lab Performance	25	CLO1	PLO5
Lab Final	40	CLO2	PLO7
		CLO3	PLO10
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

John J. Craig, Introduction to Robotics: Mechanics and Control, Global edition, 4th edition, Pearson, 2021, ISBN: 978-1292164953.

Reference Books/ Other Supplementary Readings

Lentin Joseph, Mastering ROS for Robotics Programming - Third Edition: Best practices and troubleshooting solutions when working with ROS, 3rd edition, Packt Publishing, 2021, ISBN 978-1801071024

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Arduino, Google Colab, Raspberry

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

CS 335 Ethical Hacking and Countermeasure Lab

Course Code: CS 335		CIE Marks: 60
ISCED: 0613-335		SEE Marks: 40
Course Title: Ethical Hacking and Countermeasure LAB		Total Marks: 100
Semester:		
Credit Value: 2 (Lab)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Major Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

The ethical hacking and countermeasures course with LAB intends to provide students with hands-on practice to identify information system security vulnerabilities and exploit those and implement countermeasures to prevent unauthorized use of corporate information. This course will give students a practical experience of acquiring information from various sources and targeted organizations using various tools and techniques, assessing and detecting vulnerabilities, and exploiting the identified weaknesses using different ethical hacking methodologies. The tools and techniques covered in class will practically prepare students for performing hacking ethically to protect corporate information. Furthermore, it would prepare them to understand and implement appropriate countermeasures to prevent unauthorized corporate information systems.

Rationale of the Course

This course provides the practical skills and ethical mindset required to defend corporate information systems against modern cyber threats. Through interactive lab sessions, students gain hands-on experience in identifying, analyzing, and mitigating security vulnerabilities. The curriculum covers real-world simulation scenarios, including network reconnaissance, vulnerability scanning, and web application penetration testing. Additionally, learners explore the mechanics of social engineering, password cracking, and software exploitation to understand adversarial tactics.

Course Objectives

To provide a solid conceptual understanding of the fundamentals Ethical Hacking, more specifically

- To know how to perform Footprinting and reconnaissance, the network scanning techniques and scanning countermeasures, enumeration techniques, and enumeration countermeasures;
- To know about packet sniffing techniques, social engineering techniques, session hijacking techniques, and web server attack methodology with their uses for discovering network vulnerabilities, plus countermeasures against these attacks.
- To know about Wireless encryption, wireless hacking methodology, wireless hacking tools, and Wi-Fi security tools;
- To know about Mobile platform attack vectors, android vulnerability exploitations, and mobile security guidelines and tools;
- To know about Firewall, IDS, honeypot evasion techniques, evasion tools, and techniques to audit a network perimeter for weaknesses and countermeasures;

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLO's	CLO Descriptions	PLO's	Learning Domains (C,P,A)	BNQF Skill
CLO1	Demonstrate the ability to conduct cybersecurity operations by utilizing various tools and techniques to apply ethical hacking methodologies.	PLO4	P3	Fundamental Domain
CLO2	Integrate the skills in network security to perform activities in network devices to understand the advanced cybersecurity techniques and implications in real-world scenarios.	PLO5	P4	Fundamental Domain
CLO3	Simulate the integrated ability to execute and analyze hacking of web servers and their applications to identify vulnerabilities and countermeasures.	PLO5	P3	Fundamental Domain
CLO4	Specify and apply the offensive warfare techniques in IoT Mobile Platforms.	PLO4	P5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1				√								
CLO2					√							
CLO3					√							
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLO	Assessment Plan
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Discussion on OBE of Lab, Preparing lab Environment for Setting up VMware, KaliLinux.	Hypervisor Isolation Configuration, Kali OS Provisioning	CLO1	Lab Performance, Viva, Lab Final
	Lesson 2: Footprinting tools and techniques, Scanning tools, Port and Service discovery, Enumeration tools and techniques, System hacking, Gaining access, Maintaining access, Clearing logs	Active Nmap Reconnaissance, Active Banner Grabbing Scripting	CLO1	Lab Performance, Viva, Lab Final
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Sniffing Sniffing Concept, Sniffing technique: MAC attacks, Sniffing technique: DHCP attacks Sniffing technique: ARP poisoning, Spoofing attacks	Wireshark Packet Interception, Ettercap ARP Poisoning Manipulation	CLO1	Lab Performance, Viva, Lab Final
	Lesson 2: Social Engineering Social engineering concepts Social engineering techniques Insider threats	SET Phishing Vector Construction, Credential Harvester Profiling	CLO1	Lab Performance, Viva, Lab Final
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Social Engineering Impersonation on social networking sites Identity theft Countermeasures	Spoofed Metadata Footprinting, Defensive Header Analysis	CLO1	Lab Performance, Viva, Lab Final
	Lesson 2: Denial-of-Service DoS/DDoS concepts Botnets DoS/DDoS attack techniques DDoS case study Countermeasures	Hping3 Flood Stress Testing, Network Traffic Rate Throttling	CLO3	Lab Performance, Viva, Lab Final
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Session Hijacking, Session hijacking concepts, Application-level session hijacking, Network-level Session hijacking, Session Hijacking tools, Countermeasures	XSS Cookie Interception, Blind TCP Sequence Prediction	CLO3	Lab Performance, Viva, Lab Final

	Lesson 2: Evading IDS, Firewalls and Honeypots IDS, IPS, Firewalls and Honeypot attack It's solutions	Snort Alert Trigger Testing, Firewall Rule Map Scanning	CLO3	Lab Performance, Viva, Lab Final
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Evading IDS, Firewalls and Honeypots Evading IDS Evading Firewall Evading NAC and Endpoint security	Packet Fragmentation Obfuscation, Decoy Scan IP Masking	CLO3	Lab Performance, Viva, Lab Final
	Lesson 2: Evading IDS, Firewalls and Honeypots IDS/Firewall Evading tools Detecting Honeypots Countermeasures	Nmap Scripting Engine (NSE) Honeypot Probing, Rule Auditing	CLO3	Lab Performance, Viva, Lab Final
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Hacking Web Servers, Web server concepts, Web server attack	HTTP Header Exploitation, Directory Traversal Payload Ingestion	CLO3	Lab Performance, Viva, Lab Final
	Lesson 2: Hacking Web Servers, Web server attack methodology	Nikto Vulnerability Assessment, Exploit Payload Customization	CLO3	Lab Performance, Viva, Lab Final
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 2: Hacking Web Servers, Web server attack countermeasure, Patch Management	OWASP Top 10 Mapping, Input Validation Auditing	CLO3	Lab Performance, Viva, Lab Final
	Lesson 1: Hacking web Application Web application concepts Web application threats	Configuration Hardening Review, Vulnerability Remediation Patching	CLO2	Lab Performance, Viva, Lab Final
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 2: Hacking web Application Web application hacking methodology	Burp Suite Proxy Interception, Target Site Spider Mapping	CLO2	Lab Performance, Viva, Lab Final
	Lesson 2: Hacking web Application Web API, Webhooks, Web Shell	REST API Parameter Tampering, PHP Reverse-Shell Injection	CLO2	Lab Performance, Viva, Lab Final

Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Hacking web Application Web application security	Security Headers Hardening, Content Security Policy Validation	CLO2	Lab Performance, Viva, Lab Final
	Lesson 2: SQL Injection SQL Injection Types Methodology	Blind Error-Based Boolean Ingestion, DVWA Database Union Enumeration	CLO3	Lab Performance, Viva, Lab Final
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: SQL Injection SQL Injection tools Evasion techniques Countermeasure	SQLmap Database Automation Extraction, Parameter Encoding Obfuscation	CLO3	Lab Performance, Viva, Lab Final
	Lesson 2: Hacking Wireless Networks Wireless concepts Wireless encryption Wireless Threats Wireless Hacking methodology	Aircrack-ng Handshake Capture, WPA2 WPA3 Dictionary Attacks	CLO2	Lab Performance, Viva, Lab Final
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Hacking Wireless Networks Wireless Hacking tools Bluetooth hacking Wireless attack countermeasures Wireless security tools	Rogue Access Point Setup, Bettercap Bluetooth Low Energy Scanning	CLO2	Lab Performance, Viva, Lab Final
	Lesson 2: Wireless Mobile Platforms Mobile platform attack vector Hacking Android OS	MSFvenom Malicious APK Generation, Meterpreter Session Handler Binding	CLO4	Lab Performance, Viva, Lab Final
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Wireless Mobile Platforms Hacking iOS Mobile device management	IPA Package Decryption Auditing, Privilege Escalation Review	CLO4	Lab Performance, Viva, Lab Final
	Lesson 2: Wireless Mobile Platforms Mobile security guidelines and tools	Static Code Analysis Profiling, Containerization Guardrail Enforcement	CLO4	Lab Performance, Viva, Lab Final
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: IoT and OT Hacking IoT Concepts IoT Attacks	Firmware Binary Extraction, Shodan Connected Appliance Mapping	CLO4	Lab Performance, Viva, Lab Final

	Lesson 2: IoT and OT Hacking IoT Hacking methodology IoT Attack Countermeasures	Hardware Protocol Interface Sniffing, Default Credential Hardening	CLO4	Lab Performance, Viva, Lab Final
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: IoT and OT Hacking OT Concepts	SCADA Topology Mapping, Industrial Protocol Stack Identification	CLO4	Lab Performance, Viva, Lab Final
	Lesson 2: OT Attacks OT hacking methodology OT Attack countermeasures	Modbus Packet Injection Simulation, Air-Gapped Perimeter Hardening	CLO4	Lab Performance, Viva, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO4
Lab Performance	25	CLO1	PLO4
		CLO3	PLO5
Lab Final	40	CLO1	PLO4
		CLO2	PLO5
		CLO3	PLO5
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Matt Walker, All-in-One CEH Certified Ethical Hacker Exam Guide, 5th edition, McGraw Hill Education, 2021, ISBN: 978-1259836565.

Reference Books/ Other Supplementary Readings

1. Daniel G. Graham, Ethical Hacking: A Hands-on Introduction to Breaking In, 1st edition, No Starch Press, 2021

2. Lab Manual.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Virtual Box, Linux, Wireshark

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

DS 423 Machine Learning Driven Data Analysis II and Communicating Data Insights

Course Code: DS 423		CIE Marks: 60
ISCED: 0613-423		SEE Marks: 40
Course Title: Machine Learning Driven Data Analysis II and Communicating Data Insights		Total Marks: 100
Semester:		
Credit Value: 2 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite: DS 411, DS 412, SE 544		
Course Type: Major		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

Introduction to deep learning, create and train neural network architectures like Basic of Neural Network, Action Function, Forward Propagation, Back Propagation, Chain Model Convolutional Neural Networks, Recurrent Neural Networks, LSTMs, and Transformers, as well as how to improve them with tactics like Dropout, BatchNorm, Xavier/He initialization, and more. Students will be able to grasp theoretical principles and their industry applications using Python and TensorFlow, and to take on real-world problems like object detection, speech recognition, music synthesis, and chat bots, and more.

Rationale of the Course

This advanced course delves into deep learning architectures and their practical applications in complex data analysis. Students build a solid foundation in neural network fundamentals, including activation functions, forward and backward propagation, and advanced models such as convolutional neural networks, recurrent neural networks, LSTMs, and transformers. The curriculum emphasizes performance optimization using practical techniques like dropout, batch normalization, and weight initialization, implemented hands-on through Python and TensorFlow. By applying these models to real-world challenges like object detection, speech recognition, and chatbots, learners develop the technical expertise required to execute advanced machine learning pipelines and effectively communicate data insights.

Course Objectives

To provide a solid conceptual understanding of the fundamentals of data communications. More specifically,

- To learn deep neural networks, implement vectorized neural networks, identify architecture parameters, and apply DL to real world applications.
- To learn test sets and analyze bias/variance for building DL applications, use standard NN techniques, apply optimization algorithms, and implement a neural network in TensorFlow.
- To learn strategies for reducing errors in ML systems, understand complex ML settings, and apply end-to-end, transfer, and multi-task learning.
- To learn a Convolutional Neural Network, apply it to visual detection and recognition tasks, use neural style transfer to generate art, and apply these algorithms to image, video, and other 2D/3D data.

- To learn the Recurrent Neural Networks and its variants (GRUs, LSTMs), apply RNNs to character-level language modeling, work with NLP and Word Embeddings, and use HuggingFace tokenizers and transformers to perform Named Entity Recognition and Question Answering.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Interpret the fundamental challenges of a deep learning system	PLO1	C2, C3	Fundamental Domain
CLO2	Evaluate the underlying mathematics of each deep learning algorithm	PLO3	C5	Fundamental Domain
CLO3	Experiment the deep learning algorithms working path and find out errors properly.	PLO4	C5	Fundamental Domain
CLO4	Analyze the appropriate deep learning algorithms & techniques at appropriate situations.	PLO4	C4	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2			√									
CLO3				√								
CLO4				√								

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of OBE (CLO, PLO, Knowledge level)	Syllabus Walkthrough, Taxonomy Mapping Q&A	N/A	CLO1
	Lesson 2: Understand the significant technological trends driving deep learning development, Where and how deep learning is applied.	Scale and Hardware Driver Analysis, Industry Use Case Mapping	Class Test, Midterm exam	CLO1

Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Set up a machine learning problem with a neural network mindset.	Logistic Regression Formulations, Single Neuron Computational Structuring	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Use vectorization to speed up deep learning models.	Matrix Vectorization Calculations, SIMD Parallel Processing Review	Class Test, Midterm exam	CLO1, CLO2
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Shallow Neural Networks	Multi-Layer Node Graphing, Hidden Layer Concept Mapping	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Build a neural network with one hidden layer using forward propagation and backpropagation.	Activation Derivation Math, Gradient Chain-Rule Calculations	Class Test, Midterm exam	CLO1, CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Understand the key computations underlying deep learning.	Deep Hidden State Matrix Dimensions, Forward-Backward Cache Analysis	Class Test, Assignment, Midterm exam	CLO1, CLO2
	Lesson 2: Use them to build and train deep neural networks, and apply them to computer vision. Class Test-1	Multi-tier Neural Tuning, Computer Vision Pixel Tensor Ingestion	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Discover and experiment with various initialization methods of Deep Learning.	Xavier/He Parameter Balance Math, Exploding/Vanishing Gradient Diagnostics	Class Test, Assignment, Midterm exam	CLO1, CLO2, CLO3
	Lesson 2: Apply L1 & L2 regularization and dropout to avoid model overfitting, Use gradient checking to identify errors in a fraud detection model of Deep Learning.	Frobenius Norm Regularization, Numerical vs Analytical Gradient Cross-Check	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2, CLO3
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Discover the introductory part of the Optimization Algorithm.	Stochastic vs Batch Gradient Tracking, Convex Optimization Visualizations	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2, CLO3
	Lesson 2: Develop deep learning toolbox by adding more advanced optimizations, random mini-batching, and learning rate decay scheduling to speed up models. Class Test-2	Momentum/Adam Vector Velocity Math, Mini-Batch Shuffling Diagnostics	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2, CLO3

Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Explanation part of Hyperparameter tuning, Batch Normalization, and Programming Frameworks.	Random Grid Hyperparameter Sampling, Covariate Hidden Shift Normalization	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2, CLO3
	Lesson 2: Explore TensorFlow, a deep learning framework that allows one to build neural networks quickly and easily.	Static/Dynamic Computation Graphing, Tensor API Architecture Walkthrough	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2, CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Train a neural network on a TensorFlow dataset.	Pipeline Data Prefetching, Batch Ingestion Optimization	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Learn to Interpret the result.	Loss-Accuracy Convergence Curve Analysis, Overfitting Graph Verification	Class Test, Assignment, Midterm exam, Final exam	CLO1, CLO2, CLO3, CLO4
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Use a machine learning flight simulator to learn how machine learning achieves human-level performance.	Simulation Agent Feedback Loop Analysis, Policy Evaluation Review	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Become familiar with the concepts of end-to-end learning, transfer learning, and multi-task learning.	Feature Weight Freezing Mapping, Shared Layer Topology Review	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3, CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Discuss the foundations of Convolutional Neural Networks	Spatial Invariance Grid Analysis, Receptive Field Concept Mapping	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Implement the foundational layers of CNNs (pooling, convolutions) and stack them properly in a deep network to solve multi-class image classification problems.	Filter Kernel Stride Matrix Math, Max-Average Pooling Volume Downsampling	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3, CLO4
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Learn the basic part of Object Detection.	Bounding Box Regression Coordinates Math, Anchor Box IoU Optimization	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Apply your knowledge of CNNs to computer vision: object detection and semantic segmentation. Class Test-3	YOLO Grid Anchor Processing, Transposed Convolution Pixel Upsampling	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Discover how CNNs can be applied to multiple fields, including art generation and face recognition.	Content-Style Loss Matrix Dissection, Siamese Network Triplet Loss Analysis	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3, CLO4

	Lesson 2: Implement an algorithm to generate art and recognize faces.	Gram Matrix Latent Feature Activation, Face Embedding Vector Comparisons	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3, CLO4
Week-13 Lesson 1 & 2 [2 Hours]	Lesson 1: Learn the basic part of Recurrent Neural Networks, Discover recurrent neural networks (RNNs) and several of their variants	Unrolled Time-Step Hidden Sequence Mapping, Vanishing Temporal Gradient Analysis	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Discover RNNs and several of their variants, including LSTMs, GRUs, and Bidirectional RNNs.	Memory Cell Gate Component Vector Calculations, Hidden Forward-Backward Passes	Class Test, Assignment, Final exam	CLO1, CLO2, CLO3, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Industry Expert Talk, Cloud Infrastructure Scaling Review	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class	Final Syllabus Blueprint Cross-Mapping, Core Architecture Troubleshooting	Final exam	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =78)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO3	PLO4	
Attendance	--	--	--	--	7				
Class Test 1	5(15)				Avg 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment					5				
Presentation					8				
Midterm Exam	10	15			25	10	15		25
Final exam		10	20	10	40		10	30	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40

Total	100
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Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Géron, Aurélien. Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems. O'Reilly Media, 2019

Reference Books/ Other Supplementary Readings

1. Weidman, Seth. Deep Learning from Scratch: Building with Python from First Principles. " O'Reilly Media, Inc.", 2019.
2. Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. Deep learning. MIT press, 2016.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

1. TensorFlow documentations, tutorials, and learning paths (<https://www.tensorflow.org/>).
2. scikit-learn documentations and tutorials (<https://scikit-learn.org/stable/>).

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

CS 334 Ethical Hacking and Countermeasure

Course Code: CS 334	CIE Marks: 60
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ISCED: 0613-334		SEE Marks: 40
Course Title: Ethical Hacking and Countermeasure		Total Marks: 100
Semester:		
Credit Value: 1 (Theory)	Contact Hours: 1.5 (Total weeks: 14)	
Prerequisite: N/A		
Course Type: Major Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Course Content (from syllabus)

In this course students will learn about Necessity of ethical hacking knowledge to protect the infrastructure, foot printing & reconnaissance process, vulnerability assessment, analyze them to prepare attack vector, exploiting vulnerabilities to hack the system, web application attack, different attack methodology, sniffing, malware analysis, cryptography & cloud computing etc.

Rationale of the Course

This course introduces the principles of information system security, focusing on the identification, assessment, and exploitation of software and system vulnerabilities. Students gain hands-on experience gathering reconnaissance data from target organizations using diverse tools and ethical hacking methodologies. The curriculum emphasizes the practical application of these techniques to defend corporate infrastructure and implement effective countermeasures against unauthorized access.

Course Objectives

To provide a solid conceptual understanding of the fundamentals Ethical Hacking, more specifically

- To know about ethical hacking, information security controls, related laws & standards, and countermeasures to ensure the System's security.
- To know how to perform footprinting and reconnaissance, the network scanning techniques and scanning countermeasures, enumeration techniques and enumeration countermeasures,
- To know about System hacking methodology, steganography, steganalysis attacks, and covering tracks to discover System and network vulnerabilities.
- Know about Different types of malwares (Trojan, Virus, worms, etc.), system auditing for malware attacks, malware analysis, and countermeasures.
- Learn about encryption algorithms, cryptography tools, Public Key Infrastructure (PKI), email encryption, disk encryption, cryptography attacks, and cryptanalysis tools.
- Learn different cloud computing concepts, various cloud computing threats, attacks, hacking methodologies, and cloud security techniques and tools.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLO's	CLO Descriptions	PLO's	Learning Domains (C,P,A)	BNQF Skill
CLO1	Articulate the Ethical Hacking process, methodologies, countermeasures and ethics to protect information.	PLO8	C3	Fundamental Domain

CLO2	Contextualize the sources of information gathered during the reconnaissance and foot printing phases to interpret and assess information in a context-specific manner.	PLO2	C4	Fundamental Domain
CLO3	Evaluate the vulnerabilities discovered during the scanning and enumeration phases to make informed decisions in complex cybersecurity scenarios.	PLO3	C5	Fundamental Domain
CLO4	Formulate strategies for gaining and maintaining access by clearing tracks to develop effective solutions addressing real-world challenges.	PLO3	C6	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1								√				
CLO2		√										
CLO3			√									
CLO4			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson [1.5 Hours]	Lesson 1: Discussion on OBE of Ethical Hacking and Countermeasure	Syllabus Walkthrough, Field Taxonomy Mapping		
Week-2 Lesson [1.5 Hours]	Lesson 1: Information security overview, Hacking methodologies and frameworks, Hacking concept, Ethical hacking concept, Information security controls, Information security laws and standards	Cyber Kill-Chain Analysis, Compliance Rule Auditing	Class Test, Mid Term Exam	CLO1

Week-3 Lesson [1.5 Hours]	Lesson 1: Malware Concepts, APT Concepts, Trojan Concepts, Virus and Worm Concepts	Attack Vector Dissection, Persistent Threat Profiling	Class Test 2, Mid Term Exam	CLO1
Week-4 Lesson [1.5 Hours]	Lesson 1: Malware Analysis, Malware Countermeasures, Anti malware software, Fileless Malware Concepts Class Test-1	Dynamic Sandboxing Code Diagnostics, Memory Injection Analysis	Assignment, Mid Term Exam	CLO1, CLO2
Week-5 Lesson [1.5 Hours]	Lesson 1: Footprinting concept, Reconnaissance concept	OSINT Attack Surface Mapping, Passive Domain Ingestion	Mid Term Exam	CLO2
Week-6 Lesson [1.5 Hours]	Lesson 1: Learn about the techniques and countermeasures. Class Test-2	Social Engineering Vector Review, Defensive Guardrail Mapping	Mid Term Exam	CLO3
Week-7 Lesson [1.5 Hours]	Lesson 1: Denial of Service, Network scanning concepts, Network scanning countermeasures	Stress Testing Syn-Flood Simulation, Firewall State Filtering Review	Final Exam	CLO3
Week-8 Lesson [1.5 Hours]	Lesson 1: Enumeration concepts, Enumeration techniques overview	Service Banner Interrogation, RPC Namespace Auditing	Class test 3, Final exam	CLO3
Week-9 Lesson [1.5 Hours]	Lesson 1: Other enumeration techniques, Enumeration countermeasures.	SNMP MIB Dictionary Parsing, Access Control Hardening	Final exam	CLO4
Week-10 Lesson [1.5 Hours]	Lesson 1: Vulnerability assessment concepts, Vulnerability classification and assessment types	CVE Database Cross-Referencing, CVSS Scoring Metric Analysis	Final exam	CLO4
Week-11 Lesson [1.5 Hours]	Lesson 1: Gaining Access, Escalating privileges, Maintaining Access, Clearing Logs	Privilege Boundary Exploitation, Log Tampering Auditing	Final exam	CLO4

	Class Test-3			
Week-12 Lesson [1.5 Hours]	Lesson 1: Cryptography Encryption algorithms, Cryptography tools, Public key, Infrastructure (PKI), Countermeasures	Block Cipher Substitution Math, Trust Chain Validation Analysis	Final exam	CLO4
Week-13 Lesson [1.5 Hours]	Lesson 1: Cloud Computing Concepts, Container technology, Cloud Hacking, Cloud security	Shared Responsibility Boundary Mapping, IAM Escape Vector Auditing	Final exam	CLO4
Week-14 Lesson [1.5 Hours]	Review class	Comprehensive Blueprint Mapping, Core Problem Troubleshooting	Final Exam	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO8	PLO2	PLO3	
Attendance			--	--	7				
Class Test 1	5(15)				Avg 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment			5		5		5		5
Presentation					8				
Midterm Exam	15	10			25	15	10		25
Final exam			20	20	40			40	40

Marks Distribution

Class attendance	7
Assignment	5

Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

Matt Walker, All-in-One CEH Certified Ethical Hacker Exam Guide, 5th edition, McGraw Hill Education, 2021, ISBN: 978-1259836565.

Reference Books/ Other Supplementary Readings

Daniel G. Graham, Ethical Hacking: A Hands-on Introduction to Breaking In, 1st edition, No Starch Press, 2021

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Virtual Box, Linux, WireShark

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 3)

DS 424 Machine Learning Driven Data Analysis II and Communication Data Insights Lab

Course Code: DS 424	CIE Marks: 60
ISCED: 0613-424	LF Marks: 40
Course Title: Machine Learning Driven Data Analysis II and Communication Data Insights Lab	Total Marks: 100
Semester:	
Credit: 1 (Lab)	Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: DS 411, DS 412, SE 544	

Course Type: DS Major

CIE: Continuous Internal Evaluation

LF: Lab Final

Course Content (from syllabus)

This course provides a broad introduction to Data Science: (i) Learn about data science engineer and data analyst, Careers in Data Science, Application of machine learning (ii) Introduction about data science tools (iii) and data set collection and data visualization (iv) data set analysis and model apply (v) statistical basics (vi) Time series data analysis and ETL

Rationale of the Course

Data Science is a combination of multiple disciplines that uses statistics, data analysis, and machine learning to analyze data and to extract knowledge and insights from it. The course will Describe what data science and machine learning are, their applications & use cases, and various types of tasks performed by data scientists. Students will Gain hands-on familiarity with common data science tools including Jupyter Lab, R Studio. The course will develop the mindset to work like a data scientist, and follow a methodology to tackle different types of data science problems.

Course Objectives

The goal of this course is to introduce the students to the concept of Data Science. The main objectives of this course are-

- To Learn the introduction of data science.
- To Learn and understand the data engineering tools.
- To learn and demonstrate data visualization and algorithms.
- To learn and apply data science methodology and understand the ETL.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Implementation of different types of Artificial Neural Network Architectures. (Coding)	PLO5	C6, P5	Fundamental Domain
CLO2	Implementation of different types of Convolutional and Recurrent Neural Network architectures. (Coding)	PLO5	C4, P2	Fundamental Domain
CLO3	Compare among various deep learning models and apply the appropriate one for solving a given problem. (Coding)	PLO2	C3, P2	Fundamental Domain

CLO4	Apply Skill to solve real life problems. (Coding). Present and Defend results of deep learning models.	PLO6	C5	Fundamental Domain
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Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1					√							
CLO2					√							
CLO3		√										
CLO4						√						

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Discussion on course outline, course learning outcomes, assessment methods, deadlines, and other administrative information.	Syllabus Walkthrough, Virtual Workspace Initialization	None	None
Week-2 [1.5 Hours]	Work with tensor data structures of TensorFlow, Build complex models using Keras functional API, save, and restore models.	Tensor Array Ingestion, Serialization H5/SavedModel Formatting	CLO1	Lab Performance, Viva, Lab Final
Week-3 [1.5 Hours]	Build a multilayer perceptron classifier and regressor using Keras sequential API.	Dense Layer Weight Compiling, Categorical Cross-Entropy Verification	CLO2	Lab Performance, Viva, Lab Final
Week-4 [1.5 Hours]	Build complex models using Keras functional API, save, and restore models.	Multi-Input Non-Linear Graph Routing, Model State Checkpointing	CLO1	Lab Performance, Viva, Assignment, Lab Final
Week-5 [1.5 Hours]	Build models using callbacks and use TensorBoard to visualize training performance.	EarlyStopping Event Hooking, Loss Scalar Telemetry Graphing	CLO4	Lab Performance, Viva, Assignment, Lab Final
Week-6 [1.5 Hours]	Use pre trained neural networks using Keras and use and evaluate different types of optimizers.	Backbone Feature Freezing, Adam/RMSprop Gradient Decay Tracking	CLO3	Lab Performance, Viva, Assignment, Lab Final

Week-7 [1.5 Hours]	Build models using learning rate schedulers and apply different types of regularization techniques.	Exponential Decay Step-Scheduling, Dropout Spatial Matrix Masking	CLO2	Lab Performance, Viva, Assignment, Lab Final
Week-8 [1.5 Hours]	Write custom loss functions, custom metrics, and custom layers.	Backpropagation Subclass Structuring, Matrix Tensor Mathematical Operations	CLO4	Lab Performance, Viva, Assignment, Lab Final
Week-9 [1.5 Hours]	Write custom training loops, activation functions, initializers, and regularizers.	GradientTape Step Auditing, He/Xavier Parameter Ingestion	CLO4	Lab Performance, Viva, Assignment, Lab Final
Week-10 [1.5 Hours]	Use TensorFlow Data API, chain transformations, shuffle data, preprocess data, and prefetch data for training.	Pipeline Prefetch Buffer Optimization, Row-Wise Transformation Interleaving	CLO4	Lab Performance, Viva, Assignment, Lab Final
Week-11 [1.5 Hours]	Work with TFRecord files and TensorFlow Datasets project.	Protocol Buffer Binary Serialization, Dataset Gzip Stream Ingestion	CLO4	Lab Performance, Viva, Assignment, Lab Final
Week-12 [1.5 Hours]	Implement ResNet architecture using Keras API, and use pretrained weights, Mask-RCNN algorithm to detect objects and segment images.	Residual Skip-Connection Compiling, RoIAlign Tensor Pixel Masking	CLO2	Lab Performance, Viva, Assignment, Lab Final
Week-13 [1.5 Hours]	Implement You Only Look Once (YOLO) algorithm to detect objects and segment images and Generate Shakespearean text using a character level RNN.	Bounding Box Regression Coordinate Anchoring, Stateful Temporal Character Slicing	CLO2	Lab Performance, Viva, Assignment, Lab Final
Week-14 [1.5 Hours]	Analyze sentiments using RNN and Implement model development through API in deep learning real life projects	Embedding Sequence Token Parsing, FastAPI Endpoint Inference Integration	CLO4	Lab Performance, Viva, Assignment, Lab Final

Overall Assessment Scheme & Marks Distribution

Criteria	Assigned Marks	Sub Criteria	Sub Criteria Marks	CLO	PLO
Attendance	10		10		

Lab Report	25	Understanding/ Analysis	7	CLO1	PLO2
		Implementation	8	CLO2	PLO5
		Report Writing	10	CLO3	PLO3
Lab Performance	25	Daily Lab Work	10	CLO2	PLO5
		Lab Assignment	10	CLO2	PLO5
		Viva	5	CLO4	PLO5
Lab Final	40	Understanding/ Analysis	10	CLO1	PLO2
				CLO2	PLO5
		Implementation	15	CLO2	PLO5
		Accuracy	10	CLO3	PLO3
		Task Efficiency	5	CLO3	PLO3
Total	100		100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Reading

1.Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 3rd edition, O'Reilly Media, 2023, ISBN: 978-1492032649.

2.David Foster, Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play, 2nd edition, O'Reilly Media, 2023, ISBN: 978-1492041948.

3.Jeremy Howard, Deep Learning for Coders with Fastai and PyTorch: AI Applications Without a PhD, 1st edition, O'Reilly Media, 2020, ISBN: 978-1492045519.

Reference Books/ Other Supplementary Readings

1.Seth Weidman, Deep Learning from Scratch: Building with Python from First Principles, 1st edition, Independently published, 2019, ISBN: 978-1693953484.

2.Lab Manual

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools Used

Google Colab, TensorFlow, PyTorch, Numpy, Pandas

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

ST 411 Agile Testing

Course Code: ST 411		CIE Marks: 60
ISCED: 0613-411		SEE Marks: 40
Course Title: Agile Testing		Total Marks: 100
Semester:		
Credit Value: 2 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite:		
Course Type: Testing Major		

CIE: Continuous Internal Evaluation,
SEE: Semester End Examination

Course Content (from syllabus)

Agile Software Development Fundamentals, Agile values and principles, Whole-team approach, Early and frequent feedback, Agile development approaches, Collaborative user story creation, Retrospectives, Continuous integration, Release and iteration planning, Differences between traditional and agile testing, Testing and development activities, Agile work products, Agile test levels, Agile configuration management, Independent testing options, Communicating test status and progress, Managing regression risk, Agile tester skills, Role of a tester in agile teams, Test-driven development, Acceptance test-driven development, Behavior-driven development, Agile test pyramid, Testing quadrants, Agile test levels and testing types, Assessing quality risks, Estimating testing effort, Acceptance criteria and adequate test coverage, Applying acceptance test-driven development, Functional black-box test design, Non-functional test design, Exploratory testing in agile, Task management tools, Communication and collaboration tools, Build and distribution tools, Configuration management tools, Test design tools, Test implementation tools, Test execution tools, Cloud-based testing tools, Virtualization tools.

Rationale of the Course

This course introduces the principles, practices, and processes of software testing within Agile development environments. Students explore how iterative delivery and continuous feedback reshape traditional testing, emphasizing team collaboration, early test involvement, and adaptive planning. The curriculum covers Agile-specific methodologies including test-driven development (TDD), acceptance test-driven development (ATDD), and behavior-driven development (BDD), alongside structural concepts like the test pyramid and testing quadrants. Additionally, learners examine risk-based estimation, exploratory testing, and the integration of tools for automation, configuration management, and continuous delivery (CI/CD).

Course Objectives

By the end of this course, students will be able to:

- Understand the fundamentals of Agile software development and the significance of testing within Agile environments.
- Explain Agile testing principles, processes, and how they differ from traditional testing models.
- Apply Agile testing methods including TDD, ATDD, BDD, exploratory testing, and risk-based test planning.
- Assess quality risks and estimate testing effort in iterative projects.
- Use Agile tools for task management, communication, configuration management, and automated testing.
- Understand the role, responsibilities, and skills of a tester in Agile teams.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Explain the fundamentals of software testing and justify the necessity of SQA and testing in the software life cycle	PLO1	C2	Fundamental Domain
CLO2	Analyze and compare black-box and white-box testing approaches	PLO2	C4	Fundamental Domain
CLO3	Design effective test cases for diverse functional and non-functional scenarios	PLO3	C6	Fundamental Domain
CLO4	Select and apply appropriate testing strategies and tools in agile environments	PLO5	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3			√									
CLO4					√							

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of Agile Software Development	Manifestu Core Principle Analysis, Lifecycle Comparison	Class Test, Midterm exam	CLO1
	Lesson 2: Whole-Team Approach in Agile Projects	Cross-Functional Roles Matrix, Shared Quality Discussion	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Agile Development Approaches	Scrum vs Kanban Mechanics, Iteration Cadence Review	Class Test, Midterm exam	CLO1
	Lesson 2: Collaborative User Story Creation	INVEST Criteria Blueprinting, Three Cs Decomposition	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Retrospectives	Root-Cause Analysis, Continuous Improvement Action Items	Class Test, Midterm exam	CLO1
	Lesson 2: Release and Iteration Planning	Velocity Metrics Calculation, Capacity Mapping Planning	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Comparison of Testing Approaches	Traditional V-Model vs Agile Testing Matrix	Class Test, Midterm exam	CLO1
	Lesson 2: Agile Project Work Products Class Test -1	Burndown Chart Profiling, Definition of Done (DoD) Auditing	Class Test, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Test Levels in Agile	Story-Level Acceptance Verification, Feature Integration Reviews	Class Test, Midterm exam	CLO1
	Lesson 2: Independent Testing Options	Outsource Vulnerability Auditing, Integrated QA Guardrail Design	Class Test, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Communicating Test Progress	Automation Pass-Fail Metric Tracing, Defect Density Profiling	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Regression Testing in Agile Class Test -2	Impact Radius Matrix Selection, CI/CD Pipeline Automation	Class Test, Midterm exam	CLO1, CLO2

Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Agile Tester Skills	Analytical Soft-Skills Mapping, T-Shaped Competency Review	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Tester Responsibilities in Agile	Daily Standup Impediment Reporting, Sprint Backlog Refining	Class Test, Midterm exam	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Test-Driven Development (TDD)	Red-Green-Refactor Logic Loops, Unit- Assertion Design	Class Test, Assignment, Midterm exam	CLO1, CLO3
	Lesson 2: Acceptance Test-Driven Development (ATDD) & Behavior- Driven Development (BDD)	Gherkin Given-When- Then Syntax Mapping, Shared Requirement Synthesis	Class Test, Assignment, Midterm exam	CLO1, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 2: The Test Pyramid	Layered Inversion ROI Analysis, Test Suite Balance Review	Class Test, Midterm exam	CLO1, CLO3
	Lesson 2: Testing Quadrants & Testing Types	Marick Quadrant Functional/Non- functional Classification, Automation Tool Alignment	Class Test, Midterm exam	CLO1, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Assessing Quality Risks	Agile Risk Matrix Likelihood Matrix, Failure Mode Impact Review	Class Test, Final exam	CLO1, CLO3
	Lesson 2: Estimating Testing Effort	Planning Poker Story Estimation, Delphi Iteration Sizing	Class Test, Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Acceptance Criteria & Adequate Coverage	Boundary Value Validation Mapping, Traceability Matrix Assessment	Class Test, Final exam	CLO1, CLO3
	Lesson 1: Exploratory Testing & Agile Class Test -3	Session-Based Charter Parsing, Time-Boxed Defect Hunting	Class Test, Final exam	CLO1, CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Task Management & Communication Tools	Jira Kanban Board Swimlane Mapping, Slack Alert Webhook Layout	Class Test, Assignment, Final exam	CLO1, CLO4
	Lesson 2: Software Build, Distribution & Configuration Management Tools	Jenkins Pipeline Artifact Archiving, Git Flow Branch Tracking	Class Test, Final exam	CLO1, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Agile Test Design Techniques (Functional & Non- Functional)	Equivalence Partitioning Matrix, Security/Performance Profile Modeling	Class Test, Final exam	CLO1, CLO4

	Lesson 2: Applying Acceptance Test-Driven Development Practically	Executable Specification Mapping, Live Framework Cross-Check	Class Test, Presentation Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Release Blueprint Review	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class	Final Syllabus Cross-Mapping, Sprint QA Strategy Troubleshooting	Final exam	CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO1	PLO's			Mark (Total =65)
	CLO1	CLO2	CLO3	CLO 4			PLO 2	PLO 3	PLO5	
Attendance	--	--	--	–	7					
Class Test 1	5(15)				15					
Class Test 2		5(15)								
Class Test 3			5(15)							
Assignment					5					
Presentation					8					
Midterm Exam	10	15			25	10	15			25
Final exam			20	20	40			20	20	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

Rex Black, Gerry Coleman, Bertrand Cornanguer, Marie Walsh, Istvan Forgacs, Kari Kakkonen, Jan Sabak, Agile Testing: An ISTQB-BCS Foundation Guide, BCS Learning & Development Limited, 2017, ISBN 1780173369, 9781780173368

Reference Books/ Other Supplementary Readings

1. Manfred Baumgartner, Martin Klonk, Christian Mastnak, Helmut Pichler, Richard Seidl, Siegfried Tanczos, Agile Testing The Agile Way to Quality, Springer International Publishing, September 9, 2021, ISBN:9783030732097, 3030732096
2. Crispin, Agile Testing: A Practical Guide For Testers And Agile Teams, Pearson Education, September 2010, ISBN:9788131730683, 8131730689

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

NA

Exam Dates

According to the Examination Schedule



**Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU) (Version
1.0)**

ST 412 Agile Testing Lab

Course Code: ST 412		CIE Marks: 60
ISCED: 0613-412		LF Marks: 40
Course Title: Agile Testing Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)	Contact Hours: 1 hour 30 minutes (14 weeks)	
Prerequisite: N/A		
Course Type: Testing Major		

CIE: Continuous Internal Evaluation

LF: Lab Final Exam

Course Content (from syllabus)

This course provides a practical, hands-on understanding of Agile Testing concepts, practices, and tools within an Outcome-Based Education (OBE) framework. Learners will explore agile values, principles, roles, and ceremonies, and apply testing activities across agile iterations (sprints). The course emphasizes user stories, acceptance criteria, TDD, BDD, exploratory testing, risk-based testing, and agile test metrics. Students will gain practical experience with agile testing tools, automation basics, CI/CD pipelines, and structured agile test documentation. Through continuous lab activities, tool-based exercises, sprint-level reporting, and evaluations, this course prepares learners for real-world agile QA environments

Rationale of the Course

Agile methodologies dominate modern software development, making agile testing skills essential for QA professionals. Unlike traditional testing, agile testing requires continuous collaboration, rapid feedback, adaptive planning, and integration with development pipelines. This course equips learners with the mindset and practical skills needed to test effectively in agile teams. By combining agile testing theory, hands-on tool usage, automation basics, CI/CD integration, and professional documentation practices, the course prepares students for roles such as Agile QA Engineer, Software Tester, Test Automation Engineer, and DevOps-aware QA professional.

Course Objectives

By the end of the course, learners will be able to:

- Understand agile testing principles, roles, ceremonies, and workflows.
- Plan, design, and execute testing activities within agile sprints.
- Apply TDD, BDD, exploratory testing, and risk-based testing techniques.
- Use agile testing and collaboration tools effectively.
- Apply basic test automation and CI/CD concepts in agile environments.
- Prepare high-quality agile test documentation, reports, and sprint test summaries.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Develop performance test plans & workload models	PLO3	C6	Fundamental Domain
CLO2	Design & execute performance tests; analyze bottlenecks	PLO4	C4, P3	Fundamental Domain
CLO3	Apply IT audit techniques: risk assessment, controls, security checks	PLO6	C3	Fundamental Domain
CLO4	Write effective reports and design test case documentation for the real life project applying a wide variety of testing techniques at various testing levels	PLO10	C5, A3	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1			√									
CLO2				√								
CLO3						√						
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

Week/ Less on (hou r)	Lesson Topic	Teaching Learning Strategy	Correspon ding CLO	Assessment Strategy
Week-1 [1.30 Hours]	Introduction to Agile Testing: Agile Manifesto, Values, Principles, Roles & Ceremonies	Agile Project Setup, Board Lifecycle Initialization	CLO1	Lab Performance
Week-2 [1.30 Hours]	Agile Testing Quadrants: Functional & Non-functional testing with hands-on examples	Test Suite Segmentation, Automation Tool Alignment	CLO1	Lab Performance
Week-3 [1.30 Hours]	User Stories & Acceptance Criteria: INVEST Model, Writing testable stories	Gherkin Requirement Slicing, Validation Bounds Definition	CLO1, CLO2	Lab Performance
Week-4 [1.30 Hours]]	Test-Driven Development (TDD): Writing tests before code (basic exercises)	Red-Green-Refactor Compilation, Unit Assert Testing	CLO2	Lab Performance
Week-5 [1.30 Hours]	Behavior-Driven Development (BDD): Gherkin syntax, Given- When-Then	Feature File Scripting, Step Definition Glue Compilation	CLO2	Viva, Lab Performance
Week-6 [1.30 Hours]	Exploratory Testing: Session- based Test Management	Time-Boxed Charter Parsing, Target Defect Log Ingestion	CLO2	Viva, Lab Performance
Week-7 [1.30 Hours]	Risk-Based Testing: Risk identification, prioritization techniques	Failure Mode Severity Matrixing, High-Risk Regression Scoping	CLO2	Lab Performance
Week-8 [1.30 Hours]	Mid-Lab Evaluation	Practical Runtime Assessment, Script Debugging Challenge	CLO1, CLO2	Mid-Lab Exam
Week-9 [1.30 Hours]	Agile Test Metrics: Burndown chart, velocity calculation	Sprint Velocity Aggregation, Defect Burn-Down Querying	CLO3	Lab Performance
Week-10 [1.30 Hours]	Continuous Integration (CI) Testing: Jenkins / GitHub Actions demo	YAML Configuration Scripting, Build Pipeline Execution	CLO3	Lab Performance

Week-11 [1.30 Hours]	Test Automation in Agile: Selenium / Python basics	DOM Element Querying, Webdriver Event Ingestion	CLO3	Viva, Lab Performance
Week-12 [1.30 Hours]	Agile Test Documentation: Test reports, Sprint review test summary	Quality Release Sheet Generation, Defect Telemetry Exporting	CLO3	Lab Reports
Week-13 [1.30 Hours]	Final Lab Revision	Integrated Framework Smoke Testing, Multi-Fault System Troubleshooting	CLO4	Lab Performance
Week-14 [1.30 Hours]	Lab Final Examination	Independent Automation Challenge, Production Code Retrospective	CLO1, CLO2, CLO3, CLO4	Viva, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO3
Lab Final	40	CLO2	PLO4
		CLO3	PLO6
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour following the procedure of the university.

Textbook/Recommended Readings

Rex Black, Gerry Coleman, Bertrand Cornanguer, Marie Walsh, Istvan Forgacs, Kari Kakkonen, Jan Sabak, Agile Testing: An ISTQB-BCS Foundation Guide, BCS Learning & Development Limited, 2017, ISBN 1780173369, 9781780173368

Reference Books/ Other Supplementary Readings

1. Manfred Baumgartner, Martin Klonk, Christian Mastnak, Helmut Pichler, Richard Seidl, Siegfried Tanczos, Agile Testing The Agile Way to Quality, Springer International Publishing, September 9, 2021, ISBN:9783030732097, 3030732096

2. Crispin, Agile Testing: A Practical Guide For Testers And Agile Teams, Pearson Education, September 2010, ISBN:9788131730683, 8131730689

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

- Jira / Trello / ClickUp
- Selenium / Playwright (Python)
- Postman
- Git & GitHub
- Jenkins / GitHub Actions

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

ST 421 Testing with Generative AI

Course Code: ST 421		CIE Marks: 60
ISCED: 0613-421		SEE Marks: 40
Course Title: Testing with Generative AI		Total Marks: 100
Semester:		
Credit Value: 2 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite:		
Course Type: Testing Major		

CIE: Continuous Internal Evaluation,
SEE: Semester End Examination

Course Content (from syllabus)

Testing with Generative AI extends the Foundation Level and provides testing and quality-engineering professionals with a structured, hands-on path to master how, when, and why to use Large Language Models (LLMs) and other generative-AI techniques across the entire test process – from requirements analysis and test design to automation, reporting, and continuous improvement. It not only explains the fundamentals of GenAI but also builds practical skills through real-world prompt-engineering patterns and applied testing use cases. Testers will learn how to integrate GenAI capabilities responsibly while managing key risks such as hallucinations, bias, security, privacy, and environmental impact.

Rationale of the Course

This course introduces the principles and practical applications of Generative Artificial Intelligence in the field of software testing. Students will learn how Large Language Models (LLMs) transform test analysis, test design, test automation, risk management, and test management activities. The course emphasizes prompt engineering, risk mitigation, ethical AI usage, and real-world deployment of GenAI in software testing organizations.

Course Objectives

By the end of this course, students will be able to:

- Understand the fundamentals of Generative AI and Large Language Models for software testing.
- Apply prompt engineering techniques for effective test design and automation.
- Analyze risks of Generative AI including hallucinations, bias, and data privacy.

- Implement LLM-powered test infrastructure and AI-driven test strategies.
- Contribute to organizational adoption and governance of Generative AI in testing.

Course Learning Outcomes (CLOs) with Mapping

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Understand the fundamentals of Generative AI and its applications in software testing.	PLO1	C2	Fundamental Domain
CLO2	Apply prompt engineering techniques for test analysis and design.	PLO5	C3	Fundamental Domain
CLO3	Analyze AI-generated test outputs and mitigate related risks.	PLO2	C4	Fundamental Domain
CLO4	Implement AI-powered test infrastructure and support organizational AI adoption.	PLO5	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2					√							
CLO3		√										
CLO4					√							

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Introduction to Generative AI & the AI Spectrum LLM Architecture, Embeddings & Multimodal Models for Testing	Deep Learning Spectrum Analysis, Generative vs Discriminative Comparison	Class Test, Midterm exam	CLO1
		Industry Automation Roadmap Review, Shift-Left AI Strategy Discussion	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Prompt Engineering Fundamentals	Transformer Attention Mechanics, Byte-Pair Encoding Breakdown	Class Test, Midterm exam	CLO1
		Vector Cosine Similarity Math, Semantic Space Evaluation	Class Test, Midterm exam	CLO1

Week-3 Lesson 1 & 2 [3 Hours]	Introduction to Generative AI & the AI Spectrum LLM Architecture, Embeddings & Multimodal Models for Testing	Cross-Modal Processing Dissection, UI Screenshot Tokenization	Class Test, Midterm exam	CLO1
		Textual Crash Log Categorization, Bug Report Ingestion Analysis	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Prompt Engineering Fundamentals	Context Window Bounds Management, Inference Paradigm Review	Class Test, Midterm exam	CLO1
		Temperature & Top-P Variability Tuning, Deterministic Output Audit	Class Test, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Prompt Engineering for Requirement & Test Analysis Prompt Engineering for Test Design (Test Case & Scenario Generation)	Few-Shot Exemplar Layout Mapping, System Role Definition Structuring	Class Test, Midterm exam	CLO1
		Chain-of-Thought Logical Progression, Self-Consistency Logic Check	Class Test, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	GenAI for Test Automation — Functional & Regression Scripting Midterm Examination	Requirement Ambiguity Identification, Boundary Condition Slicing	Class Test, Midterm exam	CLO1, CLO2
		Acceptance Criteria Dissection, BDD Gherkin Ingestion Mapping	Class Test, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	GenAI for Non-Functional Testing — Load, Performance & Security Testing Evaluating & Safeguarding GenAI Outputs: Quality, Hallucination, Bias & Security Risk	Test Case Scenario Generation, Equivalence Partition Synthesis	Class Test, Midterm exam	CLO1, CLO2
		Playwright/Selenium Script Compiling, DOM Locators Structural Mapping	Class Test, Midterm exam	CLO1, CLO2
Week-8 Lesson 1 & 2 [3 Hours]	AI Agents & MCP — Building Agentic, Tool- Using Test Workflows Reusable Skills, LLM- Powered Test Infrastructure & LLMOps	Foundational Architecture Cross- Mapping, Core Prompting Troubleshooting	Class Test, Assignment, Midterm exam	CLO1, CLO3
		Mock Infrastructure Blueprint Review, Pipeline Integration Assessment	Class Test, Assignment, Final exam	CLO1, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	AI Strategy, Roadmap & Organizational Governance	LLM-as-a-Judge Evaluation Design, Code Syntax Compilability Auditing	Class Test, Final exam	CLO1, CLO3
		Format Enforcer JSON Mapping, Output Verification Cascading	Class Test, Final exam	CLO1, CLO3
Week-10		Grounded Source Verification, Model Toxic Bias Assessment	Class Test, Final exam	CLO1, CLO3

Lesson 1 & 2 [3 Hours]	Prompt Engineering for Requirement & Test Analysis Prompt Engineering for Test Design (Test Case & Scenario Generation)	Prompt Injection Vulnerability Testing, Jailbreak Payload Analysis	Class Test, Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	GenAI for Test Automation — Functional & Regression Scripting Midterm Examination	Retrieval-Augmented Generation (RAG) Architecture, Vector DB Embedding Ingestion	Class Test, Final exam	CLO1, CLO3
		Multi-Agent Loop Orchestration, Self-Healing Code Pipeline Mapping	Class Test, Final exam	CLO1, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	GenAI for Non-Functional Testing — Load, Performance & Security Testing Evaluating & Safeguarding GenAI Outputs: Quality, Hallucination, Bias & Security Risk	ROI Cost Benefit Matrixing, Enterprise Automation Sizing	Class Test, Assignment, Final exam	CLO1, CLO4
		Commercial vs Open-Source Model Evaluation, Accuracy Metric Review	Class Test, Final exam	CLO1, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	AI Agents & MCP — Building Agentic, Tool-Using Test Workflows Reusable Skills, LLM-Powered Test Infrastructure & LLMOps	Corporate Compliance Rules Audit, Data Privacy Perimeter Review	Class Test, Final exam	CLO1, CLO4
		Air-Gapped Network Topology Design, Proprietary Code Leak Isolation	Class Test, Presentation Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	AI Strategy, Roadmap & Organizational Governance	Guest Lecture / Industry Expert Talk, Production AI Blueprint Review	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
		Whole-Syllabus Parameter Consolidation, System Boundary Defect Auditing	Final exam	CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =70)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO5	
Attendance	--	--	--	--	7				
Class Test 1	5(15)				15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment					5				
Presentation					8				
Midterm Exam	10	15			25	10		15	25
Final exam			20	20	40		20	20	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

- Testing AI Systems: Machine Learning and Generative AI, *Mark Winteringham*
- ISTQB® Certified Tester Specialist Level – Testing with Generative AI (CT-GenAI) Syllabus v1.0

Reference Books/ Other Supplementary Readings

- Schulhoff, Prompt Engineering Best Practices (2024).
- OpenAI, Generative AI Documentation.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

NA

Exam Dates

According to the Examination Schedule



**Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU) (Version
1.0)**

ST 422 Testing with Generative AI Lab

Course Code: ST 422		CIE Marks: 60
ISCED: 0613-422		LF Marks: 40
Course Title: Testing with Generative AI Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)		Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: N/A		
Course Type: Testing Major		

CIE: Continuous Internal Evaluation LF:
Lab Final Exam

Course Content (from syllabus)

This laboratory course provides hands-on experience in applying Generative AI and Agentic AI concepts to real-world software testing activities. Students will use LLM-powered tools to support test analysis, test design, test automation, defect reporting, and test management. The lab emphasizes AI-assisted testing workflows, prompt engineering for testing tasks, autonomous test agents, and integration with modern automation and CI/CD tools.

Rationale of the Course

With the rapid adoption of Generative AI and Agentic AI in the software industry, testing professionals are expected to leverage AI-driven tools for faster, smarter, and more reliable quality assurance. This lab course bridges theory and practice by enabling students to experiment with LLM-based testing tools, AI-powered automation frameworks, and intelligent test agents aligned with current industry practices. It prepares students for modern QA roles where AI-assisted decision-making and autonomous testing agents are increasingly prevalent.

Course Objectives

By the end of this laboratory course, students will be able to:

- Apply Generative AI tools for test analysis, test case generation, and test automation.
- Design and execute AI-assisted functional, regression, and exploratory tests.
- Use Agentic AI concepts to create semi-autonomous testing workflows.
- Analyze AI-generated test results and ensure quality, reliability, and ethical use.
- Document and report testing outcomes using industry-standard tools.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcome's (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Develop AI-assisted performance test plans and workload models using generative AI tools	PLO3	C6, P3	Fundamental Domain
CLO2	Design, execute, and investigate AI-supported performance tests and analyze system bottlenecks	PLO4	C4,P4	Fundamental Domain
CLO3	Apply AI-related IT audit techniques, including risk assessment, controls, security, and compliance	PLO6	C3,A3	Fundamental Domain
CLO4	Prepare AI-generated and human-validated test reports and documentation for real-life projects	PLO10	C5, A3, P4	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1			√									
CLO2				√								
CLO3						√						
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLO

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.30 Hours]	Introduction to GenAI Tools for Software Testing	Playground Studio Initialization, Token Context Testing	CLO1	Lab Performance
Week-2 [1.30 Hours]	Prompt Engineering for Test Analysis	System Role Parameter Configuration, Few-Shot Pattern Structuring	CLO1	Lab Performance

Week-3 [1.30 Hours]	AI-Based Test Case Generation	Equivalence Class Partition Compilation, Boundary Matrix Mapping	CLO1	Lab Performance
Week-4 [1.30 Hours]]	AI-Assisted Test Automation Overview	Runtime Driver Hooking, Dynamic Selector Inspection	CLO2	Lab Performance
Week-5 [1.30 Hours]	Automation Script Generation Using GenAI	Code Sandbox Co-pilot Pairing, Script Syntax Verification	CLO2	Viva, Lab Performance
Week-6 [1.30 Hours]	AI-Augmented Regression Testing	Impact Radius Delta Scanning, Automated Suite Pruning	CLO2	Lab Performance
Week-7 [1.30 Hours]	Agentic AI Concepts in Testing	Loop Pipeline Event Configuration, Multi-Agent Tool Binding	CLO2, CLO3	Viva
Week-8 [1.30 Hours]	Mid Lab Evaluation	Independent Prompt Engineering Sprint, Execution Pipeline Debugging	CLO1–CLO3	Lab Test
Week-9 [1.30 Hours]	AI-Based Defect Analysis & Reporting	Log Telemetry Parsing, Jira Issue JSON Automation Mapping	CLO4	Lab Performance
Week-10 [1.30 Hours]	Risk Analysis of AI-Generated Tests	Hallucination Boundary Stress-Testing, Static Syntax Verification	CLO3	Lab Performance
Week-11 [1.30 Hours]	Security, Ethics, and Bias in AI Testing	Prompt Injection Attack Simulation, Leakage Token Verification	CLO3	Viva
Week-12 [1.30 Hours]	CI/CD Integration with AI Testing Tools	GitHub Actions Runner Event Hooking, YAML Script Validation	CLO2	Lab Performance
Week-13 [1.30 Hours]	Mini Project: AI-Assisted Testing Workflow	End-to-End Agent Execution, Self-Healing Code Pipeline Tuning	CLO1–CLO4	Project Evaluation

Week-14 [1.30 Hours]	Lab Final Examination	Independent Automation Sprint, Live Prompting Defense Showcasing	CLO1–CLO4	Lab Final
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Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO3
Lab Final	40	CLO2	PLO4
		CLO3	PLO6
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

- Testing AI Systems: Machine Learning and Generative AI, *Mark Winteringham*
- ISTQB® Certified Tester Specialist Level – Testing with Generative AI (CT-GenAI) Syllabus v1.0

Reference Books/ Other Supplementary Readings

- Schulhoff, Prompt Engineering Best Practices (2024).
- OpenAI, Generative AI Documentation.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

AI Testing Tools

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

ST 413 Test Automation & Test Management

Course Code: ST 413		CIE Marks: 60
ISCED: 0613-413		SEE Marks: 40
Course Title: Test Automation & Test Management		Total Marks: 100
Semester:		
Credit Value: 2 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite:		
Course Type: Testing Major		

CIE: Continuous Internal Evaluation,
SEE: Semester End Examination

Course Content (from syllabus)

This course covers the principles, tools, and practices of modern test automation and test management. Students will learn the fundamentals of automation, including automation scope, framework architecture, locator strategies, wait mechanisms, assertions, and the use of industry-standard tools such as Selenium, Cypress, Postman, and JMeter. The course also introduces API test automation, performance testing, and Behavior-Driven Development (BDD) through Cucumber or similar tools. In the second part, the course focuses on core test management activities. Topics include test planning, test strategy development, estimation techniques, test documentation (test cases, scenarios, RTM, summary report), and defect management using JIRA. Students explore test monitoring and control, quality metrics, regression management, and reporting practices used by QA teams. The course concludes with integrating automation into CI/CD pipelines, emphasizing continuous testing, version control, pipeline execution, and automation maintenance. By the end of the course, learners will be able to design automation scripts, build automation frameworks, and manage complete testing processes aligned with industry standards.

Rationale of the Course

This course integrates the technical design of test automation frameworks with the strategic principles of test management within modern software quality environments. Students explore how automated testing supports Agile and DevOps practices to facilitate continuous delivery and rapid feedback loops. The curriculum covers defect and risk management, lifecycle documentation, and compliance with ISTQB and IEEE standards. Through hands-on experience with industry-essential tools like Selenium, Cypress, Postman, JMeter, and JIRA, learners build the skills necessary to plan, execute, and monitor comprehensive quality workflows.

Course Objectives

By the end of this course, students will be able to:

- Understand the fundamental concepts, benefits, and limitations of test automation and test management.
- Build and execute automated test scripts using industry-standard tools for UI, API, and performance testing.
- Design scalable automation frameworks and apply best practices for maintainability and efficiency.

- Develop comprehensive test plans, strategies, estimates, and documentation aligned with IEEE and ISTQB standards.
- Apply defect management processes, including defect reporting, classification, analysis, and tracking.
- Use test management and collaboration tools such as JIRA, Git, Jenkins, and CI/CD platforms to support continuous testing.
- Monitor test progress using quality metrics, prepare professional test reports, and communicate results effectively to stakeholders.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Understand the fundamental concepts of software testing, automation, and test management processes	PLO1	C2	Fundamental Domain
CLO2	Analyze and select appropriate test automation tools and frameworks for different types of software applications.	PLO2	C4	Fundamental Domain
CLO3	Design and implement automated test scripts using industry-standard tools	PLO3	C6	Fundamental Domain
CLO4	Apply the right strategy and tools for testing	PLO5	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3			√									
CLO4					√							

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Test Automation & Test Management	Lifecycle Automation ROI Analysis, Framework Feasibility Evaluation	Class Test, Midterm exam	CLO1

	Lesson 2: Understanding roles in testing teams, Test Manager responsibilities, and types of automated tests.	Resource Responsibility Matrixing, Regression vs Sanity Classification	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to the automation pyramid	Cohn Pyramid Cost Layer Analysis, Unit-to-UI Ratio Balancing	Class Test, Midterm exam	CLO1
	Lesson 2: Basics of scripting with an overview of Python and Java for test automation	Object-Oriented Syntax Selection, Typology Mapping Comparison	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of Selenium WebDriver architecture and introduction to Cypress	JSON Wire Protocol Dissection, Browser Engine Runtime Review	Class Test, Midterm exam	CLO1
	Lesson 2: Introduction to Appium for mobile testing and understanding the automation workflow.	Mobile JSON Wire Protocol Brokerage, Mobile Emulator Capabilities Audit	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Understanding locators, performing actions, and handling waits in automated scripts.	XPath/CSS Selector Strategy Optimization, Explicit/Implicit Wait Collision Analysis	Class Test, Midterm exam	CLO1
	Lesson 2: Writing assertions in tests and applying reusability principles for efficient scripting. Class Test -1	Assert Boolean Evaluation Matrixing, Modular Action Component Design	Class Test, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to data-driven, keyword-driven, and hybrid frameworks	CSV-Excel Input Mapping, Action String Keyword Dissection	Class Test, Midterm exam	CLO1
	Lesson 2: Organizing folder structure and ensuring modularity in test automation projects	Page Object Model Layout Design, Package Namespace Segregation	Class Test, Midterm exam	CLO1, CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Creating feature files and writing step definitions using Gherkin syntax	Given-When-Then Regex Parsing, Scenario Outline Parametrization	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Comparing BDD and TDD approaches and implementing BDD frameworks in automation. Class Test -2	Lifecycle Velocity Matrixing, Shared Collaboration Specification Mapping	Class Test, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Understanding REST API concepts and exploring Postman collections	HTTP Verb Response Payload Validation, JSON Schema Format Analysis	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Automating API tests using REST-Assured (Java) or Python Requests	Response Body Chain Assertions, Payload	Class Test, Midterm exam	CLO1, CLO2

		Authentication Token Injection		
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to JMeter and basic concepts of load, stress, and endurance testing	Thread Group Ramp-Up Vector Setup, Performance Boundary Profiling	Class Test, Assignment, Midterm exam	CLO1, CLO3
	Lesson 2: Analyzing performance test results and interpreting performance reports.	Latency Error-Rate Log Graphing, Throughput Saturation Inversion Review	Class Test, Assignment, Midterm exam	CLO1, CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 2: Test planning, strategy, and the test process life cycle.	QA Milestone Timeline Scheduling, Environment Scope Mapping	Class Test, Midterm exam	CLO1, CLO3
	Lesson 2: Understanding test entry and exit criteria and their importance in test management.	Quality Gate Threshold Boundary Analysis, Release Readiness Rule Formulation	Class Test, Midterm exam	CLO1, CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Creating a Work Breakdown Structure (WBS) and understanding estimation techniques (FP, WBS, TPA)	Test Function Point Matrixing, Three-Point Sizing Calculations	Class Test, Final exam	CLO1, CLO3
	Lesson 2: Building test schedules and planning resources effectively for testing projects.	Critical Path Network Dependency Analysis, Human Resource Utilization Allocation	Class Test, Final exam	CLO1, CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Writing a test plan, test scenarios, and test cases.	IEEE-829 Specification Structuring, Boundary Case Logic Design	Class Test, Final exam	CLO1, CLO3
	Lesson 1: Creating traceability matrix and preparing test summary reports. Class Test -3	Requirement-to-Test Forward Bi-directional Mapping, Execution Quality Sign-Off Reporting	Class Test, Final exam	CLO1, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Understanding the defect life cycle, severity vs priority, and defect reporting in JIRA.	Defect State Machine Routing, Severity-Priority Matrix Calibration	Class Test, Assignment, Final exam	CLO1, CLO4
	Lesson 2: performing root cause analysis (RCA) and managing defect resolution effectively	Ishikawa Fishbone Fault Graphing, 5-Whys Contributory Factor Auditing	Class Test, Final exam	CLO1, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Learning quality metrics, test coverage, defect density, and defect removal efficiency (DRE).	Code Block Coverage Metric Auditing, Injected-vs-Escaped Defect Mathematical Tracing	Class Test, Final exam	CLO1, CLO4
	Lesson 2: Using burn-down charts and status dashboards for test monitoring and control.	Sprint Quality Telemetry Tracking, Real-Time	Class Test, Presentation Final exam	CLO1, CLO4

		Dashboard KPI Verification		
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production QA Pipeline Blueprint Review	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class	Whole-Syllabus Framework Parameter Consolidation, System Boundary Defect Auditing	Final exam	CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO1	PLO's			Mark (Total =65)
	CLO1	CLO2	CLO3	CLO 4			PLO 2	PLO 3	PLO5	
Attendance	--	--	--	–	7					
Class Test 1	5(15)				15					
Class Test 2		5(15)								
Class Test 3			5(15)							
Assignment					5					
Presentation					8					
Midterm Exam	10	15			25	10	15			25
Final exam			20	20	40			20	20	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Evaluation Policy (Grading Policy)

Marks obtained out of 100	Grade	Grade point equivalent	Remarks
80% and above	A+	4.00	Outstanding
75% to less than 80%	A	3.75	Excellent

70% to less than 75%	A-	3.50	Very Good
65% to less than 70%	B+	3.25	Good
60% to less than 65%	B	3.00	Satisfactory
55% to less than 60%	B-	2.75	Above Average
50% to less than 55%	C+	2.50	Average
45% to less than 50%	C	2.25	Below Average
40% to less than 45%	D	2.00	Pass
Less than 40%	F	0.00	Fail

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

Rex Black, Gerry Coleman, Bertrand Cornanguer, Marie Walsh, Istvan Forgacs, Kari Kakkonen, Jan Sabak, Agile Testing: An ISTQB-BCS Foundation Guide, BCS Learning & Development Limited, 2017, ISBN 1780173369, 9781780173368

Reference Books/ Other Supplementary Readings

1. Selenium WebDriver Practical Guide – Unmesh Gundecha
2. Foundations of Software Testing – Dorothy Graham
3. Continuous Testing for DevOps Professionals – Eran Kinsbruner

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

Selenium, Cypress, Postman, JMeter, and JIRA

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU) (Version 1)

ST 414 Test Automation & Test Management Lab

Course Code: ST 414		CIE Marks: 60
ISCED: 0613-414		LF Marks: 40
Course Title: Test Automation & Test Management Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)	Contact Hours: 1 hour 30 minutes (14 weeks)	
Prerequisite: N/A		
Course Type: Testing Major		

CIE: Continuous Internal Evaluation

LF: Lab Final Exam

Course Content (from syllabus)

This lab course provides hands-on experience in test automation and test management practices used in modern software development. Students will work with automation tools such as Selenium, API testing tools, and CI/CD pipelines while also learning test planning, estimation, documentation, defect management, and test metrics. The lab emphasizes real-life project scenarios, automation framework development, test execution, reporting, and integration with DevOps workflows to ensure industry readiness.

Rationale of the Course

With the increasing complexity of software systems, automation and structured test management have become essential for ensuring software quality. This lab course enables students to bridge theory with practice by implementing automated tests, managing test artifacts, tracking defects, and evaluating quality metrics. The course prepares students for professional roles such as Test Automation Engineer, QA Engineer, and Test Analyst.

Course Objectives

By the end of the course, students will be able to:

- Develop automated test scripts using industry-standard tools
- Design and execute structured test plans and test cases
- Manage defects, test artifacts, and reports using test management tools
- Analyze test results and quality metrics to improve software quality
- Integrate test automation with CI/CD pipelines

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Develop and execute automated test scripts for web and API-based applications	PLO3	C6, P4	Fundamental Domain

CLO2	Design automation frameworks and integrate tests within CI/CD pipelines	PLO4	C6, P4	Fundamental Domain
CLO3	Apply test management practices including test planning, estimation, and defect tracking	PLO11	C3,A3	Fundamental Domain
CLO4	Write effective reports and design test case for real life project applying a wide variety of testing techniques at various testing levels	PLO10	C5, A3, P4	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1			√									
CLO2				√								
CLO3											√	
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Correspondin g CLOs	Assessment Strategy
Week-1 [1.30 Hours]	Lab orientation, environment setup, Selenium & API tools installation	SDK Path Configuration, WebDriver Binary Dependency Provisioning	CLO1, CLO2,CLO3	Lab Performance
Week-2 [1.30 Hours]	Automation basics & automation pyramid	Test Suite Tier Segmentation, UI-vs-API Execution Costing	CLO1	Lab Performance
Week-3 [1.30 Hours]	Selenium WebDriver architecture & basic scripts	Browser Driver Socket Binding, Session Instantiation Controls	CLO1	Lab Performance
Week-4 [1.30 Hours]]	Locators, waits, assertions	Dynamic DOM Element Querying, Explicit Condition Exception Polling	CLO2	Lab Performance

Week-5 [1.30 Hours]	Automation frameworks (data-driven, hybrid)	Data-Provider CSV Ingestion, Decoupled Page Object Architecture Design	CLO2	Viva, Lab Final
Week-6 [1.30 Hours]	BDD using Cucumber / Behave	Step Definition Regex Mapping, Feature Scenario Glue Compiling	CLO2	Viva, Lab Final
Week-7 [1.30 Hours]	API automation using Postman & scripting	Sandbox HTTP Request Chaining, JSON Schema Validation Scripting	CLO2	Viva, Lab Final
Week-8 [1.30 Hours]	Midterm automation project	Independent Suite Engineering Sprint, Automated Regression Run Defense	CLO1 & CLO2	Viva, Lab Final
Week-9 [1.30 Hours]	Test planning & test strategy preparation	QA Milestone Gantt Charting, Defect Management Pipeline Configuration	CLO4	Lab Performance, Viva, Lab Final

Week-10 [1.30 Hours]	Test estimation & scheduling	Three-Point Estimation Modeling, Jira Sprint Planning Resource Allocation	CLO4	Lab Performance
Week-11 [1.30 Hours]	Test documentation & RTM	Forward-Backward Matrix Linking, QA Summary Exporting	CLO3	Lab Performance, Viva, Lab Final
Week-12 [1.30 Hours]	Defect life cycle & JIRA defect tracking	Workflow State Machine Transitions, Severity-Priority Matrix Calibration	CLO3	Lab Performance, Viva, Lab Final
Week-13 [1.30 Hours]	Test metrics, dashboards & reports	Quality Release Metric Telemetry, Real-Time JQL Dashboard Component Layout	CLO4	Lab Performance

Week-14 [1.30 Hours]	Lab Final	Hands-on exam	CLO1, CLO2, CLO3, CLO4	Viva, Lab Final
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Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO3
Lab Final	40	CLO2	PLO4
		CLO3	PLO11
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

Rex Black, Gerry Coleman, Bertrand Cornanguer, Marie Walsh, Istvan Forgacs, Kari Kakkonen, Jan Sabak, Agile Testing: An ISTQB-BCS Foundation Guide, BCS Learning & Development Limited, 2017, ISBN 1780173369, 9781780173368

Reference Books/ Other Supplementary Readings

4. Selenium WebDriver Practical Guide – Unmesh Gundecha
5. Foundations of Software Testing – Dorothy Graham
6. Continuous Testing for DevOps Professionals – Eran Kinsbruner

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

Selenium WebDriver, Postman, JUnit / TestNG, JIRA ,Git / GitHub, Jenkins

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)

Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

ST 423 Software Performance Engineering & Security Testing

Course Code: ST 423		CIE Marks: 60
ISCED: 0613-423		SEE Marks: 40
Course Title: Software Performance Engineering & Security Testing		Total Marks: 100
Semester:		
Credit Value: 2 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite:		
Course Type: Testing Major		

CIE: Continuous Internal Evaluation,

SEE: Semester End Examination

Course Content (from syllabus)

This course integrates Software Performance Engineering and IT Audit & Software Quality Assurance, providing students with a strong foundation in evaluating system performance, governance, and controls. Topics include performance testing fundamentals, workload modeling, SLAs/SLOs/KPIs, performance risks, test planning, monitoring, bottleneck analysis, and the impact of system architecture and environments on performance. Students will learn to interpret performance data, conduct root-cause analysis, and design effective performance test strategies. The IT audit component introduces software quality concepts, IEEE standards, verification and validation, IT and data governance, lifecycle frameworks such as IEEE 12207, and the role of audit checkpoints in SDLC. Additional areas include test design, estimation, defect and incident management, risk and vulnerability assessment, information security frameworks (ISO, COBIT, OCTAVE), and the conceptual application of IT audit processes, controls, and evidence collection. The course prepares learners to understand performance fundamentals, analyze performance issues, design audit frameworks, and apply audit procedures within software environments.

Rationale of the Course

This course bridges the disciplines of software performance engineering and IT audit practices to ensure system reliability, operational efficiency, and regulatory compliance. Students explore the core mechanics of performance testing, including bottleneck identification, workload modeling, and system behavior analysis under varying load conditions. In the IT audit segment, the curriculum details governance frameworks, risk management strategies, and security controls essential for organizational accountability. Through hands-on evaluations and conceptual audits, learners acquire the practical skills necessary to design performance tests, interpret execution metrics, and implement IT governance structures that safeguard system effectiveness.

Course Objectives

By the end of this course, students will be able to:

- Understand the fundamentals of software performance testing, including risks, KPIs, workload models, and architectural impacts.
- Analyze performance data, identify bottlenecks, and interpret system behavior across diverse environments.

- Design IT audit frameworks and governance models, applying standards such as ISO, COBIT, IEEE 12207, and security controls.
- Apply audit procedures conceptually, including risk assessment, control evaluation, incident documentation, and audit reporting.
- Connect software performance principles with organizational security, quality assurance, and IT governance practices.
- Evaluate testing and audit documentation, apply conceptual reasoning, and understand the relationship between quality, risk, and compliance.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Demonstrate fundamentals of performance testing	PLO1	C2	Fundamental Domain
CLO2	Analyze performance techniques & bottlenecks	PLO2	C4	Fundamental Domain
CLO3	Explain IT audit frameworks, governance, controls	PLO5	C3	Fundamental Domain
CLO4	Apply audit procedures conceptually	PLO6	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3					√							
CLO4						√						

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of Software Performance Engineering & IT Audit	SPE Framework Analysis, IT Audit Scope Mapping	Class Test, Midterm exam	CLO1
	Lesson 2: Introduction to Performance Testing – Need, Risks, Failures	Degradation Risk Vectoring, Outage Cost Estimation	Class Test, Midterm exam	CLO1

Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Performance Failures, Risks, Historical Cases	Cascade Failure Topology Analysis, Production Bug Post- Mortem	Class Test, Midterm exam	CLO1
	Lesson 2: Performance Testing Fundamentals – Objectives, SLAs, KPIs	Latency Percentile Formula Calculations, Saturation Target Definition	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Workload Modeling, SLOs/SLAs/KPIs	Concurrent User Volume Distribution, Transaction Throughput Math	Class Test, Midterm exam	CLO1
	Lesson 2: Identifying Performance Requirements & Transactions	Critical Production Path Traversal, Business Step Frequency Analysis	Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Performance Testing Process – Overview	Test Cycle Execution Sequencing, Baseline Benchmarking Calibration	Class Test, Midterm exam	CLO1
	Lesson 2: Workload & Transaction Modeling Best Practices Class Test -1	Think-Time Distribution Profiling, Pacing Parameter Calculations	Class Test, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Designing Test Strategy & Test Plan	Spike-Stress Scenario Strategy Configuration, Performance Scope Bounds Definition	Class Test, Midterm exam	CLO1
	Lesson 2: Sample Test Plan Review	Document Structure Quality Verification, Environment Resource Cross-Check	Class Test, Midterm exam	CLO1
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Interpreting Results – KPIs, Monitoring	Resource Telemetry Log Graphing, Memory Leak Trend Analysis	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Root-Cause Analysis – Bottleneck Detection Class Test -2	CPU Thread Contention Diagnostics, Database I/O Locking Analysis	Class Test, Midterm exam	CLO1, CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Technology Impact on Performance	Network Stack Overhead Evaluation, Garbage Collection Ingestion Review	Class Test, Midterm exam	CLO1, CLO2
	Lesson 2: Environment & Architecture Influences	Microservice Network Hop Delay Tracking, Load Balancer Policy Audit	Class Test, Midterm exam	CLO1, CLO2

Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Quality Concepts, Characteristics, IEEE Standards	Software Integrity Attribute Assessment, Standard Compliance Rule Auditing	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Verification, Validation, Reviews, Audits	Formal Inspection Process Verification, Independence Level Risk Mapping	Class Test, Assignment, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Software Management – IT Governance, Data Governance, IEEE 12207	IT Control Frame Mapping, Lifecycle Process Boundary Analysis	Class Test, Final exam	CLO3
	Lesson 2: SDLC Models – Waterfall, Spiral, Agile	Risk Profile Process Variability Review, Pipeline QA Gate Comparison	Class Test, Final exam	CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Primary & Supporting Life Cycle Processes	Configuration Management Audit Control, Supply Chain Traceability Tracking	Class Test, Final exam	CLO3
	Lesson 2: Testing in SDLC & STLC	Test Requirement Phase Injection, Verification V- Model Layering	Class Test, Final exam	CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Types of Testing & QA Role in IT Audit	Control Effectiveness Evidence Verification, Audit Trail Logging Assessment	Class Test, Final exam	CLO3
	Lesson 2: Test Planning, Strategy, Approach Class Test -3	Enterprise Test Program Scheduling, Boundary Scope Risk Prioritization	Class Test, Final exam	CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Test Design Factors, Requirement Analysis, Estimation	Test Sizing Function Point Matrixing, Complexity Defect Density Sizing	Class Test, Assignment, Final exam	CLO4
	Lesson 2: Test Case Specification & Examples	Input Parameter Boundary Matrix Mapping, Expected Outcome Rule Blueprinting	Class Test, Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Test Execution, Reporting, RTM, Defects	Bi-directional Requirements Linking Matrix, Flaky Defect Escape Tracing	Class Test, Final exam	CLO3, CLO4
	Lesson 2: Incident Definition, Classification, Documentation	Defect State Machine Triage Classification,	Class Test, Presentation Final exam	CLO3, CLO4

		Core Impact Severity Calibration		
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Guest Lecture / Industry Expert Talk, Production Infrastructure Audit Review	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class	Final Syllabus Parameter Consolidation, System Compliance Defect Auditing	Final exam	CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4		PLO1	PLO2	PLO6	PLO5	
Attendance	--	--	--	--	7					
Class Test 1	5(15)				15					
Class Test 2		5(15)								
Class Test 3			5(15)							
Assignment					5					
Presentation					8					
Midterm Exam	10	15			25	10	15			25
Final exam			20	20	40			20	20	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

1. *The Art of Application Performance Testing*, Ian Molyneaux, O'Reilly Media, 2nd Edition, ISBN: 978-1491900540, Published: November 2014.
2. *Software Quality Assurance: Integrating Testing, Security, and Audit*, Abu Sayed Mahfuz, Apress, ISBN: 978-1484280474, Published: July 2022.

Reference Books/ Other Supplementary Readings

Systems Performance: Enterprise and the Cloud, Brendan Gregg, Prentice Hall, 2nd Edition, ISBN: 978-0136820154, Published: November 2020.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

Debugging, Junit, Selenium

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

ST 424 Software Performance Engineering & Security Testing Lab

Course Code: ST 424	CIE Marks: 60
ISCED: 0613-424	LF Marks: 40
Course Title: Software Performance Engineering & Security Testing Lab	Total Marks: 100
Semester:	
Credit: 1 (Lab)	Contact Hours: 1 hour 30 minutes (14 weeks)
Prerequisite: N/A	
Course Type: Testing Major	

CIE: Continuous Internal Evaluation LF:
Lab Final

Course Content (from syllabus)

This course provides an in-depth, hands-on understanding of software performance testing and IT audit practices. Learners will acquire both technical and analytical skills by designing performance test plans, executing workload models, analyzing system bottlenecks, and preparing performance reports. Alongside performance engineering, students will explore IT governance, audit methodologies, security controls, risk assessment, and audit documentation techniques aligned with international frameworks such as ISO and COBIT. Through a blend of lab activities, projects, root-cause analysis, and structured audit documentation, this course prepares learners to contribute effectively to industry-level QA and IT audit environments.

Rationale of the Course

Performance testing and IT auditing are two critical pillars of modern software quality assurance. Organizations today rely on scalable systems and stringent regulatory compliance frameworks, making it essential for QA engineers to understand both system behavior under load and the audit processes governing software operations. This course integrates these two domains so that learners gain a holistic understanding of software quality—covering reliability, performance efficiency, risk management, security evaluation, and audit reporting. The goal is to prepare students for roles such as Performance Engineer, QA Analyst, Security Tester, and IT Auditor.

Course Objectives

By the end of the course, learners will be able to:

- Understand the principles and processes of software performance engineering.
- Develop workload models, test strategies, and performance test plans.

- Execute performance tests and analyze results to identify bottlenecks.
- Apply IT audit concepts including risk assessment, control evaluation, and security auditing.
- Prepare structured documentation such as test plans, RTMs, defect logs, incident reports, and audit reports.
- Integrate performance testing outcomes with audit findings to assess overall system quality.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Develop performance test plans & workload models	PLO3	C6, P4	Fundamental Domain
CLO2	Design & execute performance tests; analyze bottlenecks	PLO4	C4, P4	Fundamental Domain
CLO3	Apply IT audit techniques: risk assessment, controls, security checks	PLO6	C3, A3	Fundamental Domain
CLO4	Write effective reports and design test case documentation for the real life project applying a wide variety of testing techniques at various testing levels	PLO10	C5, A3, P4	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1			√									
CLO2				√								
CLO3						√						
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.30 Hours]	Lab Orientation & Tool Setup: Introduction to lab tools, Install JMeter/Locust/Postman, Run simple HTTP request	Host SDK Path Provisioning, Web Server Socket Verification	CLO1, CLO2, CLO3	Lab Performance

Week-2 [1.30 Hours]	Developing Basic Performance Test Plans: Create Test Plan, Add Thread Group & Samplers, Identify load conditions	Thread Group Parameter Setup, HTTP Request Sampler Mapping	CLO1	Lab Performance
Week-3 [1.30 Hours]	Workload Modeling: Open vs closed models, ramp-up/thread count/duration, Define transactions & scenarios	Concurrency Ramp-Up Vector Setup, Pacing-Think Time Boundary Throttling	CLO1	Lab Performance
Week-4 [1.30 Hours]]	SLA-Based & KPI-Driven Test Design: Assertions, Timers, Define KPIs & SLAs, Build KPI-driven scenarios	Response-Time Duration Assertion Mapping, Transaction Error Threshold Configuration	CLO2	Lab Performance
Week-5 [1.30 Hours]	Performance Test Execution: Execute load tests, Collect response time/throughput/error %, Generate HTML report	Headless CLI Engine Execution, Runtime Log Ingestion & HTML Report Parsing	CLO2	Viva, Lab Final
Week-6 [1.30 Hours]	Bottleneck Identification: Analyze logs/errors/graphs, Identify CPU/DB/Network/I/O bottlenecks	Hardware Telemetry Log Profiling, Saturation Inversion Curve Diagnostics	CLO2	Viva, Lab Final
Week-7 [1.30 Hours]	Environment & Architecture Testing: API vs UI performance, Multi-environment testing, DB load/caching/CDN effects	Headless Protocol vs DOM Render Profiling, CDN Edge Layer Caching Invalidation Auditing	CLO2	Viva, Lab Final
Week-8 [1.30 Hours]	Midterm Performance Project: Develop full Test Plan, Execute workload, Prepare performance findings	Multi-Scenario Load Injection Sprint, Capacity Bounds Verification Reporting	CLO1 & CLO2	Viva, Lab Final
Week-9 [1.30 Hours]	Test Documentation Preparation: Create test cases, Write RTM, Prepare defect logs,	Bi-directional Requirement Traceability Mapping, Defect Log Schema Construction	CLO4	Lab Performance, Viva, Lab Final
Week-10 [1.30 Hours]	Incident Documentation & RCA: Create incident logs, Perform RCA, Write incident report	Ishikawa Fishbone Fault Graphing, Triage Defect Lifecycle Schema Design	CLO4	Lab Performance

Week-11 [1.30 Hours]	Applying IT Audit Techniques: Prepare risk register, Identify preventive/detective/corrective controls, Map risks to COBIT/ISO controls	Compliance Matrix Risk Cataloging, Security Control Capability Mapping	CLO3	Lab Performance, Viva, Lab Final
Week-12 [1.30 Hours]	Security & Audit Testing: Check security controls, Identify vulnerabilities, Apply threat & risk evaluation,	Static Application Code Scanning, CVE Vulnerability Ledger Auditing	CLO3	Lab Performance, Viva, Lab Final
Week-13 [1.30 Hours]	Audit Report Writing: Prepare audit observations, Write audit report, Document evidence/findings/recommendations	Deficiency Non-Conformance Ledger Writing, Audit Trail Evidence Compilation	CLO4	Lab Performance
Week-14 [1.30 Hours]	Lab Final	Multi-Tier Performance Injection Challenge, Comprehensive System Attestation Defense	CLO1, CLO2, CLO3, CLO4	Viva, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO10
Lab Performance	25	CLO1	PLO3
Lab Final	40	CLO2	PLO4
		CLO3	PLO6
Total	100		

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

3. *The Art of Application Performance Testing*, Ian Molyneaux, O'Reilly Media, 2nd Edition, ISBN: 978-1491900540, Published: November 2014.
4. *Software Quality Assurance: Integrating Testing, Security, and Audit*, Abu Sayed Mahfuz, Apress, ISBN: 978-1484280474, Published: July 2022.

Reference Books/ Other Supplementary Readings

Systems Performance: Enterprise and the Cloud, Brendan Gregg, Prentice Hall, 2nd Edition, ISBN: 978-0136820154, Published: November 2020.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

Debugging, Junit, Selenium

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

GED S237 Entrepreneurship in IT Business

Course Code: GEDS237		CIE Marks: 60
ISCED:		SEE Marks: 40
Course Title: Entrepreneurship in IT Business		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite:		
Course Type: GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Instructor Details

Name:	
Employee ID:	
Designation:	
Department:	
Office Address:	
Telephone/Extension:	
Mobile:	
Website:	
GTA/UTA(If Any):	

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 9:45 AM	9:45 AM- 11:00 AM	11:00 AM- 12:15 PM	12:15 PM- 1:30 PM	1:30 PM- 2:45 PM	2:45 PM- 4:00 PM	4:00 PM- 5:15 PM	5:15 PM- 6:30 PM
Saturday								
Sunday								
Monday								
Tuesday								
Wednesday								
Thursday								

Course Content (from syllabus)

Entrepreneurial mindset and traits, Pre-assignment review, Finding business ideas, Considering options, Outlining and assessing ideas, Evaluating capacity for risk, Developing product ideas, Making business connections,

Identifying target market and customers, Developing value proposition, Creating financial projections, Types of business ownership, Purchasing a franchise, Purchasing an existing business, Business plan preparation, Executive summary, Pitch deck, 30-second pitch, Practicing the pitch, White papers, Basics of financial statements, Brainstorming funding solutions, Funding options, Product development steps, Protecting intellectual property, Introduction to marketing, Consumer and market analysis, Competition and self-analysis, Distribution channel analysis, Creating marketing plan, Implementing, evaluating, reviewing and revising marketing strategy, Leveraging social media, Building sales force, Branding fundamentals, Brand name development, Slogan development, Logo design, Creative branding, Office setup and location selection, Pros and cons of home-based businesses, Business launch strategies, Organizational chart setup, Recruitment and team engagement, Business operations and tracking, Business growth strategies, Entrepreneurial leadership perspectives, Case studies on entrepreneurship, Sustaining and scaling IT ventures

Rationale of the Course

This course equips students with the knowledge and skills to develop, launch, and manage technology-based ventures. It covers entrepreneurial mindset, opportunity identification, business models, product development, market analysis, financial planning, and funding strategies. Students learn to prepare business plans, pitch decks, and branding strategies while exploring intellectual property, digital marketing, organizational management, and business growth. Through case studies and project-based learning, the course develops leadership, innovation, and problem-solving skills, preparing students to create sustainable and scalable IT enterprises in a competitive business environment.

Course Objectives

The main objectives of this course are to provide students with a strong foundation in entrepreneurship, focusing on developing, managing, and scaling technology-driven ventures. More specifically,

- To understand the traits and mindset of successful entrepreneurs.
- To identify and evaluate viable business ideas and opportunities.
- To develop product concepts, value propositions, and financial projections.
- To learn strategies for marketing, branding, and customer engagement.
- To explore business ownership models, essential documentation, and funding options.
- To acquire skills in operational management, team building, and organizational setup.
- To apply entrepreneurial best practices for launching, sustaining, and growing IT-based businesses.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Analyze entrepreneurial traits and assess personal readiness to undertake business ventures.	PLO1	C4	Fundamental Domain
CLO2	Generate and evaluate innovative business ideas and opportunities using critical thinking and problem-solving approaches.	PLO2	C5, A3	Fundamental Domain
CLO3	Develop actionable business plans and financial strategies that demonstrate decision-making and strategic planning skills.	PLO2	C6, P3	Fundamental Domain

CLO4	Apply effective communication, teamwork, and leadership skills in collaborative business development activities.	PLO3	C3, A4, P4	Fundamental Domain
CLO5	Implement self-learning, evaluation, and reflection techniques to adapt business strategies and sustain venture growth.	PLO5	C5, A5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3		√										
CLO4			√									
CLO5					√							

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Course Orientation, OBE Discussion, CLO-PLO Mapping, Pre-assignment Review	Classroom discussion, Lecture video, Voice-over slides, Open discussion	N/A	N/A
	Lesson 2: Entrepreneurial Mindset and Traits	Lecture slides, Case studies, Group discussion	Participation, Reflection notes	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Finding Business Ideas, Considering Options, Outlining Ideas	Brainstorming sessions, Group activities	Assignment, Class discussion	CLO2
	Lesson 2: Assessing Ideas, Evaluating Capacity for Risk	Interactive discussion, Risk assessment exercises	Assignment, Peer review	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Developing Product Ideas, Making Connections	Hands-on activity, Group exercises	Assignment submission	CLO2
	Lesson 2: Identifying Target Market and Customers, Developing Value Proposition	Case study analysis, Group presentation	Quiz, Group presentation	CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Creating Financial Projections	Lecture slides, Practical exercises	Assignment, Quiz	CLO3
	Lesson 2: Types of Business Ownership, Purchasing Franchise/Existing Business	Case study, Discussion	Short assignment, Class test	CLO3
	Class Test - 1			

Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Business Plan Preparation, Executive Summary	Workshop, Group activity	Assignment, Peer review	CLO3
	Lesson 2: Pitch Deck, 30-Second Pitch, Practicing the Pitch	Presentation practice, Feedback session	Presentation evaluation	CLO4
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: White Papers, Basics of Financial Statements	Lecture, Hands-on exercises	Assignment	CLO3
	Lesson 2: Brainstorming Funding Solutions, Funding Options	Group discussion, Problem-solving activity	Quiz, Assignment	CLO3
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Product Development Steps	Interactive lecture, Case study	Assignment	CLO3
	Lesson 2: Protecting Intellectual Property Class Test -2	Lecture, Scenario-based discussion	Quiz, Case analysis, Midterm exam	CLO2
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Marketing, Consumer and Market Analysis	Lecture slides, Group work	Assignment, Midterm exam	CLO2
	Lesson 2: Competition and Self-Analysis, Distribution Channel Analysis	Case study, Group discussion.	Assignment, Class test	CLO2
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Creating Marketing Plan, Implementing, Evaluating, Reviewing Strategy	Workshop, Simulation exercise	Assignment, Project work	CLO3
	Lesson 2: Leveraging Social Media, Building Sales Force	Hands-on activities, Group discussion	Assignment, Presentation	CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Branding Fundamentals, Brand Name, Slogan, Logo Design	Interactive lecture, Creative workshop	Assignment, Peer evaluation	CLO2
	Lesson 2: Creative Branding, Office Setup, Location Selection	Discussion, Group activity	Assignment	CLO2
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Pros and Cons of Home-Based Businesses, Business Launch Strategies	Case study, Lecture	Assignment, Quiz	CLO3
	Lesson 2: Organizational Chart Setup, Recruitment and Team Engagement Class Test - 3	Group activity, Role-play.	Peer assessment	CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Business Operations and Tracking	Practical workshop, Discussion	Assignment	CLO5
	Lesson 2: Business Growth Strategies	Case study, Interactive lecture	Assignment, Quiz	CLO5

Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Entrepreneurial Leadership Perspectives	Lecture, Group discussion	Reflection notes, Class test	CLO4
	Lesson 2: Case Studies on Entrepreneurship	Group presentations, Discussion	Presentation, Peer evaluation	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Sustaining and Scaling IT Ventures, Expert Session on Emerging Trends	Guest lecture, Workshop, Q&A session	Presentation, Assignment	CLO5
	Lesson 2: Review Class, Final Exam Preparation	Brainstorming, Open discussion	Final exam	CLO2, CLO3, CLO4, CLO5

Overall Assessment Scheme

Assessment Task	CLO's					Mark (Total=100)	PLO's				Mark (Total =65)
	CLO1	CLO2	CLO3	CLO4	CLO5		PLO1	PLO2	PLO3	PLO 5	
Attendance			--	--	--	7					
Class Test 1	5(15)					Avg 15					
Class Test 2		5(15)									
Class Test 3			5(15)								
Assignment		5				5					
Presentation				8		8					
Midterm Exam	10	10	5			25	10	10	5		
Final exam		5	10	20	5	40		15	20	5	

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Evaluation Policy (Grading Policy)

Marks obtained out of 100	Grade	Grade point equivalent	Remarks
80% and above	A+	4.00	Outstanding
75% to less than 80%	A	3.75	Excellent
70% to less than 75%	A-	3.50	Very Good
65% to less than 70%	B+	3.25	Good
60% to less than 65%	B	3.00	Satisfactory
55% to less than 60%	B-	2.75	Above Average
50% to less than 55%	C+	2.50	Average

45% to less than 50%	C	2.25	Below Average
40% to less than 45%	D	2.00	Pass
Less than 40%	F	0.00	Fail

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

Thomas N. Duening, Robert A. Hisrich, Michael A. Lechter, Technology Entrepreneurship: Taking Innovation to the Marketplace, 3rd Edition, Academic Press, 2020, ISBN: 978-0128222034.

Reference Books/ Other Supplementary Readings

Thomas H. Byers, Richard C. Dorf, Andrew J. Nelson, Technology Ventures: From Idea to Enterprise, 5th Edition, McGraw-Hill Education, 2025, ISBN: 978-1264403011.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)

Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

SE 345 AI System & Application Development

AI System & Application Development		
Course Code: SE345		CIE Marks: 60
ISCED:		SEE Marks: 40
Course Title: AI System & Application Development		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite:		
Course Type: Core		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Instructor Details

Name:	
Employee ID:	
Designation:	
Department:	
Office Address:	
Telephone/Extension:	
Mobile:	
Website:	
GTA/UTA(If Any):	

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 9:45 AM	9:45 AM- 11:00 AM	11:00 AM- 12:15 PM	12:15 PM- 1:30 PM	1:30 PM- 2:45 PM	2:45 PM- 4:00 PM	4:00 PM- 5:15 PM	5:15 PM- 6:30 PM
Saturday								
Sunday								
Monday								
Tuesday								
Wednesday								
Thursday								

Course Content (from syllabus)

Mobile application ecosystem and evolution; mobile operating systems and framework comparisons (native, hybrid, and cross-platform); fundamentals of Dart programming language including syntax, data types, operators, control statements, functions, and object-oriented principles; class and object concepts applied in mobile app logic; introduction to Flutter framework, project architecture, and widget tree; stateless and stateful widget design; form handling and navigation; asynchronous programming and API communication; local storage mechanisms including shared preferences and SQLite CRUD operations; performance optimization, maps and geolocation, deployment processes, and professional practices in mobile app development with ethical and user experience considerations.

Rationale of the Course

XMobile application ecosystem and evolution; mobile operating systems and framework comparisons (native, hybrid, and cross-platform); fundamentals of Dart programming language including syntax, data types, operators, control statements, functions, and object-oriented principles; class and object concepts applied in mobile app logic; introduction to Flutter framework, project architecture, and widget tree; stateless and stateful widget design; form handling and navigation; asynchronous programming and API communication; local storage mechanisms including shared preferences and SQLite CRUD operations; performance optimization, maps and geolocation, deployment processes, and professional practices in mobile app development with ethical and user experience considerations.

Course Objectives

To provide a solid conceptual understanding of the Mobile Application and Development, more specifically,

- To understand mobile application ecosystems, operating systems, and cross-platform frameworks.
- To apply Dart programming and object-oriented principles in mobile application development.
- To explore Flutter architecture and widget-based interface design for effective UI/UX.
- To analyze performance, scalability, and ethical considerations in mobile application design and deployment.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Demonstrate contemporary technology and business trends influencing mobile application ecosystems and digital innovation.	PLO1	C2	Fundamental Domain
CLO2	Apply programming logic, object-oriented principles, and mobile application frameworks to develop cross-platform applications while considering legal, ethical, security, and societal requirements.	PLO6	C3	Fundamental Domain
CLO3	Collaborate effectively as an individual and as a member of a development team to analyze, design, and implement mobile application solutions using appropriate communication and teamwork practices.	PLO9	C4	Fundamental Domain
CLO4	Design, and evaluate mobile application development projects by applying engineering management principles, resource planning, task scheduling, and professional documentation practices..	PLO11	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12

CLO1	√											
CLO2						√						
CLO3									√			
CLO4											√	

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of OBE (CLO, PLO, Knowledge level)	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note,	N/A	N/A
	Lesson 2: Introduction to Mobile Applications – Definition, types of apps, ecosystem overview, business impact.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes.	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Mobile Framework Evolution – Native vs Hybrid vs Cross-platform comparison and industry adoption trends.	Classroom discussion, Lecture video, Lecture notes, Open discussion.	Class Test, Midterm exam	CLO1
	Lesson 2: Why Dart and Flutter? Overview of cross-platform architecture and modern development approaches.	Brainstorming sessions, Classroom discussion, Lecture video, Lecture notes, Open discussion.	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Dart Programming Introduction – Structure of a Dart program, main() method, and basic syntax.	Brainstorming sessions, Lecture video, Lecture notes,	Class Test, Midterm exam	CLO2
	Lesson 2: Variables, Constants, and Data Types in Dart with properties and examples.	Brainstorming sessions, Lecture video, Lecture notes, Open discussion.	Class Test, Midterm exam	CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Operators and Expressions in Dart – Arithmetic, logical, relational, and type test operators. Class Test -1	Classroom discussion, Voice over PPT, Lecture video, Lecture notes	Class Test, Midterm exam	CLO2
	Lesson 2: Control Flow Statements – if-else, switch-case, loops (for, while, do-while).	Voice over PPT, Lecture video, Lecture notes	Class Test, Midterm exam	CLO2
Week-5	Lesson 1: Functions in Dart – parameters, return types, anonymous and arrow functions.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes.	Class Test, Midterm exam	CLO2

Lesson 1 & 2 [3 Hours]	Lesson 2: Classes and Objects in Dart – constructors, methods, and object behavior.	Voice over PPT, Lecture video, Lecture note.	Class Test, Midterm exam	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Core OOP Concepts – Encapsulation, Inheritance, Polymorphism, Abstraction in Dart. Class Test -2	Classroom discussion, Voice over PPT, Lecture video, Lecture notes	Class Test, Midterm exam	CLO2
	Lesson 2: Advanced OOP – Method overriding, interfaces, mixins and abstract classes.	Classroom discussion, Lecture notes, Open discussion	Class Test, Midterm exam	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Flutter Framework – Project structure, Widget tree, and basic architecture.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion.	Class Test, Midterm exam	CLO2, CLO4
	Lesson 2: Stateless Widgets and 20 Common Flutter Widgets (Text, Image, Container, Row, Column, Button etc).	Voice over PPT, Lecture video, Lecture note, Open discussion.	Class Test, Midterm exam	CLO2, CLO4
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Stateful Widgets – Concept, Lifecycle (initState, dispose), and setState mechanism.	Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment	CLO3, CLO4
	Lesson 2: Form Handling – TextField, Form Widget, and Validation Logic.	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment, Midterm exam	CLO3, CLO4
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Navigation – Screen transitions and data passing between screens.	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Midterm exam	CLO3, CLO4
	Lesson 2: Routing Mechanisms – Navigator, named routes, and navigation stack.	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Midterm exam	CLO3, CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: API Integration – Understanding REST API, JSON structure, and asynchronous data fetching using Future and async/await.	Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO3
	Lesson 2: Practical Concept of API Response Handling and UI data binding.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO3
Week-11	Lesson 1: Local Storage Concepts – Shared Preferences and File Handling in Flutter.	Classroom discussions, Voice over PPT, Lecture	Class Test, Final exam	CLO3

Lesson 1 & 2 [3 Hours]		video, Lecture note, Open discussion		
	Lesson 1: Local Database (SQLite) – CRUD (Create, Read, Update, Delete) Operations conceptually. Class Test -3	Classroom discussions, Lecture video, Lecture note	Class Test, Final exam	CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Map Integration – Embedding Google Maps, Markers, and Location services.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment, Final exam	CLO3, CLO4
	Lesson 2: Performance Optimization – Reducing widget rebuilds and maintaining responsiveness.	Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO3, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: App Deployment – Build generation (APK/AAB), signing, and Play Store release workflow.	Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO1, CLO4
	Lesson 2: Professional and Ethical Practices – User data privacy, consent, and accessibility standards.	Voice over PPT, Lecture note, Open discussion	Class Test, Presentation Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Interactive Lectures and Presentations, Expert Talks, Hands-On Workshops, Site Visits and Virtual Tours	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class	Brainstorming sessions, Voice over PPT, Lecture note, Open discussion	Final exam	CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =65)
	CLO 1	CLO 2	CLO 3	CLO 4		PLO1	PLO6	PLO9	PLO11	
Attendance	--	--	--	--	7	--	--	--	--	
Class Test 1	15				AVG 15					
Class Test 2		15								
Class Test 3			15							
Assignment			5		5					
Presentation				8	8					
Midterm Exam	10	10	5		25	10	10	5		25
Final exam	10	10	10	10	40	10	10	10	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Evaluation Policy (Grading Policy)

Marks obtained out of 100	Grade	Grade point equivalent	Remarks
80% and above	A+	4.00	Outstanding
75% to less than 80%	A	3.75	Excellent
70% to less than 75%	A-	3.50	Very Good
65% to less than 70%	B+	3.25	Good
60% to less than 65%	B	3.00	Satisfactory
55% to less than 60%	B-	2.75	Above Average
50% to less than 55%	C+	2.50	Average
45% to less than 50%	C	2.25	Below Average
40% to less than 45%	D	2.00	Pass
Less than 40%	F	0.00	Fail

Class Make-up Procedure

Missed class will be taken at convenient free class hour

Textbook/Recommended Readings

Napoli, Marco L. (2019). *Beginning Flutter: A Hands-On Guide to App Development*. Wiley Publications. ISBN: 978-1-119-55082-2

Reference Books/ Other Supplementary Readings

Phillips, Bill et al. (2022). *Android Programming: The Big Nerd Ranch Guide*. Pearson.
Flutter Official Documentation – <https://flutter.dev>

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

Google Codelabs – *Flutter and Dart Tutorials* (<https://codelabs.developers.google.com>)

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

SE 346 AI Systems & Application Development Lab

Course Code: SE 346		CIE Marks: 60
ISCED:		LF Marks: 40
Course Title: AI Systems & Application Development Lab		Total Marks: 100
Semester:		
Credit: 1 (Lab)	Contact Hours: 1.5 (14 weeks)	
Prerequisite:		
Course Type: Core		

CIE: Continuous Internal Evaluation

LF: Lab Final

Instructor Details

Name:	
Employee ID:	
Designation:	
Department:	
Office Address:	
Telephone/Extension:	
Mobile:	
Website:	
GTA/UTA(If Any):	

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 9:45 AM	9:45 AM- 11:00 AM	11:00 AM- 12:15 PM	12:15 PM- 1:30 PM	1:30 PM- 2:45 PM	2:45 PM- 4:00 PM	4:00 PM- 5:15 PM	5:15 PM- 6:30 PM
Saturday								
Sunday								
Monday								
Tuesday								
Wednesday								
Thursday								

Course Content (from syllabus)

Hands-on exercises on Dart programming, Flutter basics, OOP implementation, and widget manipulation; layout design and UI development; event handling and state management; navigation and validation; backend API integration; local database (SQLite) and CRUD operations; app deployment practice.

Rationale of the Course

This laboratory course complements by providing practical experience in applying Flutter and Dart concepts. Students will gradually build mobile applications, starting from basic syntax and user interface design to database integration and final deployment. The course focuses on developing strong coding, debugging, and problem-solving skills through hands-on projects aligned with industry practices.

Course Objectives

- To apply Dart programming constructs and Flutter widgets in real code.
- To design responsive UI layouts using Flutter's widget tree.
- To integrate navigation, data validation, API communication, and local storage.
- To build and test small-scale apps reflecting real-world use cases.
- To collaborate and follow best practices in mobile app development.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Implement Dart syntax, control statements, and OOP structures in mobile app coding.	PLO2	C3, P2	Fundamental Domain
CLO2	Develop basic Flutter user interfaces using stateless and common widgets, and communicate design decisions through clear documentation and presentations	PLO10	P3	Fundamental Domain
CLO3	Construct stateful interfaces, form validation, and navigation systems.	PLO3	C4, P4	Fundamental Domain
CLO4	Integrate APIs, local databases, and shared preferences in Flutter projects.	PLO5	P5	Social Domain, Thinking Domain, Personal Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										

CLO2		√										
CLO3			√									
CLO4					√							

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning, and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Corresponding CLOs	Assessment Strategy
Week-1 [1.5 Hours]	Lab setup, Flutter & Dart installation, first app	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO1	Lab Performance
Week-2 [1.5 Hours]	Dart syntax, variables, datatypes, operators	Classroom discussion, Voice over PPT, Lecture video, Lecture notes, Open discussion.	CLO1	Lab Performance
Week-3 [1.5 Hours]	Control flow, functions, loops, conditional logic	Classroom discussion, Voice over PPT, Lecture video, Lecture notes, Open discussion	CLO1	Lab Performance
Week-4 [1.5 Hours]	Classes, objects, constructors, OOP exercises	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO1	Lab Performance
Week-5 [1.5 Hours]	Stateless widgets and 20 essential widget practices	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO2	Viva, Lab Final
Week-6 [1.5 Hours]	Layouts (Row, Column, Container, etc.) and styling	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO2	Viva, Lab Final
Week-7 [1.5 Hours]	Mid evaluation: Mini project using basic widgets	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO1,CLO2	Viva, Lab Final
Week-8 [1.5 Hours]	Stateful widgets, setState(), and forms	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO3	Lab Performance, Viva, Lab Final
Week-9 [1.5 Hours]	Navigation and routing between screens	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO3	Lab Performance

Week-10 [1.5 Hours]	API integration (GET, POST), async handling	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO4	Lab Performance, Viva, Lab Final
Week-11 [1.5 Hours]	Local database (SQLite), CRUD operations	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO4	Lab Performance, Viva, Lab Final
Week-12 [1.5 Hours]	Shared preferences and persistent data	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO4	Lab Performance, Viva, Lab Final
Week-13 [1.5 Hours]	Maps and location-based services	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO4	Lab Performance
Week-14 [1.5 Hours]	Final project presentation and viva	Classroom discussion, Lecture video, Lecture notes, Open discussion	CLO1, CLO2, CLO3, CLO4	Viva, Lab Final

Overall Assessment Scheme & Marks Distribution

Assessment Task	Assigned Mark	CLO's	PLO's
Attendance	10		
Lab Report	25	CLO4	PLO2
Lab Performance	25	CLO1	PLO2
Lab Final	40	CLO2	PLO3
		CLO3	PLO5
Total	100		

Evaluation Policy (Grading Policy)

Marks obtained out of 100	Grade	Grade point equivalent	Remarks
80% and above	A+	4.00	Outstanding
75% to less than 80%	A	3.75	Excellent
70% to less than 75%	A-	3.50	Very Good
65% to less than 70%	B+	3.25	Good
60% to less than 65%	B	3.00	Satisfactory
55% to less than 60%	B-	2.75	Above Average
50% to less than 55%	C+	2.50	Average
45% to less than 50%	C	2.25	Below Average
40% to less than 45%	D	2.00	Pass
Less than 40%	F	0.00	Fail

Class Make-up Procedure

Missed class will be taken at convenient free class hour following the procedure of the university.

Textbook/Recommended Readings

Napoli, Marco L. (2019). Beginning Flutter: A Hands-On Guide to App Development. Wiley Publications. ISBN: 978-1-119-55082-2

Reference Books/ Other Supplementary Readings

1. Phillips, Bill et al. (2022). Android Programming: The Big Nerd Ranch Guide. Pearson.
2. Flutter Official Documentation – <https://flutter.dev>
3. Lab Manual.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

Flutter, Jupyter Notebook, Google Colab.

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

SE 344 Distributed Systems and Cloud Computing

Course Code: SE 344	CIE Marks: 60
ISCED:	SEE Marks: 40
Course Title: Distributed Systems and Cloud Computing	Total Marks: 100
Semester:	
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)
Prerequisite:	
Course Type: PGC	

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Instructor Details

Name:	
Employee ID:	
Designation:	
Department:	
Office Address:	
Telephone/Extension:	
Mobile:	
Website:	
GTA/UTA(If Any):	

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 9:45 AM	9:45 AM- 11:00 AM	11:00 AM- 12:15 PM	12:15 PM- 1:30 PM	1:30 PM- 2:45 PM	2:45 PM- 4:00 PM	4:00 PM- 5:15 PM	5:15 PM- 6:30 PM
Saturday								
Sunday								
Monday								
Tuesday								
Wednesda y								
Thursday								

Course Content (from syllabus)

Concepts of Distributed Computing, fundamentals of Cloud concepts and capabilities across the various Cloud service models including Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service

(SaaS), Introduction to AWS Cloud, AWS Global Infrastructure Overview, the Cloud security model, associated challenges, and implementation, security and privacy issues in cloud computing.

Rationale of the Course

This course introduces the fundamentals of cloud computing, covering key service models including Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), and Business Process as a Service (BPaaS). Students explore virtualization, cloud infrastructure, and leading platforms such as AWS, Microsoft Azure, and Google Cloud. The course also examines cloud storage, vendor-managed applications, multi-tenant environments, cloud security, and the deployment of high-performance computing and big data solutions, providing a strong foundation in modern cloud technologies and services.

Course Objectives

The course aims to provide students with a comprehensive knowledge and understanding of a wide range of services and deployment models for implementing new applications in the cloud. Students will also gain an understanding of the cloud security issues to be considered to deploy new applications.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Analyze distributed computing concepts and cloud computing architectures to select appropriate solutions for different computing environments.	C4	PLO3	Fundamental Skills
CLO2	Design cloud-based solutions by selecting appropriate cloud service models (IaaS, PaaS, SaaS, BPaaS) to meet specified application requirements.	C6	PLO3	Thinking Skills
CLO3	Apply modern cloud platforms and tools, including AWS services and global cloud infrastructure, to deploy and manage cloud-based applications.	C3	PLO5	Fundamental Skills
CLO4	Evaluate cloud deployment models and cloud infrastructure technologies by considering performance, scalability, cost, and organizational requirements.	C5	PLO5	Thinking Skills
CLO5	Assess cloud security challenges, microservices architecture, and ethical, legal, and societal implications to recommend secure and responsible cloud computing practices.	C5	PLO6	Personal Skills

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1			√									
CLO2			√									

CLO3					√							
CLO4					√							
CLO5						√						

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessm ent Strategy	Correspon ding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Computing System, centralized system characteristics, Distributed system characteristics, Failure Handling (Fault Tolerance), Concurrency	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-1, PLO-11	Class Test, Midterm
	Lesson 2: Advantages & Disadvantages of Distributed Systems, Examples of Distributed Systems, Local Area Network, Database Management System, Automatic Teller Machine Network, Web Servers and Web Browser	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-2 Lesson 1 & 2 [3Hours s]	Lesson 1: Cloud Computing, Access to Cloud Computing, History & Architecture of cloud Computing, Cloud Components	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-2, PLO-2	Class Test, Midterm
	Lesson 2: Cloud Deployment Models, Public Cloud, Private Cloud, Hybrid Cloud	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Layered architecture of Cloud Computing, Infrastructure as a service (IaaS), Management of IaaS, Characteristics of IaaS,	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-2, PLO-2	Class Test, Midterm
	Lesson 2: When to use IaaS, Common IaaS business scenarios, Advantages of IaaS, Limitations of IaaS	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-4 Lesson 1 & 2	Lesson 1: Layered architecture of Cloud Computing, Platform as a Service, Spectrum of Cloud Users, Management of PaaS	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-3, PLO-1	Class Test, Assignment, Midterm

[3Hours]	Lesson 2: Characteristics of PaaS, When to use PaaS , Common PaaS business scenarios, Advantages of PaaS, Limitations of PaaS	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Cloud Concepts Overview, Similarities between AWS and Traditional IT, Introduction to Amazon Web Services (AWS), Categories of AWS services	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-4, PLO-10	Class Test, Assignment, Midterm
	Lesson 2: AWS Services, Interaction with AWS, Moving to the AWS Cloud – The AWS Cloud Adoption Framework (AWS CAF)	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: AWS Cloud Adoption Framework (AWS CAF), Six core perspectives of AWS Cloud Adoption Framework ,	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-4, PLO-10	Class Test, Assignment, Midterm, Final Exam
	Lesson 2: Strategies for Migrating Applications to the Cloud	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: AWS Global Infrastructure Overview, AWS service and service category overview	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-4, PLO-10	Class Test, Assignment, Midterm, Final Exam
	Lesson 2: AWS Management Console, AWS Regions, AWS data centers, AWS infrastructure feature	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: AWS foundational services, Storage service category, AWS Compute services, AWS Database services, AWS networking and content delivery services	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-4, PLO-10	

	Lesson 2: AWS networking and content delivery services,AWS cost management services , AWS management and governance services , AWS Cloud Products.	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Networking basics, Amazon VPC, VPC networking, VPC security, Amazon Route 53, Amazon CloudFront.	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-5, PLO-6	Class Test, Assignment, Midterm, Final Exam
	Lesson 2:IP addresses, Classless Inter-Domain Routing (CIDR),Open Systems Interconnection (OSI) model	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Review class on topic discussed in Week-1 – Week-5	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	---	None
	Lesson 2: Review class on topic discussed in Week-1 – Week-5	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Amazon VPC,VPCs and subnets,Reserved IP addresses	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-5, PLO-6	Class Test, Assignment, Final Exam
	Lesson 2: Public IP address types,Elastic network interface,Route tables and routes	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1:VPC networking, Internet gateway,Network address translation (NAT) gateway,VPC sharing.	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-5, PLO-6	Class Test, Assignment, Final Exam
	Lesson 2: VPC peering,AWS Site-to-Site VPN, AWS Direct Connect,VPC endpoints	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		

Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: VPC security, Security groups, Custom security groups, Network access control lists (network ACLs)	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-5, PLO-6	Class Test, Assignment, Final Exam
	Lesson 2: Security groups versus network ACLs, Amazon Route 53, Amazon Route 53 DNS resolution .	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Amazon CloudFront, Content delivery and network latency, Content delivery network (CDN)	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-5, PLO-6	Class Test, Assignment, Final Exam
	Lesson 2: Amazon CloudFront, Amazon CloudFront infrastructure, Amazon CloudFront benefits	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-15 Lesson 1 & 2 [3 Hours]	Lesson 1: Create a VPC, Create additional subnets, Create a VPC security group, Launch a web server instance	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	CLO-5, PLO-6	Class Test, Assignment, Final Exam
	Lesson 2: Security management of Cloud Computing, Transparent Cloud Protection system, Secure and Efficient access to outsourced data.	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.		
Week-16 Lesson 1 & 2 [3 Hours]	Lesson 1: Review class on topic discussed in Week-6 –Week-9	Brainstorming sessions, Classroom discussion, Voice over PPT, Lecture video, Lecture note, Open discussion.	- - -	None

Overall Assessment Scheme

Assessment Task	CLO's					Mark (Total=100)	PLO's			
	CL O1	CL O2	CL O3	CL O4	CL O5		PL O3	PL O5	PL O6	Mark (Total=65)
Attendance	--	--	--	--		7	--	--	--	
Class Test 1	15					AVG 15				
Class Test 2		15								
Class Test 3			15							
Assignment			5			5				
Presentation				8		8				
Midterm Exam	10	10	5			25	20	5		25
Final exam		10	10	10	10	40	10	20	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Evaluation Policy (Grading Policy)

Marks obtained out of 100	Grade	Grade point equivalent	Remarks
80% and above	A+	4.00	Outstanding
75% to less than 80%	A	3.75	Excellent
70% to less than 75%	A-	3.50	Very Good
65% to less than 70%	B+	3.25	Good
60% to less than 65%	B	3.00	Satisfactory
55% to less than 60%	B-	2.75	Above Average
50% to less than 55%	C+	2.50	Average
45% to less than 50%	C	2.25	Below Average
40% to less than 45%	D	2.00	Pass
Less than 40%	F	0.00	Fail

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

1. “CLOUD COMPUTING” by Nayan B. Ruparelia, Cambridge, MA : The MIT Press, [2015].
2. “Distributed systems: principles and paradigms” by Andrew S. Tanenbaum, Maarten Van Steen (Second Edition).

Reference Books/ Other Supplementary Readings

1. “A Brief Guide To Cloud Computing “- by Christopher Barnatt, Robinson, UK edition[2010].

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

N/A

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

GEDS 416 AI for Strategic Decision Making

Course Code: GEDS416

CIE Marks: 60

ISCED:		SEE Marks: 40
Course Title: AI for Strategic Decision Making		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite:		
Course Type: GED		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Instructor Details

Name:	
Employee ID:	
Designation:	
Department:	
Office Address:	
Telephone/Extension:	
Mobile:	
Website:	
GTA/UTA(If Any):	

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 9:45 AM	9:45 AM- 11:00 AM	11:00 AM- 12:15 PM	12:15 PM- 1:30 PM	1:30 PM- 2:45 PM	2:45 PM- 4:00 PM	4:00 PM- 5:15 PM	5:15 PM- 6:30 PM
Saturday								
Sunday								
Monday								
Tuesday								
Wednesday								
Thursday								

Course Content (from syllabus)

This course introduces students to the role of Artificial Intelligence (AI) in strategic business and organizational decision making. It focuses on how data-driven and AI-powered methods, including predictive analytics, machine learning, and decision-support systems, can be used to analyze complex business problems and guide strategic choices. The course begins with an overview of strategic decision-making frameworks and the evolving role of AI in organizations. Over the course of the subsequent units, it covers data-driven decision models, predictive and prescriptive analytics, machine learning for forecasting and classification, AI-based recommendation and optimization systems, and the ethical, organizational, and risk considerations of deploying AI in strategic contexts.

Rationale of the Course

The course will start with a foundational understanding of strategic decision making and how AI and analytics are transforming this process across industries. It will cover the major categories of AI-driven decision tools, including descriptive, predictive, and prescriptive analytics, machine learning models, and AI-based decision-support systems. This discussion will cover the tradeoffs between model accuracy, interpretability, and practical business value when adopting AI for strategic decisions. Particular emphasis will be given to three major areas: data-driven

strategic analysis (business intelligence, KPIs, and decision frameworks), AI/ML techniques for forecasting, classification, and optimization, and the responsible, ethical, and organizational deployment of AI systems to support executive and managerial decision making.

Course Objectives

To provide a solid conceptual understanding of how AI supports strategic decision making in organizations. More specifically,

To explore strategic decision-making frameworks and the role of data and AI in shaping organizational strategy.

- To demonstrate the use of predictive and prescriptive analytics for forecasting and resource optimization.
- To explain and apply machine learning techniques for classification, forecasting, and pattern discovery in business contexts.
- To evaluate AI-driven recommendations and decision-support systems, considering accuracy, interpretability, and business risk.
- To design and propose AI-enabled strategic solutions while addressing ethical, organizational, and governance considerations.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Interpret strategic decision-making frameworks and the role of data, analytics, and AI in organizational strategy.	PLO1	C2	Fundamental Domain
CLO2	Apply predictive and prescriptive analytics techniques to forecast outcomes and optimize strategic resource decisions.	PLO2	C3	Fundamental Domain
CLO3	Analyze business problems using machine learning techniques, including classification, regression, and clustering, to support decision making.	PLO3	C4	Fundamental Domain
CLO4	Evaluate AI-driven decision-support systems and recommendations, addressing model performance, interpretability, ethics, and organizational risk.	PLO10	C5	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3			√									
CLO4										√		

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Correspon ding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of OBE Introduction to Strategic Decision Making Types of Decisions: Strategic, Tactical, Operational Classical Decision-Making Models (Rational, Bounded Rationality)	Classroom discussion, Voice over PPT Lecture	Class Test	CLO1
	Lesson 2: The Evolving Role of Data and AI in Strategy Overview of AI, Machine Learning, and Analytics Case Study: AI-Driven Strategy in Industry.	Classroom discussion, Voice over PPT, Lecture	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Decision-Making Frameworks SWOT, PESTLE, and Porter's Five Forces in the AI Era Data-Informed Strategic Planning.	Classroom discussion, Lecture video, Open discussion.	Class Test, Midterm exam	CLO1
	Lesson 2: Business Intelligence (BI) Foundations Key Performance Indicators (KPIs) and Dashboards From Data to Strategic Insight	Brainstorming sessions, Classroom discussion, Open discussion.	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Descriptive, Predictive, and Prescriptive Analytics Analytics Maturity Model for Organizations.	Brainstorming sessions, Lecture video, Lecture notes,	Assignment, Class Test, Midterm exam	CLO2
	Lesson 2: Data Sources for Strategic Decisions Data Quality, Bias, and Limitations.	Brainstorming sessions Lecture notes, Open discussion.	Assignment, Class Test, Midterm exam	CLO1
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Predictive Analytics for Forecasting Time Series Forecasting Basics Demand and Sales Forecasting Examples. Class Test -1	Classroom discussion, Voice over PPT	Assignment, Class Test, Midterm exam	CLO2
	Lesson 2: Regression-Based Forecasting Models Evaluating Forecast Accuracy	Voice over PPT, Lecture video, Lecture notes	Class Test, Midterm exam	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Prescriptive Analytics and Optimization Linear Programming for Resource Allocation.	Classroom discussion, Voice over PPT, Lecture video	Assignment, Class Test, Midterm exam	CLO2
	Lesson 2: Decision Trees and Scenario Analysis Sensitivity Analysis for Strategic Choices.	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note.	Assignment, Class Test, Midterm exam	CLO2

Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Machine Learning for Business Supervised vs. Unsupervised Learning Class Test -2	Classroom discussion, Voice over PPT, Lecture video, Lecture notes	Class Test, Midterm exam	CLO2, CLO3
	Lesson 2: Classification Techniques (Logistic Regression, Decision Trees) Use Case: Customer Churn Prediction.	Brainstorming sessions, Classroom discussion, Lecture notes	Class Test, Midterm exam	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Clustering and Segmentation Techniques Use Case: Customer and Market Segmentation	Classroom discussions, Voice over PPT	Class Test, Midterm exam	CLO3
	Lesson 2: Association Rules and Pattern Discovery Market Basket Analysis for Strategic Insight.	Brainstorming sessions, Voice over PPT	Class Test, Midterm exam	CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Midterm Review and Strategic Case Discussion.	Brainstorming sessions, Open discussion	Assignment	CLO3
	Lesson 2: Model Evaluation Metrics (Accuracy, Precision, Recall, ROC) Choosing the Right Model for the Decision Problem.	Brainstorming sessions, Voice over PPT, Open discussion	Class Test, Assignment, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: AI-Based Decision Support Systems (DSS) Architecture of a Strategic DSS.	Brainstorming sessions, Voice over PPT	Class Test, Midterm exam	CLO3, CLO4
	Lesson 2: Recommendation Systems for Strategic Choices Use Case: Product and Investment Recommendation Engines.	Brainstorming sessions, Voice over PPT, Open discussion	Class Test, Final exam	CLO3, CLO4
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Natural Language Processing for Strategic Insight Sentiment Analysis of Market and Customer Data	Voice over PPT, Lecture note, Open discussion	Class Test, Final exam	CLO3, CLO4
	Lesson 2: AI-Driven Competitive Intelligence Text Mining for Strategic Monitoring.	Classroom discussions, Voice over PPT	Class Test, Final exam	CLO3, CLO4
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Interpretability and Explainable AI (XAI) Trust and Transparency in AI-Driven Decisions.	Classroom discussions, Voice over PPT, Open discussion	Class Test, Final exam	CLO4
	Lesson 2: Risk and Uncertainty in AI-Based Strategic Decisions Human-in-the-Loop Decision Making Class Test -3	Classroom discussions, Lecture note	Class Test, Final exam	CLO3

Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Ethics, Bias, and Fairness in AI for Strategy Algorithmic Bias and Its Organizational Impact.	Classroom discussions, Voice over PPT	Presentation, Final exam	CLO4
	Lesson 2: AI Governance and Organizational Policy Regulatory Considerations for AI in Decision Making.	Voice over PPT, Lecture video, Lecture note, Open discussion	Presentation, Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Change Management for AI Adoption in Organizations Building an AI-Ready Decision-Making Culture.	Voice over PPT, Lecture video, Lecture note, Open discussion	Presentation, Final exam	CLO1, CLO4
	Lesson 2: Industry Case Studies: AI in Finance, Retail, and Healthcare Strategy.	Brainstorming sessions, Classroom discussion	Presentation Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Interactive Lectures and Presentations, Expert Talks, Hands-On Workshops, Site Visits and Virtual Tours	Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Capstone Case Study: Designing an AI-Enabled Strategic Recommendation	Brainstorming sessions, Voice over PPT, Lecture note, Open discussion	Final exam	CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =65)
	CLO 1	CLO 2	CLO 3	CLO 4		PLO1	PLO2	PLO3	PLO10	
Attendance	--	--	--	--	7	--	--	--	--	
Class Test 1	15				AVG 15					
Class Test 2		15								
Class Test 3			15							
Assignment			5		5					
Presentation				8	8					
Midterm Exam	10	10	5		25	10	10	5		25
Final exam	10	10	10	10	40	10	10	10	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8

3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Evaluation Policy (Grading Policy)

Marks obtained out of 100	Grade	Grade point equivalent	Remarks
80% and above	A+	4.00	Outstanding
75% to less than 80%	A	3.75	Excellent
70% to less than 75%	A-	3.50	Very Good
65% to less than 70%	B+	3.25	Good
60% to less than 65%	B	3.00	Satisfactory
55% to less than 60%	B-	2.75	Above Average
50% to less than 55%	C+	2.50	Average
45% to less than 50%	C	2.25	Below Average
40% to less than 45%	D	2.00	Pass
Less than 40%	F	0.00	Fail

Class Make-up Procedure

Missed class will be taken at convenient free class hour

Textbook/Recommended Readings

Business Intelligence, Analytics, and Data Science: A Managerial Perspective” Ramesh Sharda, Dursun Delen, Efraim Turban, 4th Edition, Pearson, 2020

Reference Books/ Other Supplementary Readings

1. H. Davenport, Jeanne G. Harris: **Competing on Analytics: Updated, with a New Introduction**, Harvard Business Review Press, 2017.
2. Erik Brynjolfsson, Andrew McAfee: **Machine, Platform, Crowd: Harnessing Our Digital Future**, W. W. Norton & Company, 2017

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

Python, Jupiter Notebook, Microsoft Copilot, Claude

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
Daffodil International University (DIU)
(Version 1.0)

SE 343 Software Maintenance

Course Code: SE343		CIE Marks: 60
ISCED:		SEE Marks: 40
Course Title: Software Maintenance		Total Marks: 100
Semester:		
Credit Value: 3 (Theory)	Contact Hours: 3 (Total weeks: 14)	
Prerequisite:		
Course Type: PGC		

CIE: Continuous Internal Evaluation

SEE: Semester End Examination

Instructor Details

Name:	
Employee ID:	
Designation:	
Department:	
Office Address:	
Telephone/Extension:	
Mobile:	
Website:	
GTA/UTA(If Any):	

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 9:45 AM	9:45 AM- 11:00 AM	11:00 AM- 12:15 PM	12:15 PM- 1:30 PM	1:30 PM- 2:45 PM	2:45 PM- 4:00 PM	4:00 PM- 5:15 PM	5:15 PM- 6:30 PM
Saturday								
Sunday								
Monday								
Tuesday								
Wednesda y								
Thursday								

Course Content (from syllabus)

Software maintenance types, models, and lifecycle integration; Lehman's laws of software evolution and change phases; change management and impact analysis techniques; version control strategies (Gitflow, trunk-based) and configuration management with build automation; CI/CD pipelines and regression testing; software quality metrics (LOC, cyclomatic complexity, cohesion, coupling) and static analysis tooling (SonarQube); code smells taxonomy and anti-pattern detection; refactoring fundamentals and catalog (Extract Method, Move Method, Rename); structural design patterns (Strategy, Observer, Factory Method, Decorator) in maintenance contexts; advanced testing strategies including coverage criteria and performance profiling; dynamic analysis and code pruning; release management and deployment strategies; technical debt measurement and management in agile environments.

Rationale of the Course

Software systems spend the majority of their lifecycle in maintenance, yet most engineering education focuses on Greenfield development. This course addresses that gap by equipping students with the theoretical foundations and practical skills necessary to work confidently on evolving codebases. Students learn to navigate large, legacy systems, manage change systematically, measure and improve quality, and apply modern tooling — skills directly demanded by the software industry. The course bridges software engineering theory (evolution laws, process models) with hands-on practice (refactoring, CI/CD, testing), preparing graduates to contribute to real-world maintenance projects from day one.

Course Objectives

To provide a solid conceptual understanding of Software Maintenance, more specifically,

- To understand the types, laws, and lifecycle models of software maintenance and evolution.
- To apply version control strategies, configuration management, and CI/CD pipelines to manage software changes effectively.
- To analyze software quality using metrics, code smells, and anti-patterns, and to apply refactoring techniques to improve maintainability.
- To design testing strategies and apply design patterns appropriate to maintenance contexts, including release management and technical debt considerations.

Course Learning Outcomes (CLOs) with Mappings

At the end of the course, students will be able to:

CLOs	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Analyze software maintenance requirements, software evolution laws, and change management processes to identify and formulate maintenance solutions for software systems.	PLO2	C4	Fundamental Domain
CLO2	Design and implement software maintenance workflows using version control systems, configuration management practices, and CI/CD tools to support efficient software evolution.	PLO3	C6	Fundamental Domain
CLO3	Apply modern software quality analysis tools and metrics to evaluate maintainability, identify code smells and anti-patterns, and recommend appropriate improvements.	PLO5	C5	Fundamental Domain

CLO4	Design and evaluate refactoring solutions using software design patterns, automated testing, and emerging software engineering practices while demonstrating the ability to adapt to evolving technologies for continuous professional development.	PLO12	C6	Fundamental Domain
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Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2			√									
CLO3					√							
CLO4												√

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Overview of OBE (CLO, PLO, Knowledge level)	Classroom discussion, Voice over PPT, Lecture video, Lecture note,	Class Test, Midterm exam	CLO1
	Lesson 2: Introduction to Software Maintenance – definition, types (corrective, adaptive, perfective, preventive), and lifecycle model placement (waterfall, agile).	Classroom discussion, Voice over PPT, Lecture, video	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Lehman's laws of software evolution and change phases..	Classroom discussion, Lecture video, Lecture notes, Open discussion.	Class Test, Midterm exam	CLO1
	Lesson 2: Change Management – change request handling, bug tracking systems (Jira), corrective and adaptive maintenance in practice.	Brainstorming sessions, Classroom discussion, Lecture video, Lecture notes	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Impact Analysis – dependency graphs, traceability matrices.	Brainstorming sessions, Lecture video, Lecture notes,	Class Test, Midterm exam	CLO2
	Lesson 2: Version Control Strategies – Gitflow and trunk-based development, merge conflict handling at scale, component versioning	Brainstorming sessions, Lecture video, Lecture notes, Open discussion.	Class Test, Midterm exam	CLO2

	Class Test-1			
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Configuration Management – build automation (Maven/Gradle), managing external dependencies	Classroom discussion, Voice over PPT	Class Test, Midterm exam	CLO2
	Lesson 2: CI/CD Pipelines – Jenkins and GitHub Actions, configuration integration testing vs. functional testing	Voice over PPT, Lecture video, Lecture notes	Class Test, Midterm exam	CLO2
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Regression Testing – test selection, minimization, prioritization	Classroom discussion, Voice over PPT	Class Test, Midterm exam	CLO2
	Lesson 2: Software Metrics – LOC, cyclomatic complexity, cohesion, and coupling. Live tool demo using SonarQube	Voice over PPT, Lecture video, Lecture note.	Class Test, Midterm exam	CLO2
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Coding standards, code reviews, and static analysis.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes	Class Test, Midterm exam	CLO2
	Lesson 2: Code Smells – Bloaters, OO Abusers, Change Preventers, and Dispensable with real-world examples.	Classroom discussion, Lecture notes, Open discussion	Class Test, Midterm exam	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Anti-Patterns – Spaghetti Code, Golden Hammer, Lava Flow.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion.	Class Test, Midterm exam	CLO2
	Lesson 2: Introduction to Refactoring Class Test-2	Voice over PPT, Open discussion.	Class Test, Midterm exam	CLO3, CLO4
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Refactoring Fundamentals – definition, golden rule (tests first), catalog of core refactoring: Extract Method, Rename, Move Method, Applying refactoring to real smells.	Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment Final Term	CLO3, CLO4
	Lesson 2: Design Patterns in Maintenance – Strategy, Observer, Factory Method, Decorator for long-term maintainability; refactoring toward patterns.	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment, Final exam	CLO3, CLO4
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Case study: refactoring conditionals into State/Strategy structure.	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO3, CLO4
	Lesson 2: Advanced Testing – unit, integration, system, acceptance levels. Coverage criteria (statement, branch, path). Tool walkthrough: JaCoCo.	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO3, CLO4

Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Performance testing and profiling. Detecting non-functional regressions.	Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO3, CLO4
	Lesson 2: Code Pruning – dead code, unused dependencies, risk assessment before pruning.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO3
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Dynamic Analysis – runtime profiling, memory leak detection. Comparison with static analysis; choosing the right tool.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO3
	Lesson 2: Release Management – version numbering, change logs, release branches. Deployment strategies: rolling vs. blue- green. Class Test-3	Classroom discussions, Lecture video, Lecture note	Class Test, Final exam	CLO3, CLO4
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Hands-on simulation: plan a release for three queued change requests.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment, Final exam	CLO3, CLO4
	Lesson 2: Agile Maintenance – how maintenance works in Scrum and Kanban	Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO3, CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: technical debt measurement and management strategies.	Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Final exam	CLO1, CLO4
	Lesson 2: Real large-scale maintenance project Analysis; what went wrong and what worked.	Voice over PPT, Lecture note, Open discussion	Class Test, Presentation Final exam	CLO1, CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Interactive Lectures and Presentations, Expert Talks, Hands-On Workshops, Site Visits and Virtual Tours	Presentation, Assignment	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review session on case studies	Brainstorming sessions, Open discussion	Final exam	CLO1, CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's				Mark (Total =65)
	CLO 1	CLO 2	CLO 3	CLO 4		PLO2	PLO3	PLO5	PLO12	

Attendance	--	--	--	--	7	--	--	--	--	
Class Test 1	15				AVG 15					
Class Test 2		15								
Class Test 3			15							
Assignment			5		5					
Presentation				8	8					
Midterm Exam	10	10	5		25	10	10	5		25
Final exam	10	10	10	10	40	10	10	10	10	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Class Make-up Procedure

Missed class will be taken at convenient free class hour.

Textbook/Recommended Readings

Tripathy, P. & Naik, K. (2014). Software Evolution and Maintenance: A Practitioner's Approach. John Wiley & Sons. ISBN: 978-0-470-60382-3

Reference Books/ Other Supplementary Readings

- Fowler, M. (2018). Refactoring: Improving the Design of Existing Code (2nd ed.). Addison-Wesley.
- Mens, T. & Demeyer, S. (Eds.) (2008). Software Evolution. Springer.
- SonarQube Documentation – <https://docs.sonarqube.org>
- JaCoCo Documentation – <https://www.jacoco.org>
- GitHub Actions Documentation – <https://docs.github.com/en/actions>

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join google classroom and BLC account).

Software/Tools used

JaCoCo, Jira, SonarQube

Exam Dates

According to the Examination Schedule



Department of Software Engineering (SWE)
Faculty of Science and Information Technology (FSIT)
 (Version 3.0)

SE 431/RE 431/CS 439/DS 431 Final Year Project/Thesis/Internship Robotics & Embedded System Major Capstone Project (RE Major Only)
Cyber Security Major Capstone Project (CS Major Only)
Data Science Major Capstone Project (DS Major Only)
Software Quality Assurance & Testing Major Capstone Project (ST Major Only)

Course Code: SE 431/RE 431/CS 439/DS 431/ ST 431	
ISCED: 0613-431	
Course Title: Final Year Project/Thesis/Internship Robotics & Embedded System Major Capstone Project (RE Major Only) Cyber Security Major Capstone Project (CS Major Only) Data Science Major Capstone Project (DS Major Only) Software Quality Assurance & Testing Major Capstone Project (ST Major Only)	
Semester:	
Credit Value: 6	Contact Hours: 2 Semesters
Prerequisite:	
Course Type: Thesis/Project	

Course Overview:

As part of their fourth year, students are required to carry out a project and submit a technical report. This project is a substantial piece of work that will require creative activity and original thinking. Individual students are supervised while working on a project accounting for six credit hours, extending over a period of two semesters. The project aims to provide students with a transitional experience from the academic world to the professional world. It is designed to serve as a platform in which Software Engineering students engage in a meaningful design experience requiring the solution of complex engineering design projects.

Course Objective:

The objectives of the final year project are:

1. To allow students to demonstrate a wide range of skills learned during their course of study for B.Sc. in Software Engineering by delivering a product that has passed through the design, analysis, testing and evaluation.
2. To enable students in exhibiting multidisciplinary research through the integration learned in a number of courses.
3. To enable students in exhibiting problem solving, analysis, synthesis and evaluation skills.
4. To improve students' communication skills by asking them to produce both a professional report and to give an oral presentation.

Course Learning Outcomes (CLOs):

After completion of the first part of the course students will be able to:

CLO	CLO Descriptions	Learning Subdomains
CLO1	Integrate new and previously acquired knowledge for identifying a real-life complex computer science and engineering problem as the capstone project.	Cognitive (C2, C3)
CLO2	Examine various problem domains (literature review), define the problems, and formulate the objectives for the capstone project.	Cognitive (C4, C5, C6), Psychomotor (P2, P3, P4)
CLO3	Analyze various aspects of the objectives for designing a solution for the capstone project.	Cognitive (C2, C3, C4, C5)
CLO4	Design and develop solutions for the capstone project that meet public health and safety, cultural, societal, and environmental considerations.	Cognitive (C6), Affective (A4)
CLO5	Identify and apply modern engineering and IT tools for the design and development of the capstone project.	Psychomotor (P2, P3), Cognitive (C2)
CLO6	Assess and address societal, health, safety, legal, and cultural aspects related to the implementation of the capstone project considering the relevant professional and engineering practices and solutions.	Affective (A4), Cognitive (C3)
CLO7	Assess and address the sustainability and impact of the capstone project in societal and environmental contexts	Affective (A4), Cognitive (C5)

CLO8	Apply professional and engineering ethical principles and practices for the implementation of the capstone project.	Affective (A4, A5) Cognitive (C5)
CLO9	Write effective reports and design documentation, and make effective presentations of the outcome of the capstone project.	Psychomotor (P3, P4), Affective (A3, A4)
CLO10	Conduct economic analysis and cost estimation, and apply appropriate project management processes in the development life cycle of the capstone project.	Cognitive (C2, C3), Psychomotor (P4)
CLO11	Prepare to take part in independent and life-long learning for adapting emerging technologies for the solution of complex computer science and engineering problems.	Affective (A3), Psychomotor (P4, P5)

Mapping of CLOs to Knowledge Profile (KP), Program Outcome (PLO), Complex Engineering Problem (CEP), and Complex Engineering Activity (CEA):

CLO	CLO Descriptions	KP	PLO	CEP	CEA
CLO1	Integrate new and previously acquired knowledge for identifying a real-life complex engineering problem as the capstone project	KP1, KP2, KP3, KP4	PLO1	CEP1, CEP6	-
CLO2	Examine various problem domains (literature review), define the problems, and formulate the objectives for the capstone project	KP8	PLO4	CEP1, CEP2, CEP6, CEP7	-
CLO3	Analyze various aspects of the objectives for designing a solution of the capstone project	KP1, KP2, KP3, KP4	PLO2	CEP1, CEP2, CEP3, CEP6, CEP7	-
CLO4	Design and develop solutions for the capstone project that meet public health and safety, cultural, societal, and environmental considerations	KP5	PLO3	CEP1, CEP2, CEP4, CEP5, CEP6, CEP7	-
CLO5	Identify and apply modern engineering and	KP6	PLO5	CEP1, CEP2, CEP4, EP5	-

	IT tools for the design and development of the capstone project				
CLO6	Assess and address societal, health, safety, legal, and cultural aspects related to the implementation of the capstone project considering the relevant professional and engineering practices and solutions	KP7	PLO6	CEP2, CEP5, CEP6	-
CLO7	Assess and address the sustainability and impact of the capstone project in societal and environmental contexts	KP7	PLO7	CEP5, CEP6	-
CLO8	Apply professional and engineering ethical principles and practices for the implementation of the capstone project	KP7	PLO8	-	-
CLO9	Write effective reports and design documentation, and make effective presentations of the outcome of the capstone project	-	PLO10	-	CEA1, CEA2, CEA3, CEA4, CEA5
CLO10	Conduct economic analysis and cost estimation; and apply appropriate project management processes in the development life cycle of the capstone project	-	PLO11	-	-
CLO11	Prepare to take part in independent and life-long learning for adapting emerging technologies for the solution of the complex computer science and engineering problems	-	PLO12	-	-

Mark Distribution and Evaluation Criteria:

CLO	CLO Descriptions	PLO Descriptions	Learning Domains	Assessment Weight	Learning Subdomains	Rubrics Design	Points
CLO1	Integrate new and previously acquired knowledge for identifying a real-life complex engineering problem as the capstone project	PLO1: Engineering Knowledge	Cognitive	5%	C2	Able to understand the need of integrating new and previously acquired knowledge for identifying a real-life complex engineering problem as the project	2%
					C3	Able to apply new and previously acquired knowledge for identifying a real-life complex engineering problem	3%
CLO2	Examine various problem domains (literature review),	PLO4: Investigation	Cognitive , Psychomotor	7% 8%	C4	Able to characterize various problem domain	2%

	define the problems, and formulate the objectives for the capstone project.				C5	Able to evaluate the problem based on criteria and standards through checking and critiquing.	2%
					C6	Able to design a problem by putting elements together in a new way	3%
					P2	Able to recreate specific problem from instruction or memory	2%
					P3	Able to build a new problem and formulate the objectives for the capstone project	4%
					P4	Able to formulate a non-	2%

						standard objective for the project	
CLO3	Analyze various aspects of the objectives for designing a solution of the capstone project	PLO2: Problem Analysis	Cognitive	10%	C2	Able to understand the need of analyzing various aspects of the objectives	2%
					C3	Able to apply various problem analyzing techniques	2%
					C4	Able to analyze various aspects of the objectives	3%
					C5	Able to evaluate the objectives for designing a solution	3%
CLO4	Design and develop solutions for the capstone project that meet public health and safety, cultural, societal, and	PLO3: Design/ Development of Solutions	Cognitive , Affective	5% 5%	C6	Able to design and develop a solution that meet public health	5%

	environmental considerations					and safety, cultural, societal, and environmental considerations	
					A4	Able to relate public health, safety, cultural, societal, and environmental aspects to resolve conflicting issues while designing and developing the solution	5%
CLO5	Identify and apply modern engineering and IT tools for the design and development of the capstone project	PLO5: Modern Tool Usage	Cognitive , Psychomotor	2% 8%	C2	Able to understand the need of identifying and applying modern engineering and IT tools	2%
					P2	Able to recreate specific solution by identifying and applying	3%

						modern engineering and IT tools	
					P3	Able to build solution using modern tools	5%
CLO6	Assess and address societal, health, safety, legal, and cultural aspects related to the implementation of the capstone project considering the relevant professional and engineering practices and solutions	PLO6: The Engineer and Society	Affective Cognitive	2% 3%	A4	Able to relate societal, health, safety, legal, and cultural impact of the developed solution considering the relevant professional and engineering practices and solutions	2%
					C3	Able to apply societal, health, safety, legal, and cultural aspects considering the relevant professional and engineering	3%

						practice s and solution s	
CLO7	Assess and address the sustainability and impact of the capstone project in societal and environmental contexts	PLO7: Environment and Sustainability	Affective , Cognitive	3% 2%	A4	Able to relate the sustainability and impact of the developed solutions in societal and environmental contexts	3%
					C5	Able to evaluate the sustainability and impact of the developed solutions in societal and environmental contexts	2%
CLO8	Apply professional and engineering ethical principles and practices for the implementation of the	PLO8: Ethics	Affective , Cognitive	7% 3%	A4	Able to recognize the effects on the economy, environment, and	3%

	capstone project					socio-culture in professional practice.	
					A5	Able to practice ethically , apart from being responsible towards society.	2%
					A5	Able to behave according to the Students ' Code of Conduct .	2%
					C5	Able to analyze and make decisions in solving problems related to ethics.	3%
CLO9	Write effective reports and design documentation, and make effective presentations of the outcome of the capstone project	PLO10: Communication	Psychomotor, Affective	8% 12%	P3	Able to appear in terms of physical self-appearance (personality, dressing , and look),	2%

						body posture (eyes, hands, arms placed in the correct positions) suiting presentation session situation .	
					P3	Able to produce presentation materials or slides and visual aids in a logical, attractive, clear, systematic, and interesting sequence with proper introduction, discussions, and conclusions.	2%
					P3	Able to present in a convincing manner (speak fluently, clear	2%

						voice, correct intonation, and pronunciation) with good style.	
					P4	Able to defend questions asked (adapting answers to new questions arising)	2%
					A3	Able to adapt well to suit the type of writing in terms of content, style of presentation, structure, and flow, etc.	3%
					A4	Able to organize writing in a certain required style of writing to suit given tasks.	2%

					A4	Able to organize writing in a very structured, proper flow, and well-written manner.	2%
					A4	Able to organize writing with very minimum errors in terms of language including choice of words, grammar, punctuation, spelling, paragraphing.	2%
					A4	Able to organize relevant and presentable photos, graphs, illustrations together with descriptions.	1%

					A4	Able to organize a very neat and tidy report.	2%
CLO10	Conduct economic analysis and cost estimation; and apply appropriate project management processes in the development life cycle of the capstone project	PLO11: Project Management and Finance	Cognitive , Psychomotor	3% 2%	C2	Able to understand economic concepts in an engineering context.	1%
					C3	Able to understand project management life cycle and identify its limitations.	2%
					P4	Able to effectively apply business tools and financial principles in managing engineering projects.	2%
CLO11	Prepare to take part in independent and life-long learning for adapting	PLO12: Life-Long Learning	Affective , Psychomotor	1% 4%	A3	Able to accept new ideas and capable	1%

	Using technologies for the solution of the complex computer science and engineering problems					of autonomous learning	
					P4	Able to develop a curious mind and a thirst for knowledge to adapt to new learning strategies.	2%
					P5	Able to search and manage relevant information from different sources.	2%
Total Points							100%

Phase and appraisal entity wise CLO:

Phase	CLO-PLO Matrix		PLO Descriptions	Learning Domains	Assessment Weight	Appraisal Entity	Assessment Weight for Appraisal Entity
Phase 1	CLO1	PLO1	Engineering Knowledge	Cognitive	5%	Supervisor,	60%
						External,	20%
						Internal	20%
	CLO2	PLO4	Investigation	Cognitive, Psychomotor	7% 8%	Supervisor, External, Internal	60% 20% 20%

	CLO3	PLO2	Problem Analysis	Cognitive	10%	Supervisor, External, Internal	60% 20% 20%
	CLO4	PLO3	Design/ Development of Solutions	Cognitive, Affective	5% 5%	Supervisor, External, Internal	60% 20% 20%
Phase 2	CLO5	PLO5	Modern Tool Usage	Cognitive, Psychomot or	2% 8%	Supervisor, External, Internal	40% 30% 30%
	CLO6	PLO6	The Engineer and Society	Affective, Cognitive	2% 3%	Supervisor, External	40% 60%
	CLO7	PLO7	Environment and Sustainability	Affective, Cognitive	3% 2%	Supervisor, External	60% 40%
	CLO8	PLO8	Ethics	Affective, Cognitive	7% 3%	Supervisor, External, Internal	40% 30% 30%
	CLO9	PLO1 0	Communicati on	Psychomoto r, Affective	8% 12%	Supervisor, External, Internal	40% 30% 30%
	CLO1 0	PLO1 1	Project Management and Finance	Cognitive, Psychomot or	3% 2%	Supervisor, External	60% 40%
	CLO1 1	PLO1 2	Life-Long Learning	Affective, Psychomot or	1% 4%	Supervisor, External	40% 60%

Activity Plan:

Phase	Assessment Weight	Duration	Teaching Learning Methodology
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Phase 1	40%	1 Semester	The student will prepare a report targeting CLO1, CLO2, CLO3 and CLO4. Grading will be done based on this report and the oral presentation of the student.
Phase 2	60%	1 Semester	The student will prepare a report targeting CLO5, CLO6, CLO7, CLO8, CLO9, CLO10 and CLO11. Grading will be done based on this report and the oral presentation of the student.

Definition of the Appraisal Entities:

Appraisal Entity	Assessment Weight	Meaning
Supervisor	50%	The supervisor of a final year project is an experienced faculty member or professional overseeing and guiding students' comprehensive academic endeavors. They provide direction, mentorship, and technical advice, ensuring project adherence to academic guidelines, fostering skill development, and facilitating successful project completion within the stipulated timeframe.
Internal	20%	An internal in a final year project typically refers to a faculty member or an academic professional affiliated with Daffodil International University. They offer guidance, assessment, and academic expertise throughout the project's execution. Their role involves evaluating project progress, providing feedback, and ensuring adherence to academic guidelines for successful completion.
External	30%	An external member in a final year project typically refers to an individual external to Daffodil International University. They contribute as an expert evaluator, offering impartial assessment, industry insights, or specialized knowledge, ensuring the project's relevance, quality, and real-world applicability, enriching the assessment process for comprehensive evaluation and practicality.

Appendix

Rubrics of Complex Engineering Problems:

Level	Attribute	Characteristics of Complex Engineering Problems	Rubrics Design	1 (Very poor)	2 (Poor)	3 (Satisfactory)	4 (Good)	5 (Excellent)
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CEP1	Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6, or K8 which allows a fundamental-based, first principles analytical approach	Analyze the problem using specified knowledge profile Evaluate the problem under such circumstance towards providing an effective solution	Use 2 levels of knowledge profile but do not elaborate Evaluate 1 circumstance only	Use 2 levels of knowledge profile with acceptable elaboration Evaluate 2 circumstances with acceptable justification	3	4	>4
CEP2	Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering, and other issues	Compare the conflicting technical, engineering and other issues arising to solve the problem Assess the conflicting	Compare only 1 issue Assess but no	Compare 2 issues with acceptable discussion Assess with 1	Compare 2 issues with acceptable discussion	3	>3

			requirements and provide a satisfactory proposal towards solving the problem	proposal	proposal	Assess with 2 proposals		
CEP3	Depth of analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models	Develop the formulae to solve the problem using suitable models Justify creativity towards the achievement of the formula	Conceptualize 1 formula used Justify 1 creative development	Conceptualize 1 formula used but do not elaborate the model Justify 1 creative development used but do not elaborate the mode	Develop 1 formula used and elaborate the model Justify 1 creative development used and elaborate the model	2	3

CEP4	Familiarity of issues	Involve infrequently encountered issues	Differentiate the infrequently encountered issues in problem-solving	Compare the basis	Compare and differentiate 2 issues	Differentiate 2 issues and propose	3	>3
			Select formulae to resolve the infrequently encountered issues	Select an approach to resolve	Select 2 approaches to resolve	Select 2 approaches to resolve and justify		
CEP5	Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering	Develop solutions using standards and codes of practice for professional engineering	Use at least 1	Use at least 2	Use at least 2 and include practicing guide	3	>3

			Justify professional engineering experiences to resolve the problem solving	Justify using at least 1 experience	Justify using at least 2 experiences	Justify using 2 experiences and select at least 1		
CEP6	Extent of stakeholder involvement and conflicting requirements	Involve diverse groups of stakeholders with widely varying needs	Differentiate the diverse groups of stakeholders with widely varying needs Select stakeholder interests and requirements that give impact on the problem	Compare the basis Select a stakeholder and discuss impact	Compare and differentiate 2 groups Select 2 stakeholders and compare impacts	Differentiate 2 groups and propose 1 solution Select 2 stakeholders and justify impacts	3	>3
CEP7	Interdependence	Are high-level problems including many component	Analyze high-level problems	Use 2 sub problems but do not	Use 2 sub problems with acceptable	Use 2 sub problems		

		parts or sub-problems	including many component parts or sub-problems	elaborate	elaboration	and differentiate	3	>3
			Proposed problems broken down into smaller components or sub-problems	Propose 1 component only	Propose 2 components	Propose 2 components with acceptable justification		

*****Rubrics of Cognitive Domain:**

Level	Category	Rubrics Design	20 % (Very poor)	40% (Poor)	60% (Satisfactory)	80% (Good)	100% (Excellent)
C1	Remembering	Recognizing or recalling knowledge from memory. Remembering is when memory is used to produce or retrieve definitions, facts, or lists, or to recite previously	Identifying information when prompted or shown cues, often without full recall.	A level between recognition and full recall where there's a sense of familiarity but not complete retrieval of information.	Ability to remember details or facts with some effort or cues, but not entirely spontaneously.	A high level of ability to retrieve information from memory with ease, without many prompts or cues.	Instantly and effortlessly remembering information, including details, definitions, and facts, without any

		learned information.					prompts or cues.
C2	Understanding	Constructing meaning from different types of functions be they written or graphic messages or activities like interpreting, exemplifying, classifying, summarizing, inferring, comparing, or explaining.	Difficulty interpreting or grasping the meaning of information presented, struggling with basic comprehension or interpretation.	Grasping some aspects of the information but lacking a comprehensive or full understanding of its complexities or nuances.	A moderate level of comprehension, capable of interpreting, summarizing, and explaining information to a certain extent, but with some gaps or inconsistencies.	Proficiently grasping the content or concept, capable of effectively summarizing, inferring, comparing, and explaining it with clarity and coherence.	Mastery of the information, demonstrating a thorough grasp of its intricacies, capable of interpreting, exemplifying, classifying, summarizing, inferring, comparing, and explaining it effortlessly and accurately across various contexts or formats.
C3	Applying	Carrying out or using a procedure through executing, or implementing. Applying relates to or refers to situations where learned material	Difficulty in executing or implementing learned material in practical situations, struggling to apply theoretical	Capable of applying learned material to simple or familiar situations, but with limited adaptability or	Demonstrating the ability to apply learned concepts effectively in various contexts, showing adaptability	Skillfully applying learned material to diverse situations with flexibility, creativity	Mastery in applying learned concepts across complex or novel scenarios, exhibiting

		is used through products like models, presentations, interviews or simulations.	knowledge to real-world scenarios.	creativity in its application .	and some level of creativity in implementation.	, and a high degree of accuracy, showcasing a strong understanding of how to use this knowledge effectively.	a high level of innovation , adaptability, and originality in execution, often setting precedents or creating new approaches.
C4	Analyzing	Breaking materials or concepts into parts, determining how the parts relate to one another or how they interrelate, or how the parts relate to an overall structure or purpose. Mental actions included in this function are differentiating, organizing, and attributing, as well as being able to distinguish between the components or parts. When one is analyzing, he/she can illustrate this mental function by creating spreadsheets, surveys, charts, or diagrams, or graphic representations.	Difficulty in breaking down information into parts or recognizing relationships between components , struggling to differentiate and organize information effectively.	Capable of identifying some components or relationships within information but with limited depth or breadth of understanding, often missing subtleties or complexities.	Competent in breaking down information into its constituent parts, discerning relationships, and organizing them into a coherent structure, albeit with occasional oversights or minor inconsistencies.	Proficiently dissecting information into its elements, adeptly recognizing and explaining intricate relationships between components, displaying a high level of accuracy and depth in attributing and organizing.	Mastery in deconstructing complex information, effortlessly identifying, differentiating, attributing , and illustrating interrelationships between components, and showcasing a comprehensive and nuanced understanding across various representations or formats.

C5	Evaluating	Making judgments based on criteria and standards through checking and critiquing. Critiques, recommendations, and reports are some of the products that can be created to demonstrate the processes of evaluation.	Struggling to form judgments or critiques based on criteria, often unable to effectively assess or critique information against standards.	Capable of making some judgments or critiques based on basic criteria, but with limited depth or consistency in assessments.	Able to make sound judgments and critiques by applying relevant criteria and standards, offering reasonable assessments, but occasionally lacking in thoroughness or scope.	Skillfully assessing and critiquing information against established criteria, providing comprehensive and insightful evaluations consistently.	Mastery in evaluation, adept at critically analyzing information against multiple criteria, offering nuanced, sophisticated, and comprehensive critiques or recommendations that demonstrate a deep understanding of the subject matter.
C6	Creating	Putting elements together to form a coherent or functional whole; reorganizing elements into a new pattern or structure through generating, planning, or producing. Creating requires users to put parts together in a new way, or synthesize parts into something new and different creating a new form or product.	Difficulty in reorganizing or synthesizing elements into a new structure, struggling to generate novel ideas or products.	Capable of combining elements to form something new, yet often relying heavily on existing patterns or structures without significant innovation or originality.	Able to generate new ideas or products by reorganizing elements into a coherent whole, demonstrating some level of creativity and innovation, albeit with occasional reliance on established frameworks.	Skillfully synthesizing elements into novel and functional wholes, displaying a high level of creativity, originality, and adaptability in generating new forms or products.	Mastery in creating, consistently producing highly innovative, groundbreaking, and original ideas, products, or solutions, showcasing unparalleled creativity and the ability to redefine established

		This process is the most difficult mental function.					paradigms or create entirely new ones.
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*****Rubrics of Psychomotor Domain (Dave's Taxonomy 1975):**

Level	Category	Rubrics Design	20 % (Very poor)	40% (Poor)	60% (Satisfactory)	80% (Good)	100% (Excellent)
P1	Imitation	Copy action of another; observe and replicate.	Difficulty in replicating actions or behaviors observed from others, struggling to accurately mimic or copy the desired action.	Capable of replicating simple actions or behaviors observed from others with moderate accuracy, but with limitations in replicating complex or nuanced actions.	Competent in imitating a range of actions or behaviors observed from others, demonstrating an ability to replicate various actions with reasonable accuracy.	Skillful in replicating actions or behaviors observed from others with high accuracy and precision, displaying the ability to imitate complex or intricate actions effectively.	Mastery in imitation, effortlessly replicating a wide range of complex or nuanced actions or behaviors observed from others with exceptional accuracy and attention to detail.
P2	Manipulation	Reproduce activity from instruction or memory	Difficulty in reproducing activities based on instructions or memory, struggling to execute actions accurately.	Capable of reproducing simple activities from instructions or memory with moderate accuracy,	Competent in reproducing a variety of activities from instructions or memory, demonstrating the ability to	Skillful in reproducing activities with a high degree of accuracy and precision	Mastery in manipulation, effortlessly reproducing a wide range of complex activities from

				but facing challenges with more complex tasks.	execute tasks accurately in various contexts.	, displaying proficiency in executing complex tasks based on instructions or memory.	instructions or memory with exceptional accuracy and attention to detail.
P3	Precision	Execute skills reliably; independent of help.	Difficulty in executing skills reliably without assistance, often needing guidance or support to complete tasks accurately.	Capable of executing some skills reliably with minor errors, but occasional guidance or assistance might still be required for complex tasks.	Competent in executing a range of skills reliably and independently, demonstrating consistent accuracy in most tasks.	Skillful in executing skills consistently and reliably without assistance, displaying a high level of accuracy and precision across various tasks.	Mastery in precision, effortlessly executing skills with unparalleled accuracy and reliability, demonstrating an exceptional level of expertise and independence in completing tasks.
P4	Articulation	Adapt and integrate expertise to satisfy a non-standard objective.	Difficulty in adapting expertise to meet non-standard objectives, often struggling to integrate knowledge effectively for unconventional goals.	Capable of adapting expertise to some extent for non-standard objectives, but with limitations in integrating knowledge effectively for unique goals.	Competent in adapting and integrating expertise to meet various non-standard objectives, demonstrating the ability to apply knowledge creatively	Skillful in adapting expertise seamlessly to satisfy non-standard objectives, displaying a high level of adaptability and	Mastery in articulation, effortlessly adapting and integrating expertise to meet diverse and complex non-standard objectives, showcasing

					for unconventional goals.	integration of knowledge for unique goals.	g unparalleled creativity and adaptability in achieving unconventional goals.
P5	Naturalization	Automated, unconscious mastery of activity and related skills at strategic level.	Just beginning to engage with an activity or skill, showing little to no automation or unconscious mastery, relying heavily on conscious effort and guidance.	Developing some level of automation and unconscious mastery, but still requiring conscious effort and attention to perform the activity or skill adequately.	Achieving a level of automation and unconscious mastery at a moderate level, where the activity or skill becomes somewhat automatic but may still require periodic conscious thought or attention.	Attaining a high degree of automation and unconscious mastery at a strategic level, performing the activity or skill almost instinctively with minimal conscious effort or thought.	Mastery level where the activity or skill is fully naturalized and performed entirely automatically and unconsciously at a strategic level, requiring almost no conscious effort or attention to execute flawlessly.

*****Rubrics of Affective Domain (Krathwohl, Bloom, Masia's Taxonomy 1973):**

Level	Category	Rubrics Design	20 % (Very poor)	40% (Poor)	60% (Satisfactory)	80% (Good)	100% (Excellent)
A1	Receiving	Willingness to participate in an activity to attend to a stimulus; getting and	Demonstrating little willingness to participate or attend to	Showing some willingness to participate and attend	Displaying a good level of willingness to participate	Actively participating and readily holding attention	Exceptional willingness to participate and

		holding the attention of students.	stimuli, struggling to maintain attention or engagement in activities.	to stimuli, but with occasional distractions or difficulties in maintaining sustained attention.	and hold attention to stimuli, with occasional minor distractions that can be managed.	to stimuli, demonstrating consistent engagement with occasional minor lapses in focus.	consistently hold attention to stimuli, demonstrating unwavering engagement and focus without significant distractions.
A2	Responding	Actively participates; demonstrates interest in an object, activity or phenomena; seeks or pursues this object, activity or phenomena.	Demonstrating minimal active participation or interest in objects, activities, or phenomena, often showing reluctance or hesitance to engage.	Showing some level of active participation and interest in certain objects, activities, or phenomena, but with occasional disinterest or passivity.	Actively participating and demonstrating interest in a variety of objects, activities, or phenomena, with occasional fluctuations in intensity or focus.	Consistently demonstrating keen interest and active participation in objects, activities, or phenomena, displaying sustained engagement with minimal periods of disinterest.	Exceptionally active and enthusiastic participation, consistently demonstrating a high level of interest, curiosity, and pursuit of objects, activities, or phenomena without any noticeable disengagement.
A3	Valuing	Value or worth attached to an object, activity or phenomena; varies from simple acceptance to commitment.	Demonstrating little to no attachment or appreciation for objects, activities, or phenomena, exhibiting	Showing some level of acknowledgment or acceptance of certain objects, activities, or	Displaying a moderate level of value or worth attached to various objects, activities, or	Exhibiting a strong attachment and appreciation for specific objects, activities	Exceptional attachment, appreciation, and commitment to objects, activities,

			indifference or disinterest.	phenomena, but with limited attachment or commitment.	phenomena, expressing a noticeable but moderate commitment or interest.	, or phenomena, demonstrating a high level of commitment and investment.	or phenomena, showcasing an unwavering dedication and profound value attached to them.
A4	Organizing	Compare and contrast, and resolve conflict to build a consistent value system; emphasis on comparing and synthesizing values.	Struggling to compare, contrast, or resolve conflicts to establish a consistent value system, showing difficulty in synthesizing values.	Capable of some comparison and contrast between values but with limited ability to resolve conflicts or create a coherent value system.	Demonstrating moderate skills in comparing and synthesizing values, capable of resolving conflicts to some extent but with occasional inconsistencies.	Skillfully comparing, contrasting, and synthesizing values to build a consistent value system, effectively resolving conflicts and inconsistencies.	Mastery in organizing values, effortlessly comparing, contrasting, and synthesizing diverse values to establish a highly coherent and consistent value system, showcasing exceptional skill in resolving conflicts to build a unified framework.
A5	Internalizing	Adopt a value system for a length of time that contributes to a particular "lifestyle" (i.e. directs behavior).	Struggling to adopt a value system that significantly influences behavior or contributes	Beginning to adopt certain values that occasionally influence behavior, but with	Demonstrating moderate adoption of a value system that contributes to behavior	Skillfully internalizing a value system that consistently	Mastery in internalization, effortlessly adopting and embodying a value

			to a consistent lifestyle.	inconsistencies in maintaining them as part of a lifestyle.	and aspects of a lifestyle, with occasional deviations.	directs behavior and contributes substantially to a specific lifestyle, with minimal deviations.	system that profoundly influences behavior, deeply embedded as an integral part of a consistent and unwavering lifestyle.
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Addressing of CLOs, Knowledge Profile (KP), and Complex Engineering Problem (CEP):

CLO	CLO Descriptions	KP	CEP
CLO1	Integrate new and previously acquired knowledge for identifying a real-life complex engineering problem as the capstone project	Results and Discussion [KP7] Write here how you have addressed KP7	Results and Discussion [CEP5, CEP6] Write here how you have addressed CEP2, CEP5, CEP6
CLO2	Examine various problem domains (literature review), define the problems, and formulate the objectives for the capstone project		
CLO3	Analyze various aspects of the objectives for designing a solution of the capstone project		
CLO4	Design and develop solutions for the capstone project that meet public health and safety, cultural, societal, and		

	environmental considerations		
CLO5	Identify and apply modern engineering and IT tools for the design and development of the capstone project		
CLO6	Assess and address societal, health, safety, legal, and cultural aspects related to the implementation of the capstone project considering the relevant professional and engineering practices and solutions		
CLO7	Assess and address the sustainability and impact of the capstone project in societal and environmental contexts		
CLO8	Apply professional and engineering ethical principles and practices for the implementation of the capstone project		
CLO9	Write effective reports and design documentation, and make effective presentations of the outcome of the capstone project		
CLO10	Conduct economic analysis and cost estimation; and apply appropriate project management processes in the development life cycle of the capstone project		

CLO1 1	Prepare to take part in independent and life-long learning for adapting emerging technologies for the solution of the complex computer science and engineering problems		
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Descriptions of Learning Domains, Knowledge Profiles, and Attributes of Complex Engineering Problem:

Cognitive Domain (Anderson and Krathwohl's Taxonomy 2001):

Level	Category	Meaning	Common Keywords
C1	Remembering	Recognizing or recalling knowledge from memory. Remembering is when memory is used to produce or retrieve definitions, facts, or lists, or to recite previously learned information.	Define, describe, draw, find, identify, label, list, match, name, quote, recall, recite, tell, write
C2	Understanding	Constructing meaning from different types of functions be they written or graphic messages or activities like interpreting, exemplifying, classifying, summarizing, inferring, comparing, or explaining.	Classify, compare, exemplify, conclude, demonstrate, discuss, explain, identify, illustrate, interpret, paraphrase, predict, report
C3	Applying	Carrying out or using a procedure through executing, or implementing. Applying relates to or refers to situations where learned material is used through products like models, presentations, interviews or simulations.	Apply, change, choose, compute, dramatize, implement, interview, prepare, produce, role play, select, show, transfer, use
C4	Analyzing	Breaking materials or concepts into parts, determining how the parts relate to one another or how they interrelate, or how the parts relate to an overall structure or purpose. Mental actions included in this function are differentiating, organizing, and attributing, as well as being able to distinguish between the components or parts. When one is analyzing, he/she can illustrate this mental function by creating spreadsheets, surveys, charts,	Analyze, characterize, classify, compare, contrast, debate, deconstruct, deduce, differentiate, discriminate, distinguish, examine, organize, outline, relate, research, separate, structure

		or diagrams, or graphic representations.	
C5	Evaluating	Making judgments based on criteria and standards through checking and critiquing. Critiques, recommendations, and reports are some of the products that can be created to demonstrate the processes of evaluation.	Appraise, argue, assess, choose, conclude, critique, decide, evaluate, judge, justify, predict, prioritize, prove, rank, rate, select, monitor
C6	Creating	Putting elements together to form a coherent or functional whole; reorganizing elements into a new pattern or structure through generating, planning, or producing. Creating requires users to put parts together in a new way, or synthesize parts into something new and different creating a new form or product. This process is the most difficult mental function.	Construct, design, develop, generate, hypothesize, invent, plan, produce, compose, create, make, perform, plan, produce

Psychomotor Domain (Dave's Taxonomy 1975):

Level	Category	Meaning	Common Keywords
P1	Imitation	Copy action of another; observe and replicate.	Relate, Repeat, Choose, Copy, Follow, Show, Identify, Isolate.
P2	Manipulation	Reproduce activity from instruction or memory	Copy, response, trace, Show, Start, Perform, Execute, Recreate.
P3	Precision	Execute skills reliably; independent of help.	Assemble, Implement, Organize, Calibrate, Demonstrate, Build, Perfect, Control, Complete, Measure.
P4	Articulation	Adapt and integrate expertise to satisfy a non-standard objective.	Modify, Master, Develop, Adapt, Formulate, Coordinate, Combine, Solve, Integrate.
P5	Naturalization	Automated, unconscious mastery of activity and related skills at strategic level.	Design, Rank, Manage, Compose, Develop, Specify, Construct, Invent.

Affective Domain (Krathwohl, Bloom, Masia's Taxonomy 1973):

Level	Category	Meaning	Common Keywords
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A1	Receiving	Willingness to participate in an activity to attend to a stimulus; getting and holding the attention of students.	Locate, Give, Point to, Follow, Sit erect, Hold, Name, reply, Identify, Choose
A2	Responding	Actively participates; demonstrates interest in an object, activity or phenomena; seeks or pursues this object, activity or phenomena.	Label, Answer, Perform, Write, Conform, Assist, Recite, Report, Read, Greet, Help, Present, Compile.
A3	Valuing	Value or worth attached to an object, activity or phenomena; varies from simple acceptance to commitment.	Work, Form, Follow, Join, Invite, Justify, Study, Explain, Share, Propose, Select, Complete, Describe, read, report, Differentiate, Initiate.
A4	Organizing	Compare and contrast, and resolve conflict to build a consistent value system; emphasis on comparing and synthesizing values.	Relate, Synthesize, Identify, Prepare, Defend, Generalize, Modify, Integrate, Order, Compare, Complete, Organize, Adhere, Arrange, Combine, Explain, Alter.
A5	Internalizing	Adopt a value system for a length of time that contributes to a particular “lifestyle” (i.e. directs behavior).	Influence, Propose, Use, Quality, Revise, Serve, Solve, Modify, Display, Practice, Listen, Question, Perform, Act, Discriminate, Verify

Knowledge Profile

Knowledge Profile	Attribute
K1: Theory-based natural sciences	A systematic, theory-based understanding of the natural sciences applicable to the discipline
K2: Conceptually-based mathematics, numerical analysis, statistics, and formal aspects of computer and information science	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline
K3: Theory-based engineering fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
K4: Forefront engineering specialist knowledge for practice	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
K5: Engineering design	Knowledge that supports engineering design in a practice area

K6: Engineering practice (technology)	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
K7: Comprehension of engineering in society	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability
K8: Research literature	Engagement with selected knowledge in the research literature of the discipline

Range of Complex Engineering Problem Solving

Attribute	Characteristics of Complex Engineering Problems
EP1: Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6, or K8 which allows a fundamental-based, first principles analytical approach
EP2: Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering and other issues
EP3: Depth of analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models
EP4: Familiarity of issues	Involve infrequently encountered issues
EP5: Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering
EP 6: Extent of stakeholder involvement and conflicting requirements	Involve diverse groups of stakeholders with widely varying needs
EP7: Interdependence	Are high level problems including many component parts or sub-problems

Range of Complex Engineering Activities

Attribute	Complex Problems
EA1: Range of resources	Involve the use of diverse resources (for this purpose, resources include people, money, equipment, materials, information and technologies)
EA2: Level of interaction	Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues

EA3: Innovation	Involve creative use of Computer Science and Engineering principles and research-based knowledge in novel ways
EA4: Consequences to society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
EA5: Familiarity	Outside problems encompassed by standards and codes of practice for professional Computer Science and Engineering

PART D: GRADING/EVALUATION

1. Grading Scale, Grades

Marks	Grade	Grade Point	Remark
80-100%	A +	4	Outstanding
75-79%	A	3.75	Excellent
70-74%	A-	3.5	Very Good
65-69%	B+	3.25	Good
60-64%	B	3	Satisfactory
55-59%	B-	2.75	Above Average
50-54%	C+	2.5	Average
45-49%	C	2.25	Below Average
40-44%	D	2	Pass
00-39%	F	0	Fail

2. Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA)

Each student's Grade Point Average (GPA) is the numerical value obtained by dividing the total grade points earned by the credits attempted in a semester. Cumulative Grade Point Average (CGPA) is calculated by multiplying the grade points obtained by a student with the respective course credits, summing over all such products and finally dividing by the total number of credits obtained by the student.

Only the courses graded A+, A, A-, B+, B, B-, C+, C, D and F are used to determine credits attempted. In case students repeat courses, GPA and CGPA will be calculated on the basis of the grades obtained at the last attempt of the course(s) only. Grades obtained in course(s) in all examinations will be shown in the grade report. In addition, students who complete courses in addition to their normal credit requirements for graduation will inform the Registrar in writing about the course, which he intends to declare for consideration towards the requirements for the degree.

3. Course Withdrawal

No additional withdrawal methods are available. All rules are under course drop rules.

4. Incomplete (I) courses

The 'Incomplete' (I) grade is used in special circumstances. The 'Incomplete' may be given only at the end of a semester to a student who has completed all other requirements except the final examination.

The student has the responsibility to take the initiative in making up the Incomplete as specified by the instructor. If action is not taken within three weeks of commencement of the next semester, the 'I' grade will automatically be converted into "F". In the event, where the instructor from whom a student received an incomplete grade is not available, the disposition of the case-involving grade rests with the Dean of Faculties.

5. Retake

A course passed with a grade less than B may be repeated to improve the relevant grade. When a course is repeated, the previous grade will be automatically cancelled. Students, failing in a course, will be allowed to repeat that course twice at the most on payment of course fee.

6. Grade Improvement

No additional grade improvement methods are available. All rules are under retake policy.

7. Course Dropout

A student may drop or add new courses during the first week of the semester. To add a course, one must have the approval of the faculty adviser. The procedure is given below:

1. Pick up a Drop/Add form from the Office of the Registrar
2. Fill up the relevant section of the form
3. Obtain the signature of the course teacher/ advisor on the form and if a course is being added; make sure that space is available in the course.
4. Submit the form to the office of the Registrar.
5. Check at the Accounts office about any change of fee.

8. Common thing of course outline for theory courses

Instructor Details

Name:	
Employee ID:	
Designation:	
Department:	
Office Address:	
Telephone/Extension:	
Mobile:	
Website:	
GTA/UTA(If Any):	

Class Schedule with Counseling Hour

Time/ Date	8:30 AM- 10:00 AM	10:00 AM- 11:30 AM	11:30 AM- 01:00 PM	01:00 PM- 02:30 PM	2:30 PM- 4:00 PM	4:00 PM- 5:30 PM
Saturday						
Sunday						
Monday						
Tuesday						
Wednesday						
Thursday						

Evaluation Policy (Grading Policy)

Marks obtained out of 100	Grade	Grade point equivalent	Remarks
80% and above	A+	4.00	Outstanding
75% to less than 80%	A	3.75	Excellent

70% to less than 75%	A-	3.50	Very Good
65% to less than 70%	B+	3.25	Good
60% to less than 65%	B	3.00	Satisfactory
55% to less than 60%	B-	2.75	Above Average
50% to less than 55%	C+	2.50	Average
45% to less than 50%	C	2.25	Below Average
40% to less than 45%	D	2.00	Pass
Less than 40%	F	0.00	Fail

Academic Code of Conduct

Academic Integrity:

Academic offenses under the Academic Code of Conduct include plagiarism, personification, physical and online cheating, falsification of a document, and any other dishonest behavior related to gaining academic gain or avoiding evaluation exercises by a student. The university's Disciplinary Committee may decide to impose severe penalties for these offenses.

Special Instructions:

- Attendance at all classes and exams is required of the students. To take the final test, a student needs to have attended at least 70% of classes.
- After ten minutes of the scheduled start time, students will not be permitted to enter the classroom.
- Plagiarism will automatically result in a zero on that exam or assignment.
- There won't often be a make-up exam. However, if a student misses an exam due to a serious sickness, the death of a family member, an emergency involving the family, or humanitarian reasons, they **MUST** request permission to make up the exam in writing through the course instructor to the chairperson within 48 hours of the exam date. The application must be submitted with the appropriate supporting documentation for the reason(s) for the absence from the exam.
- There won't be a makeup exam for the final exam. However, if a student is unable to attend the final exam due to a serious illness, a family member's death, an emergency, or humanitarian reasons, they **MUST** request an incomplete grade in writing from the course instructor via the chairperson within 48 hours of the exam date. Along with the application, appropriate supporting documentation for the reason(s) for missing the final exam must be provided. It is the student's duty, in consultation with the course instructor, to schedule an incomplete exam by the deadline specified in the academic calendar.
- It is required that all cell phones be in silent mode during class and test times.
- Exam cheating is not tolerated at all. Examinees will be penalized for cheating if they are found in possession of cheat sheets, used or not; if they write on their palms, the backs of calculators, chairs, or adjacent walls; if they copy from cheat sheets or other sources; if they copy from other examiners, etc. Cheating only carries a single, multi-semester expulsion, as determined by the university's disciplinary committee

Appendix

Program Learning Outcomes (PLOs)

No.	Program Learning Outcomes
PLO1	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex software engineering problems.
PLO2	Identify, formulate, research and analyze complex software engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.
PLO3	Design solutions for complex software engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal and environmental concerns.
PLO4	Conduct investigations of complex problems, considering experimental design, data analysis and interpretation and information synthesis to provide valid conclusions.
PLO5	Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex software engineering activities with an understanding of their limitations.
PLO6	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PLO7	Understand the impact of professional software engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
PLO8	Apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PLO9	Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PLO10	Communicate effectively about complex software engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PLO11	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PLO12	Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

Learning Domain (Bloom's taxonomy)

- (a) **Cognitive Domain (Knowledge):** The cognitive domain aims to develop the mental skills and the acquisition of knowledge of the individual.

Domain Level	Definition
Remember (C1)	Retrieving, recalling, or recognizing information from memory.
Understand(C2)	Changing from one form of representation to another; illustrating a concept; drawing conclusions; determining cause and effect
Apply(C3)	Using learned materials, students can use/apply information in a new way.
Analyze(C4)	Breaking material or concepts into parts, determining how the parts related or interrelated to one another or to an overall structure or purpose.
Evaluate(C5)	Assessing, making judgments and drawing conclusions from ideas, information, or data.
Create(C6)	Developing a hypothesis; devising a procedure; inventing a product

- (b) **Affective Domain (Attitude):** The affective domain includes how learners deal with things emotionally, such as feelings, values, appreciation, enthusiasms, motivations, and attitudes.

Domain Level	Definition
Receiving (A1)	Being aware of or attending to something in the environment.
Responding(A2)	Showing some new behaviors as a result of experience.
Valuing(A3)	Showing some definite involvement or commitment.
Organization(A4)	Integrating a new value into one's general set of values, giving it some ranking among one's general priorities.
Characterization by value(A5)	Acting consistently with the new value.

(c) Psychomotor Domain (skill): Includes physical movement, coordination, and use of the motor-skill areas.

Domain Level	Definition
Imitating (P1)	Attempted copying of a physical behavior
Manipulation(P2)	Reproducing activity from instruction or memory
Precision(P3)	Fine tuning. Making minor adjustments in the physical activity in order to perfect it.
Articulation(P4)	Adapting and integrating expertise to satisfy a non- standard objective
Naturalization(P5)	Automated, unconscious mastery of activity and related skills at strategic level

BNQF Skills (4 year's Bachelors):

The Learning Outcome Domains are Fundamental Domain, Social Domain, Thinking Domain and Personal Domain. These domains are classified below.

Fundamental Skills:

- Demonstrate knowledge and critical understanding of the well-established principles of his/her field of study, and of the way in which those principles have developed.
- Apply underlying concepts and principles outside the context in which they were first studied, including, where appropriate, the application of those principles in an employment context.
- Apply knowledge and skills in addressing issues/solving problems with minimal supervision.
- Evaluate critically the appropriateness of different approaches to solving problems in his/her field of study.
- Support supervision of junior staff via a mentor or a leader/manager.
- Display advanced digital literacy which is adequate to perform complex tasks and bring about solutions.

Social Skills:

- Communicate and interact effectively and clearly, ideas, information, problems and solutions as a team to peers, experts and non-experts in Bangla and English.
- Express her/himself fluently and spontaneously in English and Bangla.
- Use language flexibly and effectively for social, academic and professional purposes.
- Produce clear, well structured, detailed text on complex subjects, showing controlled use of organizational patterns, connectors and cohesive devices in advanced proficiency level of Bangla and English.
- Demonstrate the ability to incorporate entrepreneurial skills in planning daily activities.
- Display advanced civic literacy and knowledge, exercising civic rights and obligations at all levels as well as participating in changes for the improvement of Bangladesh society.

Thinking Skills:

- Exercise a very substantial degree of autonomy and often significant responsibility in making
- Judgments/decisions towards the management of self, others and for the allocation of substantial resources.

- Demonstrate professional knowledge and practical skills in both technical and management to lead a team in an inexperienced environment.

Personal Skills:

- Engage in self-direction and self-enterprise skills.
- Demonstrate social, professional, environmental and ethical practice/ values.
- Show-case global knowledge and competencies to fulfil employment, entrepreneurial and lifelong learning skills; and contribute significantly to the society.

9. Common things at course outline for lab courses

Instructor Details

Name:	
Employee ID:	
Designation:	
Department:	
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Telephone/Extension:	
Mobile:	
Website:	
GTA/UTA(If Any):	

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